



Faculty of Science, Engineering and Built Environment

SIT292 Linear Algebra for Data Analysis

Deakin University Unit Guide

Trimester 2, 2026

CONTENTS

Welcome	2
Who is the unit team?	2
Unit chair: leads the teaching team and is responsible for overall delivery of this unit	2
Other members of the team and how to contact them	2
Help with administrative and academic queries	2
About this unit	2
Unit development in response to student feedback	2
Learning outcomes	3
Assessing your achievement of the unit learning outcomes	3
Hurdle requirements	3
Summative assessment task 1	4
Your learning experiences in this unit	5
Educator-facilitated (scheduled) learning activities - on-campus unit enrolment	5
Educator-facilitated (scheduled) learning activities - online unit enrolment	5
Typical study commitment	5
Learning activities	5
Unit learning resources and equipment	6
Bring Your Own Device (BYOD)	6
Essential learning resources	6
Recommended learning resources	6
Key dates for this study period	7
Unit weekly activities	7

Welcome

Linear algebra is the foundation for many sophisticated mathematical and computational methods. In SIT292 students will learn the basics of linear algebra, and solve systems of linear equations. This unit extends students ability apply mathematical formulae and to operate with complex mathematical objects. SIT292 introduces students to vector spaces, matrix theory, systems of linear equations and methods for solving them, eigenvalues and eigenvectors, and their application to similarity of diagonal matrices. The techniques that students learn from this unit will enable them with the analysis of complex data and ensuring its reliability in studies of cryptography.

This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Gleb Beliakov

Campus: Melbourne Burwood

Email: gleb.beliakov@deakin.edu.au

Phone: +61 3 925 17475

Other members of the team and how to contact them

Lecturers and sessional staff involved in teaching will provide you with information on how to best contact them during class. You can always reach out to the Unit Chair if you need clarification on any matters relating to the unit.

Help with administrative and academic queries

If you have any concerns about the unit during the trimester, help is available from a range of sources. It's best to seek support early to discuss your concerns as adjustments can be made if appropriate. Support is also available as follows:

- SEBE Student Hub section in your unit site – for faculty-specific support
- Unit Chair or Campus Leader – for academic matters such as content, assessments, and learning support
- [Student Central](#) – for administrative or enrolment-related queries (e.g., course advice, special consideration, or fee queries), or to make an appointment with a student adviser.

For additional support information, please see the 'Your Rights and Responsibilities' section under 'Content' in your unit site.

About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each unit. You are strongly encouraged to provide constructive feedback for this unit when eVALUate opens (you will be emailed a link).

In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- NUMBAS problems throughout the unit site for practice
- Clear expectations with the OnTrack model of assessment
- Well organised unit site.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- SIT292 was recently redeveloped to follow a similar structure (although with some differences) to SIT192. Based on feedback last year we have updated the requirements for module summaries, including mathematical definitions in a template so that students don't need to spend time writing these out.
- We have also adjusted some of the module NUMBAS assessments to better reflect the learning outcomes, and clarified the expectations for the written summaries and learning evidence.

Learning outcomes

Each unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Perform operations and transformations on matrices and vectors and apply them to solve systems of linear equations, calculate determinants, eigenvalues and eigenvectors.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Perform operations on linear vector spaces and apply them for solving approximation and regression problems.	GLO1: Discipline-specific knowledge and capabilities
ULO3	Demonstrate mathematical literacy by sourcing, interpreting and discussing mathematical texts and digital resources.	GLO3: Digital Literacy GLO4: Critical thinking

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must meet certain milestones as part of the Learning portfolio.

Brief summary of the unit milestones	Rationale
1. Pass tasks • By the end of week 4, students are required to have completed at least the week 1 and 2 Pass Tasks. • By the end of week 7 students are required to have completed at least the weeks 1 to 4 Pass Tasks.	The pass tasks in this unit provides students the opportunity to develop and demonstrate the achievement of Unit Learning Outcomes at the minimum expected standards. These tasks are included as hurdle requirements so that students are able to provide evidence of achievement of these ULOs through their portfolio.
• By the end of the unit, students are required to have submitted all pass tasks, and have sufficient Pass Tasks marked as completed to demonstrate achievement of the unit learning outcomes. Credit, Distinction, and High Distinction tasks are required for higher grades.	The portfolio artefact that students submit is used to measure their performance against the minimum standards as well as their ability to justify the outcomes that they have achieved through self-assessment and reflection. The hurdle requirement also provides a mechanism for student-staff interaction to check progress and address educational and motivational issues before it is too late in the trimester.

Summative assessment (tasks that will be graded or marked)

It is your responsibility to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read [Your rights and responsibilities as a student in this unit](#) to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST. Australian Eastern Standard Time (AEST) is observed from the first Sunday in April to the first Sunday in October, while Australian Eastern Daylight Time (AEDT) is observed from the first Sunday in October to the first Sunday in April, during daylight saving time in all states apart from Queensland, NT and WA, which remain as AEST.

Summative assessment task 1

Title	Learning portfolio
Brief description of assessment task	In this unit, assessment is designed to encourage and reward you for demonstrating achievement of the unit learning outcomes; with higher grades representing better achievement of these outcomes. Working regularly and completing the tasks on time will help you collect evidence for your portfolio. Your portfolio will consist of work that you complete in response to the unit's tasks. These tasks are designed to help you learn, and demonstrate achievement of the unit learning outcomes. Tasks will consist of the following kinds of activities: • Research learning material and summarise concepts from mathematical textbooks and online resources. • Review problem descriptions and design appropriate solving strategies. • Explain and categorize the relevant methodology to apply these strategies. • Provide solutions to routine and non-routine mathematical problems, with and without working. • Communicate your understanding. • Reflect on your progress.
Individual/Group	Individual
Detail of student output	You will work through a number of tasks throughout this unit and produce a range of artefacts including online quizzes, written and oral reports. This work will be combined together with your own critique and reflections on your learning performance into your learning portfolio for assessment.
Grading and weighting (% total mark for unit)	100% - marked and graded Each task in the unit is associated with a grade: either Pass, Credit, Distinction, or High Distinction. Complete all the Pass Tasks for a result between 50 and 59. Complete all the Pass and Credit tasks for a result between 60 and 69, complete all the Pass, Credit, and Distinction tasks for a result between 70 and 79, and complete all Pass, Credit, Distinction and High Distinction tasks for a result between 80 and 100. High Distinction tasks will be graded with the final portfolio, and a grade of 80 or above will only be achieved if they meet the HD level criteria. OnTrack lets you select a target grade, and will show you only the tasks you need to complete in order to achieve that grade.
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Perform operations and transformations on matrices and vectors and apply them to solve systems of linear equations, calculate determinants, eigenvalues and eigenvectors. ULO2 - Perform operations on linear vector spaces and apply them for solving approximation and regression problems. ULO3 - Demonstrate mathematical literacy by sourcing, interpreting and discussing mathematical texts and digital resources

This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - through applying knowledge of discrete mathematics to develop methodologies to solve given problems. GLO3 - through sourcing, interpreting and applying results from mathematical texts and digital resources. GLO4 - through analysing problems and evaluating and describing problem solving strategies.
How and when you will receive feedback on your work	You will be required to work on and submit tasks for formative feedback each week. The teaching team will then review your progress and provide you with individual feedback to assist you in completing the tasks and achieving your target grade for the unit.
When and how to submit your work	At the end of the unit you will combine together the artefacts you have created and a learning summary report into a single portfolio for assessment. This will be due by the end of Week 12, Sunday 4 October 2026, 8pm (AEDT) .

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 3 hour seminar per week, weekly meetings.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 1 x 2 hour online seminar per week, weekly meetings.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

For Campus students, this time includes 3 hours of weekly face-to-face seminars, weekly meetings, discussion boards, assessment tasks, readings and study time. For online students this time includes 2 hours of weekly online seminars, weekly meetings, online independent and collaborative learning including optional scheduled activities, discussion boards, assessment tasks, readings and study time.

This unit is divided into learning modules each targeting one or more learning outcomes. For each module you will engage with online learning activities, which will usually be based around chapters of the online textbook, with additional formative quizzes, short videos and learning activities. The material will be provided on the unit site, and any external activity will be linked from there. You will progress through a sequence of modules through the term to cover the unit's content. You are expected to progress at your own pace, under the supervision of the teaching team, and you will be allowed to repeat a module if necessary. However, you will need to demonstrate your progress at key points during the trimester, and you will be required to complete all the core modules in order to pass the unit.

Besides the core modules, there will be additional optional modules for students wishing to investigate certain topics further. Completion of the optional modules is not required for passing the unit, but you will need to demonstrate a deeper engagement with the content by completing optional modules in order to obtain higher grades.

Campus and online students will interact with the teaching team asynchronously through discussion forums and email, and will be provided with regular feedback on their progress, and additional resources based on your demand.

Learning activities

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In 'seminars', an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include 'practical experiences' such as 'laboratory', 'workshops', 'clinical skills' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

If you see a 'meeting' in your timetable, this is an optional drop-in session.

'Assessments' or 'team-based learning' indicate an evaluation of your skills or knowledge. A 'pre-assessment practice' could be scheduled to prepare you for these assessments. Visit the [About CloudDeakin – Assessment webpage](#), or the [Understanding Assessments webpage](#) for more information.

Take a look at the [Learning activities webpage](#) for a full list of learning activity types.

Generative AI

Deakin welcomes the opportunity to engage with emerging technologies in education and seeks to build your capability in the ethical and responsible use of current and emergent technology. Deakin also upholds a commitment to academic integrity and to ensuring high-quality educational outcomes that prepare you for an AI-driven future.

For more information on the acceptable use of GenAI, [visit Using GenAI at Deakin](#), read your assessment instructions and talk to your unit chair.

Unit learning resources and equipment

Your unit learning resources can be accessed from your unit site.

The texts and reading list for [SIT292](#) can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

The [Deakin Software Library](#) provides students with access to software that you may need or find useful for your study at Deakin.

Bring Your Own Device (BYOD)

To fully engage with Deakin's learning experiences, students must be able to access and use internet-connected devices as outlined in [computing requirements](#) at Deakin.

To support student success at Deakin, we have a bring-your-own-device (BYOD) learning environment that acknowledges that students and educators bring with them the digital tools they regularly use to complete academic tasks. These tools stay with you beyond the classroom, helping you to keep learning, explore ideas more deeply, and connect with knowledge in ways that matter to you.

Students requiring a loan device should visit our [Loan Laptop](#) webpage or students requiring longer-term assistance should visit our [Student Financial Assistance](#) webpage.

Essential learning resources

Prescribed text: Keith Nicholson (2021), Linear Algebra with Applications, Lyryx (freely available online).

Recommended learning resources

- H. Anton, Elementary Linear Algebra 12th edition (or any other edition)

There are many textbooks on Linear algebra in the University library, under the code 515.5

(<https://library.deakin.edu.au/search~S1/?searchtype=X&searcharg=515.5&searchscope=1&sortdropdown=-&SORT=D&exte>

[nded=0&SUBMIT=Search&searchlimits=&searchorigarg=.b1301281](#)). Most of them are suitable for this unit.

Key dates for this study period

You will find all important dates for your 2026 enrolment, including teaching periods, holidays, census dates, withdrawal dates, exam periods and deadlines to apply to graduate in the [2026 Handbook – Dates and deadlines](#)

Trimester 2 teaching period begins	Monday 6 July 2026
Census date	Saturday 15 August 2026
Intra-trimester break	Monday 10 August - Friday 14 August 2026
Trimester 2 teaching period ends	Friday 25 September 2026
Study period (end-of-unit assessment/ examination preparation period)	Monday 28 September - Friday 2 October 2026
End-of-unit assessment/examination period	Monday 5 October - Friday 16 October 2026
Inter-trimester break (the period between trimesters)	Monday 19 October - Friday 30 October 2026
Unit results released	Friday 30 October 2026 (10.30am)

Unit weekly activities

Note: this is a suggested schedule. The unit is designed for you to progress at your own pace, and includes optional content and tasks for interested students.

Week	Commencing	Topic	Assessment activity	Assessment due date
1	6 July 2026	Systems of linear equations	Learning Plan, Portfolio update	
2	13 July 2026	Matrix algebra	Portfolio update	
3	20 July 2026	Linear transformations and special matrices	Portfolio update	
4	27 July 2026	Determinants	Portfolio update, enrichment task 1 (draft)	
5^	3 August 2026	Linear vector space $\mathbb{R}^n$	Portfolio update, enrichment task 1 (peer review)	
6	17 August 2026	Eigenvalues, eigenvectors and diagonalisation	Portfolio update, enrichment task 1 (final)	
7	24 August 2026	Abstract vector spaces and orthogonality	Portfolio update	
8	31 August 2026	Least squares approximation	Portfolio update, enrichment task 2 (draft)	
9	7 September 2026	Advanced topics	Portfolio update, enrichment task 2 (peer review)	
10	14 September 2026	Advanced topics	Portfolio update, enrichment task 2 (final)	
11#	21 September 2026	Advanced topics	Portfolio update	
12 - Study period	28 September - 2 October 2026			Learning portfolio due Sunday 4 October

Week	Commencing	Topic	Assessment activity	Assessment due date
End-of-unit assessment and exam period	5-16 October 2026			

^ Intra-trimester break: **Monday 10 August - Friday 14 August 2026** (between weeks 5 and 6)

AFL Grand Final Eve public holiday: **Friday 25 September 2026** - University closed

DEAKIN UNIVERSITY

LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

Module 1 - Systems of Linear Equations

Submitted By:
Jahan GARG
s226100103
2026/07/31 19:52

Tutor:
Simon JAMES

Task Outcomes		Supports
M1.1	Apply elementary row operations to systems of linear equations represented in matrix form	ULO1 ULO1
M1.2	Apply the Gaussian Algorithm to reduce a matrix to row-echelon form	ULO1 ULO1
M1.3	Express the general solution to a homogeneous equation as a linear combination of sets of basic solutions	ULO1 ULO1
Evid	Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.	
SA	Full printout of Self-Assessment results with all questions.	
Sum	Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions.	
FB	Respond to any feedback and work with the OnTrack tutor to complete the module.	

July 31, 2026



Module 1 Summary

Response to feedback

For this first submission, I would especially like feedback on whether my explanations correctly link the Gaussian Algorithm and the rank theorems to the different types of solution sets (unique, infinite, or none). In particular, I want to know if my use of Examples from Sections 1.2 and 1.3 in Nicholson's Linear Algebra with Applications to justify my classification of systems is sufficiently clear and mathematically rigorous. I would also appreciate comments on whether my parameterised vector forms for homogeneous and non-homogeneous systems are expressed in a way that aligns with the expectations of SIT292 – Linear Algebra for Data Analysis.

I used AI assistance (Perplexity) to help me structure this Module Summary, refine my explanations of the key theorems and definitions, and check my written English. The mathematical understanding, worked examples, and NUMBAS results referenced throughout are my own.

Module Learning Objectives

I certify that I achieved the following learning objectives for the module:

- Apply elementary row operations to systems of linear equations represented in matrix form
- Apply the Gaussian Algorithm to reduce a matrix to row-echelon form
- Express the general solution to a homogeneous equation as a linear combination of sets of basic solutions

Key Definitions and Theorems

Solutions and Elementary Operations

Textbook Ref	Theorem/Definition statement	Learning Notes
Linear equations	An equation of the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$ is called a <u>linear equation</u> in the n variables $x_1, x_2, \dots, x_n$. With $a_1, a_2, \dots, a_n$ are referred to as coefficients and b as the constant term (or sometimes ‘right-hand side’).	A linear equation is one where each variable appears only to the first power and is not inside functions like squares, roots, or trigonometric expressions. This fits the module goal because it defines the class of equations for which row operations and matrix methods will preserve linearity and make sense.

System of linear equations	A finite collection of linear equations defined with respect to the same variables is called a <u>system of linear equations</u> .	A system of linear equations is several linear equations sharing the same variables, so any solution must satisfy all of them at once. Thinking in terms of systems is what justifies representing the equations by an augmented matrix and then working entirely at the matrix level.
Solutions	A set of values $s_1, s_2, \dots, s_n$ is a <u>solution</u> to a linear equation if setting $x_1=s_1, x_2=s_2, \dots, x_n=s_n$ allows the equation to hold (the left-hand side is equal to the right-hand side, b). A solution to a system of equations requires the solution to hold for all equations in the system.	A solution to a single linear equation is a specific assignment of values to the variables that balances left and right sides; for a system, the same assignment must work for every equation. This connects to the module's emphasis on using row operations that change the <i>appearance</i> of the equations without changing the underlying solution set.
Augmented matrix	A system of equations can be represented as an array of numbers consisting of the coefficients and constant terms, e.g., the system $3x_1 + 2x_2 - x_3 = 3$ $-x_1 + 4x_3 = -6$ Could be represented as the matrix, $\left[\begin{array}{ccc c} 3 & 2 & -1 & 3 \\ -1 & 0 & 4 & -6 \end{array} \right]$ Which will later be denoted $[A \mid \mathbf{b}]$ where A is the array of coefficients and $\mathbf{b}$ is the column of constants on the right-hand side.	The augmented matrix $[A \mid \mathbf{b}]$ is a compact way of storing both the coefficients and the right-hand side constants of a system. Once the system has been encoded in this form, every elementary row operation is just an algebraic shortcut for combining equations in ways that keep the same solution set, which is crucial for Gaussian elimination.
Definition 1.2 Elementary row operations	The following operations are called <u>elementary row operations</u> on a matrix. I. Interchange two rows. II. Multiply one row by a nonzero number. III. Add a multiple of one row to a different row	The three operations (row swap, scaling a row by a nonzero constant, and adding a multiple of one row to another) correspond exactly to valid algebraic manipulations of equations that preserve the set of solutions. In practice, I treat them as tools for simplifying the augmented matrix step-by-step, aiming to introduce zeros in specific positions without introducing mistakes that would change the system.

Gaussian Elimination

Textbook Ref	Theorem/Definition statement	Notes
Definition 1.3 Row-Echelon Form (Reduced)	A matrix is said to be in <u>row-echelon form</u> (and will be called a row-echelon matrix) if it satisfies the following three conditions: 1. All zero rows (consisting entirely of zeros) are at the bottom. 2. The first nonzero entry from the left in each nonzero row is a 1, called the <u>leading 1</u> for that row. 3. Each leading 1 is to the right of all leading 1s in the rows above it. A row-echelon matrix is said to be in <u>reduced row-echelon form</u> (and will be called a reduced row-echelon matrix) if, in addition, it satisfies the following condition: 4. Each leading 1 is the only nonzero entry in its column.	Row-echelon form gives a "staircase" pattern of leading 1s with zeros below each pivot, which makes back substitution straightforward. Reduced row-echelon form goes further by clearing entries above each leading 1, so every pivot column has a single 1 and zeros elsewhere, and the solution can be read off directly without further algebra.
Gaussian Algorithm	Step 1. If the matrix consists entirely of zeros, stop—it is already in row-echelon form. Step 2. Otherwise, find the first column from the left containing a nonzero entry (call it a), and move the row containing that entry to the top position. Step 3. Now multiply the new top row by $1/a$ to create a leading 1. Step 4. By subtracting multiples of that row from rows below it, make each entry below the leading 1 zero. This completes the first row, and all further row operations are carried out on the remaining rows. Step 5. Repeat steps 1–4 on the matrix consisting of the remaining rows. The process stops when either no rows remain at step 5, or the remaining rows consist entirely of zeros.	The Gaussian Algorithm is the systematic procedure that uses elementary row operations to move from an arbitrary matrix to row-echelon form. I find it helpful to think of it column by column: pick a pivot, normalise it to a leading 1, eliminate entries below, then repeat on the smaller submatrix, which matches the steps described in Nicholson, Section 1.2.
Gaussian Elimination	To solve a system of linear equations, proceed as follows: 1. Carry the augmented matrix to a reduced row-echelon matrix using elementary row operations. 2. If a row $[0 \ 0 \ 0 \ \cdots \ 0 \ 1]$ occurs, the system is inconsistent. 3. Otherwise, assign the nonleading variables (if any) as parameters, and use the equations corresponding to	Gaussian elimination applies the Gaussian Algorithm to the augmented matrix, then uses the resulting row-echelon (or reduced row-echelon) form to classify the system and read off solutions. The inconsistent case is flagged by a row of zeros on the left with a nonzero constant, while free variables appear when some columns lack leading 1s, which ties directly to the rank theorems.

	the reduced row-echelon matrix to solve for the leading variables in terms of the parameters.	
Definition 1.4 Rank of a matrix	The rank of matrix A is the number of leading 1s in any row-echelon matrix to which A can be carried by row operations.	The rank of a matrix is the number of pivot positions (leading 1s) in any row-echelon form obtained by row operations. For this module, rank acts as a numerical summary of how many independent equations we really have, and comparing rank to the number of variables tells us how many parameters appear in the general solution.
Theorem 1.2.2 Rank and solutions to a system of linear equations	Suppose a system of m equations in n variables is consistent, and that the rank of the augmented matrix is r. 1. The set of solutions involves exactly $n - r$ parameters. 2. If $r < n$, the system has infinitely many solutions. 3. If $r = n$, the system has a unique solution.	Theorem 1.2.2 formalises the observation that when a consistent system has rank r in n variables, there must be n minus r free variables and thus n minus r parameters in the solution. The "unique solution" case where r equals n and the "infinitely many solutions" case where r is less than n both appeared directly in my Gaussian elimination NUMBAS questions, so I used this theorem to interpret each row-echelon matrix.

Homogeneous Equations

Textbook Ref	Theorem/Definition statement	Notes
Homogeneous system of equations	A system of equations in the variables $x_1, x_2, \dots, x_n$ is called <u>homogeneous</u> if all the constant terms are zero—that is, if each equation of the system has the form $a_1x_1 + a_2x_2 + \dots + a_nx_n = 0$	A homogeneous system has all constant terms equal to zero, so the zero vector is always a solution (the trivial solution). This structure means the solution set is closed under linear combinations, and in later modules this is exactly what allows us to interpret homogeneous solutions as subspaces.
Trivial/non-trivial solutions	Homogeneous systems always have the solution $x_1 = x_2 = \dots = x_n = 0$, which is called the <u>trivial solution</u>. Any solution where one variable has a nonzero value is called a <u>nontrivial</u> solution.	The trivial solution, where every variable equals zero, is always present in a homogeneous system, but non-trivial solutions appear only when the rank is strictly less than the number of variables. In my practice problems, the presence of a free variable in a homogeneous system was the practical signal that non-trivial

		solutions (and basic solution vectors) exist.
Definition 1.5 Basic Solutions	The gaussian algorithm systematically produces solutions to any homogeneous linear system, called basic solutions, one for every parameter. E.g., where solutions to a system of equations can be expressed as $s\mathbf{x} + t\mathbf{y}$, then we say s and t are parameters and $\mathbf{x}$ and $\mathbf{y}$ are <u>basic solutions</u>.	Basic solutions are the special solutions produced by giving one free variable the value 1 and the others 0 and then solving for the leading variables. Any solution to the homogeneous system can be written as a linear combination of these basic solutions, which is the connection between Gaussian elimination and the idea of expressing the solution set as a span of vectors.
Theorem 1.3.2 Rank and homogeneous systems	Let A be an $m \times n$ matrix of rank r, and consider the homogeneous system in n variables with A as coefficient matrix. Then: 1. The system has exactly $n-r$ basic solutions, one for each parameter. 2. Every solution is a linear combination of these basic solutions.	Theorem 1.3.2 refines the rank idea for homogeneous systems by stating that there are exactly n minus r basic solutions and that every solution is a linear combination of them. This matches what I observed in the NUMBAS problems: whenever the coefficient matrix had rank r less than n , the number of linearly independent solution vectors matched n minus r .

Summarising your understanding:

1.1 Solutions and Elementary Operations

In Nicholson's Section 1.1, linear equations and systems are formalised and then re-expressed using augmented matrices and elementary row operations. I found it helpful to see that each elementary row operation, whether swapping rows, scaling a row, or adding a multiple of one row to another, corresponds exactly to a legitimate algebraic manipulation of the original equations, so the solution set is preserved. In my NUMBAS practice, I deliberately used simple examples to check this: for instance, swapping two rows simply reorders the equations, scaling a row by 2 multiplies both sides of an equation without changing its solutions, and adding a multiple of one row to another mimics adding equations together to eliminate a variable. These small checks built my confidence that working at the matrix level is mathematically justified, which supports the first learning objective about applying row operations to systems in matrix form.

1.2 Gaussian Elimination and strategies

Section 1.2 develops the Gaussian Algorithm as a systematic way of applying the elementary row operations to reach row-echelon form. In the NUMBAS exercise N1.3.1, I followed the textbook's pivot strategy: choose the leftmost column with a nonzero entry, bring a suitable row into the pivot position using a row swap if needed, normalise the pivot to 1, and then use it to eliminate entries below. I noticed that there are several practical strategies to make the arithmetic easier, which also appeared in N1.3.2: sometimes I can swap rows to bring a 1 or negative 1 into the pivot position instead of

dividing by a messy number, and in other cases multiplying rows to avoid fractions until the final step is more efficient. These strategies all respect the core algorithm, but they make hand calculations faster and reduce the chance of arithmetic mistakes.

Compared with a substitution-based approach, Gaussian elimination scales far better when the number of variables increases. Substitution quickly becomes unwieldy because each step introduces longer expressions, whereas the matrix approach keeps the information structured and allows me to apply the same three operations repeatedly in a predictable pattern. This is particularly important in the 4x4 NUMBAS problems and in data analysis contexts more generally, where we often need to solve large systems numerically.

1.3 Homogeneous Equations, rank, and basic solutions

Section 1.3 focuses on homogeneous systems, where all right-hand sides are zero, and shows how Gaussian elimination and rank lead to a clean description of the solution space. In my underdetermined homogeneous practice questions, I observed that whenever the coefficient matrix had rank r less than n , the row-echelon form contained at least one free variable, and I could describe all solutions as a linear combination of basic solution vectors. This matches Theorem 1.3.2 exactly: there are n minus r basic solutions, and every solution is a linear combination of them.

The relationship between Theorem 1.2.2 and Theorem 1.3.2 became clearer when I compared my non-homogeneous and homogeneous examples. Theorem 1.2.2 applies to consistent systems in general and tells me that the number of parameters in the solution is n minus r , with a unique solution when r equals n and infinitely many when r is less than n . Theorem 1.3.2 is the homogeneous special case: not only do we get n minus r parameters, but the entire solution set forms a subspace spanned by n minus r basic solutions. In other words, the general rank theorem describes the structure of solution sets, while the homogeneous rank theorem highlights that, in that special case, the solution set has a clean vector-space description in terms of linear combinations.

Reflecting on the content:

In this module, I found it particularly interesting how a few simple row operations, which at first look like mechanical matrix manipulations, actually encode very precise algebraic transformations of the original equations. The Gaussian Algorithm initially felt procedural, but working through both 3x3 and 4x4 examples helped me see the underlying structure in terms of pivots, rank, and the distinction between leading and free variables. I also noticed that the homogeneous case ties directly into ideas of spans and linear combinations, which I expect will be fundamental later when we study vector spaces and eigenvectors. The main challenge for me was keeping the arithmetic accurate during longer elimination sequences, especially when fractions appeared, but the module's strategy suggestions, such as judicious row swaps and delaying division, helped me manage this. Overall, I feel that I now meet the three learning objectives: I can perform elementary row operations in matrix form, carry out Gaussian elimination to row-echelon form, and express homogeneous solutions as linear combinations of basic solution vectors.

1.2 Solutions and Elementary Operations

Q. Write the augmented matrix for a set of equations.

Consider the following system of linear equations

$$-5x_1 + 9x_2 + 9x_4 = 30$$

$$x_3 + 6x_4 = 34$$

$$-4x_1 - 5x_2 + 7x_3 + 6x_4 = 46$$

Provide the augmented matrix of the system

Ans. The provided system of linear equations can be written as:

$$-5x_1 + 9x_2 + 0x_3 + 9x_4 = 30$$

$$0x_1 + 0x_2 + x_3 + 6x_4 = 34$$

$$-4x_1 - 5x_2 + 7x_3 + 6x_4 = 46$$

Its matrix form will be:

$$Ax = b$$

$$\begin{bmatrix} -5 & 9 & 0 & 9 \\ 0 & 0 & 1 & 6 \\ -4 & -5 & 7 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 30 \\ 34 \\ 46 \end{bmatrix}$$

Its augmented form will be:

$$[A|b] := \left[\begin{array}{cccc|c} -5 & 9 & 0 & 9 & 30 \\ 0 & 0 & 1 & 6 & 34 \\ -4 & -5 & 7 & 6 & 46 \end{array} \right]$$

② Substitute and verify whether or not a given solution satisfies a set of eq^{ns}.

Note for this question, the answer options represent a solution $x_1 = a, x_2 = b$ as a column of values, i.e.,

$$\begin{pmatrix} a \\ b \end{pmatrix}$$

For the following non-homogeneous system represented by augmented matrix

$$\left(\begin{array}{cc|c} -8 & 0 & -24 \\ 0 & 4 & -16 \\ 2 & -9 & 42 \\ 0 & -2 & 8 \end{array} \right)$$

which of the following are solutions?

- a) $\begin{pmatrix} 4 \\ -4 \end{pmatrix}$ b) $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ ✓ c) $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ d) $\begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}$

Ans: This augmented matrix can be written as the following system of linear ~~systems~~ equations:

$$-8x + 0y = -24 \quad \text{--- (1)}$$

$$0x + 4y = -16 \quad \text{--- (2)}$$

$$2x - 9y = 42 \quad \text{--- (3)}$$

$$0x - 2y = 8 \quad \text{--- (4)}$$

Solving (1) eqn,

$$\Rightarrow +8x = +24$$

$$\Rightarrow x = \frac{24}{8}$$

$$\therefore \boxed{x = 3}$$

Substituting the value of x in (3) eqn,

$$\Rightarrow 2(3) - 9y = 42$$

$$\Rightarrow 6 - 9y = 42$$

$$\Rightarrow -9y = 36$$

$$\Rightarrow y = \frac{36}{-9}$$

$$\therefore \boxed{y = -4}$$

Thus, the solution for the system of linear equations is $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

Exercise 1.1.2 Find all solutions to the following in parametric form in two ways.

a. $3x + y = 2$

Ans:

solving the equation for y in terms of x , we get $y = 2 - 3x$. If s is an arbitrary then, setting $x = s$, we get solutions

M-1

$$x = s$$

$$y = 2 - 3s \quad s \text{ is arbitrary}$$

when solving for x : $x = \frac{1}{3}(2 - y)$. Then, if we take $y = p$, the solution are represented as follows:

M-2

$$x = \frac{1}{3}(2 - y)^p$$

$$y = p \quad p \text{ is arbitrary}$$

c. $3x - y + 2z = 5$

Ans:

solving the equation for y in terms of x and z , we get $y = 3x + 2z - 5$. If s and t are arbitrary then, setting $x = s$, $z = t$, we get solutions.

M-1

$$x = s$$

$$y = 3s + 2t - 5 \quad s \text{ and } t \text{ arbitrary}$$

$$z = t$$

when solving for x : $x = \frac{1}{3}(y - 2z + 5)$. Then, if

we take $y = p$, $z = q$, the solutions are represented as follows:

M-2

$$x = \frac{1}{3}(p - 2q + 5)$$

$$y = p$$

$$z = q$$

p and q arbitrary

Exercise 1.1.7 Write the augmented matrix for each of the following systems of linear equations.

b.
$$\begin{aligned} x + 2y &= 0 \\ y &= 1 \end{aligned}$$

Ans:
$$\begin{aligned} x + 2y &= 0 \\ 0x + y &= 1 \end{aligned}$$

Its augmented form will be:

$$[A|b] := \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & 1 \end{array} \right]$$

d.
$$\begin{aligned} x + y &= 1 \\ y + z &= 0 \\ z - x &= 2 \end{aligned}$$

Ans:
$$\begin{aligned} x + y + 0z &= 1 \\ 0x + y + z &= 0 \\ -x + 0y + z &= 2 \end{aligned}$$

Its augmented form will be:

$$[A|b] := \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 1 & 2 \end{array} \right]$$

This question will allow you to practice applying the steps of the **Gaussian Algorithm** provided in the Nicholson text (p11).

You can check your results as you go by clicking on [Save Answer]. If there is an error in any of the rows, feedback will be provided to help let you know what the issue might be. The rows are checked in order and your matrix can be submitted without being complete (i.e., you can check each row as you go!).

--

Consider the following matrix representing the augmented form $[A|b]$ of a system of 3 linear equations:

$$\begin{pmatrix} 2 & -6 & 6 & 24 \\ 2 & -3 & 24 & 72 \\ 1 & 0 & 19 & 54 \end{pmatrix}$$

Step 3

In **Step 1** of the algorithm, you will have verified that the matrix does not consist entirely of zeros and then in **Step 2**, since there are no zeros in the first column, we just leave the matrix with its rows in the current order.

Now perform **Step 3** to create a leading 1 in the first row and write down the resulting augmented matrix (do not perform any other row operations)

$$\begin{pmatrix} \boxed{1} & \boxed{-3} & \boxed{3} & \boxed{12} \\ \boxed{2} & \boxed{-3} & \boxed{24} & \boxed{72} \\ \boxed{1} & \boxed{0} & \boxed{19} & \boxed{54} \end{pmatrix} \checkmark$$

Score: 1/1 

▼ Hide feedback

 Matrix is correct. You were awarded 1 mark.

You scored **1** mark for this part.

Step 4

Perform row operations with multiples of row 1 to obtain zeros in the rows below. These should be of the form:

$R'_k = R_k - a_{k1}R_1$ (i.e., don't take the negative or other multiples of rows 2 or 3)

$$\begin{pmatrix} \boxed{1} & \boxed{-3} & \boxed{3} & \boxed{12} \\ \boxed{0} & \boxed{3} & \boxed{18} & \boxed{48} \\ \boxed{0} & \boxed{3} & \boxed{16} & \boxed{42} \end{pmatrix} \checkmark$$

After performing this step, you should have a leading 1 in the first column with all 0s below it.

Score: 1/1

▼ Hide feedback

Matrix is correct. You were awarded **1** mark.

You scored **1** mark for this part.

Step 5.3

Now proceed to the next row and perform **Steps 1-3** on the matrix consisting of the remaining rows (don't apply **Step 4** yet). This should result in a leading 1 in the second row.

$$\begin{pmatrix} \boxed{1} & \boxed{-3} & \boxed{3} & \boxed{12} \\ \boxed{0} & \boxed{1} & \boxed{6} & \boxed{16} \\ \boxed{0} & \boxed{3} & \boxed{16} & \boxed{42} \end{pmatrix} \checkmark$$

Score: 1/1 ✓

► Show feedback

Step 5.4

Use the new Row 2 to make the value below its leading 1 a zero. Again, this operation should be of the form:

$$R'_3 = R_3 - a_{32}R_2$$

$$\begin{pmatrix} \boxed{1} & \boxed{-3} & \boxed{3} & \boxed{12} \\ \boxed{0} & \boxed{1} & \boxed{6} & \boxed{16} \\ \boxed{0} & \boxed{0} & \boxed{-2} & \boxed{-6} \end{pmatrix} \quad \checkmark$$

Score: 1/1 ✓

▼ Hide feedback

✓ Matrix is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Step 5.5.3

In the last step, we now look at row 3 and apply **Steps 1-3**. Enter the final matrix below, which should have a leading 1 in all rows.

$$\begin{pmatrix} \boxed{1} & \boxed{-3} & \boxed{3} & \boxed{12} \\ \boxed{0} & \boxed{1} & \boxed{6} & \boxed{16} \\ \boxed{0} & \boxed{0} & \boxed{1} & \boxed{3} \end{pmatrix} \quad \checkmark$$

Score: 1/1 ✓

► Show feedback

Solve the system

Based on the augmented form you've obtained at **Step 5.5.3**, solve the system:

$$x_1 = \boxed{-3} \quad \checkmark$$

$$x_2 = \boxed{-2} \quad \checkmark$$

$$x_3 = \boxed{3} \quad \checkmark$$

Score: 3/3 

▼ Hide feedback

x1

 Correct. You were awarded **1** mark.

x2

 Correct. You were awarded **1** mark.

x3

 Correct. You were awarded **1** mark.

You scored **3** marks for this part.

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

This question introduces two strategies that can be useful when performing **Gaussian Elimination**.

- Row swap
- Row swap with addition/subtraction of another row to help us get a leading 1.

Note that we still are essentially performing the steps of the Gaussian algorithm, potentially with a slight modification at Step 2 depending on the starting matrix.

--

Consider the following matrix representing the augmented form $[A \mid \mathbf{b}]$ of a system of 3 linear equations:

$$\left(\begin{array}{cccc} 0 & 3 & 10 & -23 \\ 1 & 2 & 8 & -16 \\ 4 & -5 & -23 & 59 \end{array} \right)$$

Steps 2-3

In **Step 1**, you will have verified that the matrix does not consist entirely of zeros.

Now in **Step 2**, we note that there is a 0 in the first position of the first row. In order to continue, we need to do a row swap.

- *Enter the original row 1 ($[0, 3, 10, -23]$) as row 2.*

If either of the original rows 2 or 3 contains a leading 1,

- *move that row to become the new row 1 and enter the remaining row as row 3.*
- *(if there is no leading 1 but there is a leading -1, do the same but multiply the new row 1 by -1 so that you obtain a leading 1).*

If there are no leading 1s or -1s,

- see if you can obtain a leading 1 by adding/subtracting a combination of rows 2 and 3.
- Enter the result (which should now have a leading 1) as the new row 1, and use either of these two original rows as row 3 (but not a combination).

Enter the resulting matrix.

$$\begin{pmatrix} \boxed{1} & \boxed{2} & \boxed{8} & \boxed{-16} \\ \boxed{0} & \boxed{3} & \boxed{10} & \boxed{-23} \\ \boxed{4} & \boxed{-5} & \boxed{-23} & \boxed{59} \end{pmatrix} \checkmark$$

Score: 1/1

▼ Hide feedback

Matrix is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Step 4

*The steps from here are the same as for NUMBAS exercise 1.3.1

Perform row operations using row 1 to obtain zeros below.

$$\begin{pmatrix} \boxed{1} & \boxed{2} & \boxed{8} & \boxed{-16} \\ \boxed{0} & \boxed{3} & \boxed{10} & \boxed{-23} \\ \boxed{0} & \boxed{-13} & \boxed{-55} & \boxed{123} \end{pmatrix} \checkmark$$

Score: 1/1

▼ Hide feedback

Matrix is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Step 5.3

Now proceed to the next row and perform **Steps 1-3** (don't apply Step 4 yet) on the matrix consisting of rows 2 and 3.

$$\begin{pmatrix} \boxed{1} & \boxed{2} & \boxed{8} & \boxed{-16} \\ \boxed{0} & \boxed{1} & \boxed{10/3} & \boxed{-23/3} \\ \boxed{0} & \boxed{-13} & \boxed{-55} & \boxed{123} \end{pmatrix} \checkmark$$

You should have a leading 1 in the second row.

Score: 1/1

▼ Hide feedback

Matrix is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Step 5.4

Use the new Row 2 to make the value below its leading 1 a zero.

$$\begin{pmatrix} \boxed{1} & \boxed{2} & \boxed{8} & \boxed{-16} \\ \boxed{0} & \boxed{1} & \boxed{10/3} & \boxed{-23/3} \\ \boxed{0} & \boxed{0} & \boxed{-35/3} & \boxed{70/3} \end{pmatrix} \checkmark$$

Score: 1/1

▼ Hide feedback

Matrix is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Step 5.5.3

Now look at row 3 and apply **Steps 1-3**. Enter the final matrix below, which should have a leading 1 in all rows.

$$\left(\begin{array}{cccc} \boxed{1} & \boxed{2} & \boxed{8} & \boxed{-16} \\ \boxed{0} & \boxed{1} & \boxed{10/3} & \boxed{-23/3} \\ \boxed{0} & \boxed{0} & \boxed{1} & \boxed{-2} \end{array} \right) \checkmark$$

Score: 1/1

► Show feedback

Solve the system

Based on the augmented form we've obtained at Step 5.5.3, solve the system:

$$x_1 = \boxed{2} \checkmark$$

$$x_2 = \boxed{-1} \checkmark$$

$$x_3 = \boxed{-2} \checkmark$$

Score: 3/3

▼ Hide feedback

x1

Correct. You were awarded 1 mark.

x2

Correct. You were awarded 1 mark.

x3

Correct. You were awarded 1 mark.

You scored 3 marks for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

This question will allow you to practice applying the steps of the **Gaussian Algorithm** when the system is *underdetermined*.

--

Consider the following matrix representing the augmented form $[A|b]$ of a system of 3 linear equations:

$$\begin{pmatrix} 0 & 3 & 12 & -33 \\ 0 & 2 & 8 & -22 \\ 0 & 2 & 8 & -22 \end{pmatrix}$$

Step 3

In **Step 1** of the algorithm, you will have verified that the matrix does not consist entirely of zeros and then in **Step 2**, we consider the first column that contains a non-zero entry (Column 2 in this case).

Now perform **Step 3** to create a leading 1 in the first row and write down the resulting augmented matrix (do not perform any other row operations)

$$\begin{pmatrix} \boxed{0} & \boxed{1} & \boxed{4} & \boxed{-11} \\ \boxed{0} & \boxed{2} & \boxed{8} & \boxed{-22} \\ \boxed{0} & \boxed{2} & \boxed{8} & \boxed{-22} \end{pmatrix} \checkmark$$

Score: 1/1 

► Show feedback

Step 4

Perform row operations with multiples of row 1 to obtain zeros in the positions below the leading 1. These should be of the form:

$R'_k = R_k - aR_1$ (where a is the required multiplier, i.e., don't take the negative or other multiples of rows 2 or 3)

$$\left(\begin{array}{|c|c|c|c|} \hline 0 & 1 & 4 & -11 \\ \hline 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \right) \checkmark$$

After performing this step, you should have a leading 1 in the first row with all 0s below it.

Score: 1/1 

► Show feedback

Step 5.3

Now proceed to the next row and perform **Steps 1-3** on the matrix consisting of the remaining rows (don't apply **Step 4** yet). This should result in a leading 1 in the second row unless you have already obtained a zero matrix (if rows 2 and 3 are all zero, you can skip to the '**Solve the system**' step below).

$$\left(\begin{array}{|c|c|c|c|} \hline 0 & 1 & 4 & -11 \\ \hline 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \right) \checkmark$$

Score: 1/1 

► Show feedback

Step 5.4

Use the new Row 2 to make the value below its leading 1 a zero. Again, this operation should be of the form:

$$R'_3 = R_3 - aR_2$$

$$\left(\begin{array}{|c|c|c|c|} \hline 0 & 1 & 4 & -11 \\ \hline 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \right) \checkmark$$

At this stage you should have obtained the final matrix, with a leading 1 in all non-zero rows, and all rows of zero at the bottom.

Score: 1/1 

► Show feedback

Solve the system

Based on the augmented form you've obtained, it's time to solve the system, but now we will have some free variables.

We can refer to the variables corresponding with columns *without* a leading 1 as free variables. If there is one free variable, let it be equal to s , if there are two, denote them by s and t (in order).

Now solve your system and enter your equations in terms of s (and t if required).

$$x_1 = \boxed{s} \quad s \quad \checkmark$$

$$x_2 = \boxed{-11-4t} \quad -11 - 4t \quad \checkmark$$

$$x_3 = \boxed{t} \quad t \quad \checkmark$$

Score: 3/3 

► Show feedback

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

Use the Gaussian Algorithm and complete the steps below to solve the following for $\mathbf{x}$.

$$\begin{pmatrix} 3 & 6 & 6 & -5 \\ -3 & -6 & -2 & -5 \\ 1 & 2 & 1 & 1 \\ 0 & 0 & -3 & 9 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 20 \\ -2 \\ 2 \\ -15 \end{pmatrix}$$

Operations with Row1

Enter the augmented matrix that results after following steps 1-4 of the Gaussian Algorithm in structions (p11 of 1.2 in the textbook). You should have a leading 1 in the first row and all entries below it equal to 0.

$$\left(\begin{array}{ccccc} 1 & 2 & 2 & -5/3 & 20/3 \\ 0 & 0 & 4 & -10 & 18 \\ 0 & 0 & -1 & 8/3 & -14/3 \\ 0 & 0 & -3 & 9 & -15 \end{array} \right) \checkmark$$

Operations with Row2

Continue the steps on the next available column and enter the resulting matrix below (remember there should be a leading 1 in the second row with all entries below it equal to 0).

$$\left(\begin{array}{ccccc} 1 & 2 & 2 & -5/3 & 20/3 \\ 0 & 0 & 1 & -5/2 & 9/2 \\ 0 & 0 & 0 & 1/6 & -1/6 \\ 0 & 0 & 0 & 3/2 & -3/2 \end{array} \right) \checkmark$$

Operations with Row3

Enter the final row-echelon augmented matrix

$$\left(\begin{array}{ccccc} 1 & 2 & 2 & -5/3 & 20/3 \\ 0 & 0 & 1 & -5/2 & 9/2 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \checkmark$$

Solution

Now use this matrix to find a vector $\mathbf{x}$ that solves the system. If you have a free variable, set it to zero (e.g., if x_3 is a free variable, then let $x_3 = 0$, and determine the remaining values based on that.

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix}$$

Operations with Row1

Score: 4/4

- This step appears correct. The matrix is in the correct form and values are correct. You were awarded 1 mark.

Operations with Row2

- This step appears correct. The matrix is in the correct form and values are correct. You were awarded 1 mark.

Operations with Row3

✓ This step appears correct. The matrix is in the correct form and values are correct. You were awarded **1** mark.

✓ You were awarded **1** mark.

You scored **4** marks for this part.

NUMBAS

Created using Numbas (<https://www.numbas.org.uk>) v10.0, developed by Newcastle University (<http://www.newcastle.ac.uk>).

Starting with the following system,

$$\begin{pmatrix} 0 & 0 & 3 & 8 \\ 3 & -6 & 3 & -6 \\ 3 & -6 & 7 & -1 \\ 1 & -2 & 2 & -1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 11 \\ -3 \\ 6 \\ 1 \end{pmatrix}$$

We can perform Gaussian elimination.

The augmented matrix is $\begin{pmatrix} 0 & 0 & 3 & 8 & 11 \\ 3 & -6 & 3 & -6 & -3 \\ 3 & -6 & 7 & -1 & 6 \\ 1 & -2 & 2 & -1 & 1 \end{pmatrix}$.

Since row 4 has 1 as the first entry, we'll use it as the top row and then subtract multiples on the rows below. We will have

$$R1' = R4$$

$$R2' = R1 - (0) \times R4$$

$$R3' = R2 - (3) \times R4$$

$$R4' = R3 - (3) \times R4$$

$$\begin{pmatrix} 1 & -2 & 2 & -1 & 1 \\ 0 & 0 & 3 & 8 & 11 \\ 0 & 0 & -3 & -3 & -6 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix}$$

Continue the steps on the next available column and enter the resulting matrix below (remember there should be a leading 1 in the second row with all entries below it equal to 0).

$$\begin{pmatrix} \boxed{1} & \boxed{-2} & \boxed{2} & \boxed{-1} & \boxed{1} \\ \boxed{0} & \boxed{0} & \boxed{1} & \boxed{2} & \boxed{3} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{2} & \boxed{2} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{3} & \boxed{3} \end{pmatrix} \quad \checkmark$$

This step appears correct. The matrix is in the correct form and values are correct. You were awarded **1** mark.



You scored **1** mark for this part.

Score: 1/1

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

Consider the following linear system.

$$\begin{pmatrix} 1 & 2 & 2 & 1 \\ -2 & -4 & -3 & -3 \\ -3 & -6 & -3 & -3 \\ 2 & 4 & 2 & 7 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 2 \\ -2 \\ -3 \\ -3 \end{pmatrix}$$

Performing Gaussian elimination leads to the following matrix in row-echelon form.

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Use this matrix to find a vector $\mathbf{x}$ that solves the system. If you have a free variable, set it to zero (e.g., if x_3 is a free variable, then let $x_3 = 0$, and determine the remaining values based on that).

$$\mathbf{x} = \begin{pmatrix} \boxed{1} \\ \boxed{0} \\ \boxed{1} \\ \boxed{-1} \end{pmatrix}$$

You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

This question will help you express solutions to systems of equations in terms of vectors when we have one parameter.

Suppose we have an underdetermined system (see N1.3.4 from the previous submodule). On the left-hand side below we have the original equations and then its row echelon form on the right.

$$\begin{pmatrix} 0 & -2 & -4 & 2 \\ 0 & -2 & -6 & 4 \\ 0 & 4 & 9 & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The set of solutions, in terms of the parameter s , is:

$$x_1 = s$$

$$x_2 = 1$$

$$x_3 = -1$$

Express this as the addition of two vectors, a constant vector and another vector multiplied by the parameter s .

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{pmatrix} \boxed{0} \\ \boxed{1} \\ \boxed{-1} \end{pmatrix} \checkmark + s \begin{pmatrix} \boxed{1} \\ \boxed{0} \\ \boxed{0} \end{pmatrix} \checkmark$$

Score: 2/2 ✓

constant

✓ Your answer is correct. You were awarded 1 mark.

coeff of s

✓ Your answer is correct. You were awarded 1 mark.

You scored 2 marks for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

This question will help you express solutions to systems of equations in terms of vectors when we have one parameter.

non-homogeneous case

Suppose we have an underdetermined system (see N1.3.4 from the previous submodule). On the left-hand side below we have the original equations and then its row echelon form on the right.

$$\begin{pmatrix} 0 & -2 & 4 & 6 \\ 0 & -4 & 8 & 12 \\ 0 & 2 & -4 & -6 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The set of solutions, in terms of the parameters s and t , is:

$$x_1 = s$$

$$x_2 = -3 + 2t$$

$$x_3 = t$$

Express this as the addition of three vectors, a constant vector and two vectors which are multiplied by s and t .

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{pmatrix} \boxed{0} \\ \boxed{-3} \\ \boxed{0} \end{pmatrix} \checkmark + s \begin{pmatrix} \boxed{1} \\ \boxed{0} \\ \boxed{0} \end{pmatrix} \checkmark + t \begin{pmatrix} \boxed{0} \\ \boxed{2} \\ \boxed{1} \end{pmatrix} \checkmark$$

Score: 3/3 ✓

► Show feedback

homogeneous case

Now suppose that we have a homogeneous system with the same coefficients matrix, in this case we have all 0s on the right-hand side and we can just perform row operations on the coefficients matrix. We will have

$$\begin{pmatrix} 0 & -2 & 4 \\ 0 & -4 & 8 \\ 0 & 2 & -4 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Let $x_1 = s$ and $x_3 = t$, then express the solution as a linear combination in terms of s and t .

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = s \begin{pmatrix} \boxed{1} \\ \boxed{0} \\ \boxed{0} \end{pmatrix} \checkmark + t \begin{pmatrix} \boxed{0} \\ \boxed{2} \\ \boxed{1} \end{pmatrix} \checkmark$$

What is the solution when $s = -1$ and $t = -3$?

$$\begin{pmatrix} \boxed{-1} \\ \boxed{-6} \\ \boxed{-3} \end{pmatrix} \checkmark$$

(You should be able to verify that this by substituting the values back into the homogeneous system)

Score: 3/3 

▼ Hide feedback

hom coeff s

 Your answer is correct. You were awarded 1 mark.

hom coeff t

 Your answer is correct. You were awarded 1 mark.

solution

 Your answer is correct. You were awarded 1 mark.

You scored **3** marks for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

Question 1

Consider the following solution vectors for a given homogenous system of linear equations,

$$\mathbf{x}_1 = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} -5 \\ 5 \\ 1 \end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$$

If it is possible, express the vector $\mathbf{v} = \begin{pmatrix} 13 \\ -22 \\ -9 \end{pmatrix}$ as a linear combination of $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$.

$$\mathbf{v} = \boxed{0} \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix} + \boxed{0} \begin{pmatrix} -5 \\ 5 \\ 1 \end{pmatrix} + \boxed{0} \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$$

If it is not possible, enter 0 for all 3 coefficients.

You're right! It is not possible to obtain this vector as a linear combination and so you have correctly entered zeros. You were awarded **3** marks. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

Consider the system of equations,

$$\begin{aligned}x + 2y - 2z &= k \\x + (3k + 2)y + (-k - 2)z &= 3 \\-x - 2y + (k^2 + k + 2)z &= 1\end{aligned}$$

where $x, y, z \in \mathbb{R}$ are variables and $k \in \mathbb{R}$ is fixed.

(a) Determine the values of k for the following solution cases:

i. The value of k for which the system has **no solutions** is $k =$ ✓

ii. A value of k for which the system has a **unique solution** is $k =$ ✓
(provide one example only)

iii. The value of k for which the system has **multiple solutions** is $k =$ ✓

(b) Find all solutions to the system in case (iii) above. Express these in the form

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{b} + a\mathbf{v} \text{ where } \mathbf{v} \text{ and } \mathbf{b} \text{ are vectors and } a \in \mathbb{R}.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \text{5/3} \\ \text{-4/3} \\ \text{0} \end{pmatrix} \checkmark + a \begin{pmatrix} \text{4/3} \\ \text{1/3} \\ \text{1} \end{pmatrix} \checkmark$$

...

For example, if your solution is $x = 3, y = 2 + 3a, z = a$, then this would be

$$\text{expressed } \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} + a \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix}$$

Gap 0

Your answer is correct. You were awarded **1** mark. ✓

Gap 1

Your answer is correct (other choices are also possible). You were awarded **1** mark. ✓

Gap 2

Your answer is correct. You were awarded **1** mark. ✓

Gap 3

Your answer is correct. This point should be a solution to the equation when $k = -1$. You were awarded **1** mark. ✓

Gap 4

Your answer is correct. Any multiple of this vector is acceptable. You were awarded **1** mark. ✓

You scored **5** marks for this part.

Score: 5/5 ✓

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

Question 1

This question is about general linear combinations.

a)

Can you write any vector $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$ as a linear combination of the vectors $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$?

If yes, fill in the coefficients. If no, write NA into the gaps.

$$\begin{pmatrix} x \\ y \end{pmatrix} = \boxed{x} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \boxed{y} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Gap 0

Your answer is numerically correct. You were awarded 1 mark. ✓

Gap 1

Your answer is numerically correct. You were awarded 1 mark. ✓

You scored 2 marks for this part.

Score: 2/2 ✓

b)

Can you write any vector $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4$ as a linear combination of the vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$?

If yes, fill in the coefficients. If no, write NA into the gaps.

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \boxed{x_1} x_1 \checkmark \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \boxed{x_2} x_2 \checkmark \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \boxed{x_3} x_3 \checkmark \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \boxed{x_4} x_4 \checkmark \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

If you decided that not every vector $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4$ can be written as a linear

combination as above, give an example of a vector which cannot be written like this. Write it into the gap as `vector(1,2,3,4)` with the numbers substituted by your chosen entries. (The marking algorithm will not understand fractions.) If you did provide general coefficients above, then write `NA` into the gap.

Counterexample: ☒

Gap 0

Your answer is numerically correct. You were awarded 1 mark. ☒

Gap 1

Your answer is numerically correct. You were awarded 1 mark. ☒

Gap 2

Your answer is numerically correct. You were awarded 1 mark. ☒

Gap 3

Your answer is numerically correct. You were awarded 1 mark. ☒

Gap 4

Your answer is numerically correct. You were awarded 1 mark. ☒

You scored **5** marks for this part.

Score: 5/5 ☒

c)

Can you write any vector $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4$ as a linear combination of just the first three vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$?

If yes, fill in the coefficients. If no, write NA into the gaps.

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \boxed{\text{NA}} \text{ NA } \checkmark \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \boxed{\text{NA}} \text{ NA } \checkmark \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \boxed{\text{NA}} \text{ NA } \checkmark \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

If you decided that not every vector $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4$ can be written as a linear

combination as above, give an example of a vector which cannot be written like this. Write it into the gap as `vector(1,2,3,4)` with the numbers substituted by your chosen entries. (The marking algorithm will not understand fractions.) If you did provide general coefficients above, then write NA into the gap.

Counter-example: `vector(0,0,0,1)` $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \checkmark$

Gap 0

Your answer is numerically correct. You were awarded 1 mark.

Gap 1

Your answer is numerically correct. You were awarded 1 mark.

Gap 2

Your answer is numerically correct. You were awarded **1** mark. ✓

Gap 3

Your answer is correct. You were awarded **1** mark. ✓

You scored **4** marks for this part.

Score: 4/4 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

SIT292 Module 1 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score
Question 1	1/1
Question 2	1/1
Question 3	4/4
Question 4	3/3
Question 5	4/4
Question 6	2/2
Total	15/15 (100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 1 Self-Assessment
Session ID:	2046032764981711
Exam Start:	Wed Jul 15 2026 21:54:02
Exam Stop:	Wed Jul 15 2026 22:32:11
Time Spent:	00:38:08

Question 1

Consider the following system of linear equations.

$$\begin{aligned} -9x_1 - 8x_2 &= -43 \\ -9x_2 - 8x_3 - 8x_4 &= -18 \\ 5x_1 - 4x_2 - 4x_3 - 5x_4 &= 10 \end{aligned}$$

Provide the **augmented matrix** of the system. Use the up and down arrows next to the row/column numbers to increase the size of your matrix.

Rows: 3 Columns: 5

$$\left(\begin{array}{cccc|c} -9 & -8 & 0 & 0 & -43 \\ 0 & -9 & -8 & -8 & -18 \\ 5 & -4 & -4 & -5 & 10 \end{array} \right) \checkmark$$

Expected answer:

$$\left(\begin{array}{cccc|c} -9 & -8 & 0 & 0 & -43 \\ 0 & -9 & -8 & -8 & -18 \\ 5 & -4 & -4 & -5 & 10 \end{array} \right)$$

✓ Your answer is correct. You were awarded 1 mark.

Score: 1/1 ✓

You scored **1** mark for this part.

Question 2

For the following (non-homogeneous) system represented by its augmented matrix,

$$\left(\begin{array}{ccc|c} 8 & 0 & 32 & \\ -8 & -8 & -48 & \\ 2 & 0 & 8 & \\ 0 & 7 & 14 & \end{array} \right)$$

Which of the following are solutions?



$$\begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 1 \\ -4 \end{pmatrix}$$



Expected answer:



$$\begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 1 \\ -4 \end{pmatrix}$$



You chose a correct answer. You were awarded **1** mark.

Score: 1/1

You scored **1** mark for this part.

Question 3

Consider the following matrix representing the augmented form $[A|\mathbf{b}]$ of a system of 3 linear equations:

$$\begin{pmatrix} -2 & 2 & -2 & -10 \\ 4 & -7 & 16 & 53 \\ -2 & 3 & -6 & -21 \end{pmatrix}$$

Row operations

After applying the first sets of Gaussian algorithm steps, the following matrix is obtained:

$$\begin{pmatrix} 1 & -1 & 1 & 5 \\ 0 & -3 & 12 & 33 \\ 0 & 1 & -4 & -11 \end{pmatrix}$$

Use division/multiplication by a scalar first to create a leading 1 in row 2, and then use the new Row 2 to make the value below its leading 1 a zero.

Enter the resulting matrix, which should be in row echelon form.

$$\begin{pmatrix} \frac{1}{-} & \frac{-1}{-} & \frac{1}{-} & \frac{5}{-} \\ \frac{0}{-} & \frac{1}{-} & \frac{-4}{-} & \frac{-11}{-} \\ \frac{0}{-} & \frac{0}{-} & \frac{0}{-} & \frac{0}{-} \end{pmatrix} \checkmark$$

Expected answer:

$$\begin{pmatrix} \frac{1}{-} & \frac{-1}{-} & \frac{1}{-} & \frac{5}{-} \\ \frac{0}{-} & \frac{1}{-} & \frac{-4}{-} & \frac{-11}{-} \\ \frac{0}{-} & \frac{0}{-} & \frac{0}{-} & \frac{0}{-} \end{pmatrix}$$

▼ Feedback

Score: 1/1 ✓

✓ Matrix is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Solve the system

Based on the augmented form you've obtained, solve the system in terms of the free variable, which you can denote by s .

Enter your equations in terms of s

$$x_1 = \underline{3s-6} \quad 3s - 6 \quad \checkmark$$

Expected answer: $\underline{-6 + 3s} \quad -6 + 3s$

$$x_2 = \underline{4s-11} \quad 4s - 11 \quad \checkmark$$

Expected answer: $\underline{-11 + 4s} \quad -11 + 4s$

$$x_3 = \underline{s} \quad s \quad \checkmark$$

Expected answer: $\underline{s} \quad s$

▼ Feedback

Score: 3/3 ✓

x1

✓ Your answer is numerically correct. You were awarded 1 mark.

x2

✓ Your answer is numerically correct. You were awarded 1 mark.

x3

✓ Your answer is numerically correct. You were awarded 1 mark.

You scored 3 marks for this part.

Question 4

On the left-hand side below we have the original equations (in augmented form $[A | \mathbf{b}]$) and then the row echelon form on the right.

$$\begin{pmatrix} -3 & -12 & -12 & 54 \\ -4 & -16 & -16 & 72 \\ -3 & -12 & -12 & 54 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 4 & 4 & -18 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The set of solutions, in terms of the parameters s and t , is:

$$x_1 = -18 - 4s - 4t$$

$$x_2 = s$$

$$x_3 = t$$

Express this as the addition of three vectors, a constant vector and two vectors which are multiplied by s and t .

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{pmatrix} \frac{-18}{0} \\ \frac{0}{0} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \frac{-18}{0} \\ \frac{0}{0} \end{pmatrix} + s$$

$$\begin{pmatrix} \frac{-4}{1} \\ \frac{0}{0} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \frac{-4}{1} \\ \frac{0}{0} \end{pmatrix} + t$$

$$\begin{pmatrix} \frac{-4}{0} \\ \frac{1}{1} \end{pmatrix} \checkmark \quad \text{Expected answer: } \begin{pmatrix} \frac{-4}{0} \\ \frac{1}{1} \end{pmatrix}$$

constant**Score: 3/3** 

 Your answer is correct. You were awarded 1 mark.

coeff of s

 Your answer is correct. You were awarded 1 mark.

coeff of t

 Your answer is correct. You were awarded 1 mark.

You scored **3** marks for this part.

Question 5

Use the Gaussian Algorithm and complete the steps below to solve the following for $\mathbf{x}$.

$$\begin{pmatrix} 0 & 1 & -2 & 0 \\ -2 & 2 & -4 & 3 \\ 1 & -1 & 2 & -1 \\ 3 & 0 & 0 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} -1 \\ 5 \\ -2 \\ -6 \end{pmatrix}$$

Enter the augmented matrix that results after following steps 1-4 of the Gaussian Algorithm in structions (p11 of 1.2 in the textbook). You should have a leading 1 in the first row and all entries below it equal to 0.

$$\left(\begin{array}{ccccc} 1 & -1 & 2 & -1 & -2 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 3 & -6 & 3 & 0 \end{array} \right) \checkmark$$

Expected answer:

$$\left(\begin{array}{ccccc} 1 & -1 & 2 & -1 & -2 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 3 & -6 & 3 & 0 \end{array} \right)$$

Continue the steps on the next available column and enter the resulting matrix below (remember there should be a leading 1 in the second row with all entries below it equal to 0).

$$\left(\begin{array}{ccccc} 1 & -1 & 2 & -1 & -2 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 & 3 \end{array} \right) \checkmark$$

Expected answer:

$$\left(\begin{array}{ccccc} 1 & -1 & 2 & -1 & -2 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 3 & 3 \end{array} \right)$$

Enter the final row-echelon augmented matrix

$$\left(\begin{array}{ccccc} 1 & -1 & 2 & -1 & -2 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \checkmark$$

Expected answer:

$$\left(\begin{array}{ccccc} 1 & -1 & 2 & -1 & -2 \\ 0 & 1 & -2 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Now use this matrix to find a vector $\mathbf{x}$ that solves the system. If you have a free variable, set it to zero (e.g., if x_3 is a free variable, then let $x_3 = 0$, and determine the remaining values based on that.

$$\mathbf{x} = \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

Gap 0**Score: 4/4** ✓

- ✓ This step appears correct. The matrix is in the correct form and values are correct. You were awarded **1** mark.

Gap 1

- ✓ This step appears correct. The matrix is in the correct form and values are correct. You were awarded **1** mark.

Gap 2

- ✓ This step appears correct. The matrix is in the correct form and values are correct. You were awarded **1** mark.
- ✓ You were awarded **1** mark.

You scored **4** marks for this part.

Question 6

Consider the following solution vectors for a given homogenous system of linear equations,

$$\mathbf{x}_1 = \begin{pmatrix} -3 \\ 1 \\ -3 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 0 \\ 3 \\ -6 \end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix} 6 \\ -5 \\ 12 \end{pmatrix}$$

If it is possible, express the vector $\mathbf{v} = \begin{pmatrix} 6 \\ -11 \\ 24 \end{pmatrix}$ as a linear combination of $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$.

$$\mathbf{v} = \underline{-2} \begin{pmatrix} -3 \\ 1 \\ -3 \end{pmatrix} + \underline{-3} \begin{pmatrix} 0 \\ 3 \\ -6 \end{pmatrix} + \underline{0} \begin{pmatrix} 6 \\ -5 \\ 12 \end{pmatrix}$$

Expected answer: -2

Expected answer: -3

Expected answer: 0

If it is not possible, enter 0 for all 3 coefficients.

✓ Correct. Your coefficients will result in a linear combination that produces the vector, $\mathbf{V}$. You were awarded **2** marks.

Score: 2/2 ✓

You scored **2** marks for this part.

Created using [Numbas](https://www.numbas.org.uk) (https://www.numbas.org.uk) v10.0, developed by [Newcastle University](https://www.newcastle.ac.uk) (https://www.newcastle.ac.uk).

DEAKIN UNIVERSITY

LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

Module 2 - Matrix Algebra

Submitted By:

Jahan GARG

s226100103

2026/07/31 19:45

Tutor:

Simon JAMES

Task Outcomes		Supports
M2.1	Perform basic operations on matrices	ULO1 ULO1
M2.2	Calculate products of matrices and identify when the operation is valid	ULO1 ULO1
M2.3	Interpret matrices and operations in context	ULO1 ULO1
Evid	Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.	
SA	Full printout of Self-Assessment results with all questions.	
Sum	Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions.	
FB	Respond to any feedback and work with the OnTrack tutor to complete the module.	

July 31, 2026



Module 2 Summary

Response to feedback

This submission is my first attempt for Module 2 – Matrix Algebra in SIT292. I have carefully aligned the module summary, learning evidence, and NUMBAS self-assessment with the unit site guidance and verbal advice from my OnTrack tutor. In particular, I have focused on linking the formal definitions and theorems in the template to worked examples from the textbook and NUMBAS practice questions, so that my summary demonstrates not only knowledge of the statements but also an integrated understanding of how they are applied to concrete problems.

I used AI assistance (Perplexity) to help me structure this Module Summary, refine my explanations of the key theorems and definitions, and check my written English. The mathematical understanding, worked examples, and NUMBAS results referenced throughout are my own.

Module Learning Objectives

I certify that I achieved the following learning objectives for the module (these objectives can be found in the introduction of the module):

- To perform basic operations on matrices, including component-wise addition, scalar multiplication, and transposition.
- To calculate products of matrices and to identify precisely when a matrix product is well-defined based on dimension compatibility.
- To interpret matrices and matrix operations in context, especially as linear transformations acting on vectors and as structured representations of data such as graphs or tabular information.

Key Definitions and Theorems

Matrix addition, scalar multiplication and transpose

Textbook Ref	Theorem/Definition statement	Learning Notes (optional)
Definition 2.1 Matrix Addition	If A and B are matrices of the same size, their <u>sum</u> $A+B$ is the matrix formed by adding corresponding entries, i.e. for the entries a_{ij} and b_{ij} in A and B, the entry in the matrix $A+B$ will be $a_{ij} + b_{ij}$	This operation behaves exactly like vector addition, entry wise, which is why the algebra "feels familiar." I confirmed this concretely in Exercise 2.1.2 and NUMBAS N2.2.1, where subtracting a scaled matrix required computing the scalar multiple first, then subtracting entry wise.

Definition 2.2 Matrix Scalar Multiplication	If A is any matrix and k is any number, the <u>scalar multiple</u> kA is the matrix obtained from A by multiplying each entry of A by k , i.e. ka_{ij} for all entries.	Scalar multiplication scales every entry of a matrix by the same real number, exactly as it would scale a vector, so I can think of kA as repeated addition when k is a positive integer. In NUMBAS N2.2.2, applying constants like -4 and -1 before combining matrices reinforced that the scalar must be distributed across every entry before any addition or subtraction takes place.
Theorem 2.1.1 Properties of matrix addition and multiplication by a scalar	Let A , B , and C denote arbitrary $m \times n$ matrices where m and n are fixed. Let k and p denote arbitrary real numbers. Then 1. $A+B = B+A$ 2. $A+(B+C) = (A+B)+C$ 3. There is an $m \times n$ matrix 0, such that $0+A = A$ for each A 4. For each A there is an $m \times n$ matrix, $-A$, such that $A+(-A) = 0$ 5. $k(A+B) = kA+kB$ 6. $(k+p)A = kA+pA$ 7. $(kp)A = k(pA)$ 8. $1A = A$	1. commutativity 2. associativity 3. additive identity element 4. additive inverse 5. distributivity for scalar times sum of two matrices 6. distributivity for sum of two scalars times a matrix 7. associativity with scalars 8. multiplicative scalar identity element These eight properties are the same field axioms that govern real-number arithmetic, applied entry wise, so commutativity, associativity, and the distributive laws all carry over directly from ordinary algebra. What stood out to me is that properties 3 and 4 (additive identity and additive inverse) guarantee that matrix addition forms a group, a structural fact that becomes important later when discussing vector spaces.
Definition 2.3 Transpose of a matrix	If A is an $m \times n$ matrix, the transpose of A , written A^T , is the $n \times m$ matrix whose rows are just the columns of A in the same order.	I think of the transpose as a reflection of the matrix across its main diagonal, where rows become columns without changing the relative order of entries. This mental picture made it straightforward to predict both the shape and the content of A^T without recomputing anything from scratch.
Theorem 2.1.2 Properties of transpose	Let A and B denote matrices of the same size, and let k denote a scalar.	The transpose commutes with scalar multiplication and distributes over addition, so

	1. If A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix. 2. $(A^T)^T = A$. 3. $(kA)^T = kA^T$. 4. $(A+B)^T = A^T + B^T$.	$(kA)^T = kA^T$ and $(A+B)^T = A^T + B^T$ hold exactly as expected. Both identities held up in Exercise 2.1.3(i), where I simplified $(B - 2E)^T$ by distributing the transpose across the subtraction first, rather than computing $B - 2E$ in full before transposing.
--	--	--

Matrix-vector multiplication

Textbook Ref	Theorem/Definition statement	Learning Notes (optional)
Definition 2.4 The set $\mathbb{R}^n$	Let $\mathbb{R}$ denote the set of all real numbers. The set of all ordered n -tuples from $\mathbb{R}$ is denoted by $\mathbb{R}^n$	This definition formalises what I already treat intuitively as a column vector: an ordered n -tuple of real numbers. Framing vectors this way clarified why matrix-vector multiplication requires the number of columns in A to exactly match the length of x .
Definition 2.5 Matrix-vector multiplication	Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ be an $m \times n$ matrix, written in terms of its columns $\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n$. If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is any n -vector (i.e., $\mathbf{x} \in \mathbb{R}^n$), the product $A\mathbf{x}$ is defined to be the m -vector in $\mathbb{R}^m$ given by: $A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$	Writing A in terms of its columns shows that $A\mathbf{x}$ is a linear combination of those columns, weighted by the entries of $\mathbf{x}$. This column-based viewpoint was the single most useful shift in perspective this module, since I used it directly in NUMBAS N2.3.1 to interpret each output vector as a weighted sum rather than a mechanical row-by-row calculation.
Theorem 2.2.2 Properties of matrix-vector multiplication	Let A and B be $m \times n$ matrices, and let $\mathbf{x}$ and $\mathbf{y}$ be n -vectors in $\mathbb{R}^n$. Then: 1. $A(\mathbf{x}+\mathbf{y}) = A\mathbf{x} + A\mathbf{y}$. 2. $A(a\mathbf{x}) = a(A\mathbf{x}) = (aA)\mathbf{x}$ for all scalars a. 3. $(A+B)\mathbf{x} = A\mathbf{x} + B\mathbf{x}$.	These three properties confirm that $A\mathbf{x}$ is linear in both $\mathbf{x}$ and A , which is exactly what justifies calling $T_A(\mathbf{x}) = A\mathbf{x}$ a linear transformation rather than just a computational shortcut. I relied on this linearity implicitly in NUMBAS N2.3.2 when computing $A\mathbf{u}$ and $A\mathbf{v}$ and then generalising to an arbitrary vector $\mathbf{x}$.
Definition 2.6 Dot product in $\mathbb{R}^n$	If $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$ are two ordered n -tuples in $\mathbb{R}^n$, their dot product is defined to be the number $a_1 b_1 + a_2 b_2 + \dots + a_n b_n$ Obtained by multiplying corresponding entries and adding the results.	The dot product sums the products of corresponding entries of two n -tuples, and it is the single computational step repeated throughout every entry of $A\mathbf{x}$ and later AB . This connected directly back to Module 1, where each equation in a linear system can be read as the dot product of a

		coefficient row with the variable vector.
Theorem 2.2.3 General solution for non-homogenous systems	Suppose $\mathbf{x}_1$ is any <u>particular solution</u> to the system $A\mathbf{x} = \mathbf{b}$ of linear equations. Then every solution $\mathbf{x}_2$ to $A\mathbf{x} = \mathbf{b}$ has the form $\mathbf{x}_2 = \mathbf{x}_0 + \mathbf{x}_1$ for some solution $\mathbf{x}_0$ of the associated <u>homogeneous system</u> $A\mathbf{x} = \mathbf{0}$.	This theorem states that every solution to $A\mathbf{x} = \mathbf{b}$ can be written as a particular solution plus some solution of the associated homogeneous system $A\mathbf{x} = \mathbf{0}$, which means the full solution set is simply a translate of the homogeneous solution space. I verified this numerically in the Module Summary using $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$, confirming that every value of the parameter t still satisfies $A\mathbf{x} = \mathbf{b}$.
Theorem 2.2.4 Relationship between rank of augmented matrix and consistency	Let $A\mathbf{x} = \mathbf{b}$ be a system of equations with augmented matrix $[A \mathbf{b}]$. Let $\text{rank } A = r$. 1. $\text{rank } [A \mathbf{b}]$ is either r or $r + 1$ 2. The system is consistent if and only if $\text{rank } [A \mathbf{b}] = r$ 3. The system is inconsistent if and only if rank $\text{rank } [A \mathbf{b}] = r + 1$	Because the augmented matrix $[A \mathbf{b}]$ contains A as a sub-matrix, its rank can only ever equal r or $r + 1$, and consistency holds precisely when no extra rank is introduced by the column $\mathbf{b}$. This is a cleaner restatement of the rank criterion I first used in Module 1's Gaussian elimination work, now expressed directly in matrix-vector form
Theorem 2.2.5 Dot product rule	Let A be an $m \times n$ matrix and let $\mathbf{x}$ be an n-vector in $\mathbb{R}^n$. Then each entry of the vector $A\mathbf{x}$ is the dot product of the corresponding row of A with $\mathbf{x}$.	This theorem confirms that each entry of $A\mathbf{x}$ is literally the dot product of the corresponding row of A with $\mathbf{x}$, tying Definition 2.6 directly to matrix-vector multiplication. When checking my hand calculations, I used this rule to verify individual entries of $A\mathbf{x}$ one row at a time, which reduced arithmetic errors in longer computations.
Theorem 2.2.6 Uniqueness of matrix-vector multiple	Let A and B be $m \times n$ matrices. If $A\mathbf{x} = B\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$, then $A = B$.	At first this result looked almost trivial, but it is actually a powerful tool: if $A\mathbf{x} = B\mathbf{x}$ holds for every vector $\mathbf{x}$, then A and B must be identical matrices. This lets me prove matrix equalities by comparing their action on vectors rather than checking every entry individually, which becomes especially useful for algebraic proofs.

Definition 2.8 Matrix Transformation T_A	T_A is called the matrix transformation induced by A , with $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ for every $\mathbf{x}$ in $\mathbb{R}^n$	This definition reframed matrices for me: instead of being just grids of numbers, each matrix A defines a linear transformation $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ via $T_A(\mathbf{x}) = A\mathbf{x}$. I made this concrete in NUMBAS N2.3.2 by deriving the explicit formula for $T_A(\mathbf{x})$ in terms of x_1, x_2 and x_3 , after first identifying the correct values of m and n .
---	---	---

Matrix multiplication

Textbook Ref	Theorem/Definition statement	Learning Notes (optional)
Definition 2.9 Matrix Multiplication	Let A be an $m \times n$ matrix, let $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_k]$ be an $n \times k$ matrix, where $\mathbf{b}_j$ is column j of B for each j . The product matrix AB is the $m \times k$ matrix defined as follows: $AB = A[\mathbf{b}_1 \ \mathbf{b}_2 \ \cdots \ \mathbf{b}_k]$ $= [A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_k]$	The product AB is defined so that its j -th column equals A applied to the j -th column of B , meaning matrix multiplication is matrix-vector multiplication applied repeatedly. I used this definition directly in NUMBAS N2.4.1 to compute AB for specific numerical matrices, checking that the resulting matrix's dimensions matched $m \times k$ as expected.
Theorem 2.3.1 Associativity with matrix-matrix-vector multiplication	Let A be an $m \times n$ matrix and let B be an $n \times k$ matrix. Then the product matrix AB is $m \times k$ and satisfies $A(B\mathbf{x}) = (AB)\mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^k$	This theorem shows that $A(B\mathbf{x}) = (AB)\mathbf{x}$, meaning it makes no difference whether I apply B then A to a vector separately, or combine A and B into a single matrix first and then apply that. This validated my intuition that matrix multiplication represents composition of linear transformations, not just an arbitrary computational rule.
Theorem 2.3.2 Dot Product Rule for matrix-matrix multiplication	Let A and B be matrices of sizes $m \times n$ and $n \times k$, respectively. Then the (i, j) -entry of AB is the dot product of row i of A with column j of B .	Generalising Theorem 2.2.5, each entry of AB is the dot product of a row of A with a column of B . I used this rule to verify individual entries in larger products by hand, which was particularly helpful when checking my NUMBAS N2.4.2 working for the 3×3 matrices A and B .
Theorem 2.3.3 Properties of matrix-matrix multiplication	Assume that a is any scalar, and that A , B , and C are matrices of sizes such that the indicated matrix products are defined and I is an identity matrix. Then: $1. IA = A \text{ and } AI = A$	This theorem confirms that identity matrices act as multiplicative identities, that multiplication is associative and distributes over addition from both sides, and that

	2. $A(BC) = (AB)C$. 3. $A(B+C) = AB+AC$. 4. $(B+C)A = BA+CA$. 5. $a(AB) = (aA)B = A(aB)$. 6. $(AB)^T = B^T A^T$	scalars can be factored through a product freely. Property 6 is the one exception to intuition, since $(AB)^T = B^T A^T$ reverses the order of the factors, which I demonstrated numerically in the Module Summary by showing $(AB)^T \neq A^T B^T$ for a specific pair of matrices.
--	---	--

Module Summary

2.1 Matrix Addition, Scalar Multiplication, and Transpose

This submodule establishes that matrices, first encountered in Module 1 purely as containers for coefficients in an augmented matrix, are objects with their own algebra. Matrix Addition (Def 2.1) and Matrix Scalar Multiplication (Def 2.2) extend real-number arithmetic entry wise, and Theorem 2.1.1 formalises exactly which real-number properties survive this extension, namely commutativity, associativity, the existence of a zero matrix and additive inverses, and the two distributive laws linking scalars and sums. These eight properties together confirm that matrices form an additive group, a structural fact that becomes important once vector spaces are introduced later in the unit. The Transpose (Def 2.3) then introduces the first operation that is not simply entry wise, since it rearranges the shape of the matrix itself, and Theorem 2.1.2 shows that despite this structural change, the transpose still interacts predictably with addition and scalar multiplication, satisfying $(kA)^T = kA^T$ and $(A + B)^T = A^T + B^T$.

The unit site's opening examples, a table of student assignment scores and a graph's adjacency matrix, were useful here, since they showed that the same entry wise rules apply whether the matrix represents marks, connections, or coefficients. I tested this entire chain of definitions and theorems directly in Exercise 2.1.2, 2.1.3, and NUMBAS N2.2.1 to N2.2.2, where every simplification relied on applying Theorem 2.1.1 or Theorem 2.1.2 before evaluating entries numerically, and both activities were completed with full marks in my Learning Evidence.

2.2 Matrix-Vector Multiplication

This submodule bridges Module 1's row-based view of linear systems with a column-based view of matrices acting on vectors in $\mathbb{R}^n$ (Def 2.4). Matrix-Vector Multiplication (Def 2.5) defines Ax as a linear combination of the columns of A , and Theorem 2.2.2 confirms that this operation is linear in both x and A , which is exactly the algebraic property that justifies Definition 2.8's claim that $T_A(x) = Ax$ is a genuine linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$. The Dot Product (Def 2.6) and the Dot Product Rule (Theorem 2.2.5) work together to explain the mechanics behind Definition 2.5, since

each entry of Ax is nothing more than a row of A dotted with x , which is the same row-based computation I already practised in Module 1's Gaussian elimination. Theorem 2.2.6 then closes the loop by showing that a matrix is uniquely determined by its action on every vector, so that A and B are equal precisely when $Ax = Bx$ for all x .

Theorem 2.2.3 and Theorem 2.2.4 reconnect this submodule to Module 1's consistency criteria, since the general solution to $Ax = b$ is a particular solution plus any solution of $Ax = 0$ (Theorem 2.2.3), while consistency itself is governed by comparing $\text{rank}[A \mid b]$ with $\text{rank}(A)$ (Theorem 2.2.4). I applied this entire chain in NUMBAS N2.3.1 and N2.3.2, where I computed Au, Av, Bu, Bv , identified the dimensions $m = 3, n = 2$ for a specific transformation, and derived $T_A(x)$ explicitly, scoring full marks across every part of this activity.

Task 3, numeric example for Theorem 2.2.3:

Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$.

A particular solution is $x_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, since $Ax_1 = \begin{bmatrix} 1(3) + 1(0) \\ 2(3) + 2(0) \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = b$.

The homogeneous system $Ax = 0$ requires $x + y = 0$, which gives the one-parameter family of solutions $x_0 = t * \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, for any real number t .

By Theorem 2.2.3, the complete solution set to $Ax = b$ is therefore

$$x_2 = x_1 + x_0 = \begin{bmatrix} 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3+t \\ -t \end{bmatrix}$$

Substituting x_2 back into $Ax = b$ confirms this holds for every value of t , since

$$Ax_2 = \begin{bmatrix} (3+t) + (-t) \\ 2(3+t) + 2(-t) \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = b$$

This verifies Theorem 2.2.3 numerically in matrix form.

2.3 Matrix Multiplication

This final submodule generalises Section 2.2's matrix-vector multiplication into full matrix-matrix multiplication. Matrix Multiplication (Def 2.9) defines AB so that its j -th column is simply A applied to the j -th column of B , which is Definition 2.5 applied repeatedly, and Theorem 2.3.2 restates this entrywise as a row-column dot product, directly generalising Theorem 2.2.5. Theorem 2.3.1 then confirms that this new operation is fully compatible with the transformation view from Definition 2.8, since $A(Bx) = (AB)x$ shows that composing two transformations matches multiplying their matrices first. Theorem 2.3.3 collects the properties that govern this composition: associativity, distributivity from both sides, scalar compatibility, the identity property $IA = A = AI$, and, most notably, the order-reversing transpose rule $(AB)^T = B^T A^T$,

which is the one property in this entire module that breaks the "matrices behave like real numbers" intuition established back in Section 2.1.

Task 1, ways of approaching matrix-matrix multiplication (Section 2.4 video):

The video presents three equivalent views of $AB = C$, each tied to a definition already covered. The dot-product view treats c_{ij} as row i of A dotted with column j of B , which matches Theorem 2.3.2 and Definition 2.9. The mixing-of-rows view treats each row of C as a linear combination of the rows of B , weighted by the corresponding row of A , which extends Definition 2.5 applied row-wise. The mixing-of-columns view treats each column of C as a linear combination of the columns of A , weighted by the corresponding column of B , which is Definition 2.5 applied directly. I confirmed the columns view explicitly in NUMBAS N2.4.2, expressing a column of A^2 as $3a_1 + 6a_2$ and a column of B^2 as $-2b_1 + b_2 + 7b_3$, both marked fully correct.

Task 2, numeric example for Theorem 2.3.3, property 6:

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Then $AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$. Computing $B^T A^T$ gives $\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$, which matches $(AB)^T$ exactly. By contrast, $A^T B^T = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$, which does not equal $(AB)^T$, confirming that the order reversal in property 6 is essential rather than optional.

Reflecting on the content:

The most conceptually rewarding shift in this module was moving from viewing matrices as static tables, which was Module 1's view, to viewing them as linear transformations through Definition 2.8. NUMBAS N2.3.2 made this concrete, since it required me to derive $T_A(x)$ explicitly rather than just compute numeric outputs. The trickiest idea was the order reversal in $(AB)^T = B^T A^T$, since it runs against the real-number intuition built up in Section 2.1, and working through Task 2 numerically is what finally made it click. Sections 2.2 and 2.3 also gave me a much clearer picture of why Module 1's rank and consistency criteria work the way they do, since Theorem 2.2.3 and Theorem 2.2.4 are the same ideas expressed through $Ax = b$ rather than row reduction directly. Completing the NUMBAS Self-Assessment with a full 10 out of 10 score across all five questions confirmed that this integrated understanding, rather than isolated memorisation of each definition, was the right way to approach the module.

References:

- W. K. Nicholson, *Linear Algebra with Applications* (Open Edition), Sections 2.1 to 2.3, SIT292 Unit Site, Deakin University.
- Own worked exercises and NUMBAS activity results, SIT292 Module 2 Learning Evidence and Self-Assessment.

PRACTICE QUESTIONS

DATE / /

PAGE

2.2 Matrix addition, scalar multiplication, and transposition

Exercise 2.1.2 Compute the following:

(a) $\begin{bmatrix} 3 & 2 & 1 \\ 5 & 1 & 0 \end{bmatrix} - 5 \begin{bmatrix} 3 & 0 & -2 \\ 1 & -1 & 2 \end{bmatrix}$

Ans: $\begin{bmatrix} 3 & 2 & 1 \\ 5 & 1 & 0 \end{bmatrix}_{2 \times 3} - \begin{bmatrix} 15 & 0 & -10 \\ 5 & -5 & 10 \end{bmatrix}_{2 \times 3}$

$= \begin{bmatrix} 3-15 & 2-0 & 1-(-10) \\ 5-5 & 1-(-5) & 0-10 \end{bmatrix}_{2 \times 3}$

$= \begin{bmatrix} -12 & 2 & 11 \\ 0 & 6 & -10 \end{bmatrix}_{2 \times 3}$

(g) $\begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}^T$

Ans: $\begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}_{2 \times 2} - \begin{bmatrix} 2 & -4 \\ 2 & 2 \end{bmatrix}_{2 \times 2}^T$

Using $k(A^T) = (kA)^T$

$= \begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}_{2 \times 2} - \begin{bmatrix} 2 & 2 \\ -4 & 2 \end{bmatrix}_{2 \times 2}$

$= \begin{bmatrix} 3-2 & -1-2 \\ 2-(-4) & 1-2 \end{bmatrix}_{2 \times 2}$

$= \begin{bmatrix} 1 & -3 \\ 6 & -1 \end{bmatrix}_{2 \times 2}$

Exercise 2.1.3 Let $A = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$,

$$D = \begin{bmatrix} 1 & 3 \\ -1 & 0 \\ 1 & 4 \end{bmatrix}, \text{ and } E = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

compute the following (where possible).

(c) $3E^T$

Ans: $3 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_{2 \times 3}^T = 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}_{3 \times 2} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \\ 3 & 0 \end{bmatrix}_{3 \times 2}$

(e) $4A^T - 3C$

Ans: $4 \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}_{2 \times 2}^T - 3 \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 8 & 0 \\ 4 & -4 \end{bmatrix}_{2 \times 2} - \begin{bmatrix} 9 & -3 \\ 6 & 0 \end{bmatrix}_{2 \times 2}$

$$= \begin{bmatrix} 8-9 & 0-(-3) \\ 4-6 & -4-0 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} -1 & 3 \\ -2 & -4 \end{bmatrix}_{2 \times 2}$$

(i) $(B-2E)^T$

Ans: $B^T - 2E^T$

Using $(A+B)^T = A^T + B^T$

$$= \begin{bmatrix} 3 & -1 & 2 \\ 0 & 1 & 4 \end{bmatrix}_{2 \times 3}^T - 2 \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}_{2 \times 3}^T$$

$$= \begin{bmatrix} 3 & 0 \\ -1 & 1 \\ 2 & 4 \end{bmatrix}_{3 \times 2} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}_{3 \times 2} = \begin{bmatrix} 3 & 0 \\ -1 & 1 \\ 2 & 4 \end{bmatrix}_{3 \times 2} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 2 & 0 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} 3-2 & 0-0 \\ -1-0 & 1-2 \\ 2-2 & 4-0 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} 1 & 0 \\ -1 & -1 \\ 0 & 4 \end{bmatrix}_{3 \times 2}$$

Exercise 2.1.14 In each case determine all s and t such that the given matrix is symmetric:

$A = A^T$ (If A is a symmetric matrix)

(b) $\begin{bmatrix} s & t \\ st & 1 \end{bmatrix}$

Ans: $A = \begin{bmatrix} s & t \\ st & 1 \end{bmatrix}_{2 \times 2}$ and $A^T = \begin{bmatrix} s & st \\ t & 1 \end{bmatrix}_{2 \times 2}$

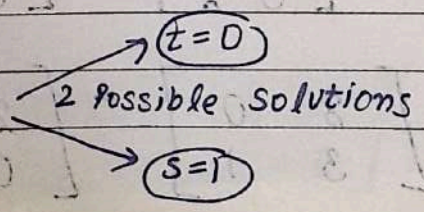
Since $A = A^T$, $\begin{bmatrix} s & t \\ st & 1 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} s & st \\ t & 1 \end{bmatrix}_{2 \times 2}$

From comparing the values in both the matrices,

$\Rightarrow t = st$

$\Rightarrow t - st = 0$

$\Rightarrow t(1-s) = 0$



(d)
$$\begin{bmatrix} 2 & s & t \\ 2s & 0 & s+t \\ 3 & 3 & t \end{bmatrix}$$

Ans: $A = \begin{bmatrix} 2 & s & t \\ 2s & 0 & s+t \\ 3 & 3 & t \end{bmatrix}_{3 \times 3}$ and $A^T = \begin{bmatrix} 2 & 2s & 3 \\ s & 0 & 3 \\ t & s+t & t \end{bmatrix}_{3 \times 3}$

From comparing the values in both the matrices,

$\Rightarrow t = 3 \quad \text{--- (1)}$
 $\Rightarrow s = 2s \quad \text{--- (2)}$
 $\Rightarrow s + t = 3 \quad \text{--- (3)}$

By substituting the value of t in (3) eqⁿ,

$\Rightarrow s + 3 = 3$
 $\Rightarrow s = 3 - 3$
 $\Rightarrow s = 0$

The value of s also satisfies (2) eqⁿ, i.e., $s = 2s$.

Exercise 2.1.15 In each case find the matrix A .

Ans: (b) $(3A^T + 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix})^T = \begin{bmatrix} 8 & 0 \\ 3 & 1 \end{bmatrix}$

L.H.S. R.H.S.

Using
 $(A+B)^T = A^T + B^T$

$\Rightarrow 3(A^T)^T + 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^T = \begin{bmatrix} 8 & 0 \\ 3 & 1 \end{bmatrix}$

$\Rightarrow 3A + 2 \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 3 & 1 \end{bmatrix}$

$\Rightarrow 3A = \begin{bmatrix} 8 & 0 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$

$$\Rightarrow 3A = \begin{bmatrix} 8-2 & 0-0 \\ 3-0 & 1-4 \end{bmatrix}$$

$$\Rightarrow 3A = \begin{bmatrix} 6 & 0 \\ 3 & -3 \end{bmatrix}$$

$$\Rightarrow 3A = 3 \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$$

$$\textcircled{d} \left(2A^T - 5 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \right)^T = 4A - 9 \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\underline{\text{Ans:}} \quad 2(A^T)^T - 5 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}^T = 4A - \begin{bmatrix} 9 & 9 \\ -9 & 0 \end{bmatrix}$$

$$\Rightarrow 2A - 5 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 4A - \begin{bmatrix} 9 & 9 \\ -9 & 0 \end{bmatrix}$$

$$\Rightarrow -4A + 2A = - \begin{bmatrix} 9 & 9 \\ -9 & 0 \end{bmatrix} + 5 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

As matrices
can be added
we subtracted like scalars.

$$\Rightarrow -2A = - \begin{bmatrix} 9 & 9 \\ -9 & 0 \end{bmatrix} + \begin{bmatrix} 5 & -5 \\ 0 & 10 \end{bmatrix}$$

$$\Rightarrow -2A = \begin{bmatrix} -9+5 & -9-5 \\ +9+0 & 0+10 \end{bmatrix}$$

$$\Rightarrow -2A = \begin{bmatrix} -4 & -14 \\ 9 & 10 \end{bmatrix}$$

$$\Rightarrow +2A = \begin{bmatrix} 2 & 7 \\ -9/2 & -5 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 2 & 7 \\ -9/2 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & | & 0 & 0 \\ 0 & 1 & | & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & | & 0 & -1 \\ 0 & 1 & | & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} p & p & | & 0 & 0 \\ 0 & p-1 & | & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} p & p & | & 0 & -1 \\ 0 & p-1 & | & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} p & p & | & 0 & -1 \\ 0 & p-1 & | & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{bmatrix} p & p & | & 0 & 0 \\ 0 & p-1 & | & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & | & 0 & 0 \\ 0 & p-1 & | & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{bmatrix} 1 & p & | & 0 & 1 \\ 0 & p-1 & | & 0 & 1 \end{bmatrix}$$

→

$$\begin{bmatrix} 2 & 2 & | & 0 & 0 \\ 0 & 0 & | & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 2 & 2 & | & 0 & 0 \\ 0 & 0 & | & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 & | & 0 & 0 \\ 0 & 0 & | & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1/2} \begin{bmatrix} 1 & 1 & | & 0 & 0 \\ 0 & 0 & | & 0 & 1 \end{bmatrix}$$

Question 1

Calculate $\begin{pmatrix} 3 & 3 & 0 \\ -3 & -5 & 5 \end{pmatrix} + \begin{pmatrix} 5 & 3 & -2 \\ 5 & -5 & -3 \end{pmatrix} =$

Rows: Columns:

$\begin{pmatrix} \boxed{8} & \boxed{6} & \boxed{-2} \\ \boxed{2} & \boxed{-10} & \boxed{2} \end{pmatrix}$ ✓

Your answer is correct. You were awarded **5** marks. ✓

You scored **5** marks for this part.

Score: 5/5 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

a)

Calculate the following (use the controls at the top to set the size of the matrix)

Rows: Columns:

$$5 \begin{pmatrix} 2 & -4 \\ -5 & -1 \\ 1 & -3 \end{pmatrix} + 4 \begin{pmatrix} 1 & -4 \\ 4 & -1 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} \boxed{14} & \boxed{-36} \\ \boxed{-9} & \boxed{-9} \\ \boxed{1} & \boxed{-23} \end{pmatrix} \quad \checkmark$$

Your answer is correct. You were awarded **5** marks.

You scored **5** marks for this part.

Score: 5/5

b)

Calculate the following

Rows: Columns:

$$\begin{pmatrix} 5 & 5 & -2 \\ -3 & 5 & -5 \end{pmatrix}^T = \begin{pmatrix} \boxed{5} & \boxed{-3} \\ \boxed{5} & \boxed{5} \\ \boxed{-2} & \boxed{-5} \end{pmatrix} \quad \checkmark$$

Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

Score: 1/1

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Calculate the following:

$$\begin{pmatrix} -3 & 3 & 3 & -3 \\ 2 & 0 & -2 & -2 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \\ -2 \\ -2 \end{pmatrix} = \begin{pmatrix} \boxed{6} \\ \boxed{4} \end{pmatrix}$$

Rows: Columns:

✓

Your answer is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Calculate the following:

Rows: Columns:

$$\begin{pmatrix} 3 & 0 & 2 \\ 3 & -1 & 0 \\ -3 & -2 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \boxed{-6} \\ \boxed{-6} \\ \boxed{6} \end{pmatrix} \quad \checkmark$$

Your answer is correct. You were awarded 1 mark. 

You scored 1 mark for this part.

Score: 1/1 

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Let $A = \begin{pmatrix} -1 & -3 & -2 \\ 1 & -3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $u = \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}$ and $v = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$.

a)

Compute:

Rows: Columns:

$$Au = \begin{pmatrix} \boxed{-7} \\ \boxed{-11} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$Av = \begin{pmatrix} \boxed{-5} \\ \boxed{5} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$Bu = \begin{pmatrix} \boxed{0} \\ \boxed{0} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$Bv = \begin{pmatrix} \boxed{0} \\ \boxed{0} \end{pmatrix} \quad \checkmark$$

Score: 4/4 

▼ Hide feedback

Gap 0

✓ Your answer is correct. You were awarded **1** mark.

Gap 1

✓ Your answer is correct. You were awarded **1** mark.

Gap 2

✓ Your answer is correct. You were awarded **1** mark.

Gap 3

✓ Your answer is correct. You were awarded **1** mark.

You scored **4** marks for this part.

b)

The matrix transformation of A is $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ for which numbers n and m ?

$n =$ ✓ and $m =$ ✓

Score: 2/2 ✓

▼ Hide feedback

m

✓ Your answer is correct. You were awarded **1** mark.

n

✓ Your answer is correct. You were awarded **1** mark.

You scored **2** marks for this part.

c)

For a general vector $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, determine $T_A(x)$. Input x_1 as x_1 or $x1$,

etc. Use the same notation for all variables, don't mix it.

$$T_A(x) = \begin{pmatrix} \boxed{-x_1 - 3x_2 - 2x_3} & -x_1 - 3x_2 - 2x_3 \checkmark \\ \boxed{x_1 - 3x_2 + 2x_3} & x_1 - 3x_2 + 2x_3 \checkmark \end{pmatrix}$$

Score: 3/3 ✓

▼ Hide feedback

✓ Your answer is correct. You were awarded **3** marks.You scored **3** marks for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

$$\text{Let } A = \begin{pmatrix} -1 & -2 \\ 2 & -2 \\ 2 & -2 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, u = \begin{pmatrix} -1 \\ -2 \end{pmatrix} \text{ and } v = \begin{pmatrix} -3 \\ 0 \end{pmatrix}.$$

a)

Compute:

Rows: Columns:

$$Au = \begin{pmatrix} \boxed{5} \\ \boxed{2} \\ \boxed{2} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$Av = \begin{pmatrix} \boxed{3} \\ \boxed{-6} \\ \boxed{-6} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$Bu = \begin{pmatrix} \boxed{0} \\ \boxed{0} \\ \boxed{0} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$Bv = \begin{pmatrix} \boxed{0} \\ \boxed{0} \\ \boxed{0} \end{pmatrix} \quad \checkmark$$

Score: 4/4 

▼ Hide feedback

Gap 0 Your answer is correct. You were awarded **1** mark.**Gap 1** Your answer is correct. You were awarded **1** mark.**Gap 2** Your answer is correct. You were awarded **1** mark.**Gap 3** Your answer is correct. You were awarded **1** mark.You scored **4** marks for this part.

b)

The matrix transformation of A is $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ for which numbers n and m ? $n = \boxed{2} \checkmark$ and $m = \boxed{3} \checkmark$ Score: 2/2 

▼ Hide feedback

m Your answer is correct. You were awarded **1** mark.**n** Your answer is correct. You were awarded **1** mark.

You scored **2** marks for this part.

c)

For a general vector $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, determine $T_A(x)$. Input x_1 as x_1 or $x1$,

etc. Use the same notation for all variables, don't mix it.

$$T_A(x) = \begin{pmatrix} \boxed{-x1-2x2} & -x_1 - 2x_2 & \checkmark \\ \boxed{2x1-2x2} & 2x_1 - 2x_2 & \checkmark \\ \boxed{2x1-2x2} & 2x_1 - 2x_2 & \checkmark \end{pmatrix}$$

Score: 3/3 

▼ Hide feedback

 Your answer is correct. You were awarded **3** marks.

You scored **3** marks for this part.

Created using [Numbas \(https://www.numbas.org.uk/\)](https://www.numbas.org.uk/), developed by [Newcastle University \(http://www.newcastle.ac.uk/\)](http://www.newcastle.ac.uk/).

Question 1

Consider the following nutrition table.

	Fat	Sodium	Carbohydrates
Banana	0.2 g	1.9mg	25 g
Cornflakes (bowl)	1.3g	200.8 mg	23g
Milk (cup)	2.1 g	105.2mg	1.9 g

The information in the table is represented as a matrix A , i.e.,

$$A = \begin{pmatrix} 0.2 & 1.9 & 25 \\ 1.3 & 200.8 & 23 \\ 2.1 & 105.2 & 10 \end{pmatrix}$$

Suppose we have a points system that scores 1.7 points per gram of fat, 0.7 per mg of sodium, and 1.4 per gram of carbohydrates. Which of the following will give the total points scored by a banana, bowl of cornflakes, and cup of milk?

☐ $\begin{pmatrix} 1.7 \\ 0.7 \\ 1.4 \end{pmatrix} A$
☐ $(1.7 \ 0.7 \ 1.4) A$
☒ $A \begin{pmatrix} 1.7 \\ 0.7 \\ 1.4 \end{pmatrix}$

☐ $A (1.7 \ 0.7 \ 1.4)$



You chose a correct answer. You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Consider the following nutrition table.

	Banana	Cornflakes (bowl)	Milk (cup)
Fat	0.2 g	0.7g	2.5 g
Sodium	1.2mg	242.4 mg	116.2mg
Carbohydrates	26 g	24g	1.2 g

The information in the table is represented as a matrix A, i.e.,

$$A = \begin{pmatrix} 0.2 & 0.7 & 2.5 \\ 1.2 & 242.4 & 116.2 \\ 26 & 24 & 12 \end{pmatrix}$$

Suppose we have a points system that scores 3 points per gram of fat, 1.5 per mg of sodium, and 1.8 per gram of carbohydrates. Which of the following will give the total points scored by a banana, bowl of cornflakes, and cup of milk?

☒ $(3 \ 1.5 \ 1.8) A$
☐ $A \begin{pmatrix} 3 \\ 1.5 \\ 1.8 \end{pmatrix}$
☐ $A (3 \ 1.5 \ 1.8)$

☐ $\begin{pmatrix} 3 \\ 1.5 \\ 1.8 \end{pmatrix} A$



You chose a correct answer. You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

$$\text{Let } A = \begin{pmatrix} 2 & 2 & -5 & -5 \\ -5 & 0 & 2 & -4 \end{pmatrix}, B = \begin{pmatrix} 2 & 5 & 4 \\ 3 & 5 & -2 \\ 4 & 1 & -2 \\ 1 & -3 & 3 \end{pmatrix}.$$

Rows: Columns:

$$\text{Compute } AB = \begin{pmatrix} \boxed{-15} & \boxed{30} & \boxed{-1} \\ \boxed{-6} & \boxed{-11} & \boxed{-36} \end{pmatrix} \quad \checkmark$$

Your answer is correct. You were awarded **3** marks. 

You scored **3** marks for this part.

Score: 3/3 

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

$$\text{Let } A = \begin{pmatrix} -1 & -1 & 2 \\ -2 & 2 & 4 \\ -4 & -3 & -5 \end{pmatrix}, B = \begin{pmatrix} 4 & -2 & 4 \\ 1 & 1 & -2 \\ 2 & 1 & -5 \end{pmatrix}.$$

Rows: Columns:

$$\text{Compute } AB = \begin{pmatrix} -1 & 3 & -12 \\ 2 & 10 & -32 \\ -29 & 0 & 15 \end{pmatrix} \quad \checkmark$$

Your answer is correct. You were awarded **3** marks. 

You scored **3** marks for this part.

Score: 3/3 

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Let $A = \begin{pmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{pmatrix}$.

a)

Compute

$$A^2 = \begin{pmatrix} \boxed{-3} & \boxed{12} & \boxed{76} \\ \boxed{48} & \boxed{29} & \boxed{98} \\ \boxed{24} & \boxed{56} & \boxed{97} \end{pmatrix} \quad \checkmark$$

$$B^2 = \begin{pmatrix} \boxed{64} & \boxed{14} & \boxed{38} \\ \boxed{21} & \boxed{22} & \boxed{18} \\ \boxed{77} & \boxed{28} & \boxed{74} \end{pmatrix} \quad \checkmark$$

A²

Your answer is correct. You were awarded 1 mark. 

B²

Your answer is correct. You were awarded 1 mark. 

You scored 2 marks for this part.

Score: 2/2 

b)

Express the first column vector of A^2 as a linear combination of the column vectors of A . You can write a_1 , a_2 , a_3 for the three columns of A , i.e.

$$a_1 = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}, a_2 = \begin{pmatrix} -2 \\ 5 \\ 4 \end{pmatrix} \text{ and } a_3 = \begin{pmatrix} 7 \\ 4 \\ 9 \end{pmatrix}.$$

First column of A^2 is $3a_1 + 6a_2$ ✓

Your answer is numerically correct. You were awarded **2** marks. ✓

You scored **2** marks for this part.

Score: 2/2 ✓

c)

Express the second column vector of B^2 as a linear combination of the column vectors of B . You can write b_1 , b_2 , b_3 for the three columns of B , i.e.

$$b_1 = \begin{pmatrix} 6 \\ 0 \\ 7 \end{pmatrix}, b_2 = \begin{pmatrix} -2 \\ 1 \\ 7 \end{pmatrix} \text{ and } b_3 = \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}.$$

second column of B^2 is $-2b_1 + b_2 + 7b_3$ ✓

Your answer is numerically correct. You were awarded **2** marks. ✓

You scored **2** marks for this part.

Score: 2/2 ✓

SIT292 Module 2 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	4/4	
Question 2	1/1	
Question 3	2/2	
Question 4	2/2	
Question 5	1/1	
Total	10/10	(100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 2 Self-Assessment
Session ID:	7280758218555642
Exam Start:	Fri Jul 31 2026 01:29:53
Exam Stop:	Fri Jul 31 2026 01:43:30
Time Spent:	0:13:37

Created using [Numbas](https://www.numbas.org.uk) (<https://www.numbas.org.uk>), developed by [Newcastle University](https://www.newcastle.ac.uk) (<https://www.newcastle.ac.uk>).

Question 1

a)

Calculate the following (use the controls at the top to set the size of the matrix)

$$-1 \begin{pmatrix} 5 & -5 \\ 1 & 0 \\ -2 & 3 \end{pmatrix} - 1 \begin{pmatrix} 2 & -5 \\ -1 & 0 \\ 3 & -5 \end{pmatrix} =$$

Rows: Columns:

$$\begin{pmatrix} -7 & 10 \\ 0 & 0 \\ -1 & 2 \end{pmatrix}$$



Expected answer: $\begin{pmatrix} -7 & 10 \\ 0 & 0 \\ -1 & 2 \end{pmatrix}$

Score: 2/2 ✓

▼ Hide feedback

✓ Your answer is correct. You were awarded **2** marks.

You scored **2** marks for this part.

b)

Calculate the following

$$\begin{pmatrix} 4 & -3 \\ 4 & 0 \\ 0 & 0 \\ -1 & 0 \end{pmatrix}^T =$$

 Rows: Columns:

$$\begin{pmatrix} \boxed{4} & \boxed{4} & \boxed{0} & \boxed{-1} \\ \boxed{-3} & \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$$

Expected answer: $\begin{pmatrix} \frac{4}{-3} & \frac{4}{0} & \frac{0}{0} & \frac{-1}{0} \end{pmatrix}$

Score: 2/2 ✓

▼ Hide feedback

✓ Your answer is correct. You were awarded **2** marks.You scored **2** marks for this part.

Question 2

Let $A = \begin{pmatrix} -2 & -1 \\ -2 & 1 \\ 3 & 0 \\ 0 & 2 \end{pmatrix}$

The matrix transformation of A is $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ for which numbers n and m ?

$n =$ ✓Expected answer: and $m =$ ✓Expected answer:

Score: 1/1 ✓

 m ✓ Your answer is correct. You were awarded **0.5** marks. n ✓ Your answer is correct. You were awarded **0.5** marks.You scored **1** mark for this part.

Question 3

Let $A = \begin{pmatrix} -1 & 0 & -2 \\ 0 & -3 & 3 \\ 4 & 0 & -1 \end{pmatrix}$

For a general vector $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, determine $T_A(x)$. Input x_1 as x_1 or $x1$, etc.

Use the same notation for all variables, don't mix it.

$$T_A(x) = \begin{pmatrix} \boxed{-x_1 - 2x_3} - x_1 - 2x_3 \checkmark \\ \text{Expected answer: } \underline{-x_1 - 2x_3} - x_1 - 2x_3 \\ \boxed{-3x_2 + 3x_3} - 3x_2 + 3x_3 \checkmark \\ \text{Expected answer: } \underline{-3x_2 + 3x_3} - 3x_2 + 3x_3 \\ \boxed{4x_1 - x_3} 4x_1 - x_3 \checkmark \\ \text{Expected answer: } \underline{4x_1 - x_3} 4x_1 - x_3 \end{pmatrix}$$

Score: 2/2 ✓

✓ Your answer is correct. You were awarded 2 marks.

You scored 2 marks for this part.

Question 4

Let $A = \begin{pmatrix} 5 & 2 & 2 \\ 4 & 0 & 4 \end{pmatrix}$, $B = \begin{pmatrix} -3 & -2 & 0 & 5 \\ -4 & -4 & -3 & -5 \\ -4 & -2 & -3 & 3 \end{pmatrix}$.

Compute $AB =$

Rows: Columns:

$$\begin{pmatrix} \boxed{-31} & \boxed{-22} & \boxed{-12} & \boxed{21} \\ \boxed{-28} & \boxed{-16} & \boxed{-12} & \boxed{32} \end{pmatrix} \checkmark$$

Expected answer: $\begin{pmatrix} \underline{-31} & \underline{-22} & \underline{-12} & \underline{21} \\ \underline{-28} & \underline{-16} & \underline{-12} & \underline{32} \end{pmatrix}$

Score: 2/2 ✓

✓ Your answer is correct. You were awarded **2** marks.

You scored **2** marks for this part.

Question 5

Consider the following nutrition table.

	Fat	Sodium	Carbohydrates
Banana	0.2 g	1.7mg	25 g
Cornflakes (bowl)	1.2g	202.8 mg	24g
Milk (cup)	2.4 g	98.7mg	1.7 g

The information in the table is represented as a matrix A, i.e.,

$$A = \begin{pmatrix} 0.2 & 1.7 & 25 \\ 1.2 & 202.8 & 24 \\ 2.4 & 98.7 & 13 \end{pmatrix}$$

Which of the following operations will provide us with the nutrition information for a breakfast that consists of 1.4 bananas, 2.5 bowls of cornflakes, and 1.3 cups of milk?

- ☐ $\begin{pmatrix} 1.4 \\ 2.5 \\ 1.3 \end{pmatrix} A$
☐ $A \begin{pmatrix} 1.4 \\ 2.5 \\ 1.3 \end{pmatrix}$
☒ $(1.4 \ 2.5 \ 1.3) A$
- ☐ $A (1.4 \ 2.5 \ 1.3)$



Expected answer:

- ☐ $\begin{pmatrix} 1.4 \\ 2.5 \\ 1.3 \end{pmatrix} A$ ☐ $A \begin{pmatrix} 1.4 \\ 2.5 \\ 1.3 \end{pmatrix}$ ☒ $(1.4 \ 2.5 \ 1.3) A$
- ☐ $A (1.4 \ 2.5 \ 1.3)$

Score: 1/1 ✓

✓ You chose a correct answer. You were awarded **1** mark.

You scored **1** mark for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<https://www.newcastle.ac.uk>).

DEAKIN UNIVERSITY

LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

Module 3 - Transformations and Special Matrices

Submitted By:
Jahan GARG
s226100103
2026/08/15 20:59

Tutor:
Simon JAMES

Task Outcomes		Supports
M3.1	Apply the inversion algorithm to find a matrix inverse	ULO1 ULO1
M3.2	Identify and work with elementary matrices	ULO1 ULO1
M3.3	Work with matrices algebraically and identify correspondence between key linear mappings and their matrices	ULO1 ULO1
Evid	Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.	
SA	Full printout of Self-Assessment results with all questions.	
Sum	Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions.	
FB	Respond to any feedback and work with the OnTrack tutor to complete the module.	

August 15, 2026



SIT292 – Linear Algebra for Data Analysis

Module 3 Summary: Transformations and Special Matrices

Response to Feedback

This is my first submission for Module 3, so no prior feedback is being addressed here. For this submission, I would particularly value feedback on whether my justification for why a non-square matrix cannot possess a two-sided inverse (addressed in the Matrix Inverses subsection below) is expressed with sufficient rigour, and whether my strategy for distinguishing rotation matrices from reflection matrices in N3.4.2 aligns with the standard I am expected to demonstrate at this stage of the unit. I have also kept in mind Simon's feedback on my Module 2 submission, namely that the summary should stay reasonably concise while still covering every definition and theorem listed in the Key Definitions and Theorems section, and that the optional Learning Notes column need not be filled in for every entry. I have applied this guidance throughout the present submission.

Module Learning Objectives

I certify that I have achieved the following learning objectives for this module, as set out on the Module 3 introduction page of the unit site [1]:

1. Apply the inversion algorithm to find a matrix inverse.
2. Identify and work with elementary matrices.
3. Work with matrices algebraically and identify correspondence between key linear mappings and their matrices.

Key Definitions and Theorems

Matrix Inverses

Textbook Reference	Theorem / Definition Statement
Definition 2.11 Matrix Inverse	If A is a square matrix, a matrix B is called an inverse of A if and only if $AB = I$ and $BA = I$. A matrix A that possesses an inverse is called invertible, and its inverse is denoted A^{-1} .
Theorem 2.4.1 Uniqueness of Inverse	If B and C are both inverses of A , then $B = C$. This guarantees that whenever an inverse exists, it is the single, well-defined matrix A^{-1} .
Matrix Inversion Algorithm	If A is an invertible $n \times n$ matrix, a sequence of elementary row operations carries A to the identity matrix I , written $A \rightarrow I$. Applying this same sequence to I carries it to A^{-1} , that is, $I \rightarrow A^{-1}$. In augmented form: $[A \mid I] \rightarrow [I \mid A^{-1}]$.
Theorem 2.4.4 Inverse Properties	For square matrices of the same size: (1) I is invertible with $I^{-1} = I$; (2) if A is invertible, so is A^{-1} , with $(A^{-1})^{-1} = A$; (3) if A and B are invertible, so is AB , with $(AB)^{-1} = B^{-1}A^{-1}$; (4) the same extends to a product of several invertible matrices, reversing their order; (5) if A is invertible, so is A^k for k greater than or equal to 1, with $(A^k)^{-1} = (A^{-1})^k$; (6) if A is invertible and a is not zero, then aA is invertible with $(aA)^{-1} = (1/a)A^{-1}$; (7) if A is invertible, so is A^T , with $(A^T)^{-1} = (A^{-1})^T$.

Textbook Reference	Theorem / Definition Statement
Theorem 2.4.5 Inverse Theorem	For an $n \times n$ matrix A , the following are equivalent: (1) A is invertible; (2) $Ax = 0$ has only the trivial solution $x = 0$; (3) A can be carried to I_n by elementary row operations; (4) $Ax = b$ has at least one solution for every column b ; (5) there exists an $n \times n$ matrix C such that $AC = I_n$.

Elementary Matrices

Textbook Reference	Theorem / Definition Statement
Definition 2.12 Elementary Matrices	An $n \times n$ matrix E is an elementary matrix if it is obtained from I_n by a single elementary row operation. E is of Type I, II or III according to whether the operation is: (I) interchanging two rows; (II) multiplying one row by a non-zero number; (III) adding a multiple of one row to a different row.
Theorem 2.5.4 Uniqueness of Reduced Row-Echelon Form	If a matrix A is carried to reduced row-echelon matrices R and S by (possibly different sequences of) row operations, then $R = S$. This is the property that ultimately guarantees the inverse produced by the inversion algorithm is unique.

Linear Transformations

Textbook Reference	Theorem / Definition Statement
Definition 2.13 Linear Transformation	A transformation $T: R^n \rightarrow R^m$ is a linear transformation if, for all vectors x, y in R^n and all scalars a : (T1) $T(x + y) = T(x) + T(y)$; (T2) $T(ax) = aT(x)$.
Theorem 2.6.1 Linearity Theorem	If $T: R^n \rightarrow R^m$ is linear, then for any scalars a_i and vectors x_i in R^n , $T(a_1x_1 + a_2x_2 + \dots + a_kx_k) = a_1T(x_1) + a_2T(x_2) + \dots + a_kT(x_k)$.
Theorem 2.6.4 Rotations	The rotation $R_{\theta}: R^2 \rightarrow R^2$ is the linear transformation with matrix $[\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}]$.
Theorem 2.6.5 Reflections	The reflection Q_m in the line $y = mx$ is the linear transformation $Q_m: R^2 \rightarrow R^2$ with matrix $\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$.
Theorem 2.6.6 Projection	The projection P_m onto the line $y = mx$ is the linear transformation $P_m: R^2 \rightarrow R^2$ with matrix $\frac{1}{1+m^2} \begin{bmatrix} 1 & m \\ m & m^2 \end{bmatrix}$.

Module Summary

3.2 Matrix Inverses

The starting point for this sub-module is the observation that although matrices support addition, subtraction and multiplication, there is no direct notion of division. Nicholson's Section 2.4 [2] resolves this gap through the concept of the matrix inverse, formalised in Definition 2.11: a square matrix B is an inverse of A precisely when $AB = BA = I$. Because the identity matrix I acts as the multiplicative identity ($IB = BI = B$), the inverse allows equations involving matrices to be rearranged in a manner analogous to ordinary algebra, provided the correct side of an equation is multiplied and provided A^{-1} actually exists. Theorem 2.4.1 establishes that whenever an inverse exists, it is unique, which is what justifies writing A^{-1} without ambiguity rather than treating it as one of several possible inverses.

The Matrix Inversion Algorithm is the practical procedure by which A^{-1} is obtained. Working with the augmented matrix $[A \mid I]$, the Gaussian Algorithm from Module 1 is applied to the left-hand block until it

reduces to the identity matrix; because the same row operations are applied simultaneously across the full augmented matrix, the block that was originally the identity is transformed into A^{-1} , summarised as $[A \mid I] \rightarrow [I \mid A^{-1}]$. I practised this directly in NUMBAS exercise N3.2.1, which walks through the inversion of a 3×3 matrix step by step, and I applied the same method by hand in Exercise 2.4.4 of my learning evidence, where I used a given inverse to solve the system $Ax = b$ via $x = A^{-1}b$ and verified the result by substitution back into the original system. I found it useful to think of the inversion algorithm as Gaussian elimination performed twice over, once to clear entries below each pivot and once above, which is essentially the Gauss-Jordan variant demonstrated in the R code example provided on the unit site and in an external tutorial I consulted for additional clarity [4].

Guiding Question 1 (summarising the inversion algorithm): in words, the algorithm augments A with the identity matrix of the same size and then performs elementary row operations exactly as in ordinary Gaussian elimination, aiming for the reduced row-echelon form of A rather than stopping at row-echelon form. Because A is square and invertible, this reduced form is necessarily the identity matrix, and whatever combination of row operations achieved this transformation, when applied to the adjoined identity block, produces A^{-1} . My handwritten solution to Exercise 2.4.4 demonstrates the algorithm's endpoint rather than the elimination itself, since the inverse was supplied, but N3.2.1 and the associated self-assessment question in my NUMBAS submission [3] (Question 1, worth 4 out of 4 marks) show the full elimination sequence, including the intermediate augmented matrices at each step.

Guiding Question 2 (rank less than n): If an $n \times n$ matrix A has rank strictly less than n , the inversion algorithm cannot succeed. Applying Gaussian elimination to A will produce at least one row of the coefficient block consisting entirely of zeros, since rank r less than n means fewer than n pivot columns are available. By Theorem 2.4.5 (the Inverse Theorem), A is invertible if and only if it can be carried to the identity matrix I_n by row operations, and a matrix with rank less than n can never be carried to I_n , because I_n itself has rank n . Equivalently, condition (2) of Theorem 2.4.5 fails, since a coefficient matrix of rank less than n corresponds to a homogeneous system $Ax = 0$ with at least one free variable and therefore infinitely many non-trivial solutions, not just the trivial solution $x = 0$. This is exactly the situation explored in Exercise 2.4.18(c) and (d) of my learning evidence, where I proved that a matrix whose rows (or columns) each sum to zero cannot be invertible, precisely because such a matrix always admits a non-trivial solution to $Ax = 0$ (or $xA = 0$), namely the vector of all ones.

Guiding Question 3 (non-square matrices and inverses): a non-square matrix cannot possess a two-sided inverse in the sense defined by Definition 2.11. If A is $m \times n$ with m not equal to n , then for a matrix B to satisfy both $AB = I$ and $BA = I$, the product AB would need to be an $m \times m$ identity and BA an $n \times n$ identity, which forces B to be $n \times m$. However, the product of an $m \times n$ matrix with an $n \times m$ matrix (AB) is $m \times m$, while BA is $n \times n$; for both of these to simultaneously equal the respective identity matrices while representing a genuine two-sided inverse relationship is inconsistent with the rank of the transformation involved, since a linear map between spaces of different dimensions cannot be bijective. Concretely, if m is less than n , the transformation $T_A(x) = Ax$ maps into a smaller space, so it cannot be injective on all of $\mathbb{R}^n$ (there must be a non-trivial vector mapped to zero, by a pigeonhole-type argument on dimensions), which by the logic of Theorem 2.4.5 already rules out invertibility for square matrices and extends naturally to the impossibility of a genuine inverse for the non-square case. This reasoning connects back to my Exercise 2.4.18 proofs, where the non-invertibility of a square matrix was established purely from a rank deficiency argument, and the same underlying idea, that invertibility requires a bijective correspondence between $\mathbb{R}^n$ and itself, explains why the non-square case is excluded from the outset.

3.3 Elementary Matrices

Section 2.5 of the textbook [2] formalises a connection that had already been implicit throughout Module 1: every elementary row operation can be represented as multiplication from the left by a specific matrix. Definition 2.12 defines an elementary matrix E as the result of applying a single row operation to the identity matrix I_n , classified as Type I (row interchange), Type II (scaling a row by a non-zero constant) or Type III (adding a multiple of one row to another). The unit-site material illustrates this clearly through the row-mixing interpretation of matrix multiplication: multiplying an $(n \times n)$ matrix from the left blends the rows of the matrix being multiplied according to the entries of each row of the left-hand matrix, so an elementary matrix's structure directly encodes the row operation it performs.

I found the worked example video for N3.3.1 particularly useful for building intuition here, since it framed each row of the product as a weighted mixture of the original rows, which made identifying the elementary matrix for a given operation, and its inverse, much more mechanical. I applied this directly in Exercise 2.5.3 of my learning evidence, where I found E_1 (a row interchange) and E_2 (a row replacement) such that $C = E_2 E_1 A$, and then proved separately that no single elementary matrix E could satisfy $C = EA$, since the transformation from A to C genuinely requires two independent row operations rather than one. I extended this reasoning further in Exercise 2.5.7(b), where I determined an invertible matrix U such that $UA = B$ by identifying the necessary row operations directly from the additive relationships between the rows of A and B , and then expressed U as the product of the corresponding elementary matrices $E_2 E_1$, verifying the result by direct multiplication.

Theorem 2.5.4 (uniqueness of reduced row-echelon form) underpins why the inversion algorithm from Section 3.2 is reliable: no matter which particular sequence of elementary row operations, and hence which particular product of elementary matrices, is used to reduce A , the resulting reduced row-echelon form is always the same. Since the inverse itself is obtained as a product of elementary matrices (once A has been carried to I_n), this theorem is what guarantees that different valid elimination paths cannot produce two different inverses for the same matrix, reinforcing the uniqueness already established in Theorem 2.4.1. I also noted, while working through Exercise 2.5.7, that the factorisation of U into elementary matrices is not unique in general (there can be more than one valid sequence of row operations achieving the same transformation), which is consistent with the textbook's own observation in Exercise 2.5.8 that Gaussian elimination sequences are not unique even though the final result is.

3.4 Linear Transformations

Section 2.6 [2] reframes matrix-vector multiplication as a transformation $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T_A(x) = Ax$, where the dimensions n and m correspond to the number of columns and rows of A , respectively. Definition 2.13 specifies the two defining properties of linearity, additivity (T1) and homogeneity (T2), and Theorem 2.6.1 (the Linearity Theorem) extends these to arbitrary linear combinations of vectors. This theorem is significant because it confirms that a linear transformation is completely determined by

its effect on the standard basis vectors $e_1, \dots, e_n$, which is precisely the method I used in Exercise 2.6.4(a) and Exercise 2.6.10(a) of my learning evidence: rather than deriving the transformation matrix from a general formula, I computed $T(e_1)$, $T(e_2)$ and $T(e_3)$ directly and assembled these images as the columns of the matrix, in line with Theorem 2.6.2.

When A is square, T_A maps $\mathbb{R}^n$ back into itself, and Theorems 2.6.4 to 2.6.6 identify three particularly important families of such transformations: rotations (Theorem 2.6.4), reflections in a line $y = mx$

(Theorem 2.6.5), and projections onto a line $y = mx$ (Theorem 2.6.6). In Exercise 2.6.4(a), I derived the reflection matrix for the xz -plane in $\mathbb{R}^3$ by observing that vectors already lying in the plane of reflection remain fixed while the perpendicular component reverses sign, giving $A = \text{diag}(1, -1, 1)$; I verified this by confirming that $Av = (x, -y, z)$ for an arbitrary vector v , which matches the geometric definition of the reflection exactly. In Exercise 2.6.10(a), I derived the rotation matrix about the x -axis by rotating e_2 and e_3 within the yz -plane through an angle θ , obtaining the block structure $[[1, 0, 0], [0, \cos(\theta), -\sin(\theta)], [0, \sin(\theta), \cos(\theta)]]$, which reduces to the familiar 2×2 rotation matrix of Theorem 2.6.4 once the fixed x -coordinate is set aside.

Guiding Question 4 (distinguishing rotations from reflections in N3.4.2): the most reliable practical strategy, and the one explicitly confirmed in the feedback for that NUMBAS question, is to compare the diagonal entries of the given 2×2 matrix. For a rotation matrix, the two diagonal entries are always equal (both equal to $\cos(\theta)$), and the two off-diagonal entries are negatives of one another ($\sin(\theta)$ and $-\sin(\theta)$). For a reflection matrix, it is the off-diagonal entries that are equal (both equal to $2m/(1+m^2)$), while the diagonal entries are related but generally distinct, following the pattern $(1-m^2)/(1+m^2)$ and $(m^2-1)/(1+m^2)$. A second, faster check is to solve for the underlying parameter (θ or m) using the off-diagonal entries rather than the diagonal ones wherever possible, since equations of the form $2m = k$ are unambiguous, whereas equations of the form $m^2 - 1 = k$ admit two solutions ($m = \pm \sqrt{k+1}$) and require an extra step to resolve the sign using a second entry. I applied exactly this approach in Question 6 of my self-assessment [3], correctly identifying a reflection through the line $y = mx$ with $\theta = -\pi/4$, and correctly constructing the combined matrix for a rotation of $-\pi/6$ followed by a reflection through $y = 0$, scoring full marks on both parts.

Observations and Connections Across Sub-modules

Working through this module reinforced how tightly the three sub-topics depend on one another. The inversion algorithm (3.2) is only Gaussian elimination applied twice, so it inherits every idea from Module 1 concerning rank, pivots and row-echelon form; a matrix fails to be invertible for the same structural reason that a linear system fails to have a unique solution, namely a rank deficiency relative to the number of variables. Elementary matrices (3.3) then supply the algebraic language that explains why the inversion algorithm works at all: each row operation performed during elimination corresponds to left-multiplication by a specific, invertible elementary matrix, so the entire elimination process is itself just multiplication by a product of invertible matrices, which is why the end result (the inverse) is also invertible and unique. Finally, linear transformations (3.4) generalise this picture by treating any matrix, invertible or not, square or not, as encoding a mapping between vector spaces; invertibility of a square matrix A then corresponds precisely to T_A being a bijection of $\mathbb{R}^n$ onto itself, which ties directly back into the rank and non-square arguments developed in 3.2.

A useful comparison that helped consolidate my understanding is the distinction between an elementary matrix (which always corresponds to exactly one of the three basic row operations and is always invertible) and a general invertible matrix U such as the one found in Exercise 2.5.7 (which may require several elementary matrices multiplied together to construct, and represents a composite of several row operations). Every invertible matrix can be written as a product of elementary matrices, but not every invertible matrix is itself elementary, which is a subtlety I had not fully appreciated before working through Exercise 2.5.3(b) and confirming, by direct comparison of rows, that a transformation involving two distinct row operations genuinely cannot be captured by a single elementary matrix.

Extra Effort and Prior Knowledge

Beyond the prescribed unit-site material, I consulted several external resources to strengthen my geometric intuition for rotations and reflections, since these concepts benefit from a visual rather than purely algebraic treatment. I watched a tutorial on the Gauss-Jordan approach to matrix inversion [4], which reinforced the connection between the textbook's inversion algorithm and the augmented-matrix method demonstrated in the R code example on the unit site, confirming that both approaches converge to the same inverse regardless of the specific row operations chosen, consistent with Theorem 2.5.4. I also watched a dedicated explanation of elementary matrices [5], which usefully distinguished genuine elementary matrices from superficially similar matrices that do not arise from a single row operation, directly supporting my reasoning in Exercise 2.5.3(b). Finally, I watched the third chapter of the Essence of Linear Algebra series [6], which visualises matrix multiplication as a transformation of the coordinate grid; this gave me a much stronger intuitive grasp of why the columns of a transformation matrix must be the images of the standard basis vectors (Theorem 2.6.2), which I relied on directly when constructing the reflection and rotation matrices in Exercises 2.6.4 and 2.6.10.

In terms of prior knowledge, I had previously encountered 2×2 rotation matrices in a high school context when studying transformations in coordinate geometry, although without the underlying linear algebra framework of basis vectors and linearity properties (T1 and T2) that this module provides. This background helped me recognise the structure of the rotation matrix in Theorem 2.6.4 quickly, but I found the extension to reflections and projections in terms of a general line $y = mx$, rather than only the coordinate axes, to be genuinely new material that required careful attention to the derivations in Nicholson's Section 2.6 [2].

Reflecting on the Content

I found the elementary matrices sub-module the most conceptually satisfying part of Module 3, because it retrospectively explained why the row operations from Module 1 always preserved the solution set of a system: each operation is simply left-multiplication by an invertible matrix, and multiplying both sides of an equation by an invertible matrix never changes the solution set. The most challenging part was Exercise 2.4.18, since constructing a proof of non-invertibility from a structural property of the matrix (row or column sums equal to zero) required me to connect several theorems together: Theorem 2.4.5, Corollary 2.4.1 and the underlying rank argument, rather than simply following a computational procedure. Working through this exercise carefully, including the alternative direct proof using $x = (1, 1, \dots, 1)$ and a proof by contradiction, has considerably strengthened my ability to reason about invertibility conceptually rather than only algorithmically, which I expect will be valuable when eigenvalues and diagonalisation are introduced in a later module, since the existence of eigenvectors is itself framed in terms of a related matrix failing to be invertible.

References

- [1] Deakin University, "Topic 3 – Transformations and Special Matrices: 3.1 Introduction, 3.2 Matrix Inverses, 3.3 Elementary Matrices, 3.4 Linear Transformations, Module 3 Assessment," SIT292 Linear Algebra for Data Analysis, CloudDeakin Unit Site, Deakin University, Trimester 2, 2026.
- [2] W. K. Nicholson, Linear Algebra with Applications, Lyryx, 2021. [Online]. Available: Nicholson-OpenLAWA.pdf, provided via the SIT292 unit site content repository.
- [3] Deakin University, "SIT292 Module 3 Self-Assessment," Numbas, developed by Newcastle University, accessed 15 Aug. 2026.
- [4] "Inverse Matrix by Gauss-Jordan Elimination | Linear Algebra," YouTube, 29 Mar. 2024. [Online video].
- [5] Wrath of Math, "What are Elementary Matrices? | Linear Algebra," YouTube, 6 Apr. 2024. [Online video].
- [6] 3Blue1Brown, "Linear Transformations and Matrices | Chapter 3, Essence of Linear Algebra," YouTube, 7 Aug. 2016. [Online video].

PRACTICE QUESTIONS

DATE / /

PAGE

3.2 Matrix Inverses

Exercise 2.4.4 Given $A^{-1} = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix}$

(a) Solve the system of equations $Ax = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$.

Concept / Theorem Applied

Theorem 2.4.2: If A is an invertible $n \times n$ matrix, then $Ax = b$

$$\Rightarrow x = A^{-1}b$$

This follows by left-multiplying $Ax = b$ by A^{-1} :

$$A^{-1}Ax = A^{-1}b$$

$$x = A^{-1}b$$

$$[\because A^{-1}A = I]$$

This is exactly the role of the inverse in solving a matrix equation.

Ans: Write the system in matrix form:

$$Ax = b \quad \text{where } b = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

$\therefore A^{-1}$ is given, $x = A^{-1}b$.

$$\therefore x = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 5 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

Multiplying row by column:

$$x_1 = (1)(1) + (-1)(-1) + (3)(3) = 1 + 1 + 9 = 11$$

$$x_2 = (2)(1) + (0)(-1) + (5)(3) = 2 + 0 + 15 = 17$$

$$x_3 = (-1)(1) + (1)(-1) + (0)(3) = -1 - 1 + 0 = -2$$

$$x = \begin{bmatrix} 11 \\ 17 \\ -2 \end{bmatrix}$$

Verification:

$$Ax = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} = b$$

$\therefore$ The solution is correct.

$$x_1 = 11, x_2 = 17, x_3 = -2.$$

Exercise 2.4.12 Verify that $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ satisfies $A^2 - 3A + 2I = 0$, and use this fact to show that $A^{-1} = \frac{1}{2}(3I - A)$.

Concept / Property Applied

The key idea is to derive an inverse from a matrix polynomial identity.

From $A^2 - 3A + 2I = 0$, we can rearrange to obtain a product involving A :

$$3A - A^2 = 2I$$

$$A(3I - A) = 2I$$

$$(3I - A)A = 2I$$

[since $3I - A$ is also a polynomial in A , it commutes with A]

$$\therefore A \left[\frac{1}{2} (3I - A) \right] = I$$

$$\left[\frac{1}{2} (3I - A) \right] A = I$$

By the definition of a matrix inverse, the bracketed matrix is A^{-1} . A matrix B is an inverse of A only when $AB = BA = I$.

Ans:

Part 1 - Verify the identity

First calculate A^2 :

$$A^2 = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1(1) + (-1)(0) & 1(-1) + (-1)(2) \\ 0(1) + 2(0) & 0(-1) + 2(2) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -3 \\ 0 & 4 \end{bmatrix}$$

$$3A = \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix}$$

$$2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Hence, $A^2 - 3A + 2I = \begin{bmatrix} 1 & -3 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$$A^2 - 3A + 2I = \begin{bmatrix} 1-3+2 & -3+3 \\ 0-0+0 & 4-6+2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

So the required identity is verified:

$$A^2 - 3A + 2I = 0$$

Part 2 - Obtain A^{-1}

Method-1

$$A^2 - 3A + 2I = 0$$

$$3A - A^2 = 2I$$

$$A(3I - A) = 2I$$

$$(3I - A)A = 2I \quad [\because A \text{ commutes with } 3I - A]$$

Dividing both the sides by 2:

$$A \left[\frac{1}{2} (3I - A) \right] = I$$

$$\left[\frac{1}{2} (3I - A) \right] A = I$$

$\therefore$ By the definition of an inverse,

$$A^{-1} = \frac{1}{2} (3I - A)$$

Method-2

$$3I - A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\text{Hence } A^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$$

Exercise 2.4.18 show that A has no inverse when:

- (c) each row of A sums to 0.
[Hint: Theorem 2.4.5(2).]

Concept / Theorem Applied

Use Theorem 2.4.5 (Inverse Theorem):

A is invertible $\Leftrightarrow Ax=0$ has only the trivial solution $x=0$.

Therefore, to prove that A is not invertible, it is enough to find a non-zero vector x satisfying $Ax=0$.

Ans: Suppose A is an $n \times n$ matrix and every row of A sums to zero.

Consider the non-zero vector $x = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$

Multiplying A by x gives the row sums of A

$$Ax = \begin{bmatrix} \text{sum of row 1} \\ \text{sum of row 2} \\ \vdots \\ \text{sum of row } n \end{bmatrix}$$

$$Ax = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0 \quad [\because \text{every row sums to } 0]$$

As $x \neq 0$, the homogeneous system $Ax = 0$ has a non-trivial solution.

By Theorem 2.4.5, an invertible matrix can have only the trivial solution to $Ax = 0$. Therefore A cannot be invertible.

A has no inverse.

Observation

Every row sum $= 0 \Rightarrow A \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = 0 \Rightarrow A$ is not invertible

(d) each column of A sums to 0.

[Hint: Corollary 2.4.1, Theorem 2.4.4]

Concept/Theorem Applied

Two results are useful here:

Corollary 2.4.1:

A is invertible $\Leftrightarrow A^T$ is invertible.

Also use the result from part (c): if every row of a matrix sums to zero, that matrix is not invertible.

Ans: Suppose every column of A sums to zero.

After transposing A , the columns of A become the rows of A^T .

Therefore, every row of A^T also sums to zero.

From part (c), A^T cannot be invertible.

But Corollary 2.4.1 states that:

A^T is invertible $\Leftrightarrow A$ is invertible

Since A^T is not invertible, A is also not invertible.

Alternative Method

Let $X = (1 \ 1 \ \dots \ 1)$

Because each column of A sums to zero, $XA = 0$.

Suppose, for contradiction, that A is invertible.

Right multiplying by A^{-1} ,

$$XAA^{-1} = 0A^{-1}$$

$$XI = 0$$

$$X = 0$$

$$X = (1 \ 1 \ \dots \ 1) \neq 0$$

This is a contradiction.

$\therefore A$ has no inverse.

This direct proof shows why the column-sum condition makes A non-invertible, rather than relying only on the theorem.

3.3. Elementary Matrices

Exercise 2.5.3 Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$.

(a) Find elementary matrices E_1 and E_2 such that $C = E_2 E_1 A$.

Concept / Properties Applied

- Definition 2.12: An elementary matrix is obtained from the identity matrix by one elementary row operation.
- Lemma 2.5.1: Performing a row operation on A is equivalent to left-multiplying A by the corresponding elementary matrix E .
- A sequence of row operation gives:
 $C = E_2 E_1 A$

This is the form described in Theorem 2.5.1.

Method-1

Perform the row operations directly

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$$

Step 1: Interchange the two rows ($R_1 \leftrightarrow R_2$)

$$\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

The corresponding elementary matrix is obtained by applying the same row interchange to I_2 :

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\text{Thus, } E_1 A = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 2. Replace Row 2 by $R_2 - R_1$

$$R_2 \rightarrow R_2 - R_1$$

$$\text{Hence, } \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} = C$$

The corresponding elementary matrix is:

$$E_2 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \quad \left(\begin{array}{l} \text{Because multiplying from the left} \\ \text{replaces row 2 by } -R_1 + R_2. \end{array} \right)$$

$$\therefore C = E_2 E_1 A \quad \text{with } E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and } E_2 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}.$$

Verification:

$$E_2 E_1 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

$$E_2 E_1 A = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} = C$$

Hence verified.

- (b.) Show that there is no elementary matrix E such that $C = EA$.

Concept / Properties Applied

An elementary matrix represents one elementary row operation only. The three possible row operations are:

1. Interchange two rows.
2. Multiply one row by a non-zero scalar.
3. Add a multiple of one row to another row.

PROOF

Ans:

We compare $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$

For a single row operation:

- A row scaling would leave one row unchanged, but neither row of C agrees with the corresponding row of A .
- A row replacement $R_i \rightarrow R_i + kR_j$ also leaves the other row unchanged, which again does not occur.
- A row interchange would give:

$$\begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} \neq C$$

Therefore, no single elementary row operation can transform A into C .

Hence, there is no elementary matrix E such that $C = EA$.

Observation

The transformation requires two elementary row operations:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \xrightarrow{R_2 \rightarrow R_2 - R_1} C$$

Exactly, this is why C can be written as $E_2 E_1 A$, but not as $E A$ for a single elementary matrix.

Exercise 2.5.7 In each case find an invertible matrix U such that $UA=B$, and express U as a product of elementary matrices.

$$(b) \quad A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 0 \end{bmatrix}$$

Concept / Properties Applied

Theorem 2.5.1: If A is transformed into B by elementary row operations, then $B=UA$ where U is invertible and is the product of the corresponding elementary matrices:

$$U = E_k \dots E_2 E_1.$$

Method-1

Ans: Write the rows of A as R_1, R_2

$$\text{Aim: } \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 0 \end{bmatrix}$$

Observation:

$$(2, -1, 0) + (1, 1, 1) = (3, 0, 1)$$

$\therefore$ We first perform $R_1 \rightarrow R_1 + R_2$, which gives:

$$\begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Now, we need the second row to become the original first row:

$$(2, -1, 0)$$

Using the current rows:

$$(3, 0, 1) - (1, 1, 1) = (2, -1, 0)$$

Therefore, $R_2 \rightarrow R_1 - R_2$.

$$\text{So, } \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 0 \end{bmatrix} = B.$$

Method-2

First operation: $R_1 \rightarrow R_1 + R_2$

$$E_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \left(\text{Because } E_1 A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right)$$

Second operation: $R_2 \rightarrow R_1 - R_2$

$$E_2 = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \quad \left(\text{Because its second row is } R_1 - R_2 \right)$$

Therefore, $B = E_2 E_1 A$.

Hence, $U = E_2 E_1$.

$$U = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\therefore U = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Its elementary matrix factorisation is:

$$U = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Verification:

$$UA = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \\ 2 & -1 & 0 \end{bmatrix} = B$$

Thus, $UA = B$.

Key Learning Point

A sequence of elementary row operations corresponds to left multiplication by their product.

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

3.4 Linear Transformations

Exercise 2.6.4 In each case use Theorem 2.6.2 to obtain the matrix A of the transformation T . You may assume that T is linear in each case.

(a) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is reflection in the $x-z$ plane.

Concept / Theorem Applied

Theorem 2.6.2 — Matrix of a linear transformation

If $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear, then its matrix is formed by placing the images of the standard basis vectors as the columns:

$$A = [T(e_1) \ T(e_2) \ \dots \ T(e_n)] \text{ where } e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Ans. Method: Apply the transformation to the standard basis

Step 1. Image of e_1

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

The vector lies in the $x-z$ plane, so reflection in the plane leaves it unchanged:

$$T(e_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Step 2. Image of e_2

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$[(e_1)^T \ (e_2)^T \ (e_3)^T]^T = A$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = A, \text{ correct.}$$

The x - z plane is the plane $y=0$. Reflection through this plane reverses the y -coordinate:

$$(x, y, z) \longrightarrow (x, -y, z)$$

: modification

= n. H. O. K. K. p. e. n. t. i. b. e. n. a. r. e. t.

$$\therefore T(e_2) = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

Step 3. Image of e_3

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The vector also lies in the x - z plane, so it remains unchanged:

$$T(e_3) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Step 4. Construct the matrix

By Theorem 2.6.2, place these transformed basis vectors as the columns:

$$A = [T(e_1) \quad T(e_2) \quad T(e_3)]$$

$$\text{Hence, } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Verification:

$$\text{For an arbitrary vector } v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$Av = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ -y \\ z \end{bmatrix}$$

This is exactly reflection through the x - z plane.

$$T(x, y, z) = (x, -y, z)$$

$$\text{Therefore, the required matrix } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Key Learning Point

The matrix of a linear transformation is obtained by using the transformed ^{standard} basis vectors as its columns.

Exercise 2.6.10 Find the matrix of $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ in each case:

- (a) T is rotation through θ about the x -axis (from the y -axis to the z -axis).

Concept / Theorem Applied

Use Theorem 2.6.2: $A = [T(e_1) \ T(e_2) \ T(e_3)]$

The rotation occurs about the x -axis, so the x -component is unchanged. The y - and z -components rotate in the yz -plane.

For a positive rotation through θ , the y -axis rotates towards the z -axis.

Ans: Method: Determine the images of the standard basis vectors

Step 1. Image of e_1

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Since e_1 lies along the axis of rotation, it is unchanged:

$$T(e_1) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Step 2. Image of e_2

$$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Rotating e_2 through θ about the x -axis gives

$$T(e_2) = \begin{bmatrix} 0 \\ \cos \theta \\ \sin \theta \end{bmatrix}$$

Here the y -component becomes $\cos \theta$, while the component in the z -direction is $\sin \theta$.

Step 3. Image of e_3

$$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

After the same rotation,

$$T(e_3) = \begin{bmatrix} 0 \\ -\sin \theta \\ \cos \theta \end{bmatrix}$$

The negative sign occurs because the z -axis direction rotates towards the negative y -direction.

Step 4. Construct the transformation matrix

Using the transformed basis vectors as columns,
 $A = [T(e_1) \ T(e_2) \ T(e_3)]$

$$\therefore A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}_{3 \times 3}$$

Verification:

Take an arbitrary vector $v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

$$Av = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{Thus, } T(v) = \begin{bmatrix} x \\ y \cos \theta - z \sin \theta \\ y \sin \theta + z \cos \theta \end{bmatrix}$$

The x -coordinate remains unchanged, while the y - and z -coordinates undergo a 2D rotation through θ . Therefore, this is precisely rotation about x -axis.

$$T(x, y, z) = (x, y \cos \theta - z \sin \theta, y \sin \theta + z \cos \theta).$$

Key Learning Point

For a linear transformation, finding the matrix reduces to finding the images of the standard basis vectors & placing these images as the columns of the matrix.

Use the **inversion algorithm** to find the inverse of the matrix

$$A = \begin{pmatrix} -2 & -2 & 1 \\ 1 & 1 & -1 \\ 2 & 3 & 0 \end{pmatrix}$$

The following steps will work you through the process. You can check each stage by submitting what you've done so far. If you prefer to do all your working by hand and just submit the final answer at the end (at step 4), you can do that too.

Step 1

Use the [show steps] buttons to see the rules you can follow so that your answer matches that expected.

Following your submission at each step, you will receive feedback on whether your answer is acceptable for proceeding to the next step, as well as whether it matches the expected format. Any feedback provided on errors in certain rows, as well as the numerical scores, assume that you have followed the specific guidelines on which row swaps to perform.

The original matrix is repeated here to make it easier to write in your first matrix.

$$A = \begin{pmatrix} -2 & -2 & 1 \\ 1 & 1 & -1 \\ 2 & 3 & 0 \end{pmatrix}$$

Getting the matrix ready for row operations

Construct $[A \ I]$ and then perform initial row swaps so that Row1 can be used to cancel out entries at the next step.

- Do not perform division or multiplication of rows in this step.

Answer:
$$\begin{pmatrix} \underline{1} & \underline{1} & \underline{-1} & \underline{0} & \underline{1} & \underline{0} \\ \underline{-2} & \underline{-2} & \underline{1} & \underline{1} & \underline{0} & \underline{0} \\ \underline{2} & \underline{3} & \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix} \checkmark$$

▼ Feedback

Score: 1/1

You revealed the steps.

Your answer is correct. You were awarded **1** mark.

You CAN proceed to the next step.

You scored **1** mark for this part.

Step 2

Perform row operations with Row 1.

- First divide Row 1 by the pivot (if necessary) so that $a'_{11} = 1$.
- Use multiples of the resulting row to eliminate any non-zero elements in the first column.

Answer:
$$\begin{pmatrix} \underline{1} & \underline{1} & \underline{-1} & \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{-1} & \underline{1} & \underline{2} & \underline{0} \\ \underline{0} & \underline{1} & \underline{2} & \underline{0} & \underline{-2} & \underline{1} \end{pmatrix} \checkmark$$

▼ Feedback

Score: 1/1

Your answer is correct. You were awarded **1** mark.

You CAN proceed to the next step based on your current matrix.

Step 3

You scored **1** mark for this part.

Now use row 2.

- Divide by the pivot (if necessary) and then use the new Row 2 to eliminate any remaining elements in the second column (**both above and below**).

Answer:
$$\left(\begin{array}{cc|cc|cc} 1 & 0 & -3 & 0 & 3 & -1 \\ \hline 0 & 1 & 2 & 0 & -2 & 1 \\ \hline 0 & 0 & -1 & 1 & 2 & 0 \end{array} \right) \checkmark$$

▼ Feedback

Score: 1/1 

 Your answer is correct. You were awarded **1** mark.

You CAN proceed to the next step based on your current matrix.

You scored **1** mark for this part.

Step 4

Finally, use Row 3.

Answer:
$$\left(\begin{array}{cc|cc|cc} 1 & 0 & 0 & -3 & -3 & -1 \\ \hline 0 & 1 & 0 & 2 & 2 & 1 \\ \hline 0 & 0 & 1 & -1 & -2 & 0 \end{array} \right) \checkmark$$

▼ Feedback

✓ Your answer is correct. You were awarded 1 mark.

Score: 1/1 ✓

You have obtained the correct matrix :)

You scored 1 mark for this part.

NUMBAS

Created using Numbas (<https://www.numbas.org.uk>) v10.0, developed by Newcastle University (<http://www.newcastle.ac.uk>).

Use the **inversion algorithm** to find the inverse of the matrix

$$A = \begin{pmatrix} -3 & 0 & -5 \\ 2 & -1 & -2 \\ 1 & 0 & 2 \end{pmatrix}$$

The following steps will work you through the process. You can check each stage by submitting what you've done so far. If you prefer to do all your working by hand and just submit the final answer at the end (at step 4), you can do that too.

Step 1

Use the [show steps] buttons to see the rules you can follow so that your answer matches that expected.

Following your submission at each step, you will receive feedback on whether your answer is acceptable for proceeding to the next step, as well as whether it matches the expected format. Any feedback provided on errors in certain rows, as well as the numerical scores, assume that you have followed the specific guidelines on which row swaps to perform.

The original matrix is repeated here to make it easier to write in your first matrix.

$$A = \begin{pmatrix} -3 & 0 & -5 \\ 2 & -1 & -2 \\ 1 & 0 & 2 \end{pmatrix}$$

Getting the matrix ready for row operations

Construct $[A \ I]$ and then perform initial row swaps so that Row1 can be used to cancel out entries at the next step.

- Do not perform division or multiplication of rows in this step.

Answer:
$$\begin{pmatrix} \frac{1}{2} & \frac{0}{-1} & \frac{2}{-2} & \frac{0}{0} & \frac{0}{1} & \frac{1}{0} \\ -3 & 0 & -5 & 1 & 0 & 0 \end{pmatrix} \checkmark$$

▼ Feedback

Score: 1/1 

 Your answer is correct. You were awarded 1 mark.

You CAN proceed to the next step.

You scored **1** mark for this part.

Step 2

Perform row operations with Row 1.

- First divide Row 1 by the pivot (if necessary) so that $a'_{11} = 1$.
- Use multiples of the resulting row to eliminate any non-zero elements in the first column.

Answer:
$$\begin{pmatrix} \frac{1}{0} & \frac{0}{-1} & \frac{2}{-6} & \frac{0}{0} & \frac{0}{1} & \frac{1}{-2} \\ 0 & 0 & 1 & 1 & 0 & 3 \end{pmatrix} \checkmark$$

▼ Feedback

Score: 1/1 

 Your answer is correct. You were awarded 1 mark.

You CAN proceed to the next step based on your current matrix.

You scored **1** mark for this part.

Step 3

Now use row 2.

- Divide by the pivot (if necessary) and then use the new Row 2 to eliminate any remaining elements in the second column (**both above and below**).

Answer:
$$\begin{pmatrix} \begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 2 & 0 & 0 & 1 \\ \hline 0 & 1 & 6 & 0 & -1 & 2 \\ \hline 0 & 0 & 1 & 1 & 0 & 3 \\ \hline \end{array} \end{pmatrix} \checkmark$$

▼ Feedback

Score: 1/1 

 Your answer is correct. You were awarded **1** mark.

You CAN proceed to the next step based on your current matrix.

You scored **1** mark for this part.

Step 4

Finally, use Row 3.

Answer:
$$\begin{pmatrix} \begin{array}{|c|c|c|c|c|c|} \hline 1 & 0 & 0 & -2 & 0 & -5 \\ \hline 0 & 1 & 0 & -6 & -1 & -16 \\ \hline 0 & 0 & 1 & 1 & 0 & 3 \\ \hline \end{array} \end{pmatrix} \checkmark$$

▼ Feedback

Score: 1/1 

✓ Your answer is correct. You were awarded **1** mark.

You have obtained the correct matrix :)

You scored **1** mark for this part.

NUMBAS

Created using Numbas (<https://www.numbas.org.uk>) v10.0, developed by Newcastle University (<http://www.newcastle.ac.uk>).

a)

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Row operation applied



I. Interchange two rows



II. Multiply one row by a nonzero number



III. Add a multiple of one row to a different row



Inverse matrix

Rows: 3 Columns: 3

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \checkmark$$

▼ Feedback

Score: 2/2

Gap 0

- ✔ You chose a correct answer. You were awarded **1** mark.

Gap 1

- ✔ Your answer is correct. You were awarded **1** mark.

You scored **2** marks for this part.

b)

Find an elementary matrix, E , such that $B = EA$ if

$$A = \begin{pmatrix} -4 & 5 \\ -7 & -6 \end{pmatrix}, B = \begin{pmatrix} -4 & 5 \\ -19 & 9 \end{pmatrix}$$

Rows: 2 Columns: 2

$$E = \begin{pmatrix} \underline{1} & \underline{0} \\ \underline{3} & \underline{1} \end{pmatrix} \quad \checkmark$$

▼ Feedback

Score: 1/1 ✔

- ✔ Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

NUMBAS

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk) v10.0, developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

a)

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 1 & -\frac{1}{4} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Row operation applied



I. Interchange two rows



II. Multiply one row by a nonzero number



III. Add a multiple of one row to a different row



Inverse matrix

Rows: 3 Columns: 3

$$\begin{pmatrix} \underline{1} & \underline{1/4} & \underline{0} \\ \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{1} \end{pmatrix} \quad \checkmark$$

▼ Feedback

Score: 2/2

Gap 0

- ✓ You chose a correct answer. You were awarded **1** mark.

Gap 1

- ✓ Your answer is correct. You were awarded **1** mark.

You scored **2** marks for this part.

b)

Find an elementary matrix, E , such that $B = EA$ if

$$A = \begin{pmatrix} -3 & 7 \\ 5 & 0 \end{pmatrix}, B = \begin{pmatrix} 5 & 0 \\ -3 & 7 \end{pmatrix}$$

Rows: 2 Columns: 2

$$E = \begin{pmatrix} \underline{0} & \underline{1} \\ \underline{1} & \underline{0} \end{pmatrix} \quad \checkmark$$

▼ Feedback

Score: 1/1 ✓

- ✓ Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

NUMBAS

Created using [Numbas](https://www.numbas.org.uk) (https://www.numbas.org.uk) v10.0, developed by [Newcastle University](http://www.newcastle.ac.uk) (http://www.newcastle.ac.uk).

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation.

Find $T \begin{pmatrix} -2 \\ 10 \end{pmatrix}$ if $T \begin{pmatrix} 1 \\ -4 \end{pmatrix} = \begin{pmatrix} -8 \\ -20 \end{pmatrix}$ and $T \begin{pmatrix} 0 \\ -2 \end{pmatrix} = \begin{pmatrix} -6 \\ -8 \end{pmatrix}$.

Enter your result below.

Rows:	2	Columns:	1
-------	--------------------------------	----------	--------------------------------

$$T \begin{pmatrix} -2 \\ 10 \end{pmatrix} = \begin{pmatrix} \boxed{22} \\ \boxed{48} \end{pmatrix}$$



Your answer is correct. You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation.

Find $T \begin{pmatrix} 8 \\ -5 \\ -3 \\ 12 \end{pmatrix}$ if $T \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 7 \\ 4 \\ -4 \end{pmatrix}$ and $T \begin{pmatrix} 2 \\ -2 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 26 \\ -7 \\ -22 \end{pmatrix}$.

Enter your result below.

Rows: Columns:

$$T \begin{pmatrix} 8 \\ -5 \\ -3 \\ 12 \end{pmatrix} = \begin{pmatrix} \boxed{83} \\ \boxed{-40} \\ \boxed{-76} \end{pmatrix} \quad \checkmark$$

Your answer is correct. You were awarded 1 mark. 

You scored 1 mark for this part.

Score: 1/1 

Created using [Numbas \(https://www.numbas.org.uk/\)](https://www.numbas.org.uk/), developed by [Newcastle University \(http://www.newcastle.ac.uk/\)](http://www.newcastle.ac.uk/).

Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and the effect of multiplying $T \begin{pmatrix} x \\ y \end{pmatrix}$.

a)

If

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

This linear mapping is

☐ a reflection through the line $y = mx$

☒ a rotation about the origin by θ



Give the value of θ if it is a rotation or the value of m if it is a reflection

$\frac{-\pi}{2}$

*Give values of θ in terms of 'pi', e.g., $\pi/2$, $3\pi/5$ etc and express over the domain $(-\pi, \pi]$.

**If your value involves surds, use 'sqrt', e.g. $\sqrt{2}$ for the square root of 2.

Gap 0

You chose a correct answer. You were awarded 1 mark.

Gap 1

Your answer is numerically correct. You were awarded 1 mark.

You scored 2 marks for this part.

Score: 2/2

b)

Identify a matrix that applies a rotation of $-\pi/4$ about the origin followed by a reflection through the line $y = mx$ with $m = 3$

$$\begin{pmatrix} \boxed{-0.99} & \boxed{-0.141} \\ \boxed{-0.141} & \boxed{0.99} \end{pmatrix} \checkmark$$

*Enter the entries of the matrix correct to 3 decimal places.

You have not given every cell in your answer to the same precision.

Your answer is correct. You were awarded 1 mark. 

You scored 1 mark for this part.

Score: 1/1 

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and the effect of multiplying $T \begin{pmatrix} x \\ y \end{pmatrix}$.

a)

If

$$T = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

This linear mapping is

- ☐ a rotation about the origin by θ
- ☒ a reflection through the line $y = mx$



Give the value of θ if it is a rotation or the value of m if it is a reflection

$\sqrt{3}$ ✓

*Give values of θ in terms of 'pi', e.g., $\pi/2$, $3\pi/5$ etc and express over the domain $(-\pi, \pi]$.

**If your value involves surds, use 'sqrt', e.g. sqrt(2) for the square root of 2.

Gap 0

You chose a correct answer. You were awarded 1 mark. ✓

Gap 1

Your answer is numerically correct. You were awarded 1 mark. ✓

You scored **2** marks for this part.

Score: 2/2 ✓

b)

Identify a matrix that applies a reflection through the line $y = mx$ with $m = -2$ followed by a rotation of $-\pi/3$ about the origin

$$\begin{pmatrix} \boxed{-0.993} & \boxed{0.12} \\ \boxed{0.12} & \boxed{0.993} \end{pmatrix} \quad \checkmark$$

*Enter the entries of the matrix correct to 3 decimal places.

Your answer is correct. You were awarded 1 mark. ✓

You scored 1 mark for this part.

Score: 1/1 ✓

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

SIT292 Module 3 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score
Question 1	4/4
Question 2	1/1
Question 3	1/1
Question 4	1/1
Question 5	1/1
Question 6	2/2
Total	10/10 (100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 3 Self-Assessment
Session ID:	3179543883059798
Exam Start:	Sat Aug 15 2026 18:46:27
Exam Stop:	Sat Aug 15 2026 19:18:54
Time Spent:	00:32:26

Question 1

Starting with the matrix,

$$A = \begin{pmatrix} -2 & 2 & -4 \\ 1 & -3 & -1 \\ 1 & -2 & 1 \end{pmatrix}$$

The results following the first steps of the inversion algorithm are as follows:

Augmented matrix

$$\begin{pmatrix} -2 & 2 & -4 & 1 & 0 & 0 \\ 1 & -3 & -1 & 0 & 1 & 0 \\ 1 & -2 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Step 1.

$$\begin{pmatrix} 1 & -3 & -1 & 0 & 1 & 0 \\ -2 & 2 & -4 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 1 \end{pmatrix}$$

Step 2.

$$\begin{pmatrix} 1 & -3 & -1 & 0 & 1 & 0 \\ 0 & -4 & -6 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 & -1 & 1 \end{pmatrix}$$

Write down the resulting augmented matrix after performing the next step (step 3).

$$\left(\begin{array}{cccccc} 1 & 0 & 7/2 & -3/4 & -1/2 & 0 \\ 0 & 1 & 3/2 & -1/4 & -1/2 & 0 \\ 0 & 0 & 1/2 & 1/4 & -1/2 & 1 \end{array} \right) \checkmark$$

Expected answer:

$$\left(\begin{array}{cccccc} 1 & 0 & 7/2 & -3/4 & -1/2 & 0 \\ 0 & 1 & 3/2 & -1/4 & -1/2 & 0 \\ 0 & 0 & 1/2 & 1/4 & -1/2 & 1 \end{array} \right)$$

(Note that the step by step version of this problem is available in sub-module 3.2. At Step 3, you should not perform any additional operations to obtain a leading 1 in row 3 -- that would be part of Step 4)

Based on this and the Step 4 that would follow, the inverse matrix is:

$$A^{-1} =$$

Rows: 3 Columns: 3

$$\left(\begin{array}{ccc} -5/2 & 3 & -7 \\ -1 & 1 & -3 \\ 1/2 & -1 & 2 \end{array} \right) \checkmark$$

Expected answer:

$$\left(\begin{array}{ccc} -5/2 & 3 & -7 \\ -1 & 1 & -3 \\ 1/2 & -1 & 2 \end{array} \right)$$

Gap 0

Score: 4/4

- ✓ Exact form accepted. You were awarded **2** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **2** marks.

You scored **4** marks for this part.

Question 2

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Row operation applied



I. Interchange two rows



II. Multiply one row by a nonzero number



III. Add a multiple of one row to a different row



Expected answer:



I. Interchange two rows



II. Multiply one row by a nonzero number



III. Add a multiple of one row to a different row

Inverse matrix

Rows: 3 Columns: 3

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Gap 0

Score: 1/1

- You chose a correct answer. You were awarded **0.5** marks.

Gap 1

- Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 3

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{pmatrix}$$

Row operation applied



I. Interchange two rows



II. Multiply one row by a nonzero number



III. Add a multiple of one row to a different row



Expected answer:



I. Interchange two rows



II. Multiply one row by a nonzero number



III. Add a multiple of one row to a different row

Inverse matrix

Rows: 3 Columns: 3

$$\begin{pmatrix} \underline{1} & \underline{0} & \underline{0} \\ \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{3} & \underline{1} \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} \underline{1} & \underline{0} & \underline{0} \\ \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{3} & \underline{1} \end{pmatrix}$$

Gap 0

Score: 1/1

- ✓ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 4

Find an elementary matrix, E , such that $B = EA$ if

$$A = \begin{pmatrix} 4 & 1 \\ -5 & 1 \end{pmatrix}, B = \begin{pmatrix} 4 & 1 \\ -17 & -2 \end{pmatrix}$$

$E =$

Rows: 2 Columns: 2

$$\begin{pmatrix} \underline{1} & \underline{0} \\ \underline{-3} & \underline{1} \end{pmatrix}$$



Expected answer: $\begin{pmatrix} \underline{1} & \underline{0} \\ \underline{-3} & \underline{1} \end{pmatrix}$

- ✓ Your answer is correct. You were awarded **1** mark.

Score: 1/1 ✓

You scored **1** mark for this part.

Question 5

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation.

Find $T \begin{pmatrix} 3 \\ -15 \\ 6 \end{pmatrix}$ if $T \begin{pmatrix} 2 \\ -4 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 44 \end{pmatrix}$ and $T \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ -5 \end{pmatrix}$.

Enter your result below.

$$T \begin{pmatrix} 3 \\ -15 \\ 6 \end{pmatrix} =$$

Rows: 2 Columns: 1

$$\begin{pmatrix} -12 \\ 117 \end{pmatrix}$$



Expected answer: $\begin{pmatrix} -12 \\ 117 \end{pmatrix}$

✓ Your answer is correct. You were awarded 1 mark.

Score: 1/1 ✓

You scored 1 mark for this part.

Question 6

Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and the effect of multiplying $T \begin{pmatrix} x \\ y \end{pmatrix}$.

a)

If

$$T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

This linear mapping is

- ☐ a reflection through the line $y = mx$
- ☒ a rotation about the origin by θ



Expected answer:

- ☐ a reflection through the line $y = mx$
- ☒ a rotation about the origin by θ

Give the value of θ if it is a rotation or the value of m if it is a reflection

$-\pi/4$ $-\frac{\pi}{4}$ ✓

Expected answer: $-\pi/4$ $-\frac{\pi}{4}$

*Give values of θ in terms of 'pi', e.g., $\pi/2$, $3\pi/5$ etc and express over the domain $(-\pi, \pi]$.

**If your value involves surds, use 'sqrt', e.g. $\sqrt{2}$ for the square root of 2.

▼ Feedback

Score: 1/1 ✓

Gap 0

- ✓ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

- ✓ Your answer is numerically correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

b)

Identify a matrix that applies a rotation of $-\pi/6$ about the origin followed by a reflection through the line $y = mx$ with $m = 0$

$$\begin{pmatrix} \underline{0.866} & \underline{0.5} \\ \underline{0.5} & \underline{-0.866} \end{pmatrix} \checkmark$$

Expected answer:

$$\begin{pmatrix} \underline{0.866} & \underline{0.500} \\ \underline{0.500} & \underline{-0.866} \end{pmatrix}$$

*Enter the entries of the matrix correct to 3 decimal places.

▼ Feedback

Score: 1/1 

 Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk) v10.0, developed by [Newcastle University \(https://www.newcastle.ac.uk\)](https://www.newcastle.ac.uk).

DEAKIN UNIVERSITY

LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

Module 4 - Determinants

Submitted By:
Jahan GARG
s226100103
2026/08/21 01:33

Tutor:
Simon JAMES

Task Outcomes		Supports
M4.1	Calculate determinants using the cofactor expansion	ULO1 ULO1
M4.2	Use properties of determinants (Theorem 3.1.2) to calculate determinants	ULO1 ULO1
M4.3	Calculate matrix inverses using $\text{adj } A$ and $\det(A)$	ULO1 ULO1
M4.4	Identify the inverse of an orthogonal matrix	ULO1 ULO1
M4.5	Perform LU-factorisation on a matrix and use this to solve systems of linear equations	ULO1 ULO1
Evid	Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.	
SA	Full printout of Self-Assessment results with all questions.	
Sum	Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions.	
FB	Respond to any feedback and work with the OnTrack tutor to complete the module.	

August 20, 2026



SIT292 – Linear Algebra for Data Analysis

Module 4 Summary: Determinants

Response to Feedback

This is a resubmission in response to Simon's feedback on my initial Module 4 submission. Simon noted that although he did not mind the reformatting of the document, the tables had lost their equation formatting, so matrices appeared as plain bracketed lists, and elements such as bold text were missing. He also noted that I do not need to rewrite the definitions themselves, only leave them as they are, and that for numeric examples it would be better to present these using the equation editor or hand-write them and include them in the evidence with a label, since describing numeric examples in a paragraph is not easy to follow. Finally, he asked me to update the table formatting where possible and to add self-assessment work, since the evidence provided was fine but did not yet include notes, annotations, or working for the NUMBAS questions.

In response, I have restored proper equation formatting in the Key Definitions and Theorems tables, including matrices, exponents, and fractions, while keeping the wording of each definition and theorem unchanged. I have also added handwritten working for the 4 by 4 determinant example as Exercise 4.2.6, which is referenced directly in the 4.2 sub-module discussion below and placed in the Learning Evidence between the handwritten section and the digital NUMBAS evidence. Finally, I have included full working and annotations for each question in my NUMBAS self-assessment submission, labelled and structured to show the concept applied, the method used, and a verification step for each answer.

Module Learning Objectives

I certify that I have achieved the following learning objectives for this module, as set out on the Module 4 introduction page of the unit site [1], which extend the unit learning outcome ULO1 (performing operations on matrices to calculate determinants):

1. Calculate determinants using the cofactor expansion.
2. Use properties of determinants (Theorem 3.1.2) to calculate determinants.
3. Calculate matrix inverses using $\text{adj } A$ and $\det(A)$.
4. Identify the inverse of an orthogonal matrix.
5. Perform LU factorisation on a matrix and use this to solve systems of linear equations.

Key Definitions and Theorems

The Cofactor Expansion

Textbook Reference	Theorem / Definition Statement
Determinant Calculation for a 3x3 matrix	For a 3x3 matrix A, the <u>determinant</u> is given by $\det A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ $= aei + bfg + cdh - ceg - afh - bdi$ $= a(ei - fh) - b(di - fg) + c(dh - eg)$

	$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$
Definition 3.1 Cofactors of a matrix	Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. Given the $n \times n$ matrix A, let A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j. Then the (i, j)-<u>cofactor</u> $c_{ij}(A)$ is the scalar defined by $c_{ij}(A) = (-1)^{i+j} \det(A_{ij})$ Here $(-1)^{i+j}$ is called the <u>sign</u> of the (i, j)-position.
Definition 3.2 Cofactor expansion of a matrix	If $A =$ Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. If $A = [a_{ij}]$ is $n \times n$ define, $\det A = a_{11}c_{11}(A) + a_{12}c_{12}(A) + \dots + a_{1n}c_{1n}(A)$ This is called the <u>cofactor expansion</u> of $\det A$ along row 1.
Theorem 3.1.2 Determinant Properties	Let A denote an $n \times n$ matrix. 1. If A has a row or column of zeros, $\det A = 0$. 2. If two distinct rows (or columns) of A are interchanged, the determinant of the resulting matrix is $-\det A$. 3. If a row (or column) of A is multiplied by a constant u, the determinant of the resulting matrix is $u(\det A)$. 4. If two distinct rows (or columns) of A are identical, $\det A = 0$. If a multiple of one row of A is added to a different row (or if a multiple of a column is added to a different column), the determinant of the resulting matrix is $\det A$.

Determinants and Matrix Inverses

Textbook Reference	Theorem / Definition Statement
Theorem 3.2.1 Product Theorem	If A and B are $n \times n$ matrices, then $\det(AB) = \det A \det B.$
Theorem 3.2.2 Det relationship to inverse	An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. When it is invertible, we have, $\det(A^{-1}) = \frac{1}{\det A}$
Theorem 3.2.3 Det of transpose	If A is any square matrix, $\det(A^T) = \det A$
Theorem 3.2.4 Adjugate formula	If A is any square matrix, then $A(\text{adj } A) = (\det A) I = (\text{adj } A) A$ In particular, if $\det A \neq 0$, the inverse of A is given by $A^{-1} = \frac{1}{\det A} \text{adj } A$

LU Factorisation

Textbook Reference	Theorem / Definition Statement
Theorem 2.7.1 LU Theorem	If A can be lower reduced to a row-echelon matrix U, then $A = LU$ where L is lower triangular and invertible, and U is upper triangular and row-echelon.

LU Algorithm	Let A be an $m \times n$ matrix of rank r, and suppose that A can be lower reduced to a row-echelon matrix U. Then $A = LU$, where the lower triangular, invertible matrix L is constructed as follows: 1. If $A=0$, take $L = I_m$ and $U = 0$. 2. If $A \neq 0$, write $A_1 = A$ and let $\mathbf{c}_1$ be the leading column of A_1. Use $\mathbf{c}_1$ to create the first leading 1 and create zeros below it (using lower reduction). When this is completed, let A_2 denote the matrix consisting of rows 2 to m of the matrix just created. 3. If $A_2 \neq 0$, let $\mathbf{c}_2$ be the leading column of A_2 and repeat Step 2 on A_2 to create A_3. 4. Continue in this way until U is reached, where all rows below the last leading 1 consist of zeros. This will happen after r steps. 5. Create L by placing $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_r$ at the bottom of the first r columns of I_m.
---------------------	--

Module Summary

4.2 The Cofactor Expansion

This sub-module extends the ideas of matrix algebra covered in Modules 1 to 3 by introducing the determinant as a single scalar value associated with any square matrix, calculated through the cofactor expansion (Definition 3.2). The determinant of a 3 by 3 matrix follows the expansion $aei + bfg + cdh - ceg - afh - bdi$, which is more conveniently grouped as $a(ei - fh)$ minus $b(di - fg)$ plus $c(dh - eg)$, where each 2 by 2 minor is obtained by deleting the row and column of the corresponding entry, consistent with Definition 3.1. What I found conceptually important here is that the cofactor $c_{ij}(A)$ already carries the alternating sign $(-1)^{i+j}$, so the expansion formula in Definition 3.2 does not require a separate sign pattern to be remembered once the cofactors themselves have been calculated correctly.

A convenience of the cofactor method, which I relied on throughout my Learning Evidence, is that the determinant can be evaluated by expanding along any row or column, not only the first row given in Definition 3.2. I confirmed this directly in NUMBAS N4.2.1, where I evaluated the same 3 by 3 matrix by expanding along row 2 and separately along row 1, obtaining the identical determinant in each case, which reinforces that the cofactor expansion is a genuinely well-defined property of the matrix rather than an artefact of the row chosen.

Simplifying a larger determinant with Theorem 3.1.2

Theorem 3.1.2 provides five properties that make evaluating larger determinants considerably more manageable than repeated cofactor expansion alone, since a matrix can first be manipulated into a form with more zero entries before expanding. To demonstrate this, I worked through a 4 by 4 example (following the same method as Exercise 4.2.6 in the Additional Examples and Exercises section of 4.2 on the unit site) using the matrix with rows (3, 5, 4, 6), (-2, 1, 0, 7), (-5, 4, 7, 2) and (8, -3, 1, 1). Since row 2 already contains a single 1 in column 2, I applied property 5 of Theorem 3.1.2 three times, replacing row 1 with row 1 minus 5 times row 2, row 3 with row 3 minus 4 times row 2, and row 4 with row 4 plus 3 times row 2. None of these operations changes the determinant, and the result produces a column 2 containing only one non-zero entry, namely the 1 already in row 2.

Expanding along this column then reduces the problem to a single 3 by 3 determinant with rows (13, 4, -29), (3, 7, -26) and (2, 1, 22). Applying property 5 once again, using row 3 (which contains a 1 in its

middle position) to clear the middle entries of rows 1 and 2, produces a further column with only one non-zero entry. Expanding along this second reduced column leaves only a 2 by 2 determinant to evaluate directly, giving a final value of 2187 for the original 4 by 4 determinant. The full handwritten working for this example is provided as Exercise 4.2.6 in the Learning Evidence, placed between the handwritten section and the digital NUMBAS evidence. This example illustrates precisely how property 5 of Theorem 3.1.2 can be used strategically, before any cofactor expansion, to collapse a larger determinant calculation into a much smaller one, which is the same strategy I used in my own digital evidence for NUMBAS N4.2.3 when reducing a matrix to upper triangular form and then reading the determinant directly as the product of the diagonal entries.

4.3 Determinants and Matrix Inverses

This sub-module connects the determinant back to the concept of the matrix inverse first introduced in Module 3, formalised through Theorem 3.2.2: an n by n matrix A is invertible if and only if $\det(A)$ is not equal to zero, with $\det(A^{-1})$ equal to 1 divided by $\det(A)$ whenever the inverse exists. Theorem 3.2.1, the Product Theorem, states that $\det(AB)$ equals $\det(A)$ times $\det(B)$ for any two n by n matrices, and Theorem 3.2.3 shows that transposing a matrix leaves its determinant unchanged: $\det(A^T)$ equals $\det(A)$. I applied all three of these properties directly in NUMBAS N4.3.3, where, given $\det(A)$ equal to 2 and $\det(B)$ equal to 3 for two 4 by 4 matrices, I calculated $\det(AB)$ equal to 6 using the Product Theorem, $\det(2A)$ equal to 32 using the scalar property $\det(kA)$ equals k to the power n times $\det(A)$ for an n by n matrix, and $\det(A^T B^{-1})$ equal to 2 thirds by combining Theorem 3.2.3 with the reciprocal relationship in Theorem 3.2.2.

Theorem 3.2.4, the Adjugate Formula, gives the explicit relationship $A(\text{adj } A)$ equals $(\det A)I$, which rearranges to A^{-1} equals $(1 \text{ divided by } \det A)$ times $\text{adj}(A)$ whenever $\det(A)$ is not equal to zero, where $\text{adj}(A)$ is the transpose of the cofactor matrix of A . I practised this method extensively in my Learning Evidence and digital submissions, most notably in Exercises 4.3.1 through 4.3.3 in the Additional Examples section of 4.3 on the unit site, where I computed all nine cofactors of a given matrix, arranged them into the cofactor matrix, transposed this to obtain the adjugate, and then verified that A times $\text{adj}(A)$ produced a scalar multiple of the identity matrix equal to $\det(A)$ times I in every case. This method is closely related to rank calculation using sub-matrix determinants, since a matrix is singular precisely when its largest non-zero square sub-matrix determinant has an order smaller than the full matrix, which connects directly back to the rank concept first introduced through row operations in Module 1.

Cramer's Rule

Cramer's rule provides an alternative method to Gaussian elimination for solving a system of linear equations $Ax = b$, applicable whenever A is a square coefficient matrix with $\det(A)$ not equal to zero. For a system in n unknowns, each unknown x_i is given by x_i equals $\det(A_i)$ divided by $\det(A)$, where A_i is the matrix formed by replacing the i -th column of A with the constants vector b , following directly from the invertibility condition in Theorem 3.2.2 since x equals $A^{-1}b$ can be expressed entirely in terms of determinants using the adjugate formula (Theorem 3.2.4). I applied this method in my handwritten Learning Evidence for Exercise 3.2.8(a), solving a three-equation system by first confirming $\det(A)$ equal to negative 23, then calculating three further determinants D_x , D_y and D_z by substituting the constants column into each respective position, and finally dividing each by $\det(A)$ to obtain x equal to negative 1, y equal to negative 4 and z equal to negative 2, which I verified directly by substitution into all three original equations. The condition $\det(A)$ not equal to zero is essential here, since a zero determinant means A is not invertible, so the formula x_i equals $\det(A_i)/\det(A)$ would involve division by

zero, and the system would not have a unique solution.

4.4 LU-Factorisation

LU factorisation revisits Gaussian elimination from Module 1, expressing it algebraically as a matrix product rather than a sequence of row operations. Theorem 2.7.1 states that if a matrix A can be lower reduced, meaning reduced to row-echelon form using only operations that add a multiple of one row to a row strictly below it, without any row interchanges and without scaling rows by a constant other than to create a leading 1, then $A = LU$, where L is lower triangular and invertible and U is upper triangular and in row-echelon form. This restriction to lower reduction only is essential to the method, since row interchanges would disrupt the direct correspondence between the leading columns encountered during elimination and the entries of L , which is why the LU Algorithm explicitly proceeds column by column without ever swapping rows, unlike the more general Gaussian Algorithm from Module 1.

The LU Algorithm constructs U through the same elimination steps used for row-echelon form, but records the leading column at each stage, before it is normalised to contain a leading 1, and places these recorded columns into the corresponding columns of an otherwise identity matrix to form L . I practised this directly in my handwritten Exercise 2.7.1, factorising a 4 by 5 matrix in which the third pivot step encountered a completely zero remaining row, requiring the leading column to be taken from column 5 rather than column 3, and in my digital evidence for NUMBAS N4.4.1, where I obtained L and U for a 3 by 3 matrix and confirmed A equals LU by direct multiplication. LU factorisation is particularly useful for solving $Ax = b$ when the same coefficient matrix A must be solved against multiple different vectors b , since once L and U are known, the system reduces to first solving $Ly = b$ for y by forward substitution and then solving $Ux = y$ for x by back substitution, both of which are computationally cheaper than repeating full Gaussian elimination each time, exactly as I demonstrated in Exercises 4.4.1 through 4.4.3 in the Additional Examples section of 4.4 on the unit site.

Observations and Connections Across Sub-modules

Working through Module 4 reinforced how the determinant acts as a single numerical summary that ties together concepts from every earlier module in this unit. The link to Module 1 is direct through Theorem 2.7.1 and the LU Algorithm, since LU factorisation is simply Gaussian elimination performed without row interchanges and recorded in matrix form, while the link to Module 3 comes through Theorem 3.2.2, since $\det(A) \neq 0$ is an alternative, single-number test for the same invertibility condition that was previously characterised through rank and the Inverse Theorem. I found it particularly satisfying that the adjugate formula in Theorem 3.2.4 gives a completely independent method of calculating a matrix inverse compared with the row-reduction method from Module 3, yet both methods necessarily agree whenever $\det(A)$ is not equal to zero, which is a useful cross-check when practising by hand.

A further connection worth noting is between Cramer's rule and the LU-based method of solving $Ax = b$. Cramer's rule is conceptually elegant and expresses each unknown directly as a ratio of determinants, but it requires calculating $n + 1$ separate determinants for a system of n equations, which becomes computationally expensive very quickly as n grows. The $Ly = b$ then $Ux = y$ approach used in LU factorisation avoids this cost entirely once L and U have been found, which is why, as noted in the video introduction to this module, LU factorisation is the more practical method for large or repeated systems, while Cramer's rule remains valuable for small systems or for deriving closed form expressions for the solution in terms of the coefficients.

Extra Effort and Prior Knowledge

Beyond the prescribed unit-site material, I consulted the second worked example video by Peter Huf referenced in section 4.3 of the unit site, which worked through the same style of adjugate and cofactor calculation as Exercise 4.3.1 but for a different numerical matrix, and this repetition with different numbers helped me confirm that I could apply the cofactor sign pattern and adjugate transpose consistently rather than having memorised a single specific example. I also worked through the R code examples provided in sections 4.2 and 4.4 of the unit site conceptually, tracing through the automated row reduction steps by hand for a small matrix to confirm that the property based simplification I was using by hand (particularly property 5 of Theorem 3.1.2) matched the systematic approach the code applies for reducing a matrix to upper triangular form before reading off the determinant as a product of diagonal entries.

In terms of prior knowledge, I had some previous exposure to the 2 by 2 determinant formula $ad - bc$ from high school mathematics, in the context of finding the area of a shape defined by two vectors, which matched directly with the geometric interpretation given in section 4.2 of the unit site regarding the determinant corresponding to the area of a parallelogram in two dimensions and the volume of a parallelepiped in three dimensions. However, the systematic cofactor expansion for 3 by 3 and larger matrices, the adjugate-based inverse formula, and LU factorisation were all genuinely new material for me in this module, requiring careful practice through the NUMBAS exercises and the recommended textbook problems to become confident with the sign conventions and the bookkeeping involved in larger matrices.

Reflecting on the Content

I found the connection between LU factorisation and the Gaussian elimination from Module 1 the most satisfying part of this module, since it demonstrated that a technique I had already practised extensively could be reframed as an explicit matrix factorisation with real computational advantages for solving multiple systems. The most challenging part was keeping track of signs during cofactor expansion of larger matrices, particularly in Exercise 2.7.1 where a zero row appeared partway through the LU reduction and I needed to identify the correct next leading column rather than assuming the pivot would always move to the next sequential column. Working through this carefully, together with the determinant properties in Theorem 3.1.2, has given me a much stronger appreciation for why certain matrix manipulations preserve important quantities like the determinant while others do not, which I expect will be directly relevant when eigenvalues are introduced in the next module, since the characteristic equation $\det(A - \lambda I) = 0$ relies entirely on the determinant techniques developed here.

References

- [1] Deakin University, "Topic 4 – Determinants: 4.1 Introduction, 4.2 The Cofactor Expansion, 4.3 Determinants and Matrix Inverses, 4.4 LU-Factorisation, Module 4 Assessment," SIT292 Linear Algebra for Data Analysis, CloudDeakin Unit Site, Deakin University, Trimester 2, 2026.
- [2] W. K. Nicholson, Linear Algebra with Applications, Lyryx, 2021. [Online]. Available: Nicholson-OpenLAWA.pdf, provided via the SIT292 unit site content repository.
- [3] Deakin University, "SIT292 Module 4 Self-Assessment," Numbas, developed by Newcastle University, accessed 18 Aug. 2026.
- [4] P. Huf, "SIT292 Practice Tutorial: Matrix Determinants and Inverses," video, referenced in 4.3 Determinants and Matrix Inverses, SIT292 Unit Site, Deakin University, 2026.

PRACTICE QUESTIONS

DATE / /

PAGE

4.2 The Cofactor Expansion

Exercise 3.1.14 Evaluate by first adding all other rows to the first row.

(a) $\det \begin{bmatrix} x-1 & 2 & 3 \\ 2 & -3 & x-2 \\ -2 & x & -2 \end{bmatrix}$

Concept / Theorem / Property Applied

Theorem 3.1.2 - Property 5:

Adding a multiple of one row to a different row does not change the value of the determinant.

Cofactor Expansion Theorem (Theorem 3.1.1):

A determinant may be evaluated by cofactor expansion along any row (or column).

2x2 determinant formula:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= ad - bc$$

Ans:

Method

First apply the required row operation:

$$R_1 \rightarrow R_1 + R_2 + R_3$$

Since this is a row-addition operation, the determinant remains unchanged.

Step 1. Apply the row operation

$$\det(A) = \det \begin{bmatrix} (x-1)+2-2 & 2-3+x & 3+(x-2)-2 \\ 2 & -3 & x-2 \\ -2 & x & -2 \end{bmatrix}$$

Simplify the first row:

$$\det(A) = \det \begin{bmatrix} x-1 & x-1 & x-1 \\ 2 & -3 & x-2 \\ -2 & x & -2 \end{bmatrix}$$

Step 2. Factor out the common factor from row 1

Property Applied: A common scalar factor may be taken outside the determinant.

$$\det(A) = (x-1) \det \begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & x-2 \\ -2 & x & -2 \end{bmatrix}$$

Step 3. Cofactor expansion along Row 1:

Method: Expand along Row 1 using the sign pattern.

$\begin{matrix} + & - & + \end{matrix}$

$$\therefore \det(A) = (x-1) \left[1 \begin{vmatrix} -3 & x-2 \\ x & -2 \end{vmatrix} - 1 \begin{vmatrix} 2 & x-2 \\ -2 & -2 \end{vmatrix} + 1 \begin{vmatrix} 2 & -3 \\ -2 & x \end{vmatrix} \right]$$

Step 4. Evaluate the 2×2 determinants

$$\text{First determinant: } \begin{vmatrix} -3 & x-2 \\ x & -2 \end{vmatrix} = (-3)(-2) - (x-2)(x) \\ = 6 - x^2 + 2x$$

$$\text{Second determinant: } \begin{vmatrix} 2 & x-2 \\ -2 & -2 \end{vmatrix} = (2)(-2) - (x-2)(-2) \\ = -4 + 2x - 4 \\ = 2x - 8$$

$$\text{Third determinant: } \begin{vmatrix} 2 & -3 \\ -2 & x \end{vmatrix} = (2)(x) - (-3)(-2) \\ = 2x - 6$$

Step 5. Substitute and simplify

$$\det(A) = (x-1) [(6-x^2+2x) - (2x-8) + (2x-6)] \\ = (x-1)(-x^2+2x+8)$$

Factor the quadratic:

$$-x^2 + 2x + 8 = -(x^2 - 2x - 8) = -(x-4)(x+2)$$

$$\text{Hence, } \det(A) = -(x-1)(x-4)(x+2)$$

Verification:

Using the original matrix directly, the determinant simplifies to the same expression:

$$\det(A) = -(x-1)(x-4)(x+2)$$

Therefore, the result is verified.

Determinant - preserving row operations can simplify a determinant before applying a cofactor expansion.

4.3 Determinants and Matrix Inverses

Exercise 3.2.8 Solve each of the following by Cramer's rule:

(c)
$$\begin{aligned} 5x + y - z &= -7 \\ 2x - y - 2z &= 6 \\ 3x + 2z &= -7 \end{aligned}$$

Concept / Theorem / Property Applied

Cramer's Rule

For a system $Ax = b$, where A is an $n \times n$ matrix and $\det(A) \neq 0$, each unknown is given by:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

(Here, A_i is obtained by replacing the i -th column of A with constants column.)

Important condition: $\det(A) \neq 0$ must hold for Cramer's Rule to give a unique solution.

Ans

Method

Write the coefficient matrix and constants vector:

$$A = \begin{bmatrix} 5 & 1 & -1 \\ 2 & -1 & -2 \\ 3 & 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} -7 \\ 6 \\ -7 \end{bmatrix}$$

First calculate $\det(A)$. Then calculate the determinants obtained by replacing the x -, y -, and z -columns.

Step 1. Find $\det(A)$.

Cofactor expansion along Row 1.

Using the cofactor sign pattern: $+-+$

$$\det(A) = 5 \begin{vmatrix} -1 & -2 \\ 0 & 2 \end{vmatrix} - (-1) \begin{vmatrix} 2 & -2 \\ 3 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 2 & -1 \\ 3 & 0 \end{vmatrix}$$

Using $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$,

$$\begin{aligned} \det(A) &= 5 [(-1)(2) - (-2)(0)] - [(2)(2) - (-2)(3)] + (-1) [(2)(0) - (-1)(3)] \\ &= 5(-2) - (4+6) + (-1)(3) \\ &= -10 - 10 - 3 \\ &= -23 \end{aligned}$$

Since, $\det(A) \neq 0$, the system has a unique solution and Cramer's Rule can be applied.

Step 2. Find D_x .

Replace the x -column with the constants.

$$A_x = \begin{bmatrix} -7 & 1 & -1 \\ 6 & -1 & -2 \\ -7 & 0 & 2 \end{bmatrix}$$

Cofactor expansion along Row 1

$$D_x = (-7) \begin{vmatrix} -1 & -2 \\ 0 & 2 \end{vmatrix} - 1 \begin{vmatrix} 6 & -2 \\ -7 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 6 & -1 \\ -7 & 0 \end{vmatrix}$$

$$\begin{aligned} &= (-7) [(-1)(2) - (-2)(0)] - [6(2) - (-2)(-7)] + (-1) [6(0) - (-1)(-7)] \\ &= (-7)(-2) - (12 - 14) + (-1)(-7) \\ &= 14 + 2 + 7 \\ &= 23 \end{aligned}$$

Step 3. Find D_y

Replace the y -column with the constants.

$$A_y = \begin{bmatrix} 5 & -7 & -1 \\ 2 & 6 & -2 \\ 3 & -7 & 2 \end{bmatrix}$$

Cofactor expansion along Row 1 -

$$D_y = 5 \begin{vmatrix} 6 & -2 \\ -7 & 2 \end{vmatrix} - (-7) \begin{vmatrix} 2 & -2 \\ 3 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 2 & 6 \\ 3 & -7 \end{vmatrix}$$

$$\begin{aligned} &= 5 [6(2) - (-2)(-7)] + 7 [(2)(2) - (-2)(3)] + (-1) [(2)(-7) - (6)(3)] \\ &= 5 (12 - 14) + 7 (4 + 6) + (-1) (-14 - 18) \\ &= 5(-2) + 7(10) + 32 \\ &= -10 + 70 + 32 \\ &= 92 \end{aligned}$$

Step 4. Find D_z

Replace the z -column with the constants.

$$A_z = \begin{bmatrix} 5 & 1 & -7 \\ 2 & -1 & 6 \\ 3 & 0 & -7 \end{bmatrix}$$

Cofactor expansion along Row 1

$$D_z = 5 \begin{vmatrix} -1 & 6 \\ 0 & -7 \end{vmatrix} - 1 \begin{vmatrix} 2 & 6 \\ 3 & -7 \end{vmatrix} + (-7) \begin{vmatrix} 2 & -1 \\ 3 & 0 \end{vmatrix}$$

$$= 5[(-1)(-7) - (6)(0)] - [(2)(-7) - (6)(3)] + (-7)[(2)(0) - (-1)(3)]$$

$$= 5(7) - (-14 - 18) + (-7)(3)$$

$$= 35 + 32 - 21$$

$$= 46$$

Step 5. Apply Cramer's Rule

For x ,

$$x = \frac{D_x}{D} = \frac{23}{-23} = -1$$

$$D = -23$$

For y ,

$$y = \frac{D_y}{D} = \frac{92}{-23} = -4$$

$$D = -23$$

For z ,

$$z = \frac{D_z}{D} = \frac{46}{-23} = -2$$

$$D = -23$$

Verification:

Substitute $x = -1$, $y = -4$, $z = -2$ into the original equations.

$$\text{Equation 1. } 5(-1) + (-4) - (-2) = -5 - 4 + 2 = -7$$

$$\text{Equation 2. } 2(-1) - (-4) - 2(-2) = -2 + 4 + 4 = 6$$

$$\text{Equation 3. } 3(-1) + 2(-2) = -3 - 4 = -7$$

Therefore, the solution is verified.

Key Learning Point

Cramer's Rule converts a system of linear eq^{ns} into determinant calculations; the key condition is $\det(A) \neq 0$, which guarantees a unique solution.

4.4 LU-Factorisation

Exercise 2.7.1 Find an LU-factorization of the following matrices:

(c)
$$\begin{bmatrix} 2 & 6 & -2 & 0 & 2 \\ 1 & 5 & -1 & 2 & 5 \\ 3 & 7 & -3 & -2 & 5 \\ -1 & -1 & 1 & 2 & 3 \end{bmatrix}$$

Concept / Theorem / Property Applied

LU-Factorisation

According to Theorem 2.7.1, if A can be reduced to a row-echelon matrix U without row interchanges, then $A=LU$, where L is lower triangular and U is upper triangular and row-echelon.

LU-Algorithm

The Gaussian elimination process is used to obtain U . The leading columns used during the reduction are then placed into L , replacing the corresponding columns of the identity matrix.

Important: No row interchange is used.

Ans: Method - Lower reduction to obtain U

Step 1: Use the first leading column

The first non-zero column of A is column 1:

$$\begin{bmatrix} 2 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

Normalise the first row: $R_1 \rightarrow \frac{1}{2} R_1$

$$\begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 1 & 5 & -1 & 2 & 5 \\ 3 & 7 & -3 & -2 & 5 \\ -1 & -1 & 1 & 2 & 3 \end{bmatrix}$$

Eliminate the entries below the first pivot:

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_4 \rightarrow R_4 + R_1$$

$$\begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 0 & 2 & 0 & 2 & 4 \\ 0 & -2 & 0 & -2 & 2 \\ 0 & 2 & 0 & 2 & 4 \end{bmatrix}$$

Leading columns recorded to L:

$$C_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \\ -1 \end{bmatrix}$$

Step 2. Use the next leading column

After removing the completed first row, the next leading column is Column 2.

Normalise the second pivot: $R_2 \rightarrow \frac{1}{2} R_2$

$$\begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & -2 & 0 & -2 & 2 \\ 0 & 2 & 0 & 2 & 4 \end{bmatrix}$$

Eliminate below the second pivot:

$$R_3 \rightarrow R_3 + 2R_2$$

$$R_4 \rightarrow R_4 - 2R_2$$

$$\begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Leading column recorded for L:

From the remaining matrix before normalising the pivot, $c_2 = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}$.

Step 3. Identify the next leading column

The ~~stomains~~ remaining non-zero row is:

$$[0 \ 0 \ 0 \ 0 \ 6]$$

Therefore, the next leading column is Column 5.

Normalise the third pivot: $R_3 \rightarrow \frac{1}{6}R_3$

$$: \begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \text{ pivots}$$

Leading column recorded for L :

$$c_3 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

There are no further non-zero rows, so the reduction is complete.

Step 4. Write the matrix U .

The row-echelon matrix obtained is:

$$U = \begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 5. Construct the matrix L .

LU-Algorithm: Start with the 4×4 identity matrix and replace the bottom portions of its first three columns with the recorded leading columns c_1, c_2, c_3 .

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Using $c_1 = \begin{bmatrix} 2 \\ 1 \\ 3 \\ -1 \end{bmatrix}$, $c_2 = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}$, $c_3 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$, we get:

$$L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 3 & -2 & 6 & 0 \\ -1 & 2 & 0 & 1 \end{bmatrix} \quad (\text{Here, } L \text{ is lower triangular})$$

Step 6. Verify $A = LU$

Multiply $L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 3 & -2 & 6 & 0 \\ -1 & 2 & 0 & 1 \end{bmatrix}$ by $U = \begin{bmatrix} 1 & 3 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

For example, $(2,1)$ entry $= 1(1) + 0 = 2$, $(2,2)$ entry $= 1(3) + 2(1) = 5$, $(3,4)$ entry $= 3(0) + (-2)(1) + 6(0) = -2$, and similarly for the remaining entries.

$$\text{Thus, } LU = \begin{bmatrix} 2 & 6 & -2 & 0 & 2 \\ 1 & 5 & -1 & 2 & 5 \\ 3 & 7 & -3 & -2 & 5 \\ -1 & -1 & 1 & 2 & 3 \end{bmatrix} = A$$

$\therefore A = LU$ is verified.

Learning Point

DATE	/	/
PAGE		

Key

In the LU-algorithm, Gaussian elimination produces U , while the leading columns used during the reduction were retained to construct L .

Additional Worked Example — Exercise 4.2.6

Simplifying a 4×4 Determinant using Theorem 3.1.2.

Question. Evaluate $\det(A)$, $A = \begin{bmatrix} 3 & 5 & 4 & 6 \\ -2 & 1 & 0 & 7 \\ -5 & 4 & 7 & 2 \\ 8 & -3 & 1 & 1 \end{bmatrix}$.

Concept / Theorem Applied:

Theorem 3.1.2 — Property 5:

Adding a multiple of one row to another row does not change the determinant.

We use this property to create zeros in Column 2 and simplify the determinant before cofactor expansion.

Step 1. Clear Column 2

Use R_2 , as its second entry is 1:

$$R_1 \rightarrow R_1 - 5R_2$$

$$R_3 \rightarrow R_3 - 4R_2$$

$$R_4 \rightarrow R_4 + 3R_2$$

Therefore, $\det(A) = \det \begin{bmatrix} 13 & 0 & 4 & -29 \\ -2 & 1 & 0 & 7 \\ 3 & 0 & 7 & -26 \\ 2 & 0 & 1 & 22 \end{bmatrix}$

Step 2. Expanded Along Column 2

Cofactor Expansion Theorem: Expanded along Column 2.

The only non-zero entry is $a_{22}=1$, and $(-1)^{2+2} = +1$.

Hence, $\det(A) = 1 \begin{vmatrix} 13 & 4 & -29 \\ 3 & 7 & -26 \\ 2 & 1 & 22 \end{vmatrix}$

Step 3. Clear Column 2 of the 3×3 Determinant
Use the third row, whose second entry is 1.

$$R_1 \rightarrow R_1 - 4R_3$$

$$R_2 \rightarrow R_2 - 7R_3$$

Thus, $\det(A) = \begin{vmatrix} 5 & 0 & -117 \\ -11 & 0 & -180 \\ 2 & 1 & 22 \end{vmatrix}$

Step 4. Expand Along Column 2 Again

The only non-zero entry is $a_{32} = 1$.

Its cofactor sign is $(-1)^{3+2} = -1$.

$$\text{Therefore, } \det(A) = (-1) \begin{vmatrix} 5 & -117 \\ -11 & -180 \end{vmatrix}$$

Step 5. Evaluate the 2×2 Determinant

$$\text{Using } \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc, \quad \begin{vmatrix} 5 & -117 \\ -11 & -180 \end{vmatrix} = 5(-180) - (-117)(-11)$$

$$= -900 - 1287$$

$$= -2187$$

$$\text{Hence, } \det(A) = (-1)(-2187) = 2187$$

$$\boxed{\det(A) = 2187}$$

Verification:

Direct evaluation of the original 4×4 determinant also gives $\det(A) = 2187$.

Therefore, the result is verified.

Key Learning Point:

Determinant preserving row-operations can be used strategically to create zeros and reduce a large determinant.

Question 1

Consider the matrix

$$A = \begin{pmatrix} -4 & -2 & -2 \\ 2 & -1 & 1 \\ -1 & 4 & -4 \end{pmatrix}$$

Evaluate the determinant by expanding along row 2.

$$\begin{aligned} \det(A) &= \boxed{-2} \checkmark \times \det \begin{pmatrix} \boxed{-2} & \boxed{-2} \\ \boxed{4} & \boxed{-4} \end{pmatrix} \checkmark + \boxed{-1} \checkmark \times \det \begin{pmatrix} \boxed{-4} & \boxed{-2} \\ \boxed{-1} & \boxed{-4} \end{pmatrix} \checkmark + \boxed{-1} \checkmark \times \det \begin{pmatrix} \boxed{-4} & \boxed{-2} \\ \boxed{-1} & \boxed{4} \end{pmatrix} \checkmark \\ &= \boxed{-2} \checkmark \times \boxed{16} \checkmark + \boxed{-1} \checkmark \times \boxed{14} \checkmark + \boxed{-1} \checkmark \times \boxed{-18} \checkmark \\ &= \boxed{-28} \checkmark \end{aligned}$$

Note: If the coefficient of any sub-matrix determinant is zero, you will still need to include the submatrix in the first line of working and include its determinant in the second line of working.

Gap 0

Your answer is correct. You were awarded 1 mark. ✓

Gap 1

Your answer is correct. You were awarded 1 mark. ✓

Gap 2

Your answer is correct. You were awarded 1 mark. ✓

Gap 3

Your answer is correct. You were awarded 1 mark. ✓

Gap 4

Your answer is correct. You were awarded 1 mark. ✓

Gap 5

Your answer is correct. You were awarded 1 mark. ✓

Gap 6

Your answer is correct. You were awarded 1 mark. ✓

Gap 7

Your answer is correct. You were awarded 1 mark. ✓

Gap 8

Your answer is correct. You were awarded 1 mark. ✓

Gap 9

Your answer is correct. You were awarded 1 mark. ✓

Gap 10

Your answer is correct. You were awarded 1 mark. ✓

Gap 11

Your answer is correct. You were awarded 1 mark. ✓

Gap 12

Your answer is correct. You were awarded 1 mark. ✓

You scored **13** marks for this part.

Score: 13/13 ✓

Question 1

Consider the matrix

$$A = \begin{pmatrix} 2 & -3 & 3 \\ -5 & 4 & 5 \\ 1 & -5 & -5 \end{pmatrix}$$

Evaluate the determinant by expanding along row 1.

$$\begin{aligned} \det(A) &= \boxed{2} \checkmark \times \det \begin{pmatrix} \boxed{4} & \boxed{5} \\ \boxed{-5} & \boxed{-5} \end{pmatrix} \checkmark + \boxed{3} \checkmark \times \det \begin{pmatrix} \boxed{-5} & \boxed{5} \\ \boxed{1} & \boxed{-5} \end{pmatrix} \checkmark + \boxed{3} \checkmark \times \det \begin{pmatrix} \boxed{-5} & \boxed{4} \\ \boxed{1} & \boxed{-5} \end{pmatrix} \checkmark \\ &= \boxed{2} \checkmark \times \boxed{5} \checkmark + \boxed{3} \checkmark \times \boxed{20} \checkmark + \boxed{3} \checkmark \times \boxed{21} \checkmark \\ &= \boxed{133} \checkmark \end{aligned}$$

Note: If the coefficient of any sub-matrix determinant is zero, you will still need to include the submatrix in the first line of working and include its determinant in the second line of working.

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

Gap 2

Your answer is correct. You were awarded 1 mark.

Gap 3

Your answer is correct. You were awarded 1 mark.

Gap 4

Your answer is correct. You were awarded 1 mark. ✓

Gap 5

Your answer is correct. You were awarded 1 mark. ✓

Gap 6

Your answer is correct. You were awarded 1 mark. ✓

Gap 7

Your answer is correct. You were awarded 1 mark. ✓

Gap 8

Your answer is correct. You were awarded 1 mark. ✓

Gap 9

Your answer is correct. You were awarded 1 mark. ✓

Gap 10

Your answer is correct. You were awarded 1 mark. ✓

Gap 11

Your answer is correct. You were awarded 1 mark. ✓

Gap 12

Your answer is correct. You were awarded 1 mark. ✓

You scored **13** marks for this part.

Score: 13/13 ✓

Question 1

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = 3$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☒ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☐ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column



What is the value of the resulting determinant?

0



Gap 0

Your answer is correct. You were awarded **1** mark. ✓

Gap 1

You chose a correct answer. You were awarded **1** mark. ✓

You scored **2** marks for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = -4$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☒ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column



What is the value of the resulting determinant?

0



Gap 0

Your answer is correct. You were awarded **1** mark. ✓

Gap 1

You chose a correct answer. You were awarded **1** mark. ✓

You scored **2** marks for this part.

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

For the matrix,

$$A = \begin{pmatrix} -2 & -3 & 0 & 3 \\ 4 & 5 & 0 & -2 \\ 0 & 1 & -1 & -2 \\ -2 & -6 & 1 & 9 \end{pmatrix}$$

Evaluate the determinant by reducing it to upper triangular form. Show your working step by step.

Step 1. Use row 1 to make all other entries in column 1 equal to 0.

$$\begin{pmatrix} \boxed{-2} & \boxed{-3} & \boxed{0} & \boxed{3} \\ \boxed{0} & \boxed{-1} & \boxed{0} & \boxed{4} \\ \boxed{0} & \boxed{1} & \boxed{-1} & \boxed{-2} \\ \boxed{0} & \boxed{-3} & \boxed{1} & \boxed{6} \end{pmatrix} \quad \checkmark$$

Step 2. Now use row 2 to make all entries in column 2 (rows 3 and 4 only) equal to 0.

$$\begin{pmatrix} \boxed{-2} & \boxed{-3} & \boxed{0} & \boxed{3} \\ \boxed{0} & \boxed{-1} & \boxed{0} & \boxed{4} \\ \boxed{0} & \boxed{0} & \boxed{-1} & \boxed{2} \\ \boxed{0} & \boxed{0} & \boxed{1} & \boxed{-6} \end{pmatrix} \quad \checkmark$$

Step 3. Finally, use row 3 to eliminate the entry in column 3, row 4, equal to 0.

$$\begin{pmatrix} \boxed{-2} & \boxed{-3} & \boxed{0} & \boxed{3} \\ \boxed{0} & \boxed{-1} & \boxed{0} & \boxed{4} \\ \boxed{0} & \boxed{0} & \boxed{-1} & \boxed{2} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{-4} \end{pmatrix} \quad \checkmark$$

Calculate the value of the determinant by multiplying the diagonal entries.

8



*Note that in this case you shouldn't have needed to multiply any of the base rows/columns by a constant (in which case we would need to also apply Theorem 3.1.2 property 3), rather we are just adding a multiple of other rows (i.e., using property 5 only).

Gap 0

Your answer is correct. You were awarded 1 mark. ✓

Gap 1

Your answer is correct. You were awarded 1 mark. ✓

Gap 2

Your answer is correct. You were awarded 1 mark. ✓

Gap 3

Your answer is correct. You were awarded 1 mark. ✓

You scored 4 marks for this part.

Score: 4/4 ✓

Question 1

For the matrix,

$$A = \begin{pmatrix} 1 & 3 & -2 & 2 \\ 1 & 5 & 1 & 1 \\ 2 & 6 & -2 & 8 \\ 1 & 1 & -1 & 6 \end{pmatrix}$$

Evaluate the determinant by reducing it to upper triangular form. Show your working step by step.

Step 1. Use row 1 to make all other entries in column 1 equal to 0.

$$\begin{pmatrix} \boxed{1} & \boxed{3} & \boxed{-2} & \boxed{2} \\ \boxed{0} & \boxed{2} & \boxed{3} & \boxed{-1} \\ \boxed{0} & \boxed{0} & \boxed{2} & \boxed{4} \\ \boxed{0} & \boxed{-2} & \boxed{1} & \boxed{4} \end{pmatrix} \quad \checkmark$$

Step 2. Now use row 2 to make all entries in column 2 (rows 3 and 4 only) equal to 0.

$$\begin{pmatrix} \boxed{1} & \boxed{3} & \boxed{-2} & \boxed{2} \\ \boxed{0} & \boxed{2} & \boxed{3} & \boxed{-1} \\ \boxed{0} & \boxed{0} & \boxed{2} & \boxed{4} \\ \boxed{0} & \boxed{0} & \boxed{4} & \boxed{3} \end{pmatrix} \quad \checkmark$$

Step 3. Finally, use row 3 to eliminate the entry in column 3, row 4, equal to 0.

$$\begin{pmatrix} \boxed{1} & \boxed{3} & \boxed{-2} & \boxed{2} \\ \boxed{0} & \boxed{2} & \boxed{3} & \boxed{-1} \\ \boxed{0} & \boxed{0} & \boxed{2} & \boxed{4} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{-5} \end{pmatrix} \quad \checkmark$$

Calculate the value of the determinant by multiplying the diagonal entries.

-20



*Note that in this case you shouldn't have needed to multiply any of the base rows/columns by a constant (in which case we would need to also apply Theorem 3.1.2 property 3), rather we are just adding a multiple of other rows (i.e., using property 5 only).

Gap 0

Your answer is correct. You were awarded 1 mark. ✓

Gap 1

Your answer is correct. You were awarded 1 mark. ✓

Gap 2

Your answer is correct. You were awarded 1 mark. ✓

Gap 3

Your answer is correct. You were awarded 1 mark. ✓

You scored 4 marks for this part.

Score: 4/4 ✓

Question 1

For the following matrix,

$$A = \begin{pmatrix} 5 & -1 & -9 \\ 5 & 6 & 0 \\ -2 & -1 & 3 \end{pmatrix}$$

First, calculate the cofactors of A ,

$$\begin{array}{lll} c_{11} = 18 & \checkmark & c_{12} = -15 & \checkmark & c_{13} = 7 & \checkmark \\ c_{21} = 12 & \checkmark & c_{22} = -3 & \checkmark & c_{23} = 7 & \checkmark \\ c_{31} = 54 & \checkmark & c_{32} = -45 & \checkmark & c_{33} = 35 & \checkmark \end{array}$$

Hence give the adjugate of A ,

Rows: 3 Columns: 3

$$\text{adj}A = \begin{pmatrix} 18 & 12 & 54 \\ -15 & -3 & -45 \\ 7 & 7 & 35 \end{pmatrix} \quad \checkmark$$

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

Gap 2

Your answer is correct. You were awarded 1 mark.

Gap 3

Your answer is correct. You were awarded 1 mark.

Gap 4

Your answer is correct. You were awarded 1 mark.

Gap 5

Your answer is correct. You were awarded **1** mark. ✓

Gap 6

Your answer is correct. You were awarded **1** mark. ✓

Gap 7

Your answer is correct. You were awarded **1** mark. ✓

Gap 8

Your answer is correct. You were awarded **1** mark. ✓

Gap 9

Your answer is correct. You were awarded **1** mark. ✓

You scored **10** marks for this part.

Score: 10/10 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

For the following matrix,

$$A = \begin{pmatrix} 0 & -9 & 7 \\ -8 & 5 & 7 \\ -7 & -9 & -7 \end{pmatrix}$$

First, calculate the cofactors of A ,

$$\begin{array}{lll} c_{11} = 28 & \checkmark & c_{12} = -105 & \checkmark & c_{13} = 107 & \checkmark \\ c_{21} = -126 & \checkmark & c_{22} = 49 & \checkmark & c_{23} = 63 & \checkmark \\ c_{31} = -98 & \checkmark & c_{32} = -56 & \checkmark & c_{33} = -72 & \checkmark \end{array}$$

Hence give the adjugate of A ,

Rows: 3 Columns: 3

$$\text{adj}A = \begin{pmatrix} 28 & -126 & -98 \\ -105 & 49 & -56 \\ 107 & 63 & -72 \end{pmatrix} \checkmark$$

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

Gap 2

Your answer is correct. You were awarded 1 mark.

Gap 3

Your answer is correct. You were awarded 1 mark.

Gap 4

Your answer is correct. You were awarded 1 mark.

Gap 5

Your answer is correct. You were awarded **1** mark. ✓

Gap 6

Your answer is correct. You were awarded **1** mark. ✓

Gap 7

Your answer is correct. You were awarded **1** mark. ✓

Gap 8

Your answer is correct. You were awarded **1** mark. ✓

Gap 9

Your answer is correct. You were awarded **1** mark. ✓

You scored **10** marks for this part.

Score: 10/10 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Use determinants to find a value of a for which the matrix is not invertible.

$$\begin{pmatrix} -5 & 7 & -5 \\ -4 & -9 & -6 \\ 2 & a & -6 \end{pmatrix}$$

Enter the your value of a exactly, using fractions if required.

$a =$ ✓

Your answer is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Use determinants to find a value of a for which the matrix is not invertible.

$$\begin{pmatrix} -5 & -9 & -5 \\ -4 & 6 & -3 \\ -1 & a & 9 \end{pmatrix}$$

Enter the your value of a exactly, using fractions if required.

$a =$ ✓

Your answer is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Let $A, B \in \mathbf{M}_{4,4}(\mathbb{R})$. Suppose that $\det(A) = 2$ and $\det(B) = 3$.

i) $\det(AB) =$ ✓

ii) $\det(2A) =$ ✓

iii) $\det(A^T B^{-1}) =$ ✓

Gap 0

Your answer is correct. You were awarded **1** mark. ✓

Gap 1

Your answer is correct. You were awarded **1** mark. ✓

Gap 2

Your answer is correct. You were awarded **1** mark. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Question 1

Let $A, B \in \mathbf{M}_{5,5}(\mathbb{R})$. Suppose that $\det(A) = 5$ and $\det(B) = 6$.

i) $\det(AB) =$ ✓

ii) $\det(2A) =$ ✓

iii) $\det(A^T B^{-1}) =$ ✓

Gap 0

Your answer is correct. You were awarded **1** mark. ✓

Gap 1

Your answer is correct. You were awarded **1** mark. ✓

Gap 2

Your answer is correct. You were awarded **1** mark. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University
(<http://www.newcastle.ac.uk>).

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 2 & 0 & 4 \\ -3 & 0 & -4 \\ 2 & 0 & 5 \end{pmatrix}$$

Give the matrices L and U that result.

Rows: Columns:

$$L = \begin{pmatrix} \boxed{2} & \boxed{0} & \boxed{0} \\ \boxed{-3} & \boxed{2} & \boxed{0} \\ \boxed{2} & \boxed{1} & \boxed{1} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$U = \begin{pmatrix} \boxed{1} & \boxed{0} & \boxed{2} \\ \boxed{0} & \boxed{0} & \boxed{1} \\ \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix} \quad \checkmark$$

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

You scored **2** marks for this part.

Score: 2/2

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 2 & -2 & -6 \\ 1 & -1 & -2 \\ 3 & -3 & -12 \\ 3 & -3 & -10 \end{pmatrix}$$

Give the matrices L and U that result.

Rows: Columns:

$$L = \begin{pmatrix} \boxed{2} & \boxed{0} & \boxed{0} & \boxed{0} \\ \boxed{1} & \boxed{1} & \boxed{0} & \boxed{0} \\ \boxed{3} & \boxed{-3} & \boxed{1} & \boxed{0} \\ \boxed{3} & \boxed{-1} & \boxed{0} & \boxed{1} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$U = \begin{pmatrix} \boxed{1} & \boxed{-1} & \boxed{-3} \\ \boxed{0} & \boxed{0} & \boxed{1} \\ \boxed{0} & \boxed{0} & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix} \quad \checkmark$$

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

You scored **2** marks for this part.

Score: 2/2

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 3 & -6 & -3 & 6 \\ 2 & -4 & 1 & 10 \\ 2 & -4 & -3 & 2 \\ -1 & 2 & 2 & 0 \end{pmatrix}$$

Give the matrices L and U that result.

Rows: Columns:

$$L = \begin{pmatrix} \boxed{3} & \boxed{0} & \boxed{0} & \boxed{0} \\ \boxed{2} & \boxed{3} & \boxed{0} & \boxed{0} \\ \boxed{2} & \boxed{-1} & \boxed{1} & \boxed{0} \\ \boxed{-1} & \boxed{1} & \boxed{0} & \boxed{1} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$U = \begin{pmatrix} \boxed{1} & \boxed{-2} & \boxed{-1} & \boxed{2} \\ \boxed{0} & \boxed{0} & \boxed{1} & \boxed{2} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix} \quad \checkmark$$

Gap 0

Your answer is correct. You were awarded 1 mark. 

Gap 1

Your answer is correct. You were awarded 1 mark. 

You scored 2 marks for this part.

Score: 2/2 

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 2 & -2 & -2 & -2 & -2 \\ 3 & -3 & -3 & -3 & -6 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Give the matrices L and U that result.

Rows: Columns:

$$L = \begin{pmatrix} \boxed{2} & \boxed{0} & \boxed{0} \\ \boxed{3} & \boxed{-3} & \boxed{0} \\ \boxed{0} & \boxed{1} & \boxed{1} \end{pmatrix} \quad \checkmark$$

Rows: Columns:

$$U = \begin{pmatrix} \boxed{1} & \boxed{-1} & \boxed{-1} & \boxed{-1} & \boxed{-1} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{1} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix} \quad \checkmark$$

Gap 0

Your answer is correct. You were awarded 1 mark.

Gap 1

Your answer is correct. You were awarded 1 mark.

You scored **2** marks for this part.

Score: 2/2

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).

SIT292 Module 4 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	2/2	
Question 2	1/1	
Question 3	1/1	
Question 4	2/2	
Question 5	1/1	
Question 6	1/1	
Question 7	2/2	
Total	10/10	(100%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 4 Self-Assessment
Session ID:	6888122577893973
Exam Start:	Tue Aug 18 2026 04:09:52
Exam Stop:	Tue Aug 18 2026 04:36:50
Time Spent:	0:26:58

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<https://www.newcastle.ac.uk>).

Question 1

Consider the matrix

$$A = \begin{pmatrix} 3 & -1 & 4 \\ -4 & 3 & -3 \\ -1 & 0 & 4 \end{pmatrix}$$

Evaluate the determinant by expanding along column 2.

$$\det(A) = \boxed{1} \checkmark$$

Expected answer: 1 $\times \det$

$$\begin{pmatrix} \boxed{-4} & \boxed{-3} \\ \boxed{-1} & \boxed{4} \end{pmatrix} \checkmark$$

Expected answer: $\begin{pmatrix} \frac{-4}{-1} & \frac{-3}{4} \end{pmatrix} +$

$$\boxed{3} \checkmark$$

Expected answer: 3 $\times \det$

$$\begin{pmatrix} \boxed{3} & \boxed{4} \\ \boxed{-1} & \boxed{4} \end{pmatrix} \checkmark$$

Expected answer: $\begin{pmatrix} \frac{3}{-1} & \frac{4}{4} \end{pmatrix} +$

$$\boxed{0} \checkmark$$

Expected answer: 0 $\times \det$

$$\begin{pmatrix} \boxed{3} & \boxed{4} \\ \boxed{-4} & \boxed{-3} \end{pmatrix} \checkmark$$

Expected answer: $\begin{pmatrix} \frac{3}{-4} & \frac{4}{-3} \end{pmatrix}$

$$= \boxed{1} \checkmark$$

Expected answer: 1 $\times$

$$\boxed{-19} \checkmark$$

Expected answer: -19 $+$

$$\boxed{3} \checkmark$$

Expected answer: 3

$$\times \boxed{16} \checkmark$$

Expected answer: 16 $+$

$$\begin{array}{c}
 \boxed{0} \checkmark \quad \text{Expected answer: } \underline{0} \times \boxed{7} \checkmark \quad \text{Expected answer: } \underline{7} \\
 = \boxed{29} \checkmark \quad \text{Expected answer: } \underline{29}
 \end{array}$$

Note: If the coefficient of any sub-matrix determinant is zero, you will still need to include the submatrix in the first line of working and include its determinant in the second line of working.

Score: 2/2 ✓

Gap 0

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 2

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 3

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 4

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 5

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 6

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 7

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 8

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 9

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 10

- ✓ Your answer is correct. You were awarded **0.15** marks.

Gap 11

✓ Your answer is correct. You were awarded **0.15** marks.

Gap 12

✓ Your answer is correct. You were awarded **0.2** marks.

You scored **2** marks for this part.

Question 2

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = 5$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{21} & a_{22} & a_{23} & a_{24} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
☐ 3. Row/Column multiplied by a constant ☒ 4. Two identical Rows/Columns
☐ 5. Multiple of row/column added to another row/column



Expected answer:

- ☐ 1. Row/Column of Zeros
- ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☒ 4. Two identical Rows/Columns
- ☐ 5. Multiple of row/column added to another row/column

What is the value of the resulting determinant?

0



Expected answer: 0

Score: 1/1 ✓

Gap 0

- ✓ You chose a correct answer. You were awarded **0.3** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.7** marks.

You scored **1** mark for this part.

Question 3

Consider the following matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

If $\det(A) = 9$, use Theorem 3.1.2 to determine the value of

$$\det \begin{pmatrix} a_{11} + 5a_{21} & a_{12} + 5a_{22} & a_{13} + 5a_{23} & a_{14} + 5a_{24} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{pmatrix}$$

Which property can be used?

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant ☐ 4. Two identical Rows/Columns
- ☒ 5. Multiple of row/column added to another row/column



Expected answer:

- ☐ 1. Row/Column of Zeros ☐ 2. Two rows/columns interchanged
- ☐ 3. Row/Column multiplied by a constant
- ☐ 4. Two identical Rows/Columns
- ☒ 5. Multiple of row/column added to another row/column

What is the value of the resulting determinant?

9



Expected answer: 9

Score: 1/1 ✓

Gap 0

- ✓ You chose a correct answer. You were awarded **0.3** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.7** marks.

You scored **1** mark for this part.

Question 4

For the following matrix,

$$A = \begin{pmatrix} 5 & 5 & -3 \\ 1 & 3 & 0 \\ 6 & -3 & 0 \end{pmatrix}$$

First, calculate the cofactors of A ,

$c_{11} =$ 0 ✓ Expected answer: 0	$c_{12} =$ 0 ✓ Expected answer: 0	$c_{13} =$ -21 ✓ Expected answer: -21
$c_{21} =$ 9 ✓ Expected answer: 9	$c_{22} =$ 18 ✓ Expected answer: 18	$c_{23} =$ 45 ✓ Expected answer: 45
$c_{31} =$ 9 ✓ Expected answer: 9	$c_{32} =$ -3 ✓ Expected answer: -3	$c_{33} =$ 10 ✓ Expected answer: 10

Hence give the adjugate of A ,

adjA =

Rows: Columns:

$$\begin{pmatrix} 0 & 9 & 9 \\ 0 & 18 & -3 \\ -21 & 45 & 10 \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} 0 & 9 & 9 \\ 0 & 18 & -3 \\ -21 & 45 & 10 \end{pmatrix}$$

Score: 2/2 **Gap 0** Your answer is correct. You were awarded **0.15** marks.**Gap 1** Your answer is correct. You were awarded **0.15** marks.**Gap 2** Your answer is correct. You were awarded **0.15** marks.**Gap 3** Your answer is correct. You were awarded **0.15** marks.**Gap 4** Your answer is correct. You were awarded **0.15** marks.**Gap 5** Your answer is correct. You were awarded **0.15** marks.**Gap 6** Your answer is correct. You were awarded **0.15** marks.**Gap 7** Your answer is correct. You were awarded **0.15** marks.**Gap 8** Your answer is correct. You were awarded **0.15** marks.**Gap 9** Your answer is correct. You were awarded **0.65** marks.

You scored **2** marks for this part.

Question 5

Use determinants to find a value of a for which the matrix is **not** invertible.

$$\begin{pmatrix} 4 & 3 & a \\ 3 & 5 & 4 \\ -2 & -2 & 7 \end{pmatrix}$$

Enter the your value of a exactly, using fractions if required.

$a =$ ✓

Expected answer: -85/4

Score: 1/1 ✓

✓ Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

Question 6

Let $A, B \in \mathbf{M}_{5,5}(\mathbb{R})$. Suppose that $\det(A) = 7$ and $\det(B) = -4$.

i) $\det(AB) =$ ✓

Expected answer: -28

ii) $\det(2A) =$ ✓

Expected answer: 224

iii) $\det(A^T B^{-1}) =$ ✓

Expected answer: -7/4

Gap 0

- ✓ Your answer is correct. You were awarded **0.3** marks.

Gap 1

- ✓ Your answer is correct. You were awarded **0.35** marks.

Gap 2

- ✓ Your answer is correct. You were awarded **0.35** marks.

You scored **1** marks for this part.

Question 7

Apply the LU-Algorithm to factorise the following matrix.

$$A = \begin{pmatrix} 2 & 4 & 6 & -2 \\ -2 & -2 & -8 & 6 \\ -3 & -4 & -10 & 6 \\ 1 & 4 & 3 & 2 \end{pmatrix}$$

Give the matrices L and U that result.

L =

Rows: 4 Columns: 4

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 \\ -3 & 2 & 1 & 0 \\ 1 & 2 & 2 & 1 \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} 2 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 \\ -3 & 2 & 1 & 0 \\ 1 & 2 & 2 & 1 \end{pmatrix}$$

U =

Rows: 4 Columns: 4

$$\begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \checkmark$$

Expected answer:

$$\begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Score: 2/2 **Gap 0** Your answer is correct. You were awarded 1 mark.**Gap 1** Your answer is correct. You were awarded 1 mark.

You scored 2 marks for this part.

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(https://www.newcastle.ac.uk\)](https://www.newcastle.ac.uk).

Module 4 Self Assessment - Working & Annotations

Question 1. Cofactor Expansion

Concept / Theorem Applied:

Cofactor Expansion Theorem (Theorem 3.1.1): Expanded the determinant along column 2.

$$A = \begin{bmatrix} 3 & -1 & 4 \\ -4 & 3 & -3 \\ -1 & 0 & 4 \end{bmatrix}$$

Using the cofactor signs for column 2:

$-, +, -$

$$\det(A) = 1 \begin{vmatrix} -4 & -3 \\ -1 & 4 \end{vmatrix} + 3 \begin{vmatrix} 3 & 4 \\ -1 & 4 \end{vmatrix} + 0 \begin{vmatrix} 3 & 4 \\ -4 & -3 \end{vmatrix}$$

$$= 1[(-4)(4) - (-3)(-1)] + 3[(3)(4) - (-4)(-1)] + 0[(3)(-3) - (-4)(-4)]$$

$$= 1(-19) + 3(16) + 0(7)$$

$$= -19 + 48$$

$$= 29$$

$$\therefore \boxed{\det(A) = 29}$$

Question 2. Two Identical Rows

Theorem 3.1.2 - Property 4:

If two rows or columns are identical, the determinant is zero.

The resulting matrix has:

$$R_1 = R_2$$

$$\therefore \boxed{\det = 0}$$

Question 3. Row Addition

Theorem 3.1.2 - Property 5:

Adding a multiple of one row to another does not change the determinant.

The transformation is:

$$R_1 \rightarrow R_1 + R_2$$

$$\text{Here, } \det(A') = \det(A)$$

$$\text{Given } \det(A) = 9, \text{ therefore } \boxed{\det(A') = 9}.$$

Question 4. Cofactors and Adjugate

Concept Applied:

Cofactor matrix and adjugate

$$A = \begin{bmatrix} 5 & 5 & -3 \\ 1 & 3 & 0 \\ 6 & -3 & 0 \end{bmatrix}$$

Calculate the cofactors:

$$C_{11} = 0, \quad C_{12} = 0, \quad C_{13} = -21$$

$$C_{21} = 9, \quad C_{22} = 18, \quad C_{23} = 45$$

$$C_{31} = 9, \quad C_{32} = -3, \quad C_{33} = 10$$

Therefore, the cofactor matrix is:

$$C = \begin{bmatrix} 0 & 0 & -21 \\ 9 & 18 & 45 \\ 9 & -3 & 10 \end{bmatrix}$$

$$\text{Adjugate} = \text{adj}(A) = C^T$$

$$\text{adj}(A) = \begin{bmatrix} 0 & 9 & 9 \\ 0 & 18 & -3 \\ -21 & 45 & 10 \end{bmatrix}$$

Question 5. Non-Invertibility

Concept Applied:

A square matrix is not invertible when $\det(A)=0$.

$$A = \begin{bmatrix} 4 & 3 & a \\ 3 & 5 & 4 \\ -2 & -2 & 7 \end{bmatrix}$$

Expanding along Row 1:

$$\det(A) = 4 \begin{vmatrix} 5 & 4 \\ -2 & 7 \end{vmatrix} - 3 \begin{vmatrix} 3 & 4 \\ -2 & 7 \end{vmatrix} + a \begin{vmatrix} 3 & 5 \\ -2 & -2 \end{vmatrix}$$

$$= 4(43) - 3(29) + a(4)$$

$$= 172 - 87 + 4a$$

$$= 85 + 4a$$

For the matrix to be invertible:

$$85 + 4a = 0$$

$$4a = -85$$

$$\boxed{a = -\frac{85}{4}}$$

Question 6. Determinant Properties

Concept / Theorem Applied:

$$\det(AB) = \det(A) \det(B)$$

$$\det(kA) = k^n \det(A) \quad [k \text{ is scalar, } n \text{ is matrix dimension}]$$

$$\det(A^T) = \det(A)$$

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Given: $A, B \in M_{5,5}(\mathbb{R})$, $\det(A) = 7$, $\det(B) = -4$

ci) $\det(AB) = 7(-4)$

$$\boxed{\det(AB) = -28}$$

cii) Since A is 5×5 , $\det(2A) = 2^5 \det(A) = 32(7) = 224$

$$\boxed{\det(2A) = 224}$$

ciii) $\det((A^T)(B^{-1})) = \det(A^T) \det(B^{-1})$

$$= 7 \left(-\frac{1}{4} \right)$$

$$\boxed{\det(A^T B^{-1}) = -\frac{7}{4}}$$

Question 7. LU-Factorisation

Concept/ Theorem Applied:

LU-Algorithm/Theorem 2.7.1

Gaussian elimination is used to obtain U , while the leading columns generated during the reduction are used to construct L . No row interchange is used.

$$A = \begin{bmatrix} 2 & 4 & 6 & -2 \\ -2 & -2 & -8 & 6 \\ -3 & -4 & -10 & 6 \\ 1 & 4 & 3 & 2 \end{bmatrix}$$

Step 1. First pivot

$$R_1 \rightarrow \frac{1}{2} R_1$$

$$R_2 \rightarrow R_2 + 2R_1$$

$$R_3 \rightarrow R_3 + 3R_1$$

$$R_4 \rightarrow R_4 - R_1$$

Therefore,
$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 2 & -2 & 4 \\ 0 & 2 & -1 & 3 \\ 0 & 2 & 0 & 3 \end{bmatrix}$$

Step 2. Second pivot

$$R_2 \rightarrow \frac{1}{2} R_2$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$R_4 \rightarrow R_4 - 2R_2$$

therefore,
$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 2 & -1 \end{bmatrix}$$

Step 3. Third pivot

$$R_4 \rightarrow R_4 - 2R_3$$

Hence,
$$U = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The leading columns used in the LU-Algorithm give:

$$L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 \\ -3 & 2 & 1 & 0 \\ 1 & 2 & 2 & 1 \end{bmatrix}$$

$$\therefore \boxed{A = LU}$$

Verification:
~~~~~

$$LV = \begin{bmatrix} 2 & 4 & 6 & -2 \\ -2 & -2 & -8 & 6 \\ -3 & -4 & 10 & 6 \\ 1 & 4 & 3 & 2 \end{bmatrix} = A$$

Hence verified.

# DEAKIN UNIVERSITY

## LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

---

### Enrichment Task 1

---

*Submitted By:*

Jahan GARG

s226100103

2026/08/21 15:40

*Tutor:*

Simon JAMES

| Task Outcomes |                                                                                                                      | Supports  |
|---------------|----------------------------------------------------------------------------------------------------------------------|-----------|
| ET1           | Independently research and then summarise mathematical ideas in a way that helps communicate the key ideas to others | ULO3 ULO3 |

August 21, 2026



# Markov Chains: Using Matrices to Predict What Happens Next

*A practical introduction to state transitions, probabilities, and long-term behaviour*

SIT292 – Linear Algebra for Data Analysis | Enrichment Task 1 | Topic 2.9: An Application to Markov Chains

## 1. The Big Idea: Systems That Remember Only ‘Right Now’

Consider a student's study behaviour on any given evening. On a particular night, they might be **focused** on their work, **distracted** by something else, or simply **resting**. What they do tomorrow evening depends mainly on what they are doing tonight: a focused evening tends to be followed by another productive one, while a distracted evening is more likely to lead to another distracted one. Crucially, tomorrow's behaviour does not depend on what happened last week – only on tonight's state. This is the defining idea behind a **Markov chain**: a system that moves between a fixed set of **states**, where the next state depends only on the current state and not on the full history that led there [1, Ch. 2.9].

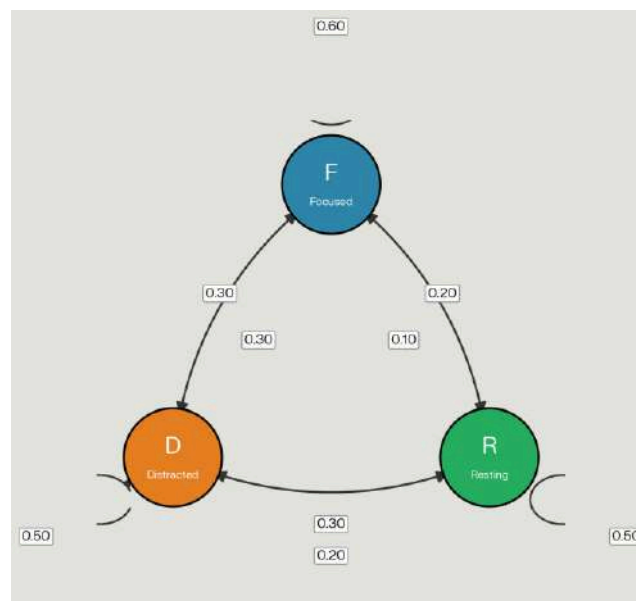
This ‘memoryless’ property is called the **Markov property**:

$$\Pr(X_{n+1} = j | X_0, \dots, X_n) = \Pr(X_{n+1} = j | X_n)$$

In words: the probability of the next state depends only on the present state, not on any earlier one. This single idea allows the entire future behaviour of the system to be captured using nothing more than a matrix.

### From States to a Diagram

We can visualise the study-behaviour example as three states connected by arrows, where each arrow carries the probability of moving from one state to another, including the probability of remaining in the same state. Figure 1 shows this network for the original example used throughout this document.



*Figure 1. Original state-transition diagram for the study-behaviour example, showing each state and its transition probabilities. (Visual created by the author using Python/Matplotlib.)*

Reading this diagram: if a student is Focused (F) tonight, there is a 60% chance they remain Focused tomorrow, a 30% chance they become Distracted, and a 10% chance they end up Resting. Every arrow leaving a state – including the loop back to itself – must add up to 100%, since the student has to end up in exactly one of the three states.

## 2. Turning the Diagram Into a Matrix

While a diagram builds intuition, it becomes unwieldy once a system has many states. Linear algebra gives us a compact way to store every transition probability at once: the **transition matrix**  $P$ . Following the convention in Nicholson [1, Ch. 2.9], the entry  $p_{ij}$  of  $P$  gives the probability of moving *into* state  $i$ , given that the system is currently in state  $j$ . Reading down any single column of  $P$  therefore tells us everything that can happen next, starting from that state.

For our study-behaviour example, with states ordered F (Focused), D (Distracted), R (Resting), the transition matrix corresponding to Figure 1 is:

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.3 & 0.5 & 0.3 \\ 0.1 & 0.2 & 0.5 \end{bmatrix}$$

Each column of  $P$  sums to exactly 1. This must be true for *any* valid transition matrix, because a system currently in a given state has to transition to *some* state (possibly itself) with total probability 1. A matrix with this property, whose entries are all non-negative and whose columns each sum to 1, is called a **stochastic matrix** [1, Ch. 2.9]:

$$\sum_i p_{ij} = 1 \quad (\text{for every column } j)$$

### Describing the System With a State Vector

We also need a way to describe how likely the system is to be in each state at a given point in time. This is done with a **state vector**  $\mathbf{s}_m$ , whose entries give the probability of being in each state after  $m$  transitions (here, after  $m$  days). If tonight the student is definitely Focused, the state vector describing today is  $\mathbf{s}_0 = [1, 0, 0]^T$ .

The key computational fact that makes Markov chains so useful is that predicting tomorrow's state vector from today's is a single matrix-vector multiplication, and more generally, the state after  $m$  transitions can be found directly by repeated multiplication without recomputing every intermediate day:

$$\mathbf{s}_{m+1} = P\mathbf{s}_m \qquad \mathbf{s}_m = P^m\mathbf{s}_0$$

## 3. Original Worked Example 1: What Happens Tomorrow (and the Day After)?

Suppose our student is definitely Focused today, so  $\mathbf{s}_0 = [1, 0, 0]^T$ . We use the transition matrix  $P$  to predict their behaviour over the next two days.

**Step 1 – Tomorrow's prediction.** Using  $\mathbf{s}_1 = P\mathbf{s}_0$ , multiplying by  $[1, 0, 0]^T$  simply extracts the first column of  $P$ :

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.3 & 0.5 & 0.3 \\ 0.1 & 0.2 & 0.5 \end{bmatrix} \times \mathbf{s}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{s}_1 = \begin{bmatrix} 0.6 \\ 0.3 \\ 0.1 \end{bmatrix}$$

**Interpretation:** after one day, there is a 60% chance the student is Focused, a 30% chance they are Distracted, and only a 10% chance they are Resting – exactly the first column of  $P$ , as expected from the F-column of arrows in Figure 1.

**Step 2 – The day after tomorrow.** We repeat the process using  $\mathbf{s}_2 = P\mathbf{s}_1$ :

$$P = \begin{bmatrix} 0.6 & 0.3 & 0.2 \\ 0.3 & 0.5 & 0.3 \\ 0.1 & 0.2 & 0.5 \end{bmatrix} \times \mathbf{s}_1 = \begin{bmatrix} 0.6 \\ 0.3 \\ 0.1 \end{bmatrix} = \mathbf{s}_2 = \begin{bmatrix} 0.47 \\ 0.36 \\ 0.17 \end{bmatrix}$$

Each entry of  $\mathbf{s}_2$  is the dot product of the corresponding row of  $P$  with  $\mathbf{s}_1$ . For example, the first entry is  $0.6(0.60) + 0.3(0.30) + 0.2(0.10) = 0.36 + 0.09 + 0.02 = 0.47$ .

**Interpretation:** two days from now, the probability of being Focused has dropped to 47%, and the probability of being Distracted has risen to 36%. We did not need to trace every possible two-step path by hand – the matrix multiplication accounted for all of them automatically. This is exactly the advantage linear algebra brings to this problem.



## 4. Original Worked Example 2: What Happens in the Long Run?

A natural question is what happens if this process runs for a very long time. Instead of multiplying by  $P$  hundreds of times, we ask whether the state vector eventually stops changing. A vector  $\mathbf{s}$  satisfying  $P\mathbf{s} = \mathbf{s}$  is called a **steady-state vector** [1, Ch. 2.9] – it represents the long-run probability of being in each state, regardless of the starting state.

The steady-state condition rearranges into a homogeneous system, solved together with the requirement that probabilities are non-negative and sum to 1:

$$P\mathbf{s} = \mathbf{s} \Leftrightarrow (P - I)\mathbf{s} = \mathbf{0} \qquad s_F + s_D + s_R = 1, \quad s_F, s_D, s_R \geq 0$$

Writing out  $(P - I)\mathbf{s} = \mathbf{0}$  as three linear equations in  $s_F, s_D, s_R$ , together with  $s_F + s_D + s_R = 1$ , gives a system solvable by Gaussian elimination exactly as in Module 1 of this unit. Solving this system for our transition matrix  $P$  gives:

$$\mathbf{s} = \begin{bmatrix} 19/48 \text{ } (\sim 0.396) \\ 18/48 = 0.375 \\ 11/48 \text{ } (\sim 0.229) \end{bmatrix}$$

**Verification:** applying  $P$  to  $\mathbf{s}$  reproduces the Focused entry of  $\mathbf{s}$  itself:

$$0.6(0.396) + 0.3(0.375) + 0.2(0.229) = 0.396$$

**Interpretation:** regardless of which state the student starts in on Day 0, after enough days pass, they will be Focused approximately 39.6% of evenings, Distracted 37.5% of evenings, and Resting 22.9% of evenings, in the long run. Figure 2 shows how the day-by-day probabilities from Worked Example 1, continued for ten days, settle toward these steady-state values (shown as dashed lines).

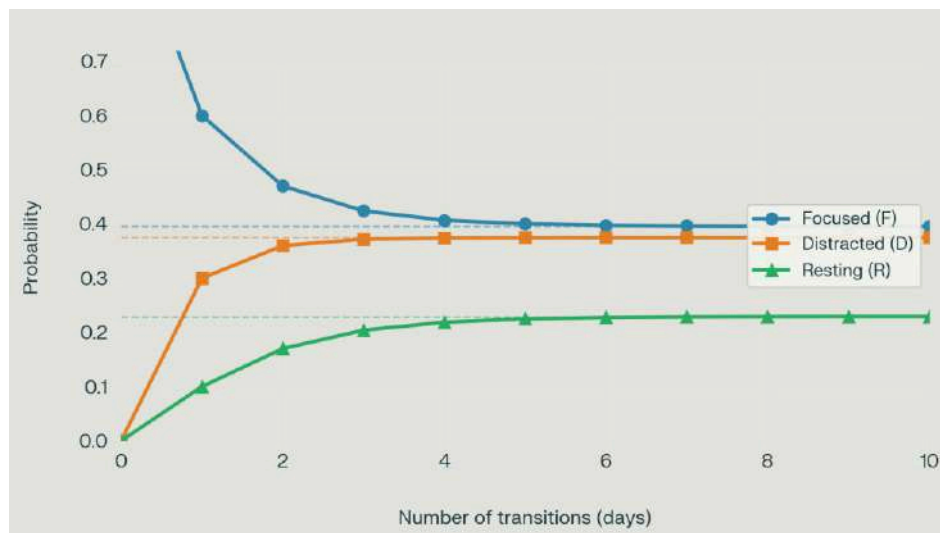


Figure 2. Convergence of the day-by-day state probabilities toward the steady-state values over 10 transitions, starting from a certain Focused state. (Visual created by the author using Python/Matplotlib.)

## 5. What the Matrix Is Really Telling Us

A transition matrix  $P$  compresses an entire network of probabilities into a single object that predicts behaviour one step ahead ( $\mathbf{s}_1 = P\mathbf{s}_0$ ), many steps ahead ( $\mathbf{s}_m = P^m\mathbf{s}_0$ ), and even infinitely far ahead (the steady-state vector) – all using the same tools, matrix multiplication and Gaussian elimination, used throughout this unit for very different-looking problems.

### A Word of Caution: Physical Meaning Matters

A solution is only useful if it makes sense in context. Because every entry of a state vector represents a probability, a valid solution must always have non-negative entries summing to 1. If solving  $(P - I)\mathbf{s} = \mathbf{0}$  together with the sum-to-1

condition ever produced a negative entry, this would signal an arithmetic error rather than a valid steady state – Gaussian elimination does not know about probability, so checking that the answer is physically sensible is the responsibility of whoever applies it.

### Where Else Could This Idea Be Useful?

The states in a Markov chain need not describe study habits – the same mathematics applies whenever a system moves between well-defined conditions, and the next condition depends only on the present one. Some illustrative examples, distinct from the worked examples above and from Nicholson's own weather and football examples [1, Ch. 2.9], include:

- **Website navigation:** modelling which page a user visits next, based only on the page they are currently viewing.
- **Customer subscription behaviour:** tracking whether a customer is Active, Paused, or Cancelled to forecast long-run subscriber numbers.
- **Machine reliability:** modelling equipment moving between Working, Under Maintenance, and Broken states.
- **Simple recommendation systems:** predicting the next item a user is likely to view, based on the item they are currently viewing.

## 6. Try It Yourself

To check your own understanding, here is a short original exercise using a different three-state system: a mobile app that a phone can be in one of three states each day – **Active (A)**, **Idle (B)**, or **Closed (C)**. The transition matrix for this system is:

$$Q = \begin{bmatrix} 0.5 & 0.2 & 0.1 \\ 0.4 & 0.6 & 0.3 \\ 0.1 & 0.2 & 0.6 \end{bmatrix}$$

**Question 1.** If the app is definitely Active today ( $\mathbf{q}_0 = [1, 0, 0]^T$ ), find the state vector  $\mathbf{q}_2$  describing the probabilities two days from now.

**Question 2.** In one sentence, explain what the entries of  $\mathbf{q}_2$  mean, in the same style used in Worked Example 1.

**Answer** – attempt the questions above first. First,  $\mathbf{q}_1 = Q \mathbf{q}_0 = [0.50, 0.40, 0.10]^T$ . Then, computing  $\mathbf{q}_2 = Q \mathbf{q}_1$ :

$$Q = \begin{bmatrix} 0.5 & 0.2 & 0.1 \\ 0.4 & 0.6 & 0.3 \\ 0.1 & 0.2 & 0.6 \end{bmatrix} \times \mathbf{q}_1 = \begin{bmatrix} 0.5 \\ 0.4 \\ 0.1 \end{bmatrix} = \mathbf{q}_2 = \begin{bmatrix} 0.34 \\ 0.47 \\ 0.19 \end{bmatrix}$$

So  $\mathbf{q}_2 = [0.34, 0.47, 0.19]^T$ : two days after being definitely Active, there is a 34% chance the app is Active, a 47% chance it is Idle, and a 19% chance it has been Closed.

### Key Takeaways

1. A Markov chain models a system moving between a fixed set of states, where the next state depends only on the present one.
2. A transition matrix  $P$  stores every transition probability; each column of  $P$  must sum to 1.
3. Repeated matrix multiplication,  $\mathbf{s}_m = P^m \mathbf{s}_0$ , predicts state probabilities any number of steps ahead.
4. A steady-state vector, found by solving  $(P - I)\mathbf{s} = \mathbf{0}$  together with the condition that entries sum to 1, describes long-run behaviour independent of the starting state.

## Tools and Originality Statement

All numerical examples (the F/D/R study-behaviour system and the A/B/C app-usage system), the transition matrices  $P$  and  $Q$ , and both figures in this document were designed specifically for this submission by Jahan Garg, and do not reproduce any example from Nicholson's textbook. The topic, structure, and all mathematical content were directed and independently verified by the author using Python (NumPy); an AI writing assistant (Perplexity) was used to help draft explanatory text and generate the Python code for the figures and equation graphics, with all content reviewed, verified, and approved by the author before inclusion. Figures 1 and 2 were generated using Python (Matplotlib), and the document was assembled using the ReportLab library in Python. No content was copied from any external source; the Markov property, transition-matrix notation, and steady-state concept follow the standard treatment given in the referenced textbook section.

## References

- [1] W. K. Nicholson, *Linear Algebra with Applications*, Open Edition, Version 2021A, Lyryx Learning Inc., Calgary, AB, Canada, 2021, Section 2.9, “An Application to Markov Chains.”

## 1. THE BIG IDEA — WHY A MATRIX CAN MOVE A PICTURE

# From Points to Pixels

*A visual guide to how matrices scale, shear, rotate, compose, and translate a picture*

A computer never stores a picture the way our eyes see it. Instead, an image built from straight edges – a logo, an icon, a game sprite – is stored as a list of coordinates, one point for every corner. This document treats a small original arrow-shaped icon as that list of points, and shows step by step how ordinary matrix multiplication is exactly the tool a computer uses to move, resize, slant, and rotate that icon on screen.

**A familiar question**

How does a phone app rotate an icon, or a game character turn to face a new direction, without recalculating every pixel by hand?

**The core idea of this document**

Every point of the icon is a column vector  $v = (x, y)^T$ . A transformation matrix  $A$  converts every point at once, using  $v' = A \cdot v$ . The whole icon – all of its vertices stored together as a data matrix  $D$  – moves according to the single rule  $D' = A \cdot D$ . This is the mathematical anchor for the whole document.

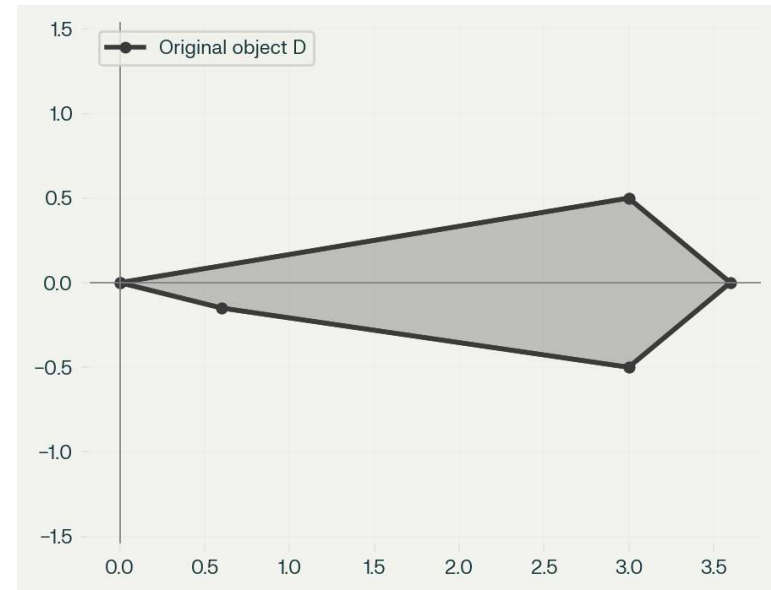


Figure 1. Original arrow-icon object  $D$ , defined by 5 original vertex coordinates chosen for this submission.  
(Created by Jahan Garg in Python/Matplotlib.)

**Why this is genuinely linear algebra**

Geometry becomes coordinates; coordinates are transformed by matrix multiplication; several transformations combine by matrix products; and homogeneous coordinates extend the same idea to translation.



## 2. THE MATRIX TOOLBOX — SCALING, SHEARING, ROTATION, TRANSLATION

**Scaling  $S(s_x, s_y)$** 

$$S = \begin{vmatrix} s_x & 0 \\ 0 & s_y \end{vmatrix}$$

Stretches or compresses the icon along each axis independently. Unequal factors distort the shape; equal factors resize it uniformly.

**Shearing  $H(k)$** 

$$H = \begin{vmatrix} 1 & k \\ 0 & 1 \end{vmatrix}$$

Slants the icon: every point is pushed sideways by an amount proportional to its height, while the horizontal axis stays fixed.

**Rotation  $R(\theta)$** 

$$R = \begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$$

Turns the icon about the origin through angle  $\theta$ , preserving every length and angle inside the shape (a rigid motion).

**Why translation is different**

$v' = v + w$  is NOT of the form  $v' = A \cdot v$

A linear map always sends the origin to itself, so an ordinary  $2 \times 2$  matrix cannot slide the icon sideways – a genuinely different tool is required.

**The fix: homogeneous coordinates**

Every point  $(x, y)$  is written as a 3-entry column  $(x, y, 1)^T$ . A translation by  $(p, q)$  then becomes an ordinary matrix product:

$$T(p, q) = \begin{vmatrix} 1 & 0 & p \\ 0 & 1 & q \end{vmatrix} \quad T(p, q) \cdot (x, y, 1)^T = (x + p, y + q, 1)^T$$

— the same matrix-multiplication language now covers moving the icon too.

*Foundation: Nicholson, Sec. 4.5 [1]*

## 3. ORIGINAL WORKED EXAMPLE 1 — SCALE, SHEAR, THEN ROTATE

## Combining three transformations on one original icon

The arrow icon from Page 1 is stored as the data matrix  $D$  below, one column per vertex. We apply three original transformations, chosen specifically for this document: scale by  $(sx, sy) = (1.3, 0.6)$ , shear by  $k = 0.3$ , then rotate by  $\theta = 40^\circ$ .

$$D = \begin{bmatrix} 0 & 3.0 & 3.6 & 3.0 & 0.6 \\ 0 & 0.5 & 0 & -0.5 & -0.15 \end{bmatrix}$$

## Step 1 – build each matrix

$$S = \begin{bmatrix} 1.3 & 0 \\ 0 & 0.6 \end{bmatrix} \quad H = \begin{bmatrix} 1 & 0.3 \\ 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 0.766 & -0.643 \\ 0.643 & 0.766 \end{bmatrix}$$

Because column vectors are used, the rightmost matrix acts first:  $M = R H S$  means “scale, then shear, then rotate.”

## Step 2 – combine, then apply to every vertex

$$M = R \cdot H \cdot S = \begin{bmatrix} 0.996 & -0.248 \\ 0.836 & 0.575 \end{bmatrix} \quad D_1 = M \cdot D$$

$$D_1 = \begin{bmatrix} 0 & 2.864 & 3.585 & 3.112 & 0.635 \\ 0 & 2.795 & 3.008 & 2.219 & 0.415 \end{bmatrix}$$

Interpretation: one matrix–vector rule repositions all five vertices consistently – no vertex needs a special case.

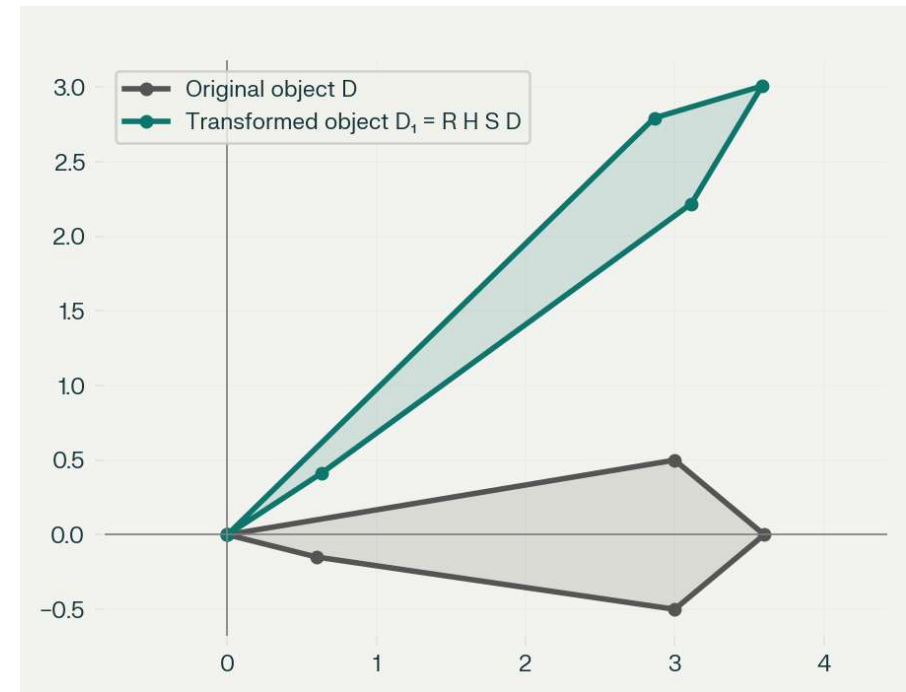


Figure 2. Original icon (grey) versus its scaled, sheared and rotated copy (teal);  $D_1 = R \cdot H \cdot S \cdot D$ . (Created by Jahan Garg in Python/Matplotlib.)

## 4. ORIGINAL WORKED EXAMPLE 2 — ROTATING ABOUT A CHOSEN PIVOT

## Rotating the icon about its own pivot, not the screen origin

A plain rotation matrix  $R(\theta)$  always turns an object about the origin  $(0, 0)$ . But a game character, an icon, or a clock hand usually needs to rotate about its own centre or a chosen pivot point instead.

This is where homogeneous coordinates genuinely earn their place.

## The three-step idea

1. Translate the pivot  $P$  to the origin:  $T(-P)$
2. Rotate about the origin:  $R(\theta)$
3. Translate the origin back to  $P$ :  $T(P)$

Combined matrix:  $M = T(P) \cdot R(\theta) \cdot T(-P)$

Worked calculation:  $P = (2.0, -0.2)$ ,  $\theta = 50^\circ$ 

$$T(-P) = \begin{vmatrix} 1 & 0 & -2.0 \\ 0 & 1 & 0.2 \\ 0 & 0 & 1 \end{vmatrix} \quad R(50^\circ) = \begin{vmatrix} 0.643 & -0.766 & 0 \\ 0.766 & 0.643 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$T(P) = \begin{vmatrix} 1 & 0 & 2.0 \\ 0 & 1 & -0.2 \\ 0 & 0 & 1 \end{vmatrix}$$

$$M = \begin{vmatrix} 0.643 & -0.766 & 0.561 \\ 0.766 & 0.643 & 0.561 \\ 0 & 0 & 1 \end{vmatrix} \quad D_2 = M \cdot [D; 1, 1, 1, 1]$$

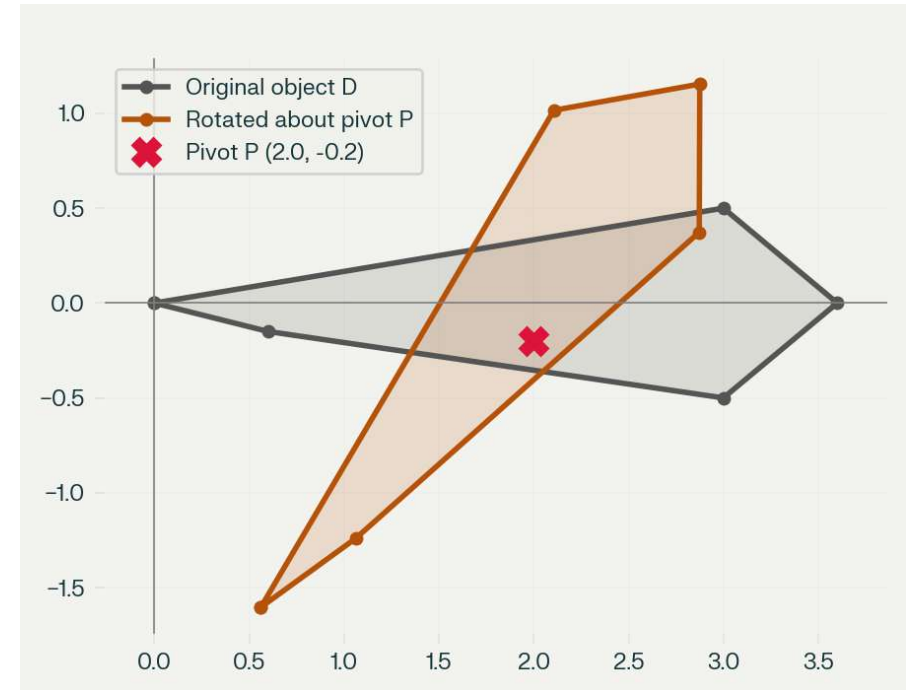


Figure 3. Icon rotated  $50^\circ$  about pivot  $P = (2.0, -0.2)$  (marked  $\times$ ), using  $T(P)R(\theta)T(-P)$  in homogeneous form. (Created by Jahan Garg in Python/Matplotlib.)

Insight: “rotate about a point” is not a new matrix type – it is a composition of three transformations you already know.

## 5. COMPOSITION ORDER MATTERS — WHY $R \cdot S \neq S \cdot R$

### The same two matrices, applied in opposite order, give different pictures

Using a fresh pair of transformations – scale  $S(1.5, 0.7)$  and rotation  $R(35^\circ)$  – the icon is transformed two different ways: first scaled, then rotated ( $M = R \cdot S$ ), and first rotated, then scaled ( $M = S \cdot R$ ). Matrix multiplication is associative but not commutative, so  $R \cdot S \neq S \cdot R$  in general, and the two resulting icons are genuinely different shapes, not just different orientations of the same shape.

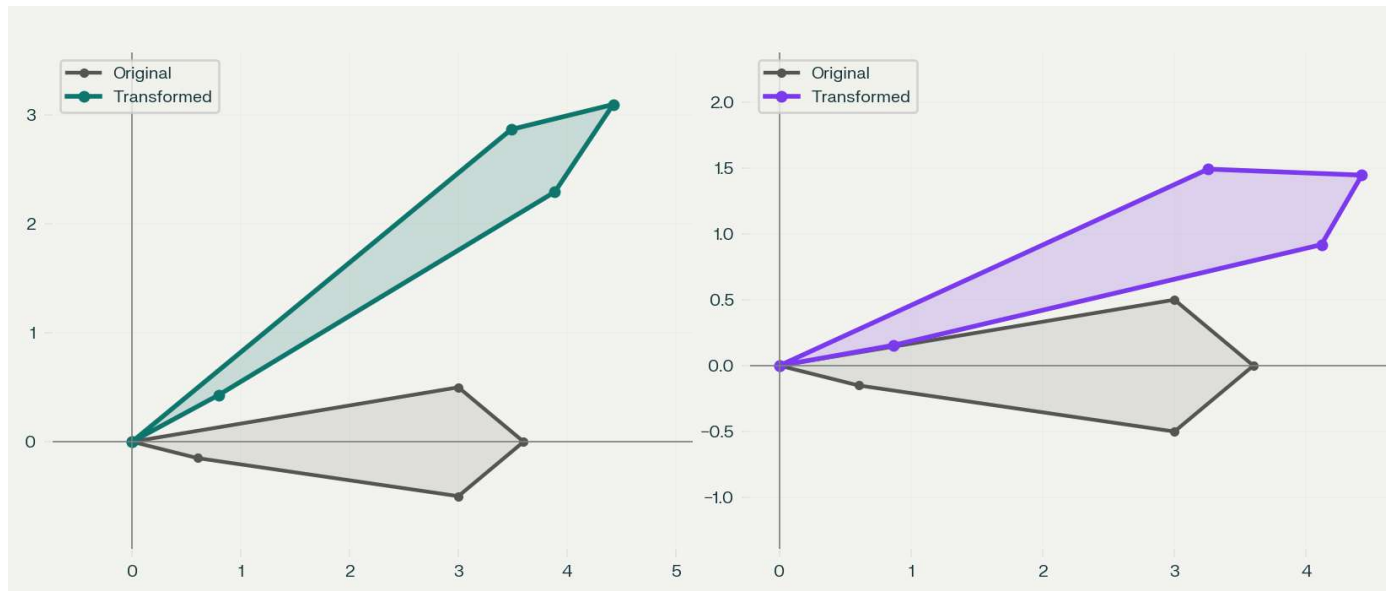


Figure 4. Left: scale, then rotate ( $R \cdot S$ ). Right: rotate, then scale ( $S \cdot R$ ). Same matrices, different final icon. (Created by Jahan Garg in Python/Matplotlib.)

#### Reading the matrices

$$R \cdot S = \begin{vmatrix} 1.229 & -0.402 \\ 0.860 & 0.573 \end{vmatrix} \quad S \cdot R = \begin{vmatrix} 1.229 & -0.860 \\ 0.402 & 0.573 \end{vmatrix}$$

Translation and rotation combine in the same order-sensitive way – this is why the pivot rotation on Page 4 had to translate first, then rotate, then translate back, in that exact order



## 6. TRY IT YOURSELF, KEY TAKEAWAYS, AND REFERENCES

### Try It Yourself

An icon has a vertex at  $v = (2, -1)$ . Apply the shear matrix  $H$  below to  $v$ .

$$H = \begin{pmatrix} 1 & 0 \\ 0.5 & 1 \end{pmatrix}$$

- 1) Compute  $H \cdot v$ .
- 2) In one sentence, describe the geometric effect this shear has on the point.
- 3) If the icon were first rotated  $90^\circ$  counter-clockwise about the origin and then this same  $H$  were applied, would you expect the same final point? Explain briefly.

#### Self-check (attempt first!)

$H \cdot v = (2, 0.5 \times 2 + (-1)) = (2, 0)$ . The shear pushes the point sideways by an amount proportional to its height, leaving its x-coordinate fixed here only because the shift exactly cancels the original y-coordinate. Since  $H \cdot R \neq R \cdot H$  in general, applying  $H$  after a rotation would give a different final point.

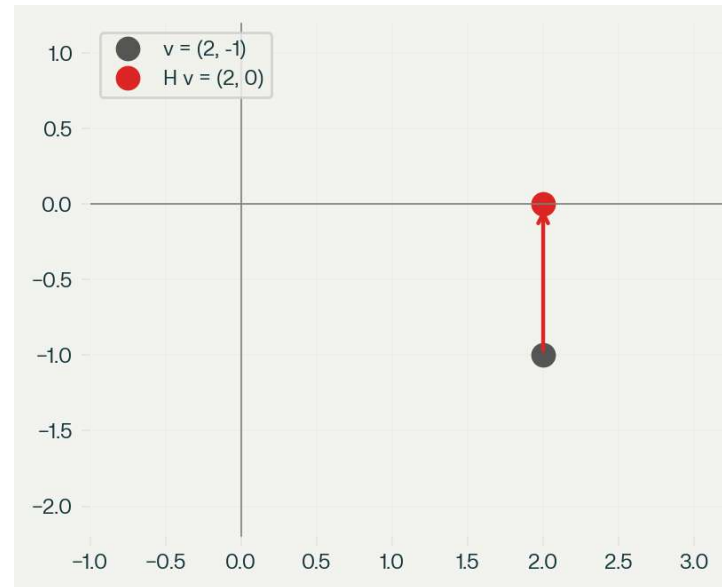


Figure 5. The shear moves  $v = (2, -1)$  to  $(2, 0)$ .

### Key Takeaways

1. An icon is stored as a data matrix  $D$  of column vectors.
2. Scaling, shearing, and rotation are all  $2 \times 2$  matrices.
3. Composing transforms multiplies matrices:  $D' = A_k \cdots A_2 A_1 D$ .
4. Matrix products are not commutative: order changes the result.
5. Homogeneous coordinates  $(x, y, 1)$  let translation – and rotation about any pivot – join the same framework.

### References

- [1] W. K. Nicholson, Linear Algebra with Applications, Open Edition, Version 2021A, Lyryx Learning Inc., Calgary, AB, Canada, 2021, Section 4.5, “An Application to Computer Graphics.”

### Tools and Originality Statement

The arrow icon, its five vertex coordinates, all scale/shear/rotation parameters, the pivot point, the composition-order example and the Try It Yourself exercise were created specifically for this submission by Jahan Garg and do not reproduce Nicholson's own worked example. All matrix calculations were performed and independently verified using Python (NumPy); Figures 1–5 were generated using Python (Matplotlib); an AI writing assistant (Perplexity) was used to help draft explanatory text, and the Python/ReportLab code used to typeset this PDF, with all mathematical content reviewed and verified by the author before inclusion.

DEAKIN UNIVERSITY

LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

Module 5 - Vector Space  $\mathbb{R}^n$

Submitted By:  
Jahan GARG  
s226100103  
2026/09/07 06:35

Tutor:  
Simon JAMES

| Task Outcomes |                                                                                                                                                       | Supports               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| M5.1          | Determine whether or not a set of objects defines a vector space, or subspace with respect to appropriate addition and multiplication operations      | ULO1 ULO1<br>ULO2 ULO2 |
| M5.2          | Find a basis that generates a given vector space                                                                                                      | ULO1 ULO1<br>ULO2 ULO2 |
| M5.3          | Calculate the dimension of a vector space                                                                                                             | ULO1 ULO1<br>ULO2 ULO2 |
| M5.4          | Identify when a set of vectors is orthogonal                                                                                                          | ULO1 ULO1              |
| SA            | Full printout of Self-Assessment results with all questions.                                                                                          |                        |
| Evid          | Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.                          |                        |
| Sum           | Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions. |                        |
| FB            | Respond to any feedback and work with the OnTrack tutor to complete the module.                                                                       |                        |

September 6, 2026



*SIT292 - Linear Algebra for Data Analysis***Module 5 Summary: Vector Space  $\mathbb{R}^n$** *Subspaces and Spanning, Independence and Dimension, Orthogonality***Response to Feedback**

This is a resubmission of my Module 5 assessment, and I am responding directly to Simon's feedback on my first attempt. Simon noted that the evidence component was fine to leave as handwritten notes and practice, since this is usually quicker for practice work, so I have kept that part of the submission unchanged. He also suggested that, since I am comfortable with LaTeX, it would be better to use it for the summary going forward, with template files provided in each module; I have taken this on board and typeset this summary using LaTeX-style formatting throughout, and intend to continue this for future modules. Most specifically, Simon pointed out that a matrix in my Module Summary was written inline as text, " $A = [1\ 2\ -1; 2\ 1\ 1; 0\ -1\ 2; -1\ 3\ 0]$ ", rather than displayed as a proper matrix, and asked me to correct this. I have now replaced every matrix in this summary, including that one in Question 1 of Section 5.2 and the coefficient matrix in Question 3, with properly typeset bracketed matrices with aligned rows and columns, rather than describing them as inline text. I have also carried forward the earlier guidance from Tanya on Module 1 to link the definitions and theorems together in the summary, which I have continued to do throughout this resubmission.

**Module Learning Objectives**

I certify that I achieved the following learning objectives for this module, as set out on the Module 5 introduction page of the unit site:

- 1 Determine whether or not a set of objects defines a vector space or subspace with respect to appropriate addition and multiplication operations.
- 2 Find a basis that generates a given vector space.
- 3 Calculate the dimension of a vector space.
- 4 Identify when a set of vectors is orthogonal.



## Key Definitions and Theorems

### Subspaces and Spanning

| Textbook Reference                                                                        | Theorem / Definition Statement                                                                                                                                                                                                                                                                    | Learning Notes                                                                                                                                                           |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 5.1</b><br><b>Subspace of <math>\mathbb{R}^n</math></b>                     | A set $U$ of vectors in $\mathbb{R}^n$ is called a subspace of $\mathbb{R}^n$ if it satisfies: S1. The zero vector $0 \in U$ . S2. If $x \in U$ and $y \in U$ , then $x + y \in U$ . S3. If $x \in U$ , then $ax \in U$ for every real number $a$ .                                               | The three conditions (S1-S3) are exactly the closure checks used throughout the enrichment and self-assessment exercises: zero vector, additive closure, scalar closure. |
| <b>Definition 5.2</b><br><b>Linear combinations and span in <math>\mathbb{R}^n</math></b> | The span of a set of vectors $\{x_1, x_2, \dots, x_k\}$ is the set of all vectors obtainable as linear combinations: $\text{span}\{x_1, \dots, x_k\} = \{t_1x_1 + t_2x_2 + \dots + t_kx_k \mid t_i \in \mathbb{R}\}$ . If $V = \text{span}\{x_1, \dots, x_k\}$ , the $x_i$ are said to span $V$ . | Span is the building block used to define both subspaces and bases; every subspace considered in this module is presented either as a solution set or as a span.         |

### Independence and Dimension

| Textbook Reference                                                                        | Theorem / Definition Statement                                                                                                                                                                                                                        | Learning Notes                                                                                                                                                      |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 5.3</b><br><b>Linear independence in <math>\mathbb{R}^n</math></b>          | A set $\{x_1, \dots, x_k\}$ is linearly independent if: $t_1x_1 + t_2x_2 + \dots + t_kx_k = 0 \Rightarrow t_1 = t_2 = \dots = t_k = 0$ .                                                                                                              | This is the Independence Test used directly in N5.3.1 and Question 3 of the self-assessment.                                                                        |
| <b>Theorem 5.2.1</b><br><b>Unique representation for independent sets</b>                 | If $\{x_1, \dots, x_k\}$ is independent in $\mathbb{R}^n$ , every vector in its span has a unique representation as a linear combination of the $x_i$ .                                                                                               | Uniqueness is what allows coefficients (such as those found for $z$ in $\text{span}\{x, y\}$ in N5.2.2) to be trusted as the only possible answer.                  |
| <b>Theorem 5.2.2</b><br><b>Independence of columns and solutions to <math>Ax=b</math></b> | For an $m \times n$ matrix $A$ with columns $\{c_1, \dots, c_n\}$ : 1. $\{c_1, \dots, c_n\}$ is independent in $\mathbb{R}^m$ iff $Ax=0$ implies $x=0$ . 2. $\mathbb{R}^m = \text{span}\{c_1, \dots, c_n\}$ iff $Ax=b$ has a solution for every $b$ . | This links to Module 5 directly back to Module 1: testing independence is the same procedure as testing whether a homogeneous system has only the trivial solution. |
| <b>Theorem 5.2.3</b><br><b>Equivalent properties related to independence</b>              | For an $n \times n$ matrix $A$ , the following are equivalent: (1) $A$ is invertible; (2) columns of $A$ are independent; (3) columns of $A$ span $\mathbb{R}^n$ ; (4) rows of $A$ are independent; (5) rows of $A$ span all $1 \times n$ rows.       | Used in Exercise 5.2.8 of the learning evidence to connect $\det A = 0$ with linear dependence of columns.                                                          |

| Textbook Reference                                                                   | Theorem / Definition Statement                                                                                                                                      | Learning Notes                                                                                                                                                      |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 5.4</b><br><b>Basis of <math>\mathbb{R}^n</math></b>                   | If $U$ is a subspace of $\mathbb{R}^n$ , $\{x_1, \dots, x_m\}$ is a basis of $U$ if: (1) it is linearly independent, and (2) $U = \text{span}\{x_1, \dots, x_m\}$ . | Applied directly in N5.3.2 and Question 4 of the self-assessment, where independence plus spanning together confirmed a full basis of $\mathbb{R}^4$ .              |
| <b>Definition 5.5</b><br><b>Dimension of a subspace of <math>\mathbb{R}^n</math></b> | If $\{x_1, \dots, x_m\}$ is any basis of subspace $U$ , the number $m$ is the dimension of $U$ , written $\dim U = m$ .                                             | Dimension is well-defined only because every basis of a given subspace contains the same number of vectors (a consequence of the Fundamental Theorem Below)         |
| <b>Theorem 5.2.4</b><br><b>Fundamental theorem</b>                                   | Let $U$ be a subspace of $\mathbb{R}^n$ . If $U$ is spanned by $m$ vectors and contains $k$ linearly independent vectors, then $k \leq m$ .                         | This theorem guarantees dimension is a well-defined, single number for a subspace, rather than depending on which spanning or independent set happens to be chosen. |

## Orthogonality

| Textbook Reference                                              | Theorem / Definition Statement                                                                                                                                                                                                                                             | Learning Notes                                                                                                                                                                     |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theorem 5.3.1</b><br><b>Properties of the dot product</b>    | For $x, y, z$ in $\mathbb{R}^n$ : 1. $x \cdot y = y \cdot x$ . 2. $x \cdot (y+z) = x \cdot y + x \cdot z$ . 3. $(ax) \cdot y = a(x \cdot y) = x \cdot (ay)$ . 4. $\ x\ ^2 = x \cdot x$ . 5. $\ x\  \geq 0$ , and $\ x\  = 0$ iff $x = 0$ . 6. $\ ax\  =  a  \cdot \ x\ $ . | Property 4 is exactly the tool used in the Learning Evidence (Exercise 5.3.13(b)) to compute squared norms and test the Pythagorean-style identity without assuming orthogonality. |
| <b>Definition 5.8</b><br><b>Orthogonal and orthonormal sets</b> | $x$ and $y$ in $\mathbb{R}^n$ are orthogonal if $x \cdot y = 0$ . A set $\{x_1, \dots, x_k\}$ is an orthogonal set if $x_i \cdot x_j = 0$ for $i \neq j$ and each $x_i \neq 0$ . It is orthonormal if it is additionally made of unit vectors: $\ x_i\  = 1$ .             | Directly checked in N5.4.1 and self-assessment Question 5, where three mutually orthogonal vectors were extended to a fourth orthogonal vector.                                    |
| <b>Definition 5.9</b><br><b>Normalising an orthogonal set</b>   | If $\{x_1, \dots, x_k\}$ is orthogonal, then $\{(1/\ x_1\ )x_1, \dots, (1/\ x_k\ )x_k\}$ is orthonormal; this is the result of normalising the orthogonal set.                                                                                                             | Applied in Exercise 5.3.1(a) of the learning evidence to convert an orthogonal basis of $\mathbb{R}^3$ into an orthonormal one by dividing each vector by its length.              |

## Module Summary

Module 5 formalises ideas that had already appeared informally in earlier modules: taking linear combinations of solution vectors in Module 1, and checking whether a set of vectors could reproduce every column of a matrix in Module 2. Chapter 5 of Nicholson [1] ties these observations together under the single framework of a vector space, using  $\mathbb{R}^n$  as the concrete setting before the more abstract treatment in Chapter 6. The three sub-modules, Subspaces and Spanning, Independence and Dimension, and Orthogonality, build on one another in a natural order: span is used to define a subspace, independence is used to trim a spanning set down to a basis, and dimension is then the single number that basis carries. Orthogonality is introduced afterwards as a particularly convenient kind of basis, one where the coefficients of any linear combination can be recovered directly by a dot product rather than by solving a system.

### 5.1 Subspaces and Spanning

A subset  $U$  of  $\mathbb{R}^n$  is a subspace precisely when it is closed under the two operations that make  $\mathbb{R}^n$  a vector space in the first place: it must contain the zero vector (S1), be closed under addition (S2), and be closed under scalar multiplication (S3), as given in Definition 5.1. In the learning evidence, Exercise 5.1.22(a) and (b) apply this three-part test to the intersection  $U \cap W$  and the sum  $U + W$  of two subspaces, and in both cases the argument follows the same pattern: write an arbitrary element of the new set in terms of elements of  $U$  and  $W$ , then invoke the corresponding subspace property of  $U$  and  $W$  separately to show the result stays inside  $U \cap W$  or  $U + W$ . This exercise made the standard structure of these proofs very clear to me: almost every subspace verification in this module reduces to the same three short checks, applied to whatever set is under consideration.

Definition 5.2 extends the informal idea of a linear combination, already used in Module 1 to describe general solutions of homogeneous systems, to the formal notion of a span. Exercise 5.1.18 in the learning evidence shows that a generator can be swapped for a suitable linear combination without changing the span, provided the coefficient of the generator being replaced is non-zero; and Exercise 5.1.12 shows that if every generator of a span is sent to the zero vector by a matrix  $A$ , then every vector in that span is also sent to zero, since  $Ay$  is itself a linear combination of the  $Ax_i$  terms. Both results reinforce that the span is entirely determined by linear combinations, so any manipulation that preserves the set of achievable linear combinations, such as replacing one generator or checking the image of every generator under  $A$ , is enough to conclude the whole span.

### 5.2 Independence and Dimension

Question 1: The Independence Test in matrix form. Definition 5.3 states that a set  $\{x_1, \dots, x_k\}$  is linearly independent exactly when the only solution of  $t_1x_1 + t_2x_2 + \dots + t_kx_k = 0$  is the trivial one,  $t_1 = t_2 = \dots = t_k = 0$ . Following the same structure as the textbook's worked example for three vectors in  $\mathbb{R}^4$ , this condition can be written directly in matrix form. Placing the vectors as the columns of a matrix  $A$  and writing the unknown coefficients as a column vector  $x = (r, s, t)^T$ , the independence condition becomes exactly  $Ax = 0$ , so testing independence is the same as solving a homogeneous system.

As an original illustration, take  $x_1=(1,2,0,-1)$ ,  $x_2=(2,1,-1,3)$  and  $x_3=(-1,1,2,0)$  in  $\mathbb{R}^4$ . Writing these as columns gives

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 0 & -1 & 2 \\ -1 & 3 & 0 \end{bmatrix}$$

and the equation  $rx_1+sx_2+tx_3=0$  is exactly  $Ax=0$  with  $x=(r,s,t)^T$ . Row-reducing  $A$  gives a reduced row-echelon form with a leading 1 in each of its three columns and no free variables, so  $\text{rank}(A) = 3$ , matching the number of vectors. By Theorem 5.2.2, since  $Ax=0$  forces  $x=0$ , the set  $\{x_1, x_2, x_3\}$  is linearly independent.

Question 2: rank, invertibility, independence and solvability. These four ideas describe the same underlying property of a matrix from different angles, and Module 5 is where they are finally drawn together explicitly, although each had already appeared separately in earlier modules. Rank, introduced in Module 1 as the number of leading ones in a row-echelon form, counts the number of linearly independent rows (equivalently columns) of a matrix; Theorem 5.2.3 states that for a square  $n \times n$  matrix  $A$ , being invertible, having independent columns, having columns that span  $\mathbb{R}^n$ , having independent rows and having rows that span all  $1 \times n$  rows are five equivalent conditions, all of which hold precisely when  $\text{rank}(A) = n$ . This means that for a square matrix, checking invertibility (via the inversion algorithm from Module 3, or via  $\det A \neq 0$  from Module 4) is the same question as checking whether its rows or columns are independent. Exercise 5.2.8 in the learning evidence makes the connection to determinants explicit:  $\det A = 0$  if and only if some column of  $A$  can be written as a linear combination of the others, which is just a restatement of linear dependence.

Solvability of  $Ax=b$  connects to the same picture through Theorem 5.2.2: the columns of  $A$  span  $\mathbb{R}^m$  exactly when  $Ax=b$  has a solution for every  $b$  in  $\mathbb{R}^m$ , and the columns are independent exactly when the only solution of  $Ax=0$  is  $x=0$ . Bringing this together with Theorem 2.2.4 from Module 2 on the rank of an augmented matrix, a system  $Ax=b$  is consistent when  $\text{rank}[A|b] = \text{rank}(A)$ , and it has a unique solution precisely when this common rank equals the number of unknowns, since that forces every column to be a pivot column and hence the columns to be independent. In short, for a square matrix, a full rank of  $n$  forces invertibility, independence of rows and columns, and a unique solution to  $Ax=b$  for every  $b$  simultaneously; any rank deficiency below  $n$  breaks all of these properties together, replacing a unique solution with either no solution or infinitely many.

Question 3: a homogeneous system with one independent row. Consider the  $3 \times 3$  matrix

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ -3 & 6 & -9 \end{bmatrix}$$

in which every row is a scalar multiple of the first row (2, respectively -3, times row 1), so  $\text{rank}(A) = 1$  and there is only one independent equation,  $x - 2y + 3z = 0$ , governing the system  $Ax=0$ . With 3 unknowns and rank 1, there are  $3 - 1 = 2$  free variables, so the solution space has dimension 2 by the relationship between rank and the number of basic solutions covered in Module 1. Setting  $z = 0$  and  $y = 1$  gives  $x = 2$ , producing the vector  $v_1 = (2, 1, 0)$ ; setting  $y = 0$  and  $z = 1$  gives  $x = -3$ , producing  $v_2 = (-3, 0, 1)$ . Direct substitution confirms  $Av_1 = 0$  and  $Av_2 = 0$ , and  $\{v_1, v_2\}$  is independent since neither is a scalar multiple of the other.

Every solution of  $Ax=0$  can be written as  $x = 2t - 3s$ ,  $y = t$ ,  $z = s$  for parameters  $s, t$ , that is, as  $tv_1 + sv_2$ , so  $U = \text{span}\{v_1, v_2\}$  exactly. By Definition 5.4,  $\{v_1, v_2\}$  is therefore a basis of  $U$ , and by Definition 5.5,  $\dim U = 2$ , consistent with the Fundamental Theorem (Theorem 5.2.4):  $U$  is spanned by 2 vectors and cannot contain more than 2 independent vectors inside a 3-variable system of rank 1. Finally,  $U$  is a subspace of  $\mathbb{R}^3$  because it satisfies Definition 5.1 directly: the zero vector solves  $A0=0$  (S1); if  $Au=0$  and  $Aw=0$  then  $A(u+w) = Au+Aw = 0$  by linearity of matrix multiplication, so the solution set is closed under addition (S2); and  $A(au) = a(Au) = a0 = 0$  for any scalar  $a$ , so it is closed under scalar multiplication (S3). This mirrors the general argument used in Exercise 5.1.12 of the learning evidence, where a matrix mapping every generator of a span to zero was shown to map the entire span to zero.

### 5.3 Orthogonality

Theorem 5.3.1 collects the algebraic properties of the dot product that had already been used informally in Module 2: symmetry, distributivity over addition, homogeneity with respect to scalars, and the identity  $\|x\|^2 = x \cdot x$  linking length to the dot product. Exercise 5.3.13(b) of the learning evidence uses property 4 directly: for three vectors  $x, y, z$  in  $\mathbb{R}^2$ , it is possible to have  $\|x+y+z\|^2 = \|x\|^2 + \|y\|^2 + \|z\|^2$  purely by numerical coincidence, even though none of the three pairwise dot products  $x \cdot y$ ,  $x \cdot z$  or  $y \cdot z$  is zero. This exercise was a useful caution: a squared-length identity that looks like a Pythagorean relationship should never be read as proof of orthogonality without checking the dot products directly.

Definition 5.8 defines an orthogonal set as one in which every pair of distinct vectors has a zero dot product, with every vector required to be non-zero, and defines an orthonormal set as an orthogonal set in which every vector additionally has unit length. In N5.4.1 and the corresponding self-assessment question, this definition was applied in reverse: given three vectors already confirmed to be mutually orthogonal, a fourth vector  $(a, b, c, d)$  was found by treating each orthogonality condition  $v_i \cdot (a, b, c, d) = 0$  as a linear equation and solving the resulting system, the same elimination technique used throughout Modules 1 and 2. Definition 5.9 then describes how any orthogonal set can be converted into an orthonormal one by dividing each vector by its own length; Exercise 5.3.1(a) of the learning evidence applies this to an orthogonal basis of  $\mathbb{R}^3$ , dividing each of the three mutually orthogonal vectors by its length ( $\sqrt{6}$ ,  $\sqrt{5}$  and  $\sqrt{30}$  respectively) to produce three unit vectors that remain mutually orthogonal, since scaling a vector by a non-zero constant cannot change whether its dot product with another vector is zero. Exercise 5.3.9 shows the same idea from the matrix side: if the columns of  $A$  are orthonormal, then  $A^T A = I$ , because each diagonal entry of  $A^T A$  is a column dotted with itself (giving 1, by the unit-length condition) and each off-diagonal entry is two distinct columns dotted together (giving 0, by orthogonality).

Question 4: orthogonal versus orthonormal. An orthogonal set only requires that every pair of distinct, non-zero vectors in the set have a dot product of zero, whereas an orthonormal set additionally requires every one of those vectors to have unit length, so that every orthonormal set is automatically orthogonal, but an orthogonal set only becomes orthonormal once each vector has been normalised by dividing it by its own length.



## Extra Effort and Prior Knowledge

Beyond the prescribed unit-site material, I revisited the informal treatment of vector spaces and subspaces given by Klemm's supplementary background summary [2], which was useful for seeing a slightly more formal, textbook-independent phrasing of the subspace and span definitions before returning to Nicholson's own notation. I also consulted a short GeeksforGeeks article on orthogonal and orthonormal vectors [4] to check that my own wording of the distinction in Question 4 above matched standard usage outside this unit's own materials, and a Mathematics Stack Exchange discussion on why  $\det A = 0$  corresponds to linearly dependent columns [5] to cross-check the direction of the argument used in Exercise 5.2.8 of the learning evidence. In terms of prior knowledge, the idea of a spanning set was already familiar to me from expressing general solutions of homogeneous systems in Module 1 as linear combinations of basic solutions; what this module added was the formal vocabulary, subspace, span, basis and dimension, needed to describe that structure precisely and to recognise the same pattern in settings other than solution sets, such as symmetric matrices or sets of continuous functions.

## Reflecting on the Content

The idea I found most satisfying in this module was seeing rank, invertibility, independence and solvability finally stated as explicitly equivalent conditions in Theorem 5.2.3, since each of these had been introduced separately in earlier modules without the connection between them being made completely explicit. Working through Question 2 above helped me realise that almost every technique covered so far in this unit, Gaussian elimination, the inversion algorithm and the determinant test, is really answering the same underlying question about a matrix from a different computational angle. The most challenging part of the module was Exercise 5.2.19 in the learning evidence, showing that a subspace  $U$  with  $\dim U = n-1$  nested inside a larger subspace  $W$  of  $\mathbb{R}^n$  forces either  $W = U$  or  $W = \mathbb{R}^n$ ; this required combining the Fundamental Theorem with the fact that two nested subspaces of equal dimension must coincide, an argument style that was new to me and felt closer to the abstract reasoning I expect to meet again in Module 6. Orthogonality was the most immediately useful sub-module, since Definition 5.9 gives such a simple recipe (dividing by length) for improving any orthogonal basis into an orthonormal one, and Exercise 5.3.9 showing  $A^T A = I$  for orthonormal columns gives a preview of why orthonormal bases will matter later when we study orthogonal diagonalisation.

## Tools and Originality Statement

All worked examples appearing in this summary that are not directly taken from the Key Definitions and Theorems table, namely the independence test example in  $\mathbb{R}^4$  (Question 1) and the rank-1 homogeneous system in  $\mathbb{R}^3$  (Question 3), were constructed specifically for this submission and independently verified using Python (NumPy and SymPy) before being written up by hand in this document. The handwritten learning evidence referenced throughout (Exercises 5.1.12, 5.1.18, 5.1.22(a)/(b), 5.2.8, 5.2.5(c), 5.2.12, 5.2.19, 5.3.1(a), 5.3.9 and 5.3.13(b)) and the NUMBAS self-assessment (10/10) were completed and submitted separately as part of this module's Learning Evidence and Self-Assessment documents, and remain unchanged from my previous submission, per Simon's feedback that this part was fine to leave as is. An AI writing assistant (Perplexity) was used to help structure this summary, refine explanations of the theorems and check written English; the mathematical reasoning, worked examples, and self-assessment results referenced throughout are my own.

## References

- [1] W. K. Nicholson, Linear Algebra with Applications, Open Edition, Version 2021A, Lyryx Learning Inc., Calgary, AB, Canada, 2021, Chapter 5: Vector Space  $\mathbb{R}^n$ , Sections 5.1-5.3.
- [2] A. D. Klemm, "Background Summary - Vector Spaces in Linear Algebra," supplementary resource, SIT292 Unit Site, Deakin University, CloudDeakin, Trimester 2, 2026.
- [3] Deakin University, "Topic 5: Vector Space  $\mathbb{R}^n$  - Introduction, 5.1 Subspaces and Spanning, 5.2 Independence and Dimension, 5.3 Orthogonality," SIT292 Linear Algebra for Data Analysis, CloudDeakin Unit Site, Trimester 2, 2026.
- [4] GeeksforGeeks, "Orthogonal and Orthonormal Vectors in Linear Algebra," GeeksforGeeks, accessed Sep. 2026. [Online]. Available: <https://www.geeksforgeeks.org>
- [5] Mathematics Stack Exchange, "Why does a zero determinant imply linearly dependent columns?" Stack Exchange Inc., accessed Sep. 2026. [Online]. Available: <https://math.stackexchange.com>
- [6] Deakin University, SIT292 Module 5 Self-Assessment, Numbas, developed by Newcastle University, accessed 3 Sep. 2026.

## Module 5: Handwritten Learning Evidence

### Section 5.1: Subspaces and Spanning

#### Exercise 5.1.12

##### Question

Suppose that

$$U = \text{span}\{x_1, x_2, \dots, x_k\},$$

where each  $x_i \in \mathbb{R}^n$ . If  $A$  is an  $m \times n$  matrix and

$$Ax_i = 0$$

for each  $i$ , show that

$$Ay = 0$$

for every vector  $y \in U$ .

##### Concepts / properties used

Definition of span; linearity of matrix multiplication.

##### Working

Take an arbitrary vector

$$y \in U.$$

Since

$$U = \text{span}\{x_1, x_2, \dots, x_k\},$$

there exist scalars  $r_1, r_2, \dots, r_k \in \mathbb{R}$  such that

$$y = r_1x_1 + r_2x_2 + \dots + r_kx_k. \quad (\text{Definition of span})$$

Applying  $A$  to both sides gives

$$Ay = A(r_1x_1 + r_2x_2 + \dots + r_kx_k).$$

Using linearity of matrix multiplication,

$$Ay = r_1Ax_1 + r_2Ax_2 + \dots + r_kAx_k. \quad (\text{Linearity})$$

Given that

$$Ax_i = 0 \quad \text{for every } i,$$

we obtain

$$Ay = r_1(0) + r_2(0) + \cdots + r_k(0) = 0.$$

### Verification

The vector  $y$  was chosen arbitrarily, so the result holds for every  $y \in U$ . Therefore,

$$\boxed{Ay = 0 \text{ for every } y \in U}.$$

Equivalently,

$$\boxed{U \subseteq \text{null}(A)}.$$

### Key Learning Point

If every generator of a span is mapped to the zero vector by  $A$ , then every linear combination of those generators is also mapped to zero.

### Exercise 5.1.18

#### Question

Suppose that  $x_1, x_2, \dots, x_k$  are vectors in  $\mathbb{R}^n$ . If

$$y = a_1x_1 + a_2x_2 + \cdots + a_kx_k, \quad a_1 \neq 0,$$

show that

$$\text{span}\{x_1, x_2, \dots, x_k\} = \text{span}\{y, x_2, \dots, x_k\}.$$

#### Concepts / properties used

Linear combination; Span Theorem; equality by proving two inclusions.

#### Method

Let

$$U = \text{span}\{x_1, x_2, \dots, x_k\}$$

and

$$V = \text{span}\{y, x_2, \dots, x_k\}.$$

We prove

$$U \subseteq V \quad \text{and} \quad V \subseteq U. \quad (\text{Span Theorem})$$

#### Part 1: Prove $V \subseteq U$

From

$$y = a_1x_1 + a_2x_2 + \cdots + a_kx_k,$$

we have

$$y \in U.$$

Also,

$$x_2, x_3, \dots, x_k \in U.$$

Thus every generator of  $V$  belongs to  $U$ . Hence,

$$\boxed{V \subseteq U}. \quad (\text{Span Theorem})$$

## Part 2: Prove $U \subseteq V$

The vectors

$$x_2, x_3, \dots, x_k$$

are already generators of  $V$ , so they belong to  $V$ .

It remains to show that  $x_1 \in V$ .

Starting with

$$y = a_1x_1 + a_2x_2 + \dots + a_kx_k,$$

rearrange to obtain

$$a_1x_1 = y - a_2x_2 - a_3x_3 - \dots - a_kx_k.$$

Since  $a_1 \neq 0$ ,

$$x_1 = \frac{1}{a_1} (y - a_2x_2 - a_3x_3 - \dots - a_kx_k). \quad (\text{Since } a_1 \neq 0)$$

Therefore  $x_1$  is a linear combination of

$$y, x_2, \dots, x_k,$$

so

$$x_1 \in V. \quad (\text{Definition of span})$$

Hence

$$x_1, x_2, \dots, x_k \in V,$$

and therefore

$$\boxed{U \subseteq V}. \quad (\text{Span Theorem})$$

## Verification

Both inclusions have been established:

$$U \subseteq V \quad \text{and} \quad V \subseteq U.$$

Therefore,

$$\boxed{\text{span}\{x_1, x_2, \dots, x_k\} = \text{span}\{y, x_2, \dots, x_k\}}.$$



**Key Learning Point**

A generator can be replaced by a suitable linear combination without changing the span, provided the coefficient of the original generator being replaced is non-zero.

**Exercise 5.1.22(a)****Question**

Let  $U$  and  $W$  be subspaces of  $\mathbb{R}^n$ . Define

$$U \cap W = \{x \in \mathbb{R}^n \mid x \text{ belongs to both } U \text{ and } W\}.$$

Show that  $U \cap W$  is a subspace of  $\mathbb{R}^n$ .

**Concepts / properties used**

Subspace conditions S1, S2 and S3.

**Method**

Let

$$S = U \cap W.$$

We verify the three subspace conditions.

**S1 — Zero vector**

Since  $U$  and  $W$  are subspaces,

$$0 \in U \quad \text{and} \quad 0 \in W.$$

Hence,

$$\boxed{0 \in U \cap W}. \quad (\text{S1})$$

**S2 — Closure under addition**

Let

$$x, y \in U \cap W.$$

Then

$$x, y \in U \quad \text{and} \quad x, y \in W.$$

Because  $U$  is a subspace,

$$x + y \in U.$$

Because  $W$  is a subspace,

$$x + y \in W.$$

Therefore,

$$\boxed{x + y \in U \cap W}. \quad (\text{S2})$$

### S3 — Closure under scalar multiplication

Let

$$x \in U \cap W, \quad a \in \mathbb{R}.$$

Then

$$x \in U \quad \text{and} \quad x \in W.$$

Since both are subspaces,

$$ax \in U \quad \text{and} \quad ax \in W.$$

Therefore,

$$\boxed{ax \in U \cap W}. \quad (\text{S3})$$

### Verification

The set  $U \cap W$  satisfies S1, S2 and S3. Hence,

$$\boxed{U \cap W \text{ is a subspace of } \mathbb{R}^n}.$$

### Key Learning Point

The intersection consists of vectors common to both subspaces, so the subspace properties are preserved simultaneously.

## Exercise 5.1.22(b)

### Question

Let  $U$  and  $W$  be subspaces of  $\mathbb{R}^n$ , and define

$$U + W = \{x \in \mathbb{R}^n \mid x \text{ is a sum of a vector in } U \text{ and a vector in } W\}.$$

Show that  $U + W$  is a subspace of  $\mathbb{R}^n$ .

Equivalently,

$$U + W = \{u + w \mid u \in U, w \in W\}.$$

### Concepts / properties used

Subspace conditions S1, S2 and S3; distributivity of scalar multiplication.

**Method**

We verify S1, S2 and S3 directly from the representation

$$u + w, \quad u \in U, w \in W.$$

**S1 — Zero vector**

Since  $U$  and  $W$  are subspaces,

$$0 \in U \quad \text{and} \quad 0 \in W.$$

Therefore,

$$0 = 0 + 0 \in U + W.$$

Hence,

$$\boxed{0 \in U + W}. \quad (\text{S1})$$

**S2 — Closure under addition**

Let

$$p, q \in U + W.$$

Then there exist

$$u_1, u_2 \in U \quad \text{and} \quad w_1, w_2 \in W$$

such that

$$p = u_1 + w_1, \quad q = u_2 + w_2.$$

Therefore,

$$\begin{aligned} p + q &= (u_1 + w_1) + (u_2 + w_2) \\ &= (u_1 + u_2) + (w_1 + w_2). \end{aligned} \quad (\text{Addition properties})$$

Since  $U$  is closed under addition,

$$u_1 + u_2 \in U.$$

Since  $W$  is closed under addition,

$$w_1 + w_2 \in W.$$

Thus  $p + q$  is again the sum of a vector from  $U$  and a vector from  $W$ . Hence,

$$\boxed{p + q \in U + W}. \quad (\text{S2})$$

**S3 — Closure under scalar multiplication**

Let

$$p \in U + W, \quad a \in \mathbb{R}.$$

Write

$$p = u + w, \quad u \in U, w \in W.$$

Then

$$\begin{aligned} ap &= a(u + w) \\ &= au + aw. \end{aligned} \quad (\text{Distributivity})$$

Because  $U$  and  $W$  are subspaces,

$$au \in U \quad \text{and} \quad aw \in W.$$

Therefore,

$$\boxed{ap \in U + W}. \quad (\text{S3})$$

**Verification**

We have shown that

$$0 \in U + W,$$

$$p, q \in U + W \Rightarrow p + q \in U + W,$$

and

$$p \in U + W, a \in \mathbb{R} \Rightarrow ap \in U + W.$$

Thus S1, S2 and S3 all hold. Therefore,

$$\boxed{U + W \text{ is a subspace of } \mathbb{R}^n}.$$

**Key Learning Point**

For  $U + W$ , writing each vector as  $u + w$  makes the closure arguments follow directly from the corresponding subspace properties of  $U$  and  $W$ .

## Section 5.2: Independence and Dimension

### Exercise 5.2.8

#### Question

If  $A$  is an  $n \times n$  matrix, show that

$$\det A = 0$$

if and only if some column of  $A$  is a linear combination of the other columns.

#### Concepts / properties used

Determinant–invertibility relationship; linear independence; linear combinations of columns; Theorem 5.2.3.

#### Method

Let the columns of  $A$  be

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}.$$

We prove both implications.

#### Part 1: Assume $\det A = 0$

Since

$$\det A = 0,$$

the matrix  $A$  is not invertible. (Determinant criterion for invertibility)

By Theorem 5.2.3, the columns of a square matrix are linearly independent if and only if the matrix is invertible. Therefore, the columns of  $A$  are linearly dependent. (Theorem 5.2.3)

Hence there exist scalars

$$c_1, c_2, \dots, c_n,$$

not all zero, such that

$$c_1 a_1 + c_2 a_2 + \cdots + c_n a_n = 0. \quad (\text{Linear dependence})$$

Because the coefficients are not all zero, choose an index  $j$  such that

$$c_j \neq 0.$$



Then

$$c_j a_j = - \sum_{i \neq j} c_i a_i.$$

Dividing by  $c_j$ ,

$$a_j = - \sum_{i \neq j} \frac{c_i}{c_j} a_i. \quad (\text{Since } c_j \neq 0)$$

Thus  $a_j$  is a linear combination of the other columns.

Therefore,

$$\boxed{\det A = 0 \implies \text{some column of } A \text{ is a linear combination of the others.}}$$

## Part 2: Assume some column is a linear combination of the others

Suppose that, for some  $j$ ,

$$a_j = \sum_{i \neq j} r_i a_i.$$

Rearranging,

$$- \sum_{i \neq j} r_i a_i + a_j = 0.$$

This is a non-trivial linear combination of the columns of  $A$ . Hence the columns are linearly dependent. (Linear dependence)

By Theorem 5.2.3,  $A$  is therefore not invertible.

$$A \text{ is not invertible.} \quad (\text{Theorem 5.2.3})$$

Hence,

$$\boxed{\det A = 0}. \quad (\text{Determinant criterion for invertibility})$$

## Verification

Both implications have been established:

$$\det A = 0 \implies \text{some column is a linear combination of the others,}$$

and

$$\text{some column is a linear combination of the others} \implies \det A = 0.$$

Therefore,

$$\boxed{\det A = 0 \iff \text{some column of } A \text{ is a linear combination of the other columns.}}$$

**Key Learning Point**

A zero determinant signals linear dependence among the columns of a square matrix: at least one column can be generated from the others.

**Exercise 5.2.5(c)****Question**

Suppose that

$$\{x, y, z, w\}$$

is a basis of  $\mathbb{R}^4$ . Show that

$$\{x, x + y, x + y + z, x + y + z + w\}$$

is also a basis of  $\mathbb{R}^4$ .

**Concepts / properties used**

Basis; linear combinations; spanning; dimension of  $\mathbb{R}^4$ .

**Method**

Let

$$u_1 = x, \quad u_2 = x + y, \quad u_3 = x + y + z, \quad u_4 = x + y + z + w.$$

Since

$$\dim(\mathbb{R}^4) = 4,$$

it is enough to show that

$$\{u_1, u_2, u_3, u_4\}$$

spans  $\mathbb{R}^4$ . (Basis and dimension)

**Working**

Take an arbitrary vector

$$v \in \mathbb{R}^4.$$

Because  $\{x, y, z, w\}$  is a basis of  $\mathbb{R}^4$ , there exist scalars

$$\alpha, \beta, \gamma, \delta \in \mathbb{R}$$

such that

$$v = \alpha x + \beta y + \gamma z + \delta w. \quad (\text{Basis representation})$$

Now consider

$$(\alpha - \beta)u_1 + (\beta - \gamma)u_2 + (\gamma - \delta)u_3 + \delta u_4.$$

Substituting the definitions of  $u_1, u_2, u_3, u_4$ ,

$$\begin{aligned} & (\alpha - \beta)x + (\beta - \gamma)(x + y) + (\gamma - \delta)(x + y + z) + \delta(x + y + z + w) \\ &= [(\alpha - \beta) + (\beta - \gamma) + (\gamma - \delta) + \delta]x \\ & \quad + [(\beta - \gamma) + (\gamma - \delta) + \delta]y \\ & \quad + [(\gamma - \delta) + \delta]z + \delta w. \end{aligned}$$

Simplifying the coefficients,

$$\begin{aligned} & (\alpha - \beta)u_1 + (\beta - \gamma)u_2 + (\gamma - \delta)u_3 + \delta u_4 \\ &= \alpha x + \beta y + \gamma z + \delta w \\ &= v. \end{aligned}$$

Therefore,

$$v \in \text{span}\{u_1, u_2, u_3, u_4\}.$$

Since  $v$  was arbitrary,

$$\mathbb{R}^4 = \text{span}\{u_1, u_2, u_3, u_4\}.$$

Thus the new set spans  $\mathbb{R}^4$ .

### Verification

The new set contains exactly four vectors and spans  $\mathbb{R}^4$ . Hence it is a basis.

Therefore,

$$\boxed{\{x, x + y, x + y + z, x + y + z + w\} \text{ is a basis of } \mathbb{R}^4}.$$

### Key Learning Point

A basis can be replaced by suitable cumulative linear combinations without changing the space it generates. The coefficients above allow the original basis representation to be recovered exactly.

**Exercise 5.2.12****Question**

If

$$\{x_1, x_2, \dots, x_k\}$$

is linearly independent, show that

$$\{x_1, x_1 + x_2, x_1 + x_2 + x_3, \dots, x_1 + x_2 + \dots + x_k\}$$

is also linearly independent.

**Concepts / properties used**

Linear independence; linear combinations; Independence Test.

**Method**

Suppose a linear combination of the new vectors is equal to the zero vector. We will show that all corresponding coefficients must be zero.

Let

$$u_1 = x_1, \quad u_2 = x_1 + x_2, \quad u_3 = x_1 + x_2 + x_3,$$

and, in general,

$$u_j = x_1 + x_2 + \dots + x_j.$$

Assume

$$t_1 u_1 + t_2 u_2 + \dots + t_k u_k = 0. \quad (\text{Independence Test})$$

Substituting the definitions of the  $u_j$ 's,

$$\begin{aligned} t_1 x_1 + t_2(x_1 + x_2) + t_3(x_1 + x_2 + x_3) + \dots \\ + t_k(x_1 + x_2 + \dots + x_k) = 0. \end{aligned}$$

Collecting like vectors,

$$\begin{aligned} (t_1 + t_2 + \dots + t_k)x_1 \\ + (t_2 + t_3 + \dots + t_k)x_2 \\ + (t_3 + \dots + t_k)x_3 \\ + \dots + t_k x_k = 0. \end{aligned}$$



Since

$$\{x_1, x_2, \dots, x_k\}$$

is linearly independent, every coefficient in this combination must be zero.

In particular,

$$t_k = 0.$$

Then the coefficient of  $x_{k-1}$  gives

$$t_{k-1} + t_k = 0,$$

so

$$t_{k-1} = 0.$$

Continuing in the same way gives

$$t_{k-2} = 0, \quad \dots, \quad t_2 = 0.$$

Finally, the coefficient of  $x_1$  gives

$$t_1 + t_2 + \dots + t_k = 0,$$

and since

$$t_2 = \dots = t_k = 0,$$

we obtain

$$t_1 = 0.$$

Hence,

$$\boxed{t_1 = t_2 = \dots = t_k = 0}.$$

### Verification

The only linear combination of the new vectors which gives the zero vector is the trivial one. Therefore, by the Independence Test,

$$\boxed{\{x_1, x_1 + x_2, x_1 + x_2 + x_3, \dots, x_1 + \dots + x_k\} \text{ is linearly independent}}.$$

### Key Learning Point

The cumulative form creates a triangular chain of coefficient equations. Working backwards forces every coefficient to vanish, so linear independence is preserved.

**Exercise 5.2.19****Question**

Let  $U$  and  $W$  be subspaces of  $\mathbb{R}^n$ , with

$$U \subseteq W.$$

If

$$\dim U = n - 1,$$

show that either

$$W = U$$

or

$$W = \mathbb{R}^n.$$

**Concepts / properties used**

Dimension of a subspace; dimension inequality for nested subspaces; equality of nested subspaces.

**Method**

Use the inclusion

$$U \subseteq W \subseteq \mathbb{R}^n$$

to restrict the possible values of  $\dim W$ .

**Working**

Since

$$U \subseteq W,$$

we have

$$\dim U \leq \dim W. \quad (\text{Dimension theorem})$$

Given

$$\dim U = n - 1,$$

it follows that

$$n - 1 \leq \dim W.$$

Also, since

$$W \subseteq \mathbb{R}^n,$$

we have

$$\dim W \leq \dim(\mathbb{R}^n) = n. \quad (\text{Dimension theorem})$$

Therefore,

$$n - 1 \leq \dim W \leq n.$$

Because  $\dim W$  is an integer,

$$\boxed{\dim W = n - 1 \quad \text{or} \quad \dim W = n}.$$

**Case 1:**  $\dim W = n - 1$

We then have

$$\dim U = \dim W.$$

Together with

$$U \subseteq W,$$

this gives

$$\boxed{U = W}. \quad (\text{Equal dimension for nested subspaces})$$

**Case 2:**  $\dim W = n$

Since

$$W \subseteq \mathbb{R}^n$$

and both spaces have dimension  $n$ ,

$$\boxed{W = \mathbb{R}^n}. \quad (\text{Dimension})$$

### Verification

The two cases

$$\dim W = n - 1 \quad \text{and} \quad \dim W = n$$

are the only possibilities.

Therefore,

$$\boxed{W = U \quad \text{or} \quad W = \mathbb{R}^n}.$$

**Key Learning Point**

A subspace of dimension  $n - 1$  is only one dimension below  $\mathbb{R}^n$ . Hence any larger subspace containing it must either have the same dimension, giving  $W = U$ , or have dimension  $n$ , giving  $W = \mathbb{R}^n$ .

## Section 5.3: Orthogonality

### Exercise 5.3.1(a)

#### Question

Obtain an orthonormal basis of  $\mathbb{R}^3$  by normalizing the following orthogonal set:

$$\{(1, -1, 2), (0, 2, 1), (5, 1, -2)\}.$$

#### Concepts / properties used

Dot product; length of a vector; normalization; orthogonal and orthonormal sets.

#### Method

First verify that the given vectors are mutually orthogonal. Then divide each non-zero vector by its length.

Let

$$v_1 = (1, -1, 2), \quad v_2 = (0, 2, 1), \quad v_3 = (5, 1, -2).$$

#### Part 1: Verify orthogonality

For the first pair,

$$\begin{aligned} v_1 \cdot v_2 &= (1)(0) + (-1)(2) + (2)(1) \\ &= 0 - 2 + 2 \\ &= 0. \end{aligned}$$

For the second pair,

$$\begin{aligned} v_1 \cdot v_3 &= (1)(5) + (-1)(1) + (2)(-2) \\ &= 5 - 1 - 4 \\ &= 0. \end{aligned}$$

For the third pair,

$$\begin{aligned} v_2 \cdot v_3 &= (0)(5) + (2)(1) + (1)(-2) \\ &= 0 + 2 - 2 \\ &= 0. \end{aligned}$$

Therefore,

|                                          |
|------------------------------------------|
| $v_1, v_2, v_3$ are mutually orthogonal. |
|------------------------------------------|



**Part 2: Normalize  $v_1$** 

The length of  $v_1$  is

$$\begin{aligned}\|v_1\| &= \sqrt{1^2 + (-1)^2 + 2^2} \\ &= \sqrt{1 + 1 + 4} && \text{(Length)} \\ &= \sqrt{6}.\end{aligned}$$

Hence the corresponding unit vector is

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{6}}(1, -1, 2). \quad \text{(Normalization)}$$

Thus,

$$u_1 = \left( \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right).$$

**Part 3: Normalize  $v_2$** 

The length of  $v_2$  is

$$\begin{aligned}\|v_2\| &= \sqrt{0^2 + 2^2 + 1^2} \\ &= \sqrt{5}. && \text{(Length)}\end{aligned}$$

Hence,

$$u_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{5}}(0, 2, 1), \quad \text{(Normalization)}$$

so

$$u_2 = \left( 0, \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right).$$

**Part 4: Normalize  $v_3$** 

The length of  $v_3$  is

$$\begin{aligned}\|v_3\| &= \sqrt{5^2 + 1^2 + (-2)^2} \\ &= \sqrt{25 + 1 + 4} && \text{(Length)} \\ &= \sqrt{30}.\end{aligned}$$

Hence,

$$u_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{30}}(5, 1, -2), \quad \text{(Normalization)}$$

so

$$u_3 = \left( \frac{5}{\sqrt{30}}, \frac{1}{\sqrt{30}}, -\frac{2}{\sqrt{30}} \right).$$

**Verification**

Each normalized vector has unit length,

$$\|u_1\| = \|u_2\| = \|u_3\| = 1,$$

and multiplying an orthogonal vector by a non-zero scalar preserves its orthogonality with the other vectors.

Therefore,

$$\left\{ \frac{1}{\sqrt{6}}(1, -1, 2), \frac{1}{\sqrt{5}}(0, 2, 1), \frac{1}{\sqrt{30}}(5, 1, -2) \right\}$$

is an orthonormal basis of  $\mathbb{R}^3$ .

**Key Learning Point**

Normalizing each non-zero vector of an orthogonal basis produces unit vectors while preserving orthogonality.

### Exercise 5.3.9

#### Question

If  $A$  is an  $m \times n$  matrix with orthonormal columns, show that

$$A^T A = I_n.$$

#### Concepts / properties used

Matrix transpose; matrix multiplication; dot product; orthonormal columns.

#### Method

Write the columns of  $A$  as

$$A = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \end{bmatrix}.$$

We determine the entries of  $A^T A$ .

#### Working

Since

$$A^T = \begin{bmatrix} c_1^T \\ c_2^T \\ \vdots \\ c_n^T \end{bmatrix},$$

matrix multiplication gives

$$A^T A = \begin{bmatrix} c_1^T c_1 & c_1^T c_2 & \cdots & c_1^T c_n \\ c_2^T c_1 & c_2^T c_2 & \cdots & c_2^T c_n \\ \vdots & \vdots & \ddots & \vdots \\ c_n^T c_1 & c_n^T c_2 & \cdots & c_n^T c_n \end{bmatrix}.$$

Using

$$c_i^T c_j = c_i \cdot c_j, \quad (\text{Dot product as a matrix product})$$

we obtain

$$A^T A = \begin{bmatrix} c_1 \cdot c_1 & c_1 \cdot c_2 & \cdots & c_1 \cdot c_n \\ c_2 \cdot c_1 & c_2 \cdot c_2 & \cdots & c_2 \cdot c_n \\ \vdots & \vdots & \ddots & \vdots \\ c_n \cdot c_1 & c_n \cdot c_2 & \cdots & c_n \cdot c_n \end{bmatrix}.$$

Because the columns are orthonormal,

$$c_i \cdot c_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (\text{Orthonormality})$$

Therefore,

$$A^T A = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Hence,

$$\boxed{A^T A = I_n}.$$

### Verification

Every diagonal entry is

$$c_i \cdot c_i = \|c_i\|^2 = 1,$$

while every off-diagonal entry is zero because distinct columns are orthogonal.

Thus the product is precisely the  $n \times n$  identity matrix  $I_n$ .

### Key Learning Point

The orthonormality of the columns of  $A$  is represented algebraically by

$$\boxed{A^T A = I_n},$$

linking the dot-product interpretation directly to matrix multiplication.

**Exercise 5.3.13(b)****Question**

Let

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad z = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$

Show that

$$\|x + y + z\|^2 = \|x\|^2 + \|y\|^2 + \|z\|^2,$$

but

$$x \cdot y \neq 0, \quad x \cdot z \neq 0, \quad y \cdot z \neq 0.$$

**Concepts / properties used**

Vector norm; dot product; squared length; Pythagorean identity.

**Method**

Calculate the squared norm of  $x + y + z$ , compare it with the sum of the individual squared norms, and then calculate the three pairwise dot products.

**Part 1: Calculate  $x + y + z$** 

$$\begin{aligned} x + y + z &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 4 \end{bmatrix}. \end{aligned}$$

Therefore,

$$\|x + y + z\|^2 = 0^2 + 4^2 = 16. \quad (\text{Squared norm})$$

**Part 2: Calculate the individual squared norms**

For  $x$ ,

$$\|x\|^2 = 1^2 + 1^2 = 2.$$

For  $y$ ,

$$\|y\|^2 = 1^2 + 0^2 = 1.$$

For  $z$ ,

$$\|z\|^2 = (-2)^2 + 3^2 = 4 + 9 = 13.$$



Hence,

$$\|x\|^2 + \|y\|^2 + \|z\|^2 = 2 + 1 + 13 = 16.$$

Thus,

$$\boxed{\|x + y + z\|^2 = \|x\|^2 + \|y\|^2 + \|z\|^2}.$$

### Part 3: Check the pairwise dot products

First,

$$\begin{aligned} x \cdot y &= (1)(1) + (1)(0) \\ &= 1. \end{aligned}$$

Therefore,

$$\boxed{x \cdot y = 1 \neq 0}.$$

Next,

$$\begin{aligned} x \cdot z &= (1)(-2) + (1)(3) \\ &= -2 + 3 \\ &= 1. \end{aligned}$$

Therefore,

$$\boxed{x \cdot z = 1 \neq 0}.$$

Finally,

$$\begin{aligned} y \cdot z &= (1)(-2) + (0)(3) \\ &= -2. \end{aligned}$$

Therefore,

$$\boxed{y \cdot z = -2 \neq 0}.$$

### Verification

We have shown that

$$\|x + y + z\|^2 = \|x\|^2 + \|y\|^2 + \|z\|^2,$$

while simultaneously

$$x \cdot y \neq 0, \quad x \cdot z \neq 0, \quad y \cdot z \neq 0.$$

Therefore, the three vectors are not pairwise orthogonal, despite satisfying the squared-norm equality.

$$\boxed{\|x + y + z\|^2 = \|x\|^2 + \|y\|^2 + \|z\|^2 \text{ does not imply pairwise orthogonality}}.$$

**Key Learning Point**

An equality involving squared norms should not be used by itself to conclude pairwise orthogonality; the dot products must be checked directly.

# Question 1

Let  $U$  be the set of vectors  $(x, y, z)$  with  $-2z = y - 1$  and  $x, y, z$  in  $\mathbb{R}$ .

Determine whether or not  $U$  is a subspace of  $\mathbb{R}^3$ .

The set  $U$ :

- ☐ includes the zero vector
- ☐ is closed under vector addition
- ☐ is closed under multiplication
- ☐ is a **subspace** of  $\mathbb{R}^3$
- ☒ is **not a subspace** of  $\mathbb{R}^3$



You chose a correct answer. You were awarded 1 mark. ✓

You scored 1 mark for this part.

---

**Score: 1/1** ✓

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

Let  $U$  be the set of vectors  $(x, y, 0)$  with  $x^2 + y^2 = 0$  and  $x, y$  in  $\mathbb{R}$ .

Determine whether or not  $U$  is a subspace of  $\mathbb{R}^3$ .

The set  $U$ :

- ☒ includes the zero vector
- ☒ is closed under vector addition
- ☒ is closed under multiplication
- ☒ is a **subspace** of  $\mathbb{R}^3$
- ☐ is **not a subspace** of  $\mathbb{R}^3$



You chose a correct answer. You were awarded 1 mark. ✓

You chose a correct answer. You were awarded 1 mark. ✓

You chose a correct answer. You were awarded 1 mark. ✓

You chose a correct answer. You were awarded 1 mark. ✓

You scored 4 marks for this part.

---

**Score: 4/4** ✓

# Question 1

Consider the vectors

$$\mathbf{x} = \begin{pmatrix} -4 \\ 3 \\ -1 \end{pmatrix}, \text{ and } \mathbf{y} = \begin{pmatrix} 4 \\ 3 \\ 6 \end{pmatrix}$$

Let  $\mathbf{z} = \begin{pmatrix} 12 \\ 1 \\ 13 \end{pmatrix}$ . Determine whether or not  $\mathbf{z}$  lies in  $U = \text{span}\{\mathbf{x}, \mathbf{y}\}$ .

If  $\mathbf{z}$  is in  $U$ , write it as a linear combination of  $\mathbf{x}$  and  $\mathbf{y}$

$$\mathbf{z} = \boxed{0} \begin{pmatrix} -4 \\ 3 \\ -1 \end{pmatrix} + \boxed{0} \begin{pmatrix} 4 \\ 3 \\ 6 \end{pmatrix}.$$

If it is not possible, enter 0 for both coefficients.

You're right! This vector is not in  $U$  and cannot be expressed as a linear combination. You were awarded **2** marks.



You scored **2** marks for this part.

**Score: 2/2** ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).



# Question 1

Consider the vectors

$$\mathbf{x} = \begin{pmatrix} -4 \\ -5 \\ 6 \end{pmatrix}, \text{ and } \mathbf{y} = \begin{pmatrix} 6 \\ 4 \\ -6 \end{pmatrix}$$

Let  $\mathbf{z} = \begin{pmatrix} -14 \\ -7 \\ 12 \end{pmatrix}$ . Determine whether or not  $\mathbf{z}$  lies in  $U = \text{span}\{\mathbf{x}, \mathbf{y}\}$ .

If  $\mathbf{z}$  is in  $U$ , write it as a linear combination of  $\mathbf{x}$  and  $\mathbf{y}$

$$\mathbf{z} = \boxed{-1} \begin{pmatrix} -4 \\ -5 \\ 6 \end{pmatrix} + \boxed{-3} \begin{pmatrix} 6 \\ 4 \\ -6 \end{pmatrix}.$$

If it is not possible, enter 0 for both coefficients.

Correct. Your coefficients will result in a linear combination that produces the vector,  $\mathbf{z}$ . You were awarded **2** marks. ✓

You scored **2** marks for this part.

---

**Score: 2/2** ✓

# Question 1

Determine whether the following subset of vectors is independent.

$$\left\{ \begin{pmatrix} -2 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix} \right\} \text{ in } \mathbb{R}^3$$

Provide your selection. Use the reveal answers button to check your reasoning.

☐ the subset of vectors is independent

☒ the vectors are linearly dependent



You chose a correct answer. You were awarded 1 mark. ✓

You scored 1 mark for this part.

**Score: 1/1** ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

Determine whether the following subset of vectors is independent.

$$\left\{ \begin{pmatrix} -2 \\ 2 \\ 4 \\ -4 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} -3 \\ 3 \\ 3 \\ -7 \end{pmatrix} \right\} \text{ in } \mathbb{R}^4$$

Provide your selection. Use the reveal answers button to check your reasoning.

☒ the subset of vectors is independent

☐ the vectors are linearly dependent



You chose a correct answer. You were awarded **1** mark. ✓

You scored **1** mark for this part.

---

**Score: 1/1** ✓

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

Find a basis and calculate the dimension for the following subspace of  $\mathbb{R}^4$ .

$$\text{span}\{(2, -1, -6, 2), (2, -3, -1, 7), (-3, 0, 6, -6), (0, 1, 2, 0)\}$$

Enter your basis as rows of a matrix.

Rows:  Columns:

|    |    |    |    |
|----|----|----|----|
| 2  | -1 | -6 | 2  |
| 2  | -3 | -1 | 7  |
| -3 | 0  | 6  | -6 |
| 0  | 1  | 2  | 0  |

What is the dimension of the subspace?

Correct. Your vectors are independent and span the entire subspace. You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

# Question 1

Find a basis and calculate the dimension for the following subspace of  $\mathbb{R}^4$ .

$$\text{span}\{(0, 0, 3, 0), (-3, -4, -7, 5), (-3, -3, -6, 3), (0, 3, 0, -3)\}$$

Enter your basis as rows of a matrix.

Rows:  Columns:

|    |    |    |    |
|----|----|----|----|
| 0  | 0  | 3  | 0  |
| -3 | -4 | -7 | 5  |
| -3 | -3 | -6 | 3  |
| 0  | 3  | 0  | -3 |

What is the dimension of the subspace?

Correct. Your vectors are independent and span the entire subspace. You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓



Find  $(a, b, c, d)$  in  $\mathbb{R}^4$  such that the given set is orthogonal.

$$\{ (-1, 0, 1, 3), (-1, 2, -1, 0), (-8, -5, -2, -2), (a, b, c, d) \}$$

a =

b =

c =

d =

Correct. The vector you've provided will make the set orthogonal. You were awarded **4** marks.



You scored **4** marks for this part.

---

**Score: 4/4**

---

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

Find  $(a, b, c, d)$  in  $\mathbb{R}^4$  such that the given set is orthogonal.

$$\{ (-2, 0, -1, -3), (-2, 2, 1, 1), (-1, -1, -1, 1), (a, b, c, d) \}$$

a =

b =

c =

d =

Correct. The vector you've provided will make the set orthogonal. You were awarded **4** marks.



You scored **4** marks for this part.

---

**Score: 4/4**

---

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).

# SIT292 Module 5 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

| Question Number | Score        | Review        |
|-----------------|--------------|---------------|
| Question 1      | 1/1          |               |
| Question 2      | 2/2          |               |
| Question 3      | 1/1          |               |
| Question 4      | 3/3          |               |
| Question 5      | 3/3          |               |
| <b>Total</b>    | <b>10/10</b> | <b>(100%)</b> |

**Congratulations! You have achieved the minimum threshold for this module's self-assessment.**

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

## Performance Summary

**Exam Name:** SIT292 Module 5 Self-Assessment

**Session ID:** 3794479807073426

**Exam Start:** Thu Sep 03 2026 20:22:15

**Exam Stop:** Thu Sep 03 2026 20:54:16

**Time Spent:** 0:32:00

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(https://www.newcastle.ac.uk\)](https://www.newcastle.ac.uk).

# Question 1

Let  $U$  be the set of vectors  $(x, y, z)$  with  $z^2 = x^2 + y^2$  and  $x, y, z$  in  $\mathbb{R}$ .

Determine whether or not  $U$  is a subspace of  $\mathbb{R}^3$ .

The set  $U$ :

- ☒ includes the zero vector
- ☐ is closed under vector addition
- ☒ is closed under multiplication
- ☐ is a **subspace** of  $\mathbb{R}^3$
- ☒ is **not a subspace** of  $\mathbb{R}^3$



Expected answer:

- ☒ includes the zero vector
- ☐ is closed under vector addition
- ☒ is closed under multiplication
- ☐ is a **subspace** of  $\mathbb{R}^3$
- ☒ is **not a subspace** of  $\mathbb{R}^3$

Score: 1/1 ✓

✓ Your answer is correct. You were awarded 1 mark.

You scored 1 mark for this part.

## Question 2

Consider the vectors

$$\mathbf{x} = \begin{pmatrix} -8 \\ 19 \\ -14 \end{pmatrix}, \text{ and } \mathbf{y} = \begin{pmatrix} -3 \\ 6 \\ -6 \end{pmatrix}$$

Let  $\mathbf{z} = \begin{pmatrix} -4 \\ 22 \\ -2 \end{pmatrix}$ . Determine whether or not  $\mathbf{z}$  lies in  $U = \text{span}\{\mathbf{x}, \mathbf{y}\}$ .

If  $\mathbf{z}$  is in  $U$ , write it as a linear combination of  $\mathbf{x}$  and  $\mathbf{y}$ . If it is not possible, enter 0 for both coefficients.

$$\mathbf{z} = \boxed{0} \text{ Expected answer: } \underline{\quad 0 \quad} \begin{pmatrix} -8 \\ 19 \\ -14 \end{pmatrix} + \boxed{0} \text{ Expected answer: } \underline{\quad 0 \quad} \begin{pmatrix} -3 \\ 6 \\ -6 \end{pmatrix}.$$

Score: 2/2 ✓

✓ You were awarded 1 mark.

✓ You were awarded 1 mark.

You scored 2 marks for this part.

## Question 3

Determine whether the following subset of vectors is independent.



$$\left\{ \begin{pmatrix} 1 \\ -2 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} -3 \\ 6 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \\ 3 \\ -2 \end{pmatrix} \right\} \text{ in } \mathbb{R}^4$$

Provide your selection. Use the reveal answers button to check your reasoning.

- ☐ the subset of vectors is independent
- ☒ the vectors are linearly dependent



Expected answer:

- ☐ the subset of vectors is independent
- ☒ the vectors are linearly dependent

Score: 1/1 ✓

✓ You chose a correct answer. You were awarded 1 mark.

You scored 1 mark for this part.

## Question 4

Find a basis and calculate the dimension for the following subspace of  $\mathbb{R}^4$ .

$$\text{span}\{(2, 2, 0, -8), (1, 2, 3, 5), (0, -1, 0, -2), (3, 6, 0, -3)\}$$

Enter your basis as rows of a matrix. Use the controls to increase the number of rows as necessary.

Rows:  Columns:

$$\begin{pmatrix} \begin{matrix} 2 \\ 1 \\ 0 \\ 3 \end{matrix} & \begin{matrix} 2 \\ 2 \\ -1 \\ 6 \end{matrix} & \begin{matrix} 0 \\ 3 \\ 0 \\ 0 \end{matrix} & \begin{matrix} -8 \\ 5 \\ -2 \\ -3 \end{matrix} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \frac{1}{0} & \frac{2}{1} & \frac{0}{0} & \frac{-1}{2} \\ \frac{0}{0} & \frac{1}{0} & \frac{0}{1} & \frac{2}{2} \\ \frac{0}{0} & \frac{0}{0} & \frac{1}{0} & \frac{2}{1} \end{pmatrix}$$

What is the dimension of the subspace?

Expected answer: 4

Score: 3/3 ✓

✓ Correct. Your vectors are independent and span the entire subspace.  
You were awarded **3** marks.

You scored **3** marks for this part.

## Question 5

Find  $(a, b, c, d)$  in  $\mathbb{R}^4$  such that the given set is orthogonal.

$$\{ (-2, 0, 2, 2), (-1, -2, -1, 0), (3, -2, 1, 2), (a, b, c, d) \}$$

a =  Expected answer: 0

b =

Expected answer: 1/2

c =

Expected answer: -1

d =

Expected answer: 1

Score: 3/3 ✓

✓ Correct. The vector you've provided will make the set orthogonal. You were awarded **3** marks.

You scored **3** marks for this part.

---

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(https://www.newcastle.ac.uk\)](https://www.newcastle.ac.uk).

## Module-5 Self-Assessment

### Working & Annotations

#### Question 1

##### Question

Let  $U$  be the set of vectors  $(x, y, z)$  satisfying

$$z^2 = x^2 + y^2, \quad x, y, z \in \mathbb{R}.$$

Determine whether or not  $U$  is a subspace of  $\mathbb{R}^3$ .

##### Working and annotations

We check the relevant subspace properties.

##### 1. Zero vector

The zero vector is

$$\mathbf{0} = (0, 0, 0).$$

Substituting into the defining equation,

$$0^2 = 0^2 + 0^2,$$

which is true. Therefore,

$$\boxed{\mathbf{0} \in U}. \quad (\text{Zero-vector condition})$$

##### 2. Closure under scalar multiplication

Take any

$$(x, y, z) \in U$$

and any scalar  $k \in \mathbb{R}$ . Since

$$z^2 = x^2 + y^2,$$

we have

$$\begin{aligned} (kz)^2 &= k^2 z^2 \\ &= k^2 (x^2 + y^2) && (\text{Scalar multiplication}) \\ &= (kx)^2 + (ky)^2. \end{aligned}$$

Hence,

$$\boxed{k(x, y, z) \in U}.$$

Therefore,  $U$  is closed under scalar multiplication.

### 3. Closure under vector addition

To test closure under addition, consider

$$u = (1, 0, 1), \quad v = (1, 0, -1).$$

Both vectors belong to  $U$ , because

$$1^2 = 1^2 + 0^2$$

and

$$(-1)^2 = 1^2 + 0^2.$$

However,

$$u + v = (2, 0, 0).$$

For this vector,

$$0^2 \neq 2^2 + 0^2,$$

since

$$0 \neq 4.$$

Therefore,

$$\boxed{u + v \notin U}. \quad (\text{Failure of closure under addition})$$

### Conclusion

Although  $U$  contains the zero vector and is closed under scalar multiplication, it is not closed under vector addition.

Therefore,

$$\boxed{U \text{ is not a subspace of } \mathbb{R}^3}.$$

### NUMBAS selections

$$\boxed{\text{includes the zero vector}}$$

$$\boxed{\text{is closed under multiplication}}$$



is not a subspace of  $\mathbb{R}^3$

## Question 2

### Question

Consider the vectors

$$\mathbf{x} = \begin{pmatrix} -8 \\ 19 \\ -14 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} -3 \\ 6 \\ -6 \end{pmatrix},$$

and let

$$\mathbf{z} = \begin{pmatrix} -4 \\ 22 \\ -2 \end{pmatrix}.$$

Determine whether or not

$$\mathbf{z} \in U = \text{span}\{\mathbf{x}, \mathbf{y}\}.$$

If  $\mathbf{z} \in U$ , write it as a linear combination of  $\mathbf{x}$  and  $\mathbf{y}$ . If this is not possible, enter 0 for both coefficients.

### Concepts / properties used

Linear combination; span; simultaneous equations.

### Working

For  $\mathbf{z}$  to belong to the span, there must exist scalars  $\alpha, \beta \in \mathbb{R}$  such that

$$\mathbf{z} = \alpha\mathbf{x} + \beta\mathbf{y}. \quad (\text{Definition of span})$$

Thus,

$$\begin{pmatrix} -4 \\ 22 \\ -2 \end{pmatrix} = \alpha \begin{pmatrix} -8 \\ 19 \\ -14 \end{pmatrix} + \beta \begin{pmatrix} -3 \\ 6 \\ -6 \end{pmatrix}.$$

Comparing corresponding components gives

$$-8\alpha - 3\beta = -4, \tag{1}$$

$$19\alpha + 6\beta = 22, \tag{2}$$

$$-14\alpha - 6\beta = -2. \tag{3}$$

From equation (1),

$$8\alpha + 3\beta = 4.$$

Multiplying by 2,

$$16\alpha + 6\beta = 8. \quad (4)$$

Subtracting equation (4) from equation (2),

$$(19\alpha + 6\beta) - (16\alpha + 6\beta) = 22 - 8,$$

so

$$3\alpha = 14,$$

and therefore

$$\alpha = \frac{14}{3}.$$

Substituting this into

$$8\alpha + 3\beta = 4,$$

we obtain

$$8\left(\frac{14}{3}\right) + 3\beta = 4.$$

Hence,

$$\frac{112}{3} + 3\beta = \frac{12}{3},$$

so

$$3\beta = -\frac{100}{3},$$

giving

$$\beta = -\frac{100}{9}.$$

We must now check equation (3):

$$\begin{aligned} -14\alpha - 6\beta &= -14\left(\frac{14}{3}\right) - 6\left(-\frac{100}{9}\right) \\ &= -\frac{196}{3} + \frac{600}{9} \\ &= -\frac{196}{3} + \frac{200}{3} \\ &= \frac{4}{3}. \end{aligned}$$

But equation (3) requires

$$-14\alpha - 6\beta = -2.$$

Since

$$\frac{4}{3} \neq -2,$$

the three equations are inconsistent.

**Conclusion**

There are no scalars  $\alpha, \beta$  satisfying all three equations simultaneously. Therefore,

$$\mathbf{z} \notin \text{span}\{\mathbf{x}, \mathbf{y}\}.$$

According to the NUMBAS instruction, the required entries are

$$\alpha = 0, \quad \beta = 0.$$

### Question 3

#### Question

Determine whether the following subset of vectors is linearly independent:

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} -3 \\ 6 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \\ 3 \\ -2 \end{pmatrix} \right\}$$

in  $\mathbb{R}^4$ .

#### Concepts / properties used

Independence Test; matrix representation; Gaussian elimination; rank.

#### Working

Let

$$v_1 = \begin{pmatrix} 1 \\ -2 \\ -2 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} -3 \\ 6 \\ 4 \\ -2 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -2 \\ 4 \\ 3 \\ -2 \end{pmatrix}.$$

To test linear independence, form a matrix using the vectors as columns:

$$A = \begin{bmatrix} 1 & -3 & -2 \\ -2 & 6 & 4 \\ -2 & 4 & 3 \\ 2 & -2 & -2 \end{bmatrix}.$$

Row-reducing,

$$\begin{aligned} A &\sim \begin{bmatrix} 1 & -3 & -2 \\ 0 & 0 & 0 \\ 0 & -2 & -1 \\ 0 & 4 & 2 \end{bmatrix} \quad \left( \begin{array}{l} R_2 \leftarrow R_2 + 2R_1 \\ R_3 \leftarrow R_3 + 2R_1 \\ R_4 \leftarrow R_4 - 2R_1 \end{array} \right) \\ &\sim \begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \left( \begin{array}{l} R_2 \leftarrow -\frac{1}{2}R_3 \\ R_4 \leftarrow R_4 - 4R_2 \end{array} \right) \end{aligned}$$



Therefore,

$$\text{rank}(A) = 2.$$

There are 3 columns but only 2 pivots, so

$$\text{rank}(A) = 2 < 3. \quad (\text{Rank test for independence})$$

Hence the vectors are linearly dependent.

To make the dependence explicit, the reduced system gives

$$c_1 - \frac{1}{2}c_3 = 0, \quad c_2 + \frac{1}{2}c_3 = 0.$$

Choose

$$c_3 = 2.$$

Then

$$c_1 = 1, \quad c_2 = -1.$$

Therefore,

$$v_1 - v_2 + 2v_3 = 0.$$

Indeed,

$$\begin{aligned} & \begin{pmatrix} 1 \\ -2 \\ -2 \\ 2 \end{pmatrix} - \begin{pmatrix} -3 \\ 6 \\ 4 \\ -2 \end{pmatrix} + 2 \begin{pmatrix} -2 \\ 4 \\ 3 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} 1 + 3 - 4 \\ -2 - 6 + 8 \\ -2 - 4 + 6 \\ 2 + 2 - 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

## Conclusion

A non-trivial linear combination of the vectors gives the zero vector. Therefore,

The vectors are linearly dependent.

## Question 4

### Question

Find a basis and calculate the dimension for the following subspace of  $\mathbb{R}^4$ :

$$\text{span} \{(2, 2, 0, -8), (1, 2, 3, 5), (0, -1, 0, -2), (3, 6, 0, -3)\}.$$

### Concepts / properties used

Determinant; linear independence; basis; dimension.

### Working

Form the matrix using the given vectors as rows:

$$A = \begin{bmatrix} 2 & 2 & 0 & -8 \\ 1 & 2 & 3 & 5 \\ 0 & -1 & 0 & -2 \\ 3 & 6 & 0 & -3 \end{bmatrix}.$$

The four vectors are linearly independent precisely when

$$\det A \neq 0. \quad (\text{Determinant test})$$

Expanding along the third row,

$$\det A = \begin{vmatrix} 2 & 0 & -8 \\ 1 & 3 & 5 \\ 3 & 6 & -3 \end{vmatrix} + 2 \begin{vmatrix} 2 & 2 & 0 \\ 1 & 2 & 3 \\ 3 & 6 & 0 \end{vmatrix}. \quad (\text{Cofactor expansion})$$

For the first determinant,

$$\begin{aligned} \begin{vmatrix} 2 & 0 & -8 \\ 1 & 3 & 5 \\ 3 & 6 & -3 \end{vmatrix} &= 2 \begin{vmatrix} 3 & 5 \\ 6 & -3 \end{vmatrix} - 8 \begin{vmatrix} 1 & 3 \\ 3 & 6 \end{vmatrix} \\ &= 2(-9 - 30) - 8(6 - 9) \\ &= -78 + 24 \\ &= -54. \end{aligned}$$

For the second determinant,

$$\begin{aligned}
\begin{vmatrix} 2 & 2 & 0 \\ 1 & 2 & 3 \\ 3 & 6 & 0 \end{vmatrix} &= 2 \begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} - 2 \begin{vmatrix} 1 & 3 \\ 3 & 0 \end{vmatrix} \\
&= 2(-18) - 2(-9) \\
&= -36 + 18 \\
&= -18.
\end{aligned}$$

Therefore,

$$\det A = -54 + 2(-18) = -90.$$

Thus,

$$\boxed{\det A = -90 \neq 0}.$$

Hence the four given vectors are linearly independent.

Since they are four linearly independent vectors in  $\mathbb{R}^4$ , they form a basis of  $\mathbb{R}^4$ , and therefore also a basis of the stated span.

## Conclusion

A suitable basis is

$$\boxed{\{(2, 2, 0, -8), (1, 2, 3, 5), (0, -1, 0, -2), (3, 6, 0, -3)\}}.$$

The dimension is the number of vectors in the basis:

$$\boxed{\dim(U) = 4}.$$

## Question 5

### Question

Find

$$(a, b, c, d) \in \mathbb{R}^4$$

such that the following set is orthogonal:

$$\{(-2, 0, 2, 2), (-1, -2, -1, 0), (3, -2, 1, 2), (a, b, c, d)\}.$$

### Concepts / properties used

Dot product; orthogonality; simultaneous linear equations.

### Working

First verify that the first three vectors are already mutually orthogonal.

Let

$$v_1 = (-2, 0, 2, 2), \quad v_2 = (-1, -2, -1, 0), \quad v_3 = (3, -2, 1, 2).$$

Then

$$\begin{aligned} v_1 \cdot v_2 &= (-2)(-1) + (0)(-2) + (2)(-1) + (2)(0) \\ &= 2 - 2 \\ &= 0, \end{aligned}$$

$$\begin{aligned} v_1 \cdot v_3 &= (-2)(3) + (0)(-2) + (2)(1) + (2)(2) \\ &= -6 + 2 + 4 \\ &= 0, \end{aligned}$$

and

$$\begin{aligned} v_2 \cdot v_3 &= (-1)(3) + (-2)(-2) + (-1)(1) + (0)(2) \\ &= -3 + 4 - 1 \\ &= 0. \end{aligned}$$

Thus the first three vectors are mutually orthogonal. Therefore,  $(a, b, c, d)$  only needs to be orthogonal to each of them.

**1. Orthogonal to  $v_1$** 

$$(a, b, c, d) \cdot (-2, 0, 2, 2) = 0.$$

Hence,

$$-2a + 2c + 2d = 0.$$

Dividing by 2,

$$-a + c + d = 0,$$

so

$$c = a - d. \quad (\text{Orthogonality}) \quad (1)$$

**2. Orthogonal to  $v_2$** 

$$(a, b, c, d) \cdot (-1, -2, -1, 0) = 0.$$

Therefore,

$$-a - 2b - c = 0. \quad (2)$$

Substituting (1),

$$-a - 2b - (a - d) = 0.$$

Hence,

$$-2a - 2b + d = 0,$$

which gives

$$b = \frac{d - 2a}{2}. \quad (\text{Orthogonality}) \quad (3)$$

**3. Orthogonal to  $v_3$** 

$$(a, b, c, d) \cdot (3, -2, 1, 2) = 0.$$



Thus,

$$3a - 2b + c + 2d = 0. \quad (4)$$

Substituting (1) and (3),

$$3a - 2\left(\frac{d - 2a}{2}\right) + (a - d) + 2d = 0.$$

Simplifying,

$$3a - (d - 2a) + a - d + 2d = 0,$$

so

$$6a = 0.$$

Therefore,

$$a = 0.$$

Using (1),

$$c = 0 - d = -d.$$

Using (3),

$$b = \frac{d}{2}.$$

Therefore the general solution is

$$(a, b, c, d) = \left(0, \frac{d}{2}, -d, d\right).$$

To obtain integer entries, choose

$$d = 2.$$

Then

$$a = 0, \quad b = 1, \quad c = -2.$$

Hence,

$$\boxed{(a, b, c, d) = (0, 1, -2, 2)}.$$

### Verification

Check the three required dot products:

$$\begin{aligned}(0, 1, -2, 2) \cdot (-2, 0, 2, 2) &= 0 + 0 - 4 + 4 \\ &= 0,\end{aligned}$$

$$\begin{aligned}(0, 1, -2, 2) \cdot (-1, -2, -1, 0) &= 0 - 2 + 2 + 0 \\ &= 0,\end{aligned}$$

and

$$\begin{aligned}(0, 1, -2, 2) \cdot (3, -2, 1, 2) &= 0 - 2 - 2 + 4 \\ &= 0.\end{aligned}$$

All three dot products are zero, so the required vector is orthogonal to each of the first three vectors.

### Conclusion

$$\boxed{a = 0, \quad b = 1, \quad c = -2, \quad d = 2}$$

DEAKIN UNIVERSITY

LINEAR ALGEBRA FOR DATA ANALYSIS

ONTRACK SUBMISSION

Module 6 - Eigenvalues, Eigenvectors and Diagonalisation

Submitted By:  
Jahan GARG  
s226100103  
2026/09/10 23:11

Tutor:  
Simon JAMES

| Task Outcomes |                                                                                                                                                       | Supports               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| M6.1          | Determine the rank, column space, row space and null space of a matrix                                                                                | ULO1 ULO1<br>ULO2 ULO2 |
| M6.2          | Find the characteristic polynomial, eigenvalues, and eigenvectors of a matrix                                                                         | ULO1 ULO1              |
| M6.3          | Identify an appropriate matrix P that can be used to diagonalise a square matrix                                                                      | ULO1 ULO1              |
| M6.4          | Use solutions of the characteristic equation to identify existence of linearly independent Eigenvectors                                               | ULO1 ULO1<br>ULO2 ULO2 |
| M6.5          | Apply the definition and properties of matrix similarity to state when a matrix A is similar to another matrix B                                      | ULO1 ULO1              |
| SA            | Full printout of Self-Assessment results with all questions.                                                                                          |                        |
| Evid          | Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.                          |                        |
| Sum           | Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions. |                        |
| FB            | Respond to any feedback and work with the OnTrack tutor to complete the module.                                                                       |                        |

September 10, 2026



*SIT292 - Linear Algebra for Data Analysis*

## Module 6 Summary: Eigenvalues, Eigenvectors and Diagonalisation

*Rank of a Matrix, Diagonalisation and Eigenvalues, Similarity and Diagonalisation*

### Response to Feedback

This is a resubmission of my Module 6 assessment, responding directly to Simon's feedback on my first attempt. Simon asked that, of the two worked diagonalisation examples in Module Summary Question 2, the first (the diagonalisable one) should have a repeated eigenvalue rather than distinct eigenvalues. My first example,  $A = \begin{bmatrix} 3 & 0 & 6 \\ 0 & -3 & 0 \\ 5 & 0 & 2 \end{bmatrix}$ , already has the eigenvalue  $-3$  with algebraic multiplicity 2 alongside the distinct eigenvalue 8, so it does satisfy this requirement; I have kept it unchanged for this reason, and have added a clear note in the working itself confirming the multiplicity, to avoid any ambiguity when it is reviewed again. Simon also asked that both worked examples carry a proper reference, since he would not expect me to have invented them myself. I have now added an explicit source label under each example: the first (repeated eigenvalue, diagonalisable) is Nicholson's Linear Algebra with Applications, Exercise 5.5.4b [1], and the second (non-diagonalisable) is SIT292 Unit Site, Topic 6.4 (Similarity and Diagonalisation), Exercise 6.4.3 [2], both of which I originally worked through by hand in my Learning Evidence before adapting them for this summary. I have also carried forward the earlier guidance from Tanya on Module 1 to link the definitions and theorems together in the summary, and from Simon on Module 5 regarding LaTeX-style matrix typesetting, both of which I have continued to apply throughout this resubmission.

### Module Learning Objectives

I certify that I achieved the following learning objectives for this module, as set out on the Module 6 introduction page of the unit site:

- 1 Determine the rank, column space, row space and null space of a matrix.
- 2 Find the characteristic polynomial, eigenvalues, and eigenvectors of a matrix.
- 3 Identify an appropriate matrix  $P$  that can be used to diagonalise a square matrix.
- 4 Use solutions of the characteristic equation to identify the existence of linearly independent eigenvectors.
- 5 Apply the definition and properties of matrix similarity to state when a matrix  $A$  is similar to another matrix  $B$ .



## Key Definitions and Theorems

### Rank

| Textbook Reference                                                | Theorem / Definition Statement                                                                                                                                                                                                                                                                                                                                                            | Learning Notes                                                                                                                                              |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 5.10</b><br><b>Column and row space of a matrix</b> | The column space, $\text{col } A$ , of $A$ is the subspace of $\mathbb{R}^m$ spanned by the columns of $A$ .<br>The row space, $\text{row } A$ , of $A$ is the subspace of $\mathbb{R}^n$ spanned by the rows of $A$ .                                                                                                                                                                    | Both spaces already appeared informally in Module 2 when $Ax$ was written as a linear combination of columns; this definition simply names that span.       |
| <b>Null and image space</b><br><b>(p291, also p262)</b>           | $\text{null}(A) = \{x \text{ in } \mathbb{R}^n \mid Ax = 0\}$<br>$\text{im}(A) = \{Ax \mid x \text{ in } \mathbb{R}^n\}$                                                                                                                                                                                                                                                                  | The null space is exactly the solution set of a homogeneous system from Module 1, now named formally as a subspace.                                         |
| <b>Theorem 5.4.1</b><br><b>Rank Theorem</b>                       | Let $A$ be any $m \times n$ matrix of rank $r$ . Then $\dim(\text{col } A) = \dim(\text{row } A) = r$ . Moreover, if $A$ is carried to a row-echelon matrix $R$ by row operations: 1. The $r$ non-zero rows of $R$ are a basis of $\text{row } A$ . 2. If the leading 1s lie in columns $j_1, \dots, j_r$ of $R$ , then columns $j_1, \dots, j_r$ of $A$ are a basis of $\text{col}(A)$ . | Applied directly in Learning Evidence 6.2 Q1: pivots landed in columns 1 and 3, so exactly those two original columns formed the basis of $\text{col}(A)$ . |
| <b>Theorem 5.4.2</b><br><b>Rank and Nullity</b>                   | Let $A$ be an $m \times n$ matrix of rank $r$ . Then: 1. The $n$ minus $r$ basic solutions to $Ax=0$ provided by the Gaussian algorithm are a basis of $\text{null}(A)$ , so $\dim[\text{null}(A)] = n$ minus $r$ . 2. Theorem 5.4.1 provides a basis of $\text{im}(A) = \text{col}(A)$ , and $\dim[\text{im}(A)] = r$ .                                                                  | Verified numerically in Learning Evidence 6.2 Q2: a $4 \times 5$ matrix of rank 3 gave exactly 5 minus 3 = 2 basic solutions spanning the null space.       |
| <b>Theorem 5.4.3</b>                                              | Equivalent for an $m \times n$ matrix $A$ : 1. $\text{rank } A = n$ . 2. Rows of $A$ span $\mathbb{R}^n$ . 3. Columns of $A$ are linearly independent in $\mathbb{R}^m$ . 4. The $n \times n$ matrix $A^T A$ is invertible. 5. $CA = I_n$ for some $n \times m$ matrix $C$ . 6. If $Ax=0$ , $x \text{ in } \mathbb{R}^n$ , then $x=0$ .                                                   | A column version of the independence criteria already met in Module 5, now stated for rectangular matrices rather than only square ones.                    |
| <b>Theorem 5.4.4</b>                                              | Equivalent for an $m \times n$ matrix $A$ : 1. $\text{rank } A = m$ . 2. Columns of $A$ span $\mathbb{R}^m$ . 3. Rows of $A$ are linearly independent in $\mathbb{R}^n$ . 4. The $m \times m$ matrix $AA^T$ is invertible. 5. $AC = I_m$ for some $n \times m$ matrix $C$ . 6. $Ax=b$ is consistent for every $b \text{ in } \mathbb{R}^m$ .                                              | The row-based mirror of Theorem 5.4.3, tying full row rank directly to solvability of $Ax=b$ for every right-hand side.                                     |

## Diagonalisation

| Textbook Reference                                                       | Theorem / Definition Statement                                                                                                                                                                                                                  | Learning Notes                                                                                                                                                                             |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theorem 3.3.1</b><br><b>Powers of diagonal matrices</b>               | If $A = PDP^{-1}$ then $A^k = PD^kP^{-1}$ for each $k = 1, 2, \dots$                                                                                                                                                                            | Since $D$ is diagonal, $D^k$ is just each diagonal entry raised to the power $k$ , so this turns an otherwise expensive repeated matrix multiplication into a single diagonal computation. |
| <b>Definition 3.4</b><br><b>Eigenvalues and Eigenvectors of a Matrix</b> | If $A$ is an $n \times n$ matrix, a number $\lambda$ is called an eigenvalue of $A$ if $Ax = \lambda x$ for some column $x \neq 0$ . Here $x$ is called an eigenvector of $A$ corresponding to $\lambda$ , or a $\lambda$ -eigenvector.         | The defining equation $Ax = \lambda x$ says the matrix acts on $x$ purely as a scalar, which is the starting point for every calculation in this module.                                   |
| <b>Definition 3.5</b><br><b>Characteristic Polynomial of a Matrix</b>    | If $A$ is an $n \times n$ matrix, the characteristic polynomial $c_A(x)$ of $A$ is defined by $c_A(x) = \det(xI - A)$ .                                                                                                                         | Practised repeatedly in Learning Evidence 6.3, both by direct cofactor expansion and by the trace/principal-minor shortcut formulas.                                                       |
| <b>Theorem 3.3.2 / Theorem 5.5.2</b>                                     | For an $n \times n$ matrix $A$ : 1. The eigenvalues $\lambda$ of $A$ are the roots of $c_A(x)$ . 2. The $\lambda$ -eigenvectors $x$ are the non-zero solutions of $(\lambda I - A)x = 0$ .                                                      | This is exactly the two-step procedure used throughout: solve $\det(\lambda I - A) = 0$ for $\lambda$ , then solve the resulting homogeneous system for each $\lambda$ in turn.            |
| <b>Definition 3.6</b><br><b>Diagonalizable Matrices</b>                  | An $n \times n$ matrix $A$ is called diagonalizable if $P^{-1}AP$ is diagonal for some invertible $n \times n$ matrix $P$ . Here $P$ is called a diagonalizing matrix for $A$ .                                                                 | The whole point of Module 6: replacing $A$ by a similar diagonal matrix $D$ whenever possible, since powers and other computations become trivial for $D$ .                                |
| <b>Theorem 3.3.4</b>                                                     | 1. $A$ is diagonalizable if and only if it has eigenvectors $x_1, \dots, x_n$ such that $P = [x_1 \dots x_n]$ is invertible. 2. When this holds, $P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)$ where $\lambda_i$ corresponds to $x_i$ . | The invertibility of $P$ (equivalently, its columns being independent) is the single condition checked in every worked example below via $\det(P) \neq 0$ .                                |

| Textbook Reference                      | Theorem / Definition Statement                                                                                                                                                                                                                                                                         | Learning Notes                                                                                                                                                                                           |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theorem 3.3.5</b>                    | A square matrix $A$ is diagonalizable if and only if every eigenvalue $\lambda$ of multiplicity $m$ yields exactly $m$ basic eigenvectors, that is, $(\lambda I - A)x = 0$ has exactly $m$ parameters.                                                                                                 | This is the theorem that distinguishes the two worked examples below: matching multiplicity to eigenspace dimension makes one diagonalisable and mismatching it makes the other not.                     |
| <b>Theorem 3.3.6 / 5.5.5</b>            | An $n \times n$ matrix with $n$ distinct eigenvalues is diagonalizable.                                                                                                                                                                                                                                | A convenient sufficient (but not necessary) condition. Learning Evidence Question 1 of Section 6.3 (Nicholson Exercise 3.3.1c) used this directly since its three eigenvalues 2, 3, 5 were all distinct. |
| <b>Diagonalization Algorithm (p179)</b> | Step 1: find the distinct eigenvalues $\lambda$ of $A$ . Step 2: compute basic eigenvectors for each $\lambda$ from $(\lambda I - A)x = 0$ . Step 3: $A$ is diagonalizable iff there are $n$ basic eigenvectors in total. Step 4: if so, $P$ (columns = basic eigenvectors) is a diagonalizing matrix. | This four-step algorithm is exactly the procedure followed in both worked examples in the Module Summary below, question by question.                                                                    |

## Similarity

| Textbook Reference                                               | Theorem / Definition Statement                                                                                                                                                     | Learning Notes                                                                                                                                                                                                                 |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition 5.11</b><br><b>Similar Matrices</b>                | If A and B are $n \times n$ matrices, A and B are similar, written $A \sim B$ , if $B = P^{-1}AP$ for some invertible matrix P.                                                    | Diagonalisation is the special case of similarity where B happens to be diagonal; every diagonalisable matrix is similar to a diagonal one.                                                                                    |
| <b>Definition 5.12</b><br><b>Trace of a Matrix</b>               | The trace $\text{tr } A$ of an $n \times n$ matrix A is the sum of the main diagonal elements of A.                                                                                | Used as the $b_{n-1}$ coefficient shortcut for the characteristic polynomial throughout Learning Evidence 6.3, avoiding a full cofactor expansion.                                                                             |
| <b>Theorem 5.5.1</b><br><b>Similar matrices share properties</b> | If A and B are similar $n \times n$ matrices, then A and B have the same determinant, rank, trace, characteristic polynomial and eigenvalues.                                      | This is exactly the test applied in NUMBAS N6.4.1: a mismatch in determinant or trace alone is sufficient to rule out similarity without ever constructing P.                                                                  |
| <b>Definition 5.13</b><br><b>Eigenspace of a Matrix</b>          | If $\lambda$ is an eigenvalue of an $n \times n$ matrix A, the eigenspace of A corresponding to $\lambda$ is $E_\lambda(A) = \{x \text{ in } \mathbb{R}^n \mid Ax = \lambda x\}$ . | The dimension of $E_\lambda(A)$ is exactly the number of basic eigenvectors referred to in Theorem 3.3.5, so comparing $\dim E_\lambda$ to the algebraic multiplicity of $\lambda$ is the deciding step for diagonalisability. |

## Module Summary

Module 6 draws together three sub-modules that, on the surface, look quite different but are all really describing the same underlying structure of a matrix from complementary angles. Rank of a Matrix (5.4) revisits the row-echelon ideas from Module 1 and formally names the row space, column space and null space as subspaces in their own right. Diagonalisation and Eigenvalues (3.3) introduces the central computational tool of the module, the eigenvalue equation  $Ax = \lambda x$ , and asks when a matrix can be replaced by a similar diagonal matrix. Similarity and Diagonalisation (5.5) then generalises this to a broader notion of similarity, of which diagonalisation is simply the most convenient special case, and introduces trace and eigenspace as further invariants that travel together with rank, determinant and characteristic polynomial under a similarity transformation.

### 6.2 Rank of a Matrix

Connecting Theorem 1.3.2 and Theorem 5.4.2. Theorem 1.3.2, met in Module 1, already established that for an  $m \times n$  coefficient matrix  $A$  of rank  $r$ , the homogeneous system  $Ax = 0$  has exactly  $n$  minus  $r$  basic solutions produced by the Gaussian algorithm, and that every solution of the system is a linear combination of these basic solutions. At that stage, this was framed purely as a computational fact about solving equations, a way of writing the general solution using free parameters. Theorem 5.4.2 restates exactly the same underlying object in the language of vector spaces: it names the solution set of  $Ax = 0$  as the null space,  $\text{null}(A)$ , confirms that the  $n$  minus  $r$  basic solutions form a basis of this subspace, and concludes  $\dim[\text{null}(A)] = n$  minus  $r$ . In other words, Theorem 5.4.2 is the vector-space formalisation of Theorem 1.3.2, taking a result that was already true and reframing the basic solutions as a genuine basis of a genuine subspace rather than just a convenient way to parameterise a solution set. This connection was made concrete in Learning Evidence Question 2 of Section 6.2, where a  $4 \times 5$  matrix of rank 3 produced exactly  $5$  minus  $3 = 2$  basic solutions, and these two vectors were confirmed, using Theorem 5.4.2 directly, to be a basis of a two-dimensional null space, matching the rank-nullity relationship exactly.

Beyond this connection, Definition 5.10 names the column space,  $\text{col } A$ , and row space,  $\text{row } A$ , as the subspaces spanned by the columns and rows of  $A$ , respectively, while the null and image spaces defined on page 291 (and page 262) capture the solution set of  $Ax = 0$  and the set of all possible outputs  $Ax$ . Theorem 5.4.1, the Rank Theorem, is what makes these definitions practically useful: it guarantees  $\dim(\text{col } A) = \dim(\text{row } A) = r$ , and shows that after row-reducing  $A$  to row-echelon form  $R$ , the non-zero rows of  $R$  form a basis of  $\text{row } A$ , while the original columns of  $A$  in the pivot positions of  $R$  form a basis of  $\text{col } A$ . This was applied directly in Learning Evidence Question 1 of Section 6.2, where the pivots after elimination landed in columns 1 and 3 of the row-echelon form, so exactly those two columns of the original matrix, not the row-reduced one, were taken as the basis of the column space. Theorems 5.4.3 and 5.4.4 then close the loop by restating, for possibly rectangular matrices, the same equivalences between full rank, spanning, independence and invertibility of  $A^T A$  or  $AA^T$  that were already familiar for square matrices from Module 5.

### 6.3 Diagonalisation and Eigenvalues

An eigenvalue  $\lambda$  and its corresponding eigenvector  $x$  satisfy Definition 3.4's condition  $Ax = \lambda x$  for some non-zero column  $x$ . Rearranging this as  $(\lambda I - A)x = 0$  shows that a non-trivial solution can only exist when  $\lambda I - A$  is singular, which is exactly Theorem 3.3.2's claim that the eigenvalues of  $A$  are the roots of the characteristic polynomial  $c_A(x) = \det(xI - A)$  from Definition 3.5, and that the corresponding eigenvectors are then the non-zero solutions of the resulting homogeneous system. Definition 3.6 calls  $A$  diagonalizable when  $P^{-1}AP$  is diagonal for some invertible  $P$ , and Theorem 3.3.4 makes this concrete:  $A$  is diagonalizable exactly when it possesses  $n$  eigenvectors whose matrix  $P = [x_1 \dots x_n]$  is invertible, in which case  $P^{-1}AP$  is the diagonal matrix of the corresponding eigenvalues in the same order. Theorem 3.3.5 refines this further for the case of repeated eigenvalues: a matrix is diagonalizable if and only if every eigenvalue of algebraic multiplicity  $m$  contributes exactly  $m$  linearly independent (basic) eigenvectors, that is, exactly  $m$  parameters in the general solution of  $(\lambda I - A)x = 0$ . Theorem 3.3.6 gives a convenient sufficient condition, namely that  $n$  distinct eigenvalues always guarantee diagonalisability, since eigenvectors for distinct eigenvalues are automatically independent.

Question 2: two worked examples of diagonalisation. The two examples below follow the Diagonalisation Algorithm exactly, and together connect to Definitions 3.4, 3.5, 3.6 and 5.13, and Theorems 3.3.2, 3.3.4 and 3.3.5. Both examples were originally worked through by hand in my Learning Evidence before being adapted here; the source of each is labelled explicitly below, as requested.

Example 1: a diagonalisable matrix with a repeated eigenvalue. Source: Nicholson, Linear Algebra with Applications, Exercise 5.5.4b [1].

$$A = \begin{bmatrix} 3 & 0 & 6 \\ 0 & -3 & 0 \\ 5 & 0 & 2 \end{bmatrix}$$

Step 1 (Definitions 3.4, 3.5; Theorem 3.3.2). Expanding  $\det(\lambda I - A)$  along the middle row, since it has only one non-zero entry, gives  $c_A(\lambda) = (\lambda+3)[(\lambda-3)(\lambda-2) - 30] = (\lambda+3)^2(\lambda-8)$ . The roots  $\lambda = -3$ , with algebraic multiplicity 2 (a genuine repeated eigenvalue, not a typographical slip), and  $\lambda = 8$ , with multiplicity 1, are therefore the eigenvalues of  $A$ .

Step 2 (Theorem 3.3.2; Definition 5.13). Solving  $(A+3I)x = 0$  gives the single independent equation  $x + z = 0$ , with  $y$  and  $z$  free, so the eigenspace  $E_{-3}(A)$  has dimension 2, spanned by  $v_1 = (0,1,0)$  and  $v_2 = (-1,0,1)$ . Solving  $(A-8I)x = 0$  gives the single eigenvector  $v_3 = (6,0,5)$ .

Step 3 (Theorem 3.3.5). Since  $\lambda = -3$  has algebraic multiplicity 2 and its eigenspace also has dimension 2, this eigenvalue contributes exactly the required number of basic eigenvectors, so the multiplicity condition of Theorem 3.3.5 is satisfied for every eigenvalue.

Step 4 (Definition 3.6; Theorem 3.3.4). Assembling  $P = [v_1 \ v_2 \ v_3] = \begin{bmatrix} 0 & -1 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 5 \end{bmatrix}$  gives

$\det(P) = 11 \neq 0$ , so  $P$  is invertible and  $P^{-1}AP = \text{diag}(-3,-3,8) = D$ , confirming  $A$  is diagonalisable.



Example 2: a matrix that is not diagonalisable. Source: SIT292 Unit Site, Topic 6.4 (Similarity and Diagonalisation), Exercise 6.4.3 [2].

$$A = \begin{bmatrix} 1 & 2 & -4 \\ 0 & -1 & 6 \\ 0 & -1 & 4 \end{bmatrix}$$

Step 1 (Definitions 3.4, 3.5; Theorem 3.3.2). Expanding along the first column,  $c_A(\lambda) = (\lambda-1)[(\lambda+1)(\lambda-4)+6] = (\lambda-1)(\lambda^2-3\lambda+2) = (\lambda-1)^2(\lambda-2)$ . The eigenvalues are  $\lambda = 1$  (algebraic multiplicity 2) and  $\lambda = 2$ .

Step 2 (Theorem 3.3.2; Definition 5.13). Solving  $(A-I)x = 0$  reduces to the two independent equations  $y - 2z = 0$  and  $z = 0$ , forcing  $y = 0$  as well, so the only free variable is  $x$ . The eigenspace  $E_1(A)$  therefore has dimension 1, spanned by  $v_1 = (1,0,0)$ , even though  $\lambda = 1$  has algebraic multiplicity 2. Solving  $(A-2I)x = 0$  gives the single eigenvector  $v_2 = (0,2,1)$ .

Step 3 (Theorem 3.3.5). The eigenvalue  $\lambda = 1$  has algebraic multiplicity 2, but its eigenspace has dimension only 1, so this eigenvalue fails to contribute the required 2 basic eigenvectors; the multiplicity condition of Theorem 3.3.5 is violated.

Step 4 (Definition 3.6; Theorem 3.3.4). Only 2 linearly independent eigenvectors,  $v_1$  and  $v_2$ , exist in total for this  $3 \times 3$  matrix, so no invertible  $3 \times 3$  matrix  $P$  of eigenvectors can be formed, and  $A$  is therefore not diagonalisable. It is, however, still similar to an upper triangular matrix  $B$  with the same eigenvalues 1, 1, 2 on its diagonal, using the triangularisation procedure covered later in Section 6.4.

## 6.4 Similarity and Diagonalisation

Definition 5.11 defines  $A$  and  $B$  as similar, written  $A \sim B$ , whenever  $B = P^{-1}AP$  for some invertible  $P$ ; diagonalisation is therefore the particular case where  $B$  happens to be diagonal. Definition 5.12 introduces the trace,  $\text{tr } A$ , as the sum of the diagonal entries, and Theorem 5.5.1 states that similar matrices always share the same determinant, rank, trace, characteristic polynomial and eigenvalues. This gives a very efficient test for ruling out similarity without ever constructing  $P$ : in NUMBAS N6.4.1, two  $2 \times 2$  matrices with different determinants (56 and -174) and different traces (-9 and -8) were immediately shown not to be similar, since a mismatch in either invariant alone is already sufficient by Theorem 5.5.1. Definition 5.13 formalises the eigenspace  $E_\lambda(A) = \{x \text{ in } \mathbb{R}^n \mid Ax = \lambda x\}$  already used implicitly in both worked examples above; the dimension of  $E_\lambda(A)$  is precisely the number of basic eigenvectors referred to in Theorem 3.3.5, so comparing  $\dim E_\lambda(A)$  against the algebraic multiplicity of  $\lambda$  in  $c_A(x)$  is the single decisive step separating a diagonalisable matrix from one that is not. For matrices that fail this test, the unit site's triangularisation procedure still guarantees similarity to an upper triangular matrix with the same eigenvalues on the diagonal, which is the fallback demonstrated at the end of Example 2 above.

## Extra Effort and Prior Knowledge

Beyond the prescribed unit-site material, I consulted a short GeeksforGeeks article on the eigenvalue-eigenvector equation [3] to check that my own phrasing of Definition 3.4 in this summary matched standard terminology used outside this unit, and a Mathematics Stack Exchange discussion on why algebraic multiplicity can exceed geometric multiplicity [4] to make sure my explanation of Example 2 above, where the eigenspace for  $\lambda = 1$  was smaller than its multiplicity, was phrased precisely. In terms of prior knowledge, the idea of a basis of free-variable solutions was already familiar to me from Module 1's treatment of homogeneous systems, and Module 5's coverage of rank, independence and dimension meant that Theorem 5.4.1's connection between pivot columns and a column-space basis felt like a natural continuation rather than an entirely new idea. What this module added was the eigenvalue equation itself and the accompanying vocabulary, characteristic polynomial, eigenspace, diagonalisable and similar, needed to describe when a matrix can be simplified into a more convenient form.

## Reflecting on the Content

The idea I found most conceptually demanding in this module was the distinction between Theorem 3.3.4 and Theorem 3.3.5, since both describe conditions for diagonalisability but from different angles: Theorem 3.3.4 speaks about the invertibility of the matrix  $P$  formed from all the eigenvectors together, while Theorem 3.3.5 speaks about each eigenvalue individually, requiring its eigenspace dimension to match its algebraic multiplicity. Working through Example 2 above, where a repeated eigenvalue produced only one independent eigenvector instead of two, is what finally made the difference between these two theorems click for me: Theorem 3.3.5 is really just Theorem 3.3.4 checked one eigenvalue at a time. The most satisfying connection was between Theorem 1.3.2 from Module 1 and Theorem 5.4.2 in this module, since it showed that the basic solutions I had already been computing since the very first week of the unit were, all along, a basis for a genuine subspace, the null space, rather than just a convenient way of writing a general solution. I also found the NUMBAS N6.4.1 similarity questions particularly efficient once I realised that determinant and trace alone could rule out similarity immediately, without needing to search for a matrix  $P$  at all. Completing the self-assessment with a full 10 out of 10 across all seven questions confirmed that connecting rank, eigenvalues and similarity together, rather than treating each sub-module in isolation, was the right way to approach this particularly large module.

## Tools and Originality Statement

Both worked examples appearing in the Module Summary above (Question 2) are sourced examples, not original constructions: the  $3 \times 3$  diagonalisable matrix with a repeated eigenvalue is Nicholson's Exercise 5.5.4b [1], and the  $3 \times 3$  non-diagonalisable matrix is Exercise 6.4.3 from the SIT292 unit site's Topic 6.4 material [2]. Both were originally worked through by hand in my own Learning Evidence for this module (Learning Evidence Question 2, Section 6.4, and Additional Exercise 6.4.3 practice respectively) before being adapted for this summary, consistent with my tutor's guidance that examples from module exercises or unit-site practice may be reused in the summary to demonstrate understanding. All matrix calculations throughout this document, including the determinants, eigenvalues and eigenvectors quoted above, were independently verified using Python (NumPy and SymPy) before being written up. The handwritten and digital learning evidence referenced throughout Sections 6.2, 6.3 and 6.4, and the NUMBAS self-assessment (10/10), were completed and submitted separately as part of this module's Learning Evidence and Self-Assessment documents. An AI writing assistant (Perplexity) was used to help structure this summary, refine explanations of the theorems and check written English; the mathematical reasoning, worked examples, and self-assessment results referenced throughout are my own.

## References

- [1] W. K. Nicholson, Linear Algebra with Applications, Open Edition, Version 2021A, Lyryx Learning Inc., Calgary, AB, Canada, 2021, Exercise 5.5.4b (Example 1, diagonalisable matrix with repeated eigenvalue), and Sections 3.3, 5.4 and 5.5 more generally.
- [2] Deakin University, "Topic 6: Eigenvalues, Eigenvectors and Diagonalisation - 6.4 Similarity and Diagonalisation, Exercise 6.4.3," SIT292 Linear Algebra for Data Analysis, CloudDeakin Unit Site, Deakin University, Trimester 2, 2026 (Example 2, non-diagonalisable matrix).
- [3] GeeksforGeeks, "Eigenvalues and Eigenvectors," GeeksforGeeks, accessed Sep. 2026. [Online]. Available: <https://www.geeksforgeeks.org>
- [4] Mathematics Stack Exchange, "Why can algebraic multiplicity be greater than geometric multiplicity?" Stack Exchange Inc., accessed Sep. 2026. [Online]. Available: <https://math.stackexchange.com>
- [5] Deakin University, SIT292 Module 6 Self-Assessment, Numbas, developed by Newcastle University, accessed 6 Sep. 2026.

# Module 6: Handwritten Learning Evidence

## Section 6.2 – Rank of a Matrix

### Question 1 – Nicholson Exercise 5.4.1(c)

**Question.**

For

$$A = \begin{bmatrix} 1 & -1 & 5 & -2 & 2 \\ 2 & -2 & -2 & 5 & 1 \\ 0 & 0 & -12 & 9 & -3 \\ -1 & 1 & 7 & -7 & 1 \end{bmatrix},$$

find bases for the row space and column space of  $A$ , and determine  $\text{rank}(A)$ .

**Concepts / properties applied:** Gaussian elimination; elementary row operations; Rank Theorem; pivot columns.

**Method.**

Reduce  $A$  to row-echelon form. Use the non-zero rows to obtain a basis for  $\text{row}(A)$ , and use the pivot positions to select the corresponding columns from the original matrix for a basis of  $\text{col}(A)$ .

### Step-by-step working

$$A = \begin{bmatrix} 1 & -1 & 5 & -2 & 2 \\ 2 & -2 & -2 & 5 & 1 \\ 0 & 0 & -12 & 9 & -3 \\ -1 & 1 & 7 & -7 & 1 \end{bmatrix}.$$

$$\begin{aligned} R_2 &\leftarrow R_2 - 2R_1, \\ R_4 &\leftarrow R_4 + R_1 \end{aligned} \quad (\text{eliminate entries below the first pivot})$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 5 & -2 & 2 \\ 0 & 0 & -12 & 9 & -3 \\ 0 & 0 & -12 & 9 & -3 \\ 0 & 0 & 12 & -9 & 3 \end{bmatrix}.$$

Now remove the dependent rows:

$$\begin{aligned} R_3 &\leftarrow R_3 - R_2, \\ R_4 &\leftarrow R_4 + R_2 \end{aligned} \quad (\text{eliminate redundant rows})$$

giving

$$R = \begin{bmatrix} 1 & -1 & 5 & -2 & 2 \\ 0 & 0 & -12 & 9 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are two non-zero rows, hence

$\text{rank}(A) = 2.$

**Basis for the row space**

The non-zero rows of  $R$  give

$$\text{row}(A) = \text{span} \{(1, -1, 5, -2, 2), (0, 0, -12, 9, -3)\}.$$

Therefore,

$$\mathcal{B}_{\text{row}(A)} = \{(1, -1, 5, -2, 2), (0, 0, -12, 9, -3)\}.$$

**Basis for the column space**

The pivot positions are in columns 1 and 3.

Therefore, take columns 1 and 3 from the *original* matrix:

$$c_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \quad c_3 = \begin{bmatrix} 5 \\ -2 \\ -12 \\ 7 \end{bmatrix}.$$

Hence,

$$\mathcal{B}_{\text{col}(A)} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \\ -12 \\ 7 \end{bmatrix} \right\}.$$

**Verification**

Let

$$v_1 = (1, -1, 5, -2, 2), \quad v_2 = (0, 0, -12, 9, -3).$$

The remaining rows satisfy

$$(2, -2, -2, 5, 1) = 2v_1 + v_2,$$

and

$$(-1, 1, 7, -7, 1) = -v_1 - v_2.$$

Thus every row of  $A$  is generated by  $v_1$  and  $v_2$ , confirming that the two displayed vectors span the entire row space.

**Final answer**

$$\text{rank}(A) = 2$$

$$\mathcal{B}_{\text{row}(A)} = \{(1, -1, 5, -2, 2), (0, 0, -12, 9, -3)\}$$

$$\mathcal{B}_{\text{col}(A)} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \\ -12 \\ 7 \end{bmatrix} \right\}.$$

**Key Learning Point.**

Gaussian elimination reveals the rank and row-space basis directly, while the pivot positions identify the corresponding columns of the original matrix that form a basis for the column space.

**Question 2 – Nicholson Exercise 5.4.7(b)****Question.**

For

$$A = \begin{bmatrix} 3 & 5 & 5 & 2 & 0 \\ 1 & 0 & 2 & 2 & 1 \\ 1 & 1 & 1 & -2 & -2 \\ -2 & 0 & -4 & -4 & -2 \end{bmatrix},$$

find a basis for the null space of  $A$ . Then calculate  $\text{rank}(A)$  and verify part (1) of Theorem 5.4.2.

**Concepts / properties applied:** Gaussian elimination; free variables; basic solutions; Theorem 5.4.2(1).

**Method.**

Solve

$$A\mathbf{x} = \mathbf{0}$$

by row reduction. Identify the free variables and express the general solution as a linear combination of basic solutions.

**Step-by-step working**

Start with

$$\begin{bmatrix} 3 & 5 & 5 & 2 & 0 \\ 1 & 0 & 2 & 2 & 1 \\ 1 & 1 & 1 & -2 & -2 \\ -2 & 0 & -4 & -4 & -2 \end{bmatrix}.$$

Interchange the first two rows:

$$\begin{bmatrix} 1 & 0 & 2 & 2 & 1 \\ 3 & 5 & 5 & 2 & 0 \\ 1 & 1 & 1 & -2 & -2 \\ -2 & 0 & -4 & -4 & -2 \end{bmatrix} \quad (\text{create the first pivot})$$

Now,

$$\begin{aligned} R_2 &\leftarrow R_2 - 3R_1, \\ R_3 &\leftarrow R_3 - R_1, \\ R_4 &\leftarrow R_4 + 2R_1 \end{aligned} \quad (\text{eliminate entries below the first pivot})$$

so

$$\begin{bmatrix} 1 & 0 & 2 & 2 & 1 \\ 0 & 5 & -1 & -4 & -3 \\ 0 & 1 & -1 & -4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Interchange  $R_2$  and  $R_3$ :



$$\begin{bmatrix} 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & -1 & -4 & -3 \\ 0 & 5 & -1 & -4 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{place the second pivot})$$

Then,

$$R_3 \leftarrow R_3 - 5R_2 \quad (\text{eliminate below the second pivot})$$

gives

$$\begin{bmatrix} 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & -1 & -4 & -3 \\ 0 & 0 & 4 & 16 & 12 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Scale the third row:

$$R_3 \leftarrow \frac{1}{4}R_3 \quad (\text{normalise the third pivot})$$

giving

$$\begin{bmatrix} 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & -1 & -4 & -3 \\ 0 & 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Finally,

$$\begin{aligned} R_1 &\leftarrow R_1 - 2R_3, \\ R_2 &\leftarrow R_2 + R_3 \end{aligned} \quad (\text{eliminate entries above the third pivot})$$

so

$$\text{RREF}(A) = \boxed{\begin{bmatrix} 1 & 0 & 0 & -6 & -5 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}.$$

Therefore,

$$x_1 - 6x_4 - 5x_5 = 0, \quad x_2 = 0, \quad x_3 + 4x_4 + 3x_5 = 0.$$

The free variables are  $x_4$  and  $x_5$ . Let

$$x_4 = s, \quad x_5 = t.$$

Then

$$x_1 = 6s + 5t, \quad x_2 = 0, \quad x_3 = -4s - 3t.$$

Hence,

$$\mathbf{x} = \begin{bmatrix} 6s + 5t \\ 0 \\ -4s - 3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\text{null}(A) = \text{span} \left\{ \begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Thus,

$$\mathcal{B}_{\text{null}(A)} = \left\{ \begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

## Rank

There are three pivot columns, so

$$\text{rank}(A) = 3.$$

## Verification of Theorem 5.4.2(1)

The matrix has  $n = 5$  columns and rank  $r = 3$ . Therefore,

$$n - r = 5 - 3 = 2.$$

Exactly two basic solutions were obtained:

$$\begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix}.$$

Hence,

$$\dim \text{null}(A) = 2 = n - \text{rank}(A) = 5 - 3,$$

which verifies Theorem 5.4.2(1).

As a direct check,

$$A \begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

and

$$A \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus both vectors are valid null-space vectors.

Final answer

$$\text{rank}(A) = 3$$

$$\mathcal{B}_{\text{null}(A)} = \left\{ \begin{bmatrix} 6 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\dim \text{null}(A) = 2 = 5 - 3$$

**Key Learning Point.**

The free variables determine the basic solutions. These solutions form a basis for the null space, and their number is  $n - \text{rank}(A)$ , linking Gaussian elimination directly to the rank–nullity relationship.

### Question 3 – Nicholson Exercise 5.4.5

**Question.**

Let  $A$  be an  $m \times n$  matrix and  $B$  an  $n \times m$  matrix. Show that

$$AB = 0 \iff \text{col}(B) \subseteq \text{null}(A).$$

**Concepts / properties applied:** Column space; null space; associativity of matrix multiplication; two-way implication.

**Method.**

Prove both implications separately.

**Step 1:**  $AB = 0 \Rightarrow \text{col}(B) \subseteq \text{null}(A)$

Assume

$$AB = 0.$$

Take any

$$\mathbf{y} \in \text{col}(B).$$

Then there exists some  $\mathbf{x} \in \mathbb{R}^m$  such that

$$\mathbf{y} = B\mathbf{x}.$$

Therefore,

$$A\mathbf{y} = A(B\mathbf{x}) = (AB)\mathbf{x} = 0\mathbf{x} = \mathbf{0} \quad (\text{associativity of matrix multiplication}).$$

Hence,

$$\mathbf{y} \in \text{null}(A).$$

Since  $\mathbf{y}$  was arbitrary,

$$\text{col}(B) \subseteq \text{null}(A).$$

**Step 2:**  $\text{col}(B) \subseteq \text{null}(A) \Rightarrow AB = 0$

Assume

$$\text{col}(B) \subseteq \text{null}(A).$$

Let  $\mathbf{e}_j$  denote the  $j$ th standard basis vector of  $\mathbb{R}^m$ .

The  $j$ th column of  $B$  is

$$B\mathbf{e}_j.$$

Since

$$B\mathbf{e}_j \in \text{col}(B),$$

the assumption gives

$$B\mathbf{e}_j \in \text{null}(A).$$

Therefore,

$$A(B\mathbf{e}_j) = \mathbf{0} \quad \text{for every } j.$$

But

$$A(B\mathbf{e}_j) = (AB)\mathbf{e}_j,$$

which is the  $j$ th column of  $AB$ .

Therefore every column of  $AB$  is zero, so

$$\boxed{AB = 0}.$$

### Verification

Both implications have been established:

$$AB = 0 \implies \text{col}(B) \subseteq \text{null}(A),$$

and

$$\text{col}(B) \subseteq \text{null}(A) \implies AB = 0.$$

Hence,

$$\boxed{AB = 0 \iff \text{col}(B) \subseteq \text{null}(A)}.$$

### Final answer

$$\boxed{AB = 0 \iff \text{col}(B) \subseteq \text{null}(A)}$$

#### Key Learning Point.

The zero product  $AB$  means that every column of  $B$  is annihilated by  $A$ . Thus matrix multiplication gives a direct connection between the column space of  $B$  and the null space of  $A$ .

## Section 6.3 – Diagonalisation and Eigenvalues

### Question 1 – Nicholson Exercise 3.3.1(c)

#### Question.

For

$$A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix},$$

find the characteristic polynomial, eigenvalues, eigenvectors and, if possible, an invertible matrix  $P$  such that  $P^{-1}AP$  is diagonal.

**Concepts / properties applied:** Characteristic polynomial; eigenvalue equation; Gaussian elimination; basic eigenvectors; diagonalisation.

**Method.** Find the eigenvalues from  $\det(\lambda I - A) = 0$ , then solve  $(\lambda I - A)\mathbf{x} = \mathbf{0}$  for each eigenvalue. Finally construct  $P$  from the corresponding eigenvectors and  $D$  from the eigenvalues.

#### 1. Characteristic polynomial

$$\lambda I - A = \begin{bmatrix} \lambda - 7 & 0 & 4 \\ 0 & \lambda - 5 & 0 \\ -5 & 0 & \lambda + 2 \end{bmatrix}.$$

$$\begin{aligned} D(\lambda) &= \det(\lambda I - A) \\ &= (\lambda - 5) \begin{vmatrix} \lambda - 7 & 4 \\ -5 & \lambda + 2 \end{vmatrix} && \text{(cofactor expansion)} \\ &= (\lambda - 5) [(\lambda - 7)(\lambda + 2) + 20] \\ &= (\lambda - 5)(\lambda^2 - 5\lambda + 6) \\ &= (\lambda - 5)(\lambda - 2)(\lambda - 3). && \text{(factorisation)} \end{aligned}$$

Hence,

$$\lambda = 2, 3, 5.$$

#### 2. Eigenvector for $\lambda = 2$

$$\begin{bmatrix} -5 & 0 & 4 \\ 0 & -3 & 0 \\ -5 & 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$-5x + 4z = 0, \quad y = 0 \quad \text{(eigenvector equation)}.$$

Thus,

$$x = \frac{4}{5}z.$$

Choose  $z = 5$ , giving  $x = 4$ . Hence,

$$\mathbf{v}_1 = \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} \quad (\lambda = 2).$$

### 3. Eigenvector for $\lambda = 3$

$$\begin{bmatrix} -4 & 0 & 4 \\ 0 & -2 & 0 \\ -5 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Hence,

$$x = z, \quad y = 0 \quad (\text{eigenvector equation}).$$

Taking  $x = 1$  gives

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad (\lambda = 3).$$

### 4. Eigenvector for $\lambda = 5$

$$\begin{bmatrix} -2 & 0 & 4 \\ 0 & 0 & 0 \\ -5 & 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus,

$$-2x + 4z = 0, \quad -5x + 7z = 0 \quad (\text{eigenvector equation}).$$

From the first equation,

$$x = 2z.$$

Substituting into the second,

$$-5(2z) + 7z = 0$$

so

$$-3z = 0 \implies z = 0 \implies x = 0.$$

Therefore  $y$  is free. Choosing  $y = 1$ ,

$$\mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (\lambda = 5).$$



## 5. Construct $P$ and $D$

Using the eigenvectors as columns in the corresponding eigenvalue order,

$$P = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

$$\det(P) = 1 \neq 0 \quad (\text{invertibility check})$$

so  $P$  is invertible.

### Verification

Using  $AP = PD$  gives a direct check:

$$A\mathbf{v}_1 = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 10 \end{bmatrix} = 2\mathbf{v}_1,$$

$$A\mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} = 3\mathbf{v}_2,$$

$$A\mathbf{v}_3 = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = 5\mathbf{v}_3.$$

Therefore,

$$AP = PD \implies \boxed{P^{-1}AP = D}.$$

### Final answer

$$\boxed{D(\lambda) = (\lambda - 2)(\lambda - 3)(\lambda - 5)}$$

$$\boxed{\lambda = 2, 3, 5}$$

$$\boxed{\mathbf{v}_1 = \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}$$

$$\boxed{P = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}}$$

#### Key Learning Point.

The eigenvectors become the columns of  $P$ , while their corresponding eigenvalues become the diagonal entries of  $D$ . The relation  $AP = PD$  provides an efficient final check of the diagonalisation.

## Question 2 – Nicholson Exercise 3.3.9(a)

### Question.

Given

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

verify that  $A$  and  $B$  are diagonalizable, but  $AB$  is not.

**Concepts / properties applied:** Diagonalisation; eigenspaces; eigenvalue multiplicity; basic eigenvectors; Theorem 3.3.5.

**Method.** Diagonalise  $A$  and  $B$ , then form  $AB$  and determine whether it has enough linearly independent eigenvectors.

### 1. Matrix $A$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -3 \\ 0 & \lambda - 2 \end{vmatrix} = (\lambda - 1)(\lambda - 2). \quad (\text{characteristic polynomial})$$

Thus,

$$\lambda_1 = 1, \quad \lambda_2 = 2.$$

For  $\lambda = 1$ ,

$$A - I = \begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix},$$

so

$$y = 0. \quad (\text{eigenvector equation})$$

Taking  $x = 1$ ,

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

For  $\lambda = 2$ ,

$$A - 2I = \begin{bmatrix} -1 & 3 \\ 0 & 0 \end{bmatrix},$$

giving

$$-x + 3y = 0 \implies x = 3y. \quad (\text{eigenvector equation})$$

Taking  $y = 1$ ,

$$\mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Therefore,

$$P_A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad D_A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

with

$$\det(P_A) = 1 \neq 0.$$

Hence,

$$\boxed{P_A^{-1}AP_A = D_A}$$

and  $A$  is diagonalizable.

## 2. Matrix $B$

Since

$$B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

is already diagonal,

$$P_B = I, \quad D_B = B.$$

Thus,

$$\boxed{P_B^{-1}BP_B = B},$$

so  $B$  is diagonalizable.

## 3. Product $AB$

$$AB = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}.$$

Its characteristic polynomial is

$$\det(\lambda I - AB) = \begin{vmatrix} \lambda - 2 & -3 \\ 0 & \lambda - 2 \end{vmatrix} = (\lambda - 2)^2. \quad (\text{characteristic polynomial})$$

Thus  $\lambda = 2$  has multiplicity 2.

Now,

$$AB - 2I = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix},$$

so

$$3y = 0 \implies y = 0.$$

Hence,

$$\mathbf{x} = \begin{bmatrix} x \\ 0 \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Therefore,

$$E_2(AB) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\},$$

and there is only one basic eigenvector.

### Verification

multiplicity of  $\lambda = 2 = 2$ ,      number of basic eigenvectors = 1.

Therefore,

$AB$  is not diagonalizable.

### Final answer

$A$  is diagonalizable

with

$$P_A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad D_A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

and

$B$  is diagonalizable

because  $B$  is already diagonal.

Finally,

$$AB = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} \text{ is not diagonalizable.}$$

#### Key Learning Point.

A repeated eigenvalue does not by itself guarantee diagonalizability. The matrix must have enough linearly independent eigenvectors to form an invertible diagonalising matrix.

## Question 3 – Nicholson Exercise 3.3.10

### Question.

Show that a square matrix  $A$  is diagonalizable if and only if its transpose  $A^T$  is diagonalizable.

**Concepts / properties applied:** Diagonalisation; transpose of a product; inverse-transpose property; invertibility.

**Method.** Prove both implications separately.

**1. Assume  $A$  is diagonalizable**

Then there exist an invertible matrix  $P$  and a diagonal matrix  $D$  such that

$$A = PDP^{-1}. \quad (\text{diagonalisation})$$

Taking transposes,

$$A^T = (PDP^{-1})^T. \quad (\text{transpose of a product})$$

Therefore,

$$A^T = (P^{-1})^T D^T P^T.$$

Since  $D$  is diagonal,

$$D^T = D,$$

and

$$(P^{-1})^T = (P^T)^{-1}.$$

Hence,

$$A^T = (P^T)^{-1} D P^T.$$

Also,

$$\det(P^T) = \det(P) \neq 0,$$

so  $P^T$  is invertible.

Thus,

$$\boxed{A^T = (P^T)^{-1} D P^T}$$

and  $A^T$  is diagonalizable.

**2. Assume  $A^T$  is diagonalizable**

Applying the same argument to  $A^T$ ,

$$(A^T)^T = A$$

is diagonalizable.

Therefore,

$$\boxed{A^T \text{ diagonalizable} \implies A \text{ diagonalizable.}}$$

## Verification

Both directions have been established:

$$A \text{ diagonalizable} \implies A^T \text{ diagonalizable},$$

and

$$A^T \text{ diagonalizable} \implies A \text{ diagonalizable}.$$

Hence,

$$A \text{ is diagonalizable} \iff A^T \text{ is diagonalizable.}$$

**Key Learning Point.**

Transposition preserves the diagonalisation structure. If  $A = PDP^{-1}$ , then  $A^T$  is diagonalized using the invertible matrix  $P^T$  and the same diagonal matrix  $D$ .



## Section 6.4 – Similarity and Diagonalisation

### Question 1 – Nicholson Exercise 5.5.2

#### Question.

Show that the following two matrices are not similar:

$$A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 4 & 3 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & 3 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & 4 & 1 \\ 5 & -1 & -1 & -4 \end{bmatrix}.$$

**Concepts / properties applied:** Similarity invariants; rank; elementary row operations.

#### Method.

Compare the ranks of  $A$  and  $B$ . A difference in rank is sufficient to rule out similarity.

#### 1. Rank of $A$

Starting with

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ 4 & 3 & 0 & 0 \end{bmatrix},$$

perform

$$\begin{aligned} R_2 &\leftarrow R_2 - 2R_1, \\ R_3 &\leftarrow R_3 - R_1, & (\text{eliminate below the first pivot}) \\ R_4 &\leftarrow R_4 - 4R_1 \end{aligned}$$

to obtain

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -4 & 3 & 1 \\ 0 & -1 & 1 & -1 \\ 0 & -5 & 4 & 0 \end{bmatrix}.$$

Next,

$$\begin{aligned} R_3 &\leftarrow 4R_3 - R_2, \\ R_4 &\leftarrow 4R_4 - 5R_2 & (\text{eliminate below the second pivot}) \end{aligned}$$

gives

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -4 & 3 & 1 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 1 & -5 \end{bmatrix}.$$

Therefore,

$$R_4 \leftarrow R_4 - R_3 \quad (\text{remove the dependent row})$$

gives

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -4 & 3 & 1 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Hence,

$$\boxed{(A) = 3}.$$

## 2. Rank of $B$

Starting with

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & -1 & 4 & 1 \\ 5 & -1 & -1 & -4 \end{bmatrix},$$

perform

$$\begin{aligned} R_2 &\leftarrow R_2 + R_1, \\ R_4 &\leftarrow R_4 - 5R_1 \end{aligned} \quad (\text{eliminate below the first pivot})$$

to obtain

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & -1 & 4 & 1 \\ 0 & -1 & 4 & 1 \\ 0 & 4 & -16 & -4 \end{bmatrix}.$$

Now,

$$\begin{aligned} R_3 &\leftarrow R_3 - R_2, \\ R_4 &\leftarrow R_4 + 4R_2 \end{aligned} \quad (\text{remove dependent rows})$$

gives

$$\begin{bmatrix} 1 & -1 & 3 & 0 \\ 0 & -1 & 4 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus,

$$\boxed{(B) = 2}.$$

## Verification

Since similarity preserves rank,

$$\boxed{(A) = 3 \neq 2 = (B)}.$$

Therefore, there is no invertible matrix  $P$  satisfying

$$B = P^{-1}AP.$$

## Final answer

$$\boxed{A \text{ and } B \text{ are not similar because } (A) = 3 \neq 2 = (B).}$$

### Key Learning Point.

Rank provides a direct similarity test: different ranks immediately show that two matrices cannot be related by an invertible similarity transformation.

## Question 2 – Nicholson Exercise 5.5.4(b)

### Question.

Decide whether

$$A = \begin{bmatrix} 3 & 0 & 6 \\ 0 & -3 & 0 \\ 5 & 0 & 2 \end{bmatrix}$$

is diagonalizable. If so, find  $P$  such that  $P^{-1}AP$  is diagonal.

**Concepts / properties applied:** Characteristic polynomial; repeated eigenvalue; eigenspace; linear independence; diagonalisation criterion.

### Method.

Find the eigenvalues, determine the corresponding eigenspaces, and check whether three linearly independent eigenvectors can be obtained.

### 1. Eigenvalues

$$\lambda I - A = \begin{bmatrix} \lambda - 3 & 0 & -6 \\ 0 & \lambda + 3 & 0 \\ -5 & 0 & \lambda - 2 \end{bmatrix}.$$

Expanding along the second row,

$$\begin{aligned} D(\lambda) &= \det(\lambda I - A) \\ &= (\lambda + 3) \begin{vmatrix} \lambda - 3 & -6 \\ -5 & \lambda - 2 \end{vmatrix} \quad (\text{cofactor expansion}) \\ &= (\lambda + 3) [(\lambda - 3)(\lambda - 2) - 30] \\ &= (\lambda + 3)(\lambda^2 - 5\lambda - 24) \\ &= (\lambda + 3)^2(\lambda - 8). \end{aligned}$$

Hence,

$$\lambda = -3 \text{ (multiplicity 2),} \quad \lambda = 8.$$

## 2. Eigenvectors for $\lambda = -3$

Solve

$$(A + 3I)\mathbf{x} = \mathbf{0}.$$

Thus,

$$\begin{bmatrix} 6 & 0 & 6 \\ 0 & 0 & 0 \\ 5 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The independent equation is

$$x + z = 0 \quad (\text{eigenvector equation})$$

so

$$x = -z.$$

The variables  $y$  and  $z$  are free. Hence,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, two linearly independent eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad (\lambda = -3).$$

Thus,

$$\dim E_{-3} = 2.$$

## 3. Eigenvector for $\lambda = 8$

Solve

$$(A - 8I)\mathbf{x} = \mathbf{0}.$$

Hence,

$$\begin{bmatrix} -5 & 0 & 6 \\ 0 & -11 & 0 \\ 5 & 0 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$-5x + 6z = 0, \quad y = 0 \quad (\text{eigenvector equation})$$

so

$$x = \frac{6}{5}z.$$

Choose  $z = 5$ , giving  $x = 6$ . Hence,

$$\mathbf{v}_3 = \begin{bmatrix} 6 \\ 0 \\ 5 \end{bmatrix} \quad (\lambda = 8).$$

#### 4. Check diagonalizability

We have obtained three linearly independent eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 6 \\ 0 \\ 5 \end{bmatrix}.$$

Form

$$P = \begin{bmatrix} 0 & -1 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 5 \end{bmatrix}.$$

Then

$$\det(P) = \begin{vmatrix} 0 & -1 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 5 \end{vmatrix} = 11 \neq 0 \quad (\text{invertibility check}).$$

Therefore  $P$  is invertible and  $A$  is diagonalizable.

#### 5. Construct $D$

The eigenvalues must correspond to the columns of  $P$  in the same order:

$$D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 8 \end{bmatrix}.$$

#### Verification

Check the eigenvector relations:

$$A\mathbf{v}_1 = \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} = -3\mathbf{v}_1,$$

$$A\mathbf{v}_2 = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix} = -3\mathbf{v}_2,$$

and

$$A\mathbf{v}_3 = \begin{bmatrix} 48 \\ 0 \\ 40 \end{bmatrix} = 8\mathbf{v}_3.$$

Hence,

$$AP = PD \implies \boxed{P^{-1}AP = D}.$$

**Final answer**

$A$  is diagonalizable.

$$\boxed{\lambda = -3, -3, 8}$$

$$P = \begin{bmatrix} 0 & -1 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 5 \end{bmatrix}, \quad D = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

with

$$\boxed{P^{-1}AP = D}.$$

**Key Learning Point.**

A repeated eigenvalue does not by itself prevent diagonalisation. The decisive step is the dimension of its eigenspace; here  $\lambda = -3$  supplies two independent eigenvectors, giving three independent eigenvectors in total.

### Question 3 – Nicholson Exercise 5.5.7

**Question.**

Let  $\lambda$  be an eigenvalue of  $A$  with corresponding eigenvector  $\mathbf{x}$ . If

$$B = P^{-1}AP,$$

where  $P$  is invertible, show that  $P^{-1}\mathbf{x}$  is an eigenvector of  $B$  corresponding to  $\lambda$ .

**Concepts / properties applied:** Similarity transformation; eigenvalue equation; associativity of matrix multiplication; invertibility.

**Method.**

Start from the eigenvalue equation for  $A$  and introduce the similarity transformation so that the resulting expression has the required eigenvector form for  $B$ .



**Step-by-step working**

Since  $\mathbf{x}$  is an eigenvector of  $A$  corresponding to  $\lambda$ ,

$$A\mathbf{x} = \lambda\mathbf{x}. \quad (\text{eigenvalue equation})$$

Multiply both sides on the left by  $P^{-1}$ :

$$P^{-1}A\mathbf{x} = \lambda P^{-1}\mathbf{x}.$$

Insert

$$I = PP^{-1}$$

between  $A$  and  $\mathbf{x}$ :

$$P^{-1}AP(P^{-1}\mathbf{x}) = \lambda P^{-1}\mathbf{x}. \quad (\text{identity insertion})$$

Since

$$B = P^{-1}AP,$$

we obtain

$$B(P^{-1}\mathbf{x}) = \lambda(P^{-1}\mathbf{x}).$$

Therefore,

$$\boxed{P^{-1}\mathbf{x} \text{ is an eigenvector of } B \text{ corresponding to } \lambda.}$$

**Verification**

Because  $P$  is invertible and  $\mathbf{x} \neq \mathbf{0}$ ,

$$P^{-1}\mathbf{x} \neq \mathbf{0}. \quad (\text{invertibility})$$

Thus the transformed vector is a valid eigenvector and the eigenvalue is unchanged:

$$\boxed{B(P^{-1}\mathbf{x}) = \lambda(P^{-1}\mathbf{x}).}$$

**Final answer**

$$\boxed{B = P^{-1}AP \implies B(P^{-1}\mathbf{x}) = \lambda(P^{-1}\mathbf{x})}$$

and therefore

$$\boxed{P^{-1}\mathbf{x} \text{ is an eigenvector of } B \text{ for the same eigenvalue } \lambda.}$$

**Key Learning Point.**

A similarity transformation changes the coordinates of an eigenvector while preserving its eigenvalue. The corresponding eigenvector of the similar matrix is obtained by applying  $P^{-1}$  to the original eigenvector.

Find a basis for the **column space** of A and determine its rank and hence the dimension of the subspace.

$$\begin{pmatrix} 3 & 0 & 6 & -3 \\ 1 & 2 & 6 & 1 \\ 2 & -1 & 2 & -6 \\ -3 & 1 & -4 & 6 \end{pmatrix}$$

Enter your basis as rows/columns of a matrix. If the basis of your vector space is empty, you can enter a zero matrix.

Rows:  Columns:

$$\begin{pmatrix} \begin{matrix} 3 \\ 1 \\ 2 \\ -3 \end{matrix} & \begin{matrix} 0 \\ 2 \\ -1 \\ 1 \end{matrix} & \begin{matrix} -3 \\ 1 \\ -6 \\ 6 \end{matrix} \end{pmatrix}$$

What is the rank of your matrix entered?

Correct. Your vectors are independent and span the entire subspace. You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

Find a basis for the **null space** of A and determine its rank and hence the dimension of the subspace.

$$\begin{pmatrix} 0 & 2 & 0 & 2 \\ 0 & -1 & 3 & -1 \\ 0 & -2 & -2 & -3 \\ 0 & -3 & 3 & 0 \end{pmatrix}$$

Enter your basis as rows/columns of a matrix. If the basis of your vector space is empty, you can enter a zero matrix.

Rows:  Columns:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

What is the rank of your matrix entered?

That's right. You have the correct dimension and your column vectors span the null space. You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

Find a basis for the **row space** of A and determine its rank and hence the dimension of the subspace.

$$\begin{pmatrix} -1 & 1 & 0 & -2 \\ 3 & -6 & -6 & 6 \\ 3 & -6 & -8 & 10 \\ -3 & 1 & -5 & -4 \end{pmatrix}$$

Enter your basis as rows/columns of a matrix. If the basis of your vector space is empty, you can enter a zero matrix.

Rows:  Columns:

$$\begin{pmatrix} \boxed{1} & \boxed{0} & \boxed{0} & \boxed{6} \\ \boxed{0} & \boxed{1} & \boxed{0} & \boxed{4} \\ \boxed{0} & \boxed{0} & \boxed{1} & \boxed{-2} \end{pmatrix}$$

What is the rank of your matrix entered?

Correct. Your vectors are independent and span the entire subspace. You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

Consider the matrix

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 5 & 0 \\ -2 & 0 & 5 \end{pmatrix}$$

Find the Eigenvalues and Eigenvectors so that you can determine an invertible matrix  $P$  and diagonal matrix  $D$  such that  $P^{-1}AP = D$ .

Rows:  Columns:

$$P = \begin{pmatrix} \boxed{1} & \boxed{1} & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{1} \\ \boxed{1} & \boxed{2} & \boxed{0} \end{pmatrix}$$

Rows:  Columns:

$$D = \begin{pmatrix} \boxed{3} & \boxed{0} & \boxed{0} \\ \boxed{0} & \boxed{4} & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{5} \end{pmatrix}$$

Use eigenvectors with integer values where possible - decimal approximations may not be marked as correct.

Correct. Your matrices satisfy  $P^{-1}AP = D$ . You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk), developed by [Newcastle University \(http://www.newcastle.ac.uk\)](http://www.newcastle.ac.uk).



Consider the matrix

$$\begin{pmatrix} -1 & 0 & 0 \\ -16 & 15 & -6 \\ -45 & 45 & -18 \end{pmatrix}$$

Find the Eigenvalues and Eigenvectors such that you can determine an invertible matrix  $P$  and diagonal matrix  $D$  such that  $P^{-1}AP = D$ .

Rows:  Columns:

$$P = \begin{pmatrix} \boxed{0} & \boxed{1} & \boxed{0} \\ \boxed{1} & \boxed{1} & \boxed{2} \\ \boxed{3} & \boxed{0} & \boxed{5} \end{pmatrix}$$

Rows:  Columns:

$$D = \begin{pmatrix} \boxed{-3} & \boxed{0} & \boxed{0} \\ \boxed{0} & \boxed{-1} & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix}$$

Do not use decimal approximations for entries in P and D.

Correct. Your matrices satisfy  $P^{-1}AP = D$ . You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

# Question 1

Use this question to practice polynomial division (see Appendix D in the textbook).

Consider the following polynomial

$$P(x) = x^3 - x^2 - 16x + 16$$

---

We want to find a linear factor. If there is no constant term, then that means  $x$  is a factor, which is fairly easy to factorise. In this case, we do have a constant, and so to find factors we look at factors of 16.

These are [ 1, 2, 4, 8, 16 ].

We need to check both the negative and positive versions of those numbers, subbing each into the polynomial until we find a factor such that  $P(x) = 0$ .

You should be able to verify that  $P(1) = 0$ , which means that  $(x - 1)$  is a factor.

We then can set out and complete our division as follows.

a)

$$\begin{array}{r}
 \boxed{x^2 - 16} \quad x^2 - 16 \quad \checkmark \\
 x - 1 \quad \boxed{x^3 - x^2 - 16x + 16} \\
 - \quad \boxed{x^3 - x^2} \quad \checkmark \\
 \quad \quad \quad \boxed{-16x + 16} \\
 \quad \quad -16x + 16 \quad \checkmark \\
 - \quad \quad \quad \boxed{0} \quad 0 \quad \checkmark \\
 \quad \quad \quad \quad \quad \boxed{-16x + 16} \\
 \quad \quad \quad \quad -16x + 16 \quad \checkmark \\
 \quad \quad \quad \quad \quad \quad \quad \quad \boxed{-16x + 16} \\
 \quad \quad \quad \quad \quad \quad -16x + 16 \quad \checkmark \\
 \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad 0
 \end{array}$$

Follow the steps below and use this box to write your final answer,  $Ax^2 + Bx + C$ , as you go.

1. Divide the first term of the polynomial by  $x$ , place the result above (this should be your  $Ax^2$  term), then multiply that by your divisor - write the expression here.
2. Subtract (1) from the starting polynomial.
3. Divide the  $x^2$  term in the polynomial at (2) by  $x$  and include the result in your final answer ( $Bx$ ), then multiply that by your divisor.
4. Subtract the result (3) from the polynomial at (2).
5. Divide the  $x$  term in the polynomial at (4) by  $x$  and include the result in your final answer ( $C$ ), then multiply that by your divisor to include here.
6. Subtract the result (5) from the polynomial at (4). This should give you a result of 0.

**Gap 0**

Your answer is numerically correct. You were awarded 1 mark.

**Gap 1**

Your answer is numerically correct. You were awarded 1 mark.

**Gap 2**

Your answer is numerically correct. You were awarded 1 mark.

**Gap 3**

Your answer is numerically correct. You were awarded 1 mark.

**Gap 4**

Your answer is numerically correct. You were awarded **1** mark. ✓

**Gap 5**

Your answer is numerically correct. You were awarded **1** mark. ✓

You scored **6** marks for this part.

---

**Score: 6/6** ✓

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

Consider the matrix

$$\begin{pmatrix} 3 & 2 & -1 \\ -2 & -2 & 2 \\ -3 & 2 & 6 \end{pmatrix}$$

Either by expanding  $(\lambda I - A)$  or using the alternative approach described in 6.3 of the online modules, find the coefficients of the characteristic equation

$$D(\lambda) = b_3 \lambda^3 + b_2 \lambda^2 + b_1 \lambda + b_0$$

$$b_3 = \boxed{1} \quad \checkmark$$

$$b_2 = \boxed{-7} \quad \checkmark$$

$$b_1 = \boxed{-3} \quad \checkmark$$

$$b_0 = \boxed{26} \quad \checkmark$$

**b3**

Your answer is correct. You were awarded 1 mark.

**b2**

Your answer is correct. You were awarded 1 mark.

**b1**

Your answer is correct. You were awarded 1 mark.

**b0**

Your answer is correct. You were awarded 1 mark.

You scored 4 marks for this part.

**Score: 4/4**

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).



# Question 1

Consider the matrices,

$$A = \begin{pmatrix} -3 & -6 \\ 8 & -8 \end{pmatrix}, B = \begin{pmatrix} -4 & 9 \\ -1 & -6 \end{pmatrix}$$

Select all statements that are true:

- ☐ A and B have the same eigenvalues and so could be similar
- ☒ A and B are not similar, because they do not have the same determinant
- ☒ A and B are not similar, because  $\text{trace}(A) \neq \text{trace}(B)$



*This feedback is based on your last submitted answer. Submit your changed answer to get updated feedback.*

You chose a correct answer. You were awarded 1 mark. ✓

You chose a correct answer. You were awarded 1 mark. ✓

You scored 2 marks for this part.

---

**Score: 2/2** ✓

# Question 1

Consider the matrices,

$$A = \begin{pmatrix} 5 & 5 \\ -2 & -1 \end{pmatrix}, B = \begin{pmatrix} 6 & 8 \\ -2 & -2 \end{pmatrix}$$

Select all statements that are true:

- ☐ A and B have the same eigenvalues and so could be similar
- ☒ A and B are not similar, because they do not have the same determinant
- ☐ A and B are not similar, because  $\text{trace}(A) \neq \text{trace}(B)$



You chose a correct answer. You were awarded **1** mark. ✓

You scored **1** mark for this part.

---

**Score: 1/1** ✓

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

Consider the following triangular matrix

$$A = \begin{pmatrix} 9 & 4 & 8 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

Is  $A$  diagonalisable?

☐ Yes ☒ No



If  $A$  is diagonalisable, provide 3 linearly independent eigenvectors as columns of the matrix  $P$  such that  $P^{-1}AP = D$  and then give the corresponding eigenvalues. If  $A$  is not diagonalisable, reduce the columns of  $P$  and give as many linearly independent eigenvectors as possible.

|       |   |          |   |
|-------|---|----------|---|
| Rows: | 3 | Columns: | 2 |
|-------|---|----------|---|

$$P = \begin{pmatrix} \underline{1} & \underline{0} \\ \underline{0} & \underline{-2} \\ \underline{0} & \underline{1} \end{pmatrix}$$

|       |   |          |   |
|-------|---|----------|---|
| Rows: | 1 | Columns: | 2 |
|-------|---|----------|---|

$$\lambda = \begin{pmatrix} \underline{9} & \underline{9} \end{pmatrix}$$

Enter the eigenvalues above corresponding with the columns of  $P$  (in order and including repeats).

✓ Correct. You have identified the eigenvalues and eigenvectors correctly. You were awarded **3** marks.

**Score: 3/3** ✓

## NUMBAS

Created using Numbas (<https://www.numbas.org.uk>) v10.0, developed by Newcastle University (<http://www.newcastle.ac.uk>).

Consider the following triangular matrix

$$A = \begin{pmatrix} 2 & 9 & -9 \\ 0 & -9 & 11 \\ 0 & 0 & 2 \end{pmatrix}$$

Is  $A$  diagonalisable?

☒ Yes    ☐ No



If  $A$  is diagonalisable, provide 3 linearly independent eigenvectors as columns of the matrix  $P$  such that  $P^{-1}AP = D$  and then give the corresponding eigenvalues. If  $A$  is not diagonalisable, reduce the columns of  $P$  and give as many linearly independent eigenvectors as possible.

|       |          |          |          |
|-------|----------|----------|----------|
| Rows: | <u>3</u> | Columns: | <u>3</u> |
|-------|----------|----------|----------|

$$P = \begin{pmatrix} \underline{1} & \underline{0} & \underline{-9} \\ \underline{0} & \underline{1} & \underline{11} \\ \underline{0} & \underline{1} & \underline{0} \end{pmatrix}$$

|       |          |          |          |
|-------|----------|----------|----------|
| Rows: | <u>1</u> | Columns: | <u>3</u> |
|-------|----------|----------|----------|

$$\lambda = \begin{pmatrix} \underline{2} & \underline{2} & \underline{-9} \end{pmatrix}$$

Enter the eigenvalues above corresponding with the columns of  $P$  (in order and including repeats).

✓ Correct. You have identified the eigenvalues and eigenvectors correctly. You were awarded 3

Score: 3/3 ✓

marks.

## NUMBAS

Created using Numbas (<https://www.numbas.org.uk>) v10.0, developed by Newcastle University (<http://www.newcastle.ac.uk>).



# SIT292 Module 6 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

| Question Number            | Score               |
|----------------------------|---------------------|
| <a href="#">Question 1</a> | 1/1                 |
| <a href="#">Question 2</a> | 1/1                 |
| <a href="#">Question 3</a> | 1/1                 |
| <a href="#">Question 4</a> | 1/1                 |
| <a href="#">Question 5</a> | 1/1                 |
| <a href="#">Question 6</a> | 3/3                 |
| <a href="#">Question 7</a> | 2/2                 |
| <b>Total</b>               | <b>10/10 (100%)</b> |

**Congratulations! You have achieved the minimum threshold for this module's self-assessment.**

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

## Performance Summary

|                    |                                 |
|--------------------|---------------------------------|
| <b>Exam Name:</b>  | SIT292 Module 6 Self-Assessment |
| <b>Session ID:</b> | 4789122529220511                |
| <b>Exam Start:</b> | Sun Sep 06 2026 14:17:15        |
| <b>Exam Stop:</b>  | Sun Sep 06 2026 14:54:03        |
| <b>Time Spent:</b> | 00:36:47                        |

Created using [Numbas \(https://www.numbas.org.uk\)](https://www.numbas.org.uk) v10.0, developed by [Newcastle University \(https://www.newcastle.ac.uk\)](https://www.newcastle.ac.uk).

## Question 1

Find a basis and the dimension for the **null space** of A.

$$\begin{pmatrix} -3 & -3 & 0 & -6 \\ -1 & -4 & -6 & -2 \\ -2 & -3 & -1 & -6 \\ 3 & 2 & -1 & 4 \end{pmatrix}$$

Enter your basis as **columns** of a matrix. Use the controls to change the size of the matrix as necessary. If your vector space is empty, you can enter a zero matrix.

Rows: 4 Columns: 1

$$\begin{pmatrix} 2 \\ -4 \\ 2 \\ 1 \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} 2 \\ -4 \\ 2 \\ 1 \end{pmatrix}$$

What is the rank of the matrix you entered? (this will be equal to the dimension of the null space).

1

Expected answer: 1

✓ That's right. You have the correct dimension and your column vectors span the null space. You were awarded 1 mark.

Score: 1/1 ✓

## Question 2

Find a basis for the **row space** of A and determine its rank and hence the dimension of the subspace.

$$\begin{pmatrix} -3 & -3 & 6 & 3 \\ 1 & 3 & -6 & -3 \\ -3 & -5 & 9 & 3 \\ 3 & 2 & -3 & -2 \end{pmatrix}$$

Enter your basis as **rows** of a matrix. If the basis of your vector space is empty, you can enter a zero matrix.

Rows: 4 Columns: 4

$$\begin{pmatrix} -3 & -3 & 6 & 3 \\ 1 & 3 & -6 & -3 \\ -3 & -5 & 9 & 3 \\ 3 & 2 & -3 & -2 \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the rank of your matrix entered?

4

Expected answer: 4

✓ Correct. Your vectors are independent and span the entire subspace. You were awarded 1 mark.

Score: 1/1 ✓

## Question 3

Consider the matrix

$$\begin{pmatrix} 3 & 0 & 0 \\ 4 & -4 & -4 \\ 1 & -2 & 0 \end{pmatrix}$$

Either by expanding  $(\lambda I - A)$  or using the alternative approach described in 6.3 of the online modules, find the coefficients of the characteristic equation

$$D(\lambda) = b_3\lambda^3 + b_2\lambda^2 + b_1\lambda + b_0$$

$b_3 =$  1 ✓

Expected answer: 1

$b_2 =$  1 ✓

Expected answer: 1

$b_1 =$  -20 ✓

Expected answer: -20

$$b_0 = \underline{24} \quad \checkmark$$

Expected answer: 24

**b3**

**Score: 1/1** 

-  Your answer is correct. You were awarded **0.25** marks.

**b2**

-  Your answer is correct. You were awarded **0.25** marks.

**b1**

-  Your answer is correct. You were awarded **0.25** marks.

**b0**

-  Your answer is correct. You were awarded **0.25** marks.

You scored **1** mark for this part.

## Question 4

Consider the following polynomial

$$P(x) = x^3 - 2x^2 - 9x + 18$$

---

We want to find a linear factor. If there is no constant term, then that means  $x$  is a factor, which is fairly easy to factorise. In this case, we do have a constant, and so to find factors we look at factors of 18.

These are [ 1, 2, 3, 6, 9, 18 ].

We need to check both the negative and positive versions of those numbers, subbing each into the polynomial until we find a factor such that  $P(x) = 0$ .

You should be able to verify that  $P(2) = 0$ , which means that  $(x - 2)$  is a factor.

We then can set out and complete our division as follows.

Complete all steps of polynomial factorisation.



a)

$$\frac{x^2 - 9}{x^2 - 9} \quad \checkmark$$

Expected  
answer:

$$\frac{x^2 - 9}{x^2 - 9}$$

Write the result of division,  $Ax^2 + Bx + C$ , as you go.

$$x - 2$$

$$x^3 - 2x^2 - 9x + 18$$

$$\frac{x^3 - 2x^2}{x^3 - 2x^2} \quad \checkmark$$

Expected  
answer:

$$\frac{x^3 - 2x^2}{x^3 - 2x^2}$$

1.

$$\frac{-9x + 18}{-9x + 18} \quad \checkmark$$

Expected  
answer:

$$\frac{-9x + 18}{-9x + 18}$$

2.

$$\frac{0}{0} \quad \checkmark$$

Expected  
answer:

$$0$$

3.

$$\begin{array}{r} -9x+18 \\ \hline -9x + 18 \end{array} \quad \checkmark$$

Expected  
answer:

4.

$$\begin{array}{r} -9x + 18 \\ \hline -9x + 18 \end{array}$$

$$\begin{array}{r} -9x+18 \\ \hline \end{array}$$

$$\begin{array}{r} -9x + 18 \end{array} \quad \checkmark$$

Expected  
answer:

5.

$$\begin{array}{r} -9x + 18 \\ \hline \end{array}$$

$$\begin{array}{r} -9x + 18 \end{array}$$

0

6.

Note: If the coefficient of  $x^2$  is 0 at step 2, then you should enter 0 at step 3. If the coefficient of  $x$  is 0 at step 4, then you should enter 0 at step 5.

▼ Feedback

Score: 1/1 **Gap 0**

 Your answer is numerically correct. You were awarded **0.5** marks.

**Gap 1**

 Your answer is numerically correct. You were awarded **0.1** marks.

**Gap 2**

 Your answer is numerically correct. You were awarded **0.1** marks.

**Gap 3**

- ✓ Your answer is numerically correct. You were awarded **0.1** marks.

**Gap 4**

- ✓ Your answer is numerically correct. You were awarded **0.1** marks.

**Gap 5**

- ✓ Your answer is numerically correct. You were awarded **0.1** marks.

You scored **1** marks for this part.

## Question 5

Consider the matrices,

$$A = \begin{pmatrix} -5 & 9 \\ -4 & -4 \end{pmatrix}, \quad B = \begin{pmatrix} 9 & 7 \\ 3 & -17 \end{pmatrix}$$

Select all statements that are true:

- ☐ A and B have the same eigenvalues and so could be similar
- ☒ A and B are not similar, because they do not have the same determinant
- ☒ A and B are not similar, because  $\text{trace}(A) \neq \text{trace}(B)$



Expected answer:

- ☐ A and B have the same eigenvalues and so could be similar
- ☒ A and B are not similar, because they do not have the same determinant
- ☒ A and B are not similar, because  $\text{trace}(A) \neq \text{trace}(B)$

✓ Your answer is correct. You were awarded 1 mark.

Score: 1/1 ✓

## Question 6

Consider the matrix

$$A = \begin{pmatrix} -6 & 0 & 1 \\ 0 & -2 & 0 \\ -18 & 0 & 5 \end{pmatrix}$$

Find the Eigenvalues and Eigenvectors so that you can determine an invertible matrix  $P$  and diagonal matrix  $D$  such that  $P^{-1}AP = D$ . Find the eigenvectors without using software and do not use decimal approximations.

$P =$

Rows: 3 Columns: 3

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 9 \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 9 & 2 \end{pmatrix}$$

$D =$ Rows: 3 Columns: 3

$$\begin{pmatrix} \underline{-4} & \underline{0} & \underline{0} \\ \underline{0} & \underline{-2} & \underline{0} \\ \underline{0} & \underline{0} & \underline{3} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{-2} & \underline{0} & \underline{0} \\ \underline{0} & \underline{3} & \underline{0} \\ \underline{0} & \underline{0} & \underline{-4} \end{pmatrix}$$

✓ Correct. Your matrices satisfy  $P^{-1} A P = D$ .  
You were awarded **3** marks.

Score: 3/3 ✓

## Question 7

Consider the following triangular matrix

$$A = \begin{pmatrix} 3 & -2 & 4 \\ 0 & 5 & -4 \\ 0 & 0 & 3 \end{pmatrix}$$

Is  $A$  diagonalisable?

☒ Yes

☐ No

Expected answer:

☒ Yes

☐ No


If  $A$  is diagonalisable, provide 3 linearly independent eigenvectors as columns of the matrix  $P$  such that  $P^{-1}AP = D$  and then give the corresponding eigenvalues. If  $A$  is not diagonalisable, reduce the columns of  $P$  and give as many linearly independent eigenvectors as possible.

$P =$

Rows: 3 Columns: 3

$$\begin{pmatrix} \underline{1} & \underline{0} & \underline{-1} \\ \underline{0} & \underline{2} & \underline{1} \\ \underline{0} & \underline{1} & \underline{0} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{-1} & \underline{1} & \underline{0} \\ \underline{1} & \underline{0} & \underline{2} \\ \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

Enter the eigenvalues corresponding with the columns of  $P$  (in order and including repeats if there are multiple linearly independent vectors corresponding with the same eigenvalue)

$\lambda =$

Rows: 1 Columns: 3

$$\begin{pmatrix} \underline{3} & \underline{3} & \underline{5} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{5} & \underline{3} & \underline{3} \end{pmatrix}$$

✓ Correct. You have identified the eigenvalues and eigenvectors correctly. You were awarded 2 marks.

Score: 2/2 ✓



---

Created using Numbas (<https://www.numbas.org.uk>) v10.0, developed by Newcastle University (<https://www.newcastle.ac.uk>).

# Module-6 Self-Assessment

## Working & Annotations

*Detailed mathematical working and annotations for Questions 1–7*

### Question 1

#### Question.

Find a basis and the dimension for the **null space** of

$$A = \begin{bmatrix} -3 & -3 & 0 & -6 \\ -1 & -4 & -6 & -2 \\ -2 & -3 & -1 & -6 \\ 3 & 2 & -1 & 4 \end{bmatrix}.$$

**Concepts / properties used:** Gaussian elimination; null space; free variable; rank–nullity.

#### Working.

We solve the homogeneous system

$$A\mathbf{x} = \mathbf{0}. \quad (\text{null-space equation})$$

Row-reducing,

$$\begin{bmatrix} -3 & -3 & 0 & -6 \\ -1 & -4 & -6 & -2 \\ -2 & -3 & -1 & -6 \\ 3 & 2 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} -3 & -3 & 0 & -6 \\ 0 & -3 & -6 & 0 \\ 0 & -1 & -1 & -2 \\ 0 & -1 & -1 & -2 \end{bmatrix}$$

$$\begin{aligned} R_2 &\leftarrow R_2 - \frac{1}{3}R_1 \\ R_3 &\leftarrow R_3 - \frac{2}{3}R_1 \\ R_4 &\leftarrow R_4 + R_1 \end{aligned} \quad (\text{eliminate the first column})$$

and then

$$\sim \begin{bmatrix} -3 & -3 & 0 & -6 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} R_3 &\leftarrow R_3 - \frac{1}{3}R_2 \\ R_4 &\leftarrow R_4 - \frac{1}{3}R_2 \\ R_4 &\leftarrow R_4 - R_3 \end{aligned} \quad (\text{echelon form})$$

which reduces to

$$\boxed{\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}} \quad (RREF).$$

Hence,

$$x_1 - 2x_4 = 0, \quad x_2 + 4x_4 = 0, \quad x_3 - 2x_4 = 0.$$

Let

$$x_4 = t \quad (\text{free variable}).$$

Then

$$x_1 = 2t, \quad x_2 = -4t, \quad x_3 = 2t.$$

Therefore,

$$\mathbf{x} = \begin{bmatrix} 2t \\ -4t \\ 2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ -4 \\ 2 \\ 1 \end{bmatrix}.$$

Thus,

$$\text{null}(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ -4 \\ 2 \\ 1 \end{bmatrix} \right\}$$

and

$$\dim(\text{null}(A)) = 1.$$

### Verification

$$A \begin{bmatrix} 2 \\ -4 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

so the proposed basis vector is indeed in the null space.

The basis matrix entered in NUMBAS is

$$\begin{bmatrix} 2 \\ -4 \\ 2 \\ 1 \end{bmatrix}$$

with rank

$$1.$$

#### Learning point.

The single free variable determines a one-dimensional null space. The corresponding basic solution gives the required basis directly.

## Question 2

### Question.

Find a basis for the row space of

$$A = \begin{bmatrix} -3 & -3 & 6 & 3 \\ 1 & 3 & -6 & -3 \\ -3 & -5 & 9 & 3 \\ 3 & 2 & -3 & -2 \end{bmatrix}$$

and determine its rank and hence the dimension of the subspace.

**Concepts / properties used:** Row space; rank; determinant; linear independence.

### Working.

A convenient way to test whether the four rows are independent is to calculate the determinant:

$$\begin{aligned} \det(A) &= \begin{vmatrix} -3 & -3 & 6 & 3 \\ 1 & 3 & -6 & -3 \\ -3 & -5 & 9 & 3 \\ 3 & 2 & -3 & -2 \end{vmatrix} \quad (\text{determinant test}) \\ &= -12 \neq 0. \end{aligned}$$

Therefore  $A$  is nonsingular, so all four rows are linearly independent.

Hence,

$$\boxed{\text{rank}(A) = 4}.$$

Since the row space is generated by the rows of  $A$ , and none of the rows is redundant, the original rows form a basis:

$$\boxed{\mathcal{B}_{\text{Row}(A)} = \{(-3, -3, 6, 3), (1, 3, -6, -3), (-3, -5, 9, 3), (3, 2, -3, -2)\}.$$

Therefore,

$$\boxed{\dim(\text{Row}(A)) = 4}.$$

### Verification

A non-zero determinant confirms four independent rows, so the basis contains exactly four vectors.

The matrix entered into NUMBAS as the basis, with the basis vectors as rows, is

$$\boxed{\begin{bmatrix} -3 & -3 & 6 & 3 \\ 1 & 3 & -6 & -3 \\ -3 & -5 & 9 & 3 \\ 3 & 2 & -3 & -2 \end{bmatrix}}$$

and its rank is

$$\boxed{4}.$$

**Learning point.**

For a square matrix with non-zero determinant, all rows are linearly independent. Consequently, the row space has full dimension and the original rows already form a non-redundant basis.

**Question 3****Question.**

For

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 4 & -4 & -4 \\ 1 & -2 & 0 \end{bmatrix},$$

find the coefficients of

$$D(\lambda) = b_3\lambda^3 + b_2\lambda^2 + b_1\lambda + b_0.$$

**Concepts / properties used:** Characteristic polynomial; trace; principal minors; determinant.

**Working.**

For a  $3 \times 3$  matrix,

$$b_3 = 1. \quad (\text{leading coefficient})$$

Next,

$$\text{tr}(A) = 3 + (-4) + 0 = -1,$$

so

$$b_2 = -\text{tr}(A) = 1. \quad (\text{trace coefficient})$$

For  $b_1$ , use the three principal  $2 \times 2$  minors:

$$\begin{aligned} b_1 &= \begin{vmatrix} 3 & 0 \\ 4 & -4 \end{vmatrix} + \begin{vmatrix} 3 & 0 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} -4 & -4 \\ -2 & 0 \end{vmatrix} \\ &= (-12) + 0 + (-8) \\ &= -20. \end{aligned} \quad (\text{principal-minor formula})$$

Finally,

$$\begin{aligned} \det(A) &= 3 \begin{vmatrix} -4 & -4 \\ -2 & 0 \end{vmatrix} \\ &= 3(0 - 8) \\ &= -24. \end{aligned} \quad (\text{cofactor expansion})$$

For a  $3 \times 3$  characteristic polynomial,

$$b_0 = (-1)^3 \det(A) = -\det(A) = 24. \quad (\text{constant-term formula})$$

Therefore,

$$b_3 = 1, \quad b_2 = 1, \quad b_1 = -20, \quad b_0 = 24.$$

Hence,

$$D(\lambda) = \lambda^3 + \lambda^2 - 20\lambda + 24.$$

### Verification

Substituting the four coefficients back into the required form gives

$$1\lambda^3 + 1\lambda^2 - 20\lambda + 24,$$

which agrees with the characteristic polynomial above.

#### Learning point.

The characteristic polynomial can be obtained efficiently from the trace, principal minors and determinant, avoiding an unnecessarily long symbolic expansion.

## Question 4

### Question.

Consider

$$P(x) = x^3 - 2x^2 - 9x + 18.$$

Find a linear factor and complete all steps of the polynomial factorisation.

**Concepts / properties used:** Factor theorem; rational-root testing; polynomial division.

### Working.

The possible integer roots come from the factors of 18:

$$\pm 1, \pm 2, \pm 3, \pm 6, \pm 9, \pm 18. \quad (\text{rational-root candidates})$$

Testing  $x = 2$ ,

$$P(2) = 8 - 8 - 18 + 18 = 0.$$

Therefore,

$$x - 2 \text{ is a factor.} \quad (\text{factor theorem})$$

Now divide

$$x^3 - 2x^2 - 9x + 18$$

by



$$x - 2.$$

The first quotient term is

$$\frac{x^3}{x} = x^2. \quad (\text{leading-term division})$$

Then

$$x^2(x - 2) = x^3 - 2x^2.$$

Subtracting,

$$\begin{aligned} (x^3 - 2x^2 - 9x + 18) - (x^3 - 2x^2) \\ = -9x + 18. \end{aligned}$$

There is no  $x^2$  term remaining, so the next coefficient is

$$\boxed{0}.$$

Continue:

$$\frac{-9x}{x} = -9.$$

Hence,

$$-9(x - 2) = -9x + 18.$$

Subtracting,

$$(-9x + 18) - (-9x + 18) = 0.$$

Thus the quotient is

$$\boxed{x^2 - 9}.$$

Therefore,

$$\boxed{P(x) = (x - 2)(x^2 - 9)}$$

and

$$\boxed{P(x) = (x - 2)(x - 3)(x + 3)}.$$

## NUMBAS entries

In the exact order of the six user-entered gaps:

$$\boxed{x^2 - 9, \quad x^3 - 2x^2, \quad -9x + 18, \quad 0, \quad -9x + 18, \quad -9x + 18}$$

## Verification

Multiplying the factors,

$$(x - 2)(x^2 - 9) = x^3 - 2x^2 - 9x + 18,$$

which reproduces the original polynomial.

### Learning point.

A root gives the corresponding linear factor, while polynomial division systematically determines the remaining factor and the zero remainder confirms the factorisation.

## Question 5

### Question.

Consider

$$A = \begin{bmatrix} -5 & 9 \\ -4 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 7 \\ 3 & -17 \end{bmatrix}.$$

Select all statements that are true:

1.  $A$  and  $B$  have the same eigenvalues and so could be similar.
2.  $A$  and  $B$  are not similar, because they do not have the same determinant.
3.  $A$  and  $B$  are not similar, because  $\text{trace}(A) \neq \text{trace}(B)$ .

**Concepts / properties used:** Similarity invariants; determinant; trace; characteristic polynomial.

### Working.

For  $A$ ,

$$\det(A) = (-5)(-4) - 9(-4) = 56.$$

For  $B$ ,

$$\det(B) = 9(-17) - 7(3) = -174.$$

Hence,

$$\boxed{\det(A) = 56 \neq -174 = \det(B)}. \quad (\text{similarity invariant})$$

Therefore  $A$  and  $B$  cannot be similar, so statement 2 is true.

Now compare the traces:

$$\text{trace}(A) = -5 - 4 = -9,$$

$$\text{trace}(B) = 9 - 17 = -8.$$

Thus,

$$\boxed{\text{trace}(A) = -9 \neq -8 = \text{trace}(B)} \quad (\text{similarity invariant})$$

so statement 3 is true.

For completeness, the characteristic polynomials are

$$D_A(\lambda) = \lambda^2 + 9\lambda + 56,$$

and

$$D_B(\lambda) = \lambda^2 + 8\lambda - 174.$$

Thus,

$$\boxed{D_A(\lambda) \neq D_B(\lambda)}$$

so the matrices do not have the same eigenvalues. Statement 1 is false.

### Final selection

Statement 1: FALSE

Statement 2: TRUE

Statement 3: TRUE

Therefore select:

A and B are not similar, because they do not have the same determinant;  
A and B are not similar, because  $\text{trace}(A) \neq \text{trace}(B)$ .

#### Learning point.

Determinant and trace are preserved by similarity. A mismatch in either gives a direct reason that two matrices cannot be similar.

## Question 6

### Question.

For

$$A = \begin{bmatrix} -6 & 0 & 1 \\ 0 & -2 & 0 \\ -18 & 0 & 5 \end{bmatrix},$$

find the eigenvalues and eigenvectors and determine an invertible matrix  $P$  and diagonal matrix  $D$  such that

$$P^{-1}AP = D.$$

**Concepts / properties used:** Characteristic polynomial; eigenvalues; eigenspaces; Gaussian elimination; linear independence; diagonalisation.

**Working.**

First,

$$\lambda I - A = \begin{bmatrix} \lambda + 6 & 0 & -1 \\ 0 & \lambda + 2 & 0 \\ 18 & 0 & \lambda - 5 \end{bmatrix}.$$

Expanding along the second row,

$$\begin{aligned} D(\lambda) &= (\lambda + 2) \begin{vmatrix} \lambda + 6 & -1 \\ 18 & \lambda - 5 \end{vmatrix} \\ &= (\lambda + 2) [(\lambda + 6)(\lambda - 5) + 18] \quad (\text{cofactor expansion and factorisation}) \\ &= (\lambda + 2)(\lambda^2 + \lambda - 12) \\ &= (\lambda + 2)(\lambda + 4)(\lambda - 3). \end{aligned}$$

Therefore,

$$\boxed{\lambda = -4, -2, 3}.$$

**Eigenvector for  $\lambda = -4$**

Solve

$$(A + 4I)\mathbf{x} = \mathbf{0}.$$

$$\begin{bmatrix} -2 & 0 & 1 \\ 0 & 2 & 0 \\ -18 & 0 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

Hence,

$$-2x + z = 0, \quad y = 0.$$

Thus,

$$z = 2x.$$

Choose  $x = 1$ :

$$\boxed{\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}} \quad (\lambda = -4).$$

**Eigenvector for  $\lambda = -2$**

Solve

$$(A + 2I)\mathbf{x} = \mathbf{0},$$

so

$$\begin{bmatrix} -4 & 0 & 1 \\ 0 & 0 & 0 \\ -18 & 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

The first and third equations give

$$-4x + z = 0, \quad -18x + 7z = 0,$$

which imply

$$x = 0, \quad z = 0.$$

Therefore  $y$  is free. Taking  $y = 1$ ,

$$\boxed{\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \quad (\lambda = -2).$$

### Eigenvector for $\lambda = 3$

Solve

$$(A - 3I)\mathbf{x} = \mathbf{0},$$

giving

$$\begin{bmatrix} -9 & 0 & 1 \\ 0 & -5 & 0 \\ -18 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

Hence,

$$-9x + z = 0, \quad y = 0,$$

so

$$z = 9x.$$

Taking  $x = 1$  gives

$$\boxed{\mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 9 \end{bmatrix}} \quad (\lambda = 3).$$

### Construct $P$ and $D$

Place the eigenvectors into the columns of  $P$  in the same order as the eigenvalues:

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 9 \end{bmatrix}.$$

Its determinant is

$$\det(P) = 7 \neq 0 \quad (\text{linear independence / invertibility}).$$

Thus  $P$  is invertible.

The corresponding diagonal matrix is

$$D = \begin{bmatrix} -4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

### Verification

$$A\mathbf{v}_1 = \begin{bmatrix} -4 \\ 0 \\ -8 \end{bmatrix} = -4\mathbf{v}_1,$$

$$A\mathbf{v}_2 = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} = -2\mathbf{v}_2,$$

and

$$A\mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ 27 \end{bmatrix} = 3\mathbf{v}_3.$$

Therefore,

$$AP = PD$$

and hence

$$P^{-1}AP = D.$$

### Final answer

$$\lambda = -4, -2, 3$$

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 9 \end{bmatrix}, \quad D = \begin{bmatrix} -4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

#### Learning point.

Once a complete independent set of eigenvectors is found, they become the columns of  $P$ , while the corresponding eigenvalues occupy the diagonal of  $D$  in exactly the same order.



## Question 7

### Question.

Consider the triangular matrix

$$A = \begin{bmatrix} 3 & -2 & 4 \\ 0 & 5 & -4 \\ 0 & 0 & 3 \end{bmatrix}.$$

Determine whether  $A$  is diagonalisable. If so, provide three linearly independent eigenvectors as columns of  $P$  and give the corresponding eigenvalues in the same order.

**Concepts / properties used:** Triangular-matrix eigenvalues; repeated eigenvalue; eigenspace; linear independence; diagonalisation criterion.

### Working.

Because  $A$  is triangular, its eigenvalues are the diagonal entries:

$$\boxed{\lambda = 3, 5, 3}. \quad (\text{triangular-matrix property})$$

Thus  $\lambda = 3$  is repeated.

### Eigenvectors for $\lambda = 3$

Solve

$$(A - 3I)\mathbf{x} = \mathbf{0},$$

giving

$$\begin{bmatrix} 0 & -2 & 4 \\ 0 & 2 & -4 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

The independent equation is

$$-2y + 4z = 0,$$

so

$$y = 2z.$$

The variables  $x$  and  $z$  are free. Hence,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}.$$

Therefore two linearly independent eigenvectors for  $\lambda = 3$  are

$$\boxed{\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}}.$$

**Eigenvector for  $\lambda = 5$** 

Solve

$$(A - 5I)\mathbf{x} = \mathbf{0},$$

so

$$\begin{bmatrix} -2 & -2 & 4 \\ 0 & 0 & -4 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{0}.$$

The final equations give

$$z = 0.$$

Then

$$-2x - 2y = 0 \implies x = -y.$$

Choose  $y = 1$ :

$$\mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

**Determine diagonalisability**

We now have three linearly independent eigenvectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Thus,

$$P = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Its determinant is

$$\det(P) = -1 \neq 0 \quad (\text{independence} / \text{invertibility}).$$

Therefore,

$$A \text{ is diagonalisable.}$$

The eigenvalues corresponding to the columns of  $P$  are

$$\lambda = \begin{bmatrix} 3 & 3 & 5 \end{bmatrix}.$$

Equivalently,

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

### Verification

$$A\mathbf{v}_1 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} = 3\mathbf{v}_1,$$

$$A\mathbf{v}_2 = \begin{bmatrix} 0 \\ 6 \\ 3 \end{bmatrix} = 3\mathbf{v}_2,$$

and

$$A\mathbf{v}_3 = \begin{bmatrix} -5 \\ 5 \\ 0 \end{bmatrix} = 5\mathbf{v}_3.$$

Hence,

$$AP = PD$$

and therefore

$$P^{-1}AP = D.$$

### Final NUMBAS entries

Yes

$$P = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\lambda = (3, 3, 5).$$

#### Learning point.

A repeated eigenvalue does not automatically prevent diagonalisation. Here the eigenspace for  $\lambda = 3$  has dimension two, supplying the two independent vectors needed to obtain three independent eigenvectors overall.

# DEAKIN UNIVERSITY

## LINEAR ALGEBRA FOR DATA ANALYSIS

### ONTRACK SUBMISSION

---

# Module 7 - Least Squares Approximation and Abstract Vector Spaces

---

*Submitted By:*

Jahan GARG

s226100103

2026/09/12 04:04

*Tutor:*

Simon JAMES

| Task Outcomes |                                                                                                                                                       | Supports  |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| M7.1          | Apply the least squares approximation method using techniques acquired in linear algebra                                                              | ULO2 ULO2 |
| M7.2          | Apply definitions and properties of vector spaces to abstract sets beyond $\mathbb{R}^n$                                                              | ULO2 ULO2 |
| SA            | Full printout of Self-Assessment results with all questions.                                                                                          |           |
| Evid          | Learning notes, hand-written attempts at exercises and practice questions, self-assessment working, NUMBAS screenshots, etc.                          |           |
| Sum           | Written summary connecting all definitions and concepts provided in the template, as well as incorporating responses to all Module Summary Questions. |           |
| FB            | Respond to any feedback and work with the OnTrack tutor to complete the module.                                                                       |           |

September 11, 2026



## Module 7 Summary

### *Least Squares Approximation and Abstract Vector Spaces*

#### Response to feedback

This is my first submission for Module 7, so no prior feedback on this specific module is being addressed here. I have, however, carried forward the guidance Simon gave me on earlier modules, since it continues to shape how I approach every summary. From Module 1, Tanya's advice to link every definition and theorem together within the same section, rather than treating each one as an isolated fact, is applied throughout this document. From Module 2, Simon's note that the Learning Notes column is optional and that the summary itself can stay reasonably concise while still covering every listed definition and theorem has kept this write-up focused on explanation rather than repetition. From Modules 3 and 4, I have made sure that every matrix, vector and equation in the Key Definitions and Theorems table is preserved exactly as given in the template, with no reformatting that would break the bold key terms or the underlying LaTeX-style notation. From Module 5, I have continued typesetting every matrix and equation properly rather than writing them inline as text. From Module 6, I have included an explicit label or reference for every worked example used, whether it is drawn from the Nicholson textbook, the SIT292 unit site, or created independently for this submission.

For this first submission, I would most value feedback on the worked least squares line example in Section 7.2 below (Module Summary Question 2), specifically whether the level of working shown, from setting up the matrix  $M$  through to solving the normal equations and plotting the fitted line, is at the standard expected for a high distinction submission, or whether further detail or a different choice of data points would demonstrate the method more clearly.

#### Module Learning Objectives

I certify that I achieved the following learning objectives for the module (these objectives can be found in the introduction of the module):

- Apply the least squares approximation method using techniques acquired in linear algebra.
- Apply definitions and properties of vector spaces to abstract sets beyond  $\mathbb{R}^n$ .

## Key Definitions and Theorems

### Least squares approximation

| Textbook Ref                                             | Theorem/Definition statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learning Notes                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Theorem 5.6.1</b><br>Best Approximation Theorem       | <p>Let <math>A</math> be an <math>m \times n</math> matrix, let <math>\mathbf{b}</math> be any column in <math>\mathbb{R}^m</math>, and consider the system</p> $A\mathbf{x} = \mathbf{b}$ <p>of <math>m</math> equations in <math>n</math> variables.</p> <ol style="list-style-type: none"> <li>Any solution <math>\mathbf{z}</math> to the normal equations <math display="block">(A^T A)\mathbf{z} = A^T \mathbf{b}</math> is a <i>best approximation</i> to a solution to <math>A\mathbf{x} = \mathbf{b}</math> in the sense that <math>\ \mathbf{b} - A\mathbf{z}\ </math> is the minimum value of <math>\ \mathbf{b} - A\mathbf{x}\ </math> as <math>\mathbf{x}</math> ranges over all columns in <math>\mathbb{R}^n</math>.</li> <li>If the columns of <math>A</math> are linearly independent, then <math>A^T A</math> is invertible, and <math>\mathbf{z}</math> is given uniquely by <math display="block">\mathbf{z} = (A^T A)^{-1} A^T \mathbf{b}.</math></li> </ol> | This is the algebraic heart of the whole module. Rather than solving $A\mathbf{x} = \mathbf{b}$ directly when it has no solution, it converts the problem into a related, always-consistent system, which I applied repeatedly in Learning Evidence 7.2 and across every N7.2.1 variant of the NUMBAS practice.         |
| <b>Theorem 5.6.2</b><br>Least squares approximation line | <p>Suppose that <math>n</math> data points <math>(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)</math> are given, where at least two of <math>x_1, x_2, \dots, x_n</math> are distinct. Put</p> $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, M = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$ <p>Then the least-squares approximating line for these data points has equation</p> $y = z_0 + z_1 x$ <p>where <math>\mathbf{z} = \begin{bmatrix} z_0 \\ z_1 \end{bmatrix}</math> is found by Gaussian elimination from the normal equations</p> $(M^T M)\mathbf{z} = M^T \mathbf{y}$ <p>The condition that at least two of <math>x_1, x_2, \dots, x_n</math> are distinct ensures that <math>M^T M</math> is an invertible matrix, so <math>\mathbf{z}</math> is unique:</p> $\mathbf{z} = (M^T M)^{-1} M^T \mathbf{y}$                                                                                                           | This is Theorem 5.6.1 specialised to the particular choice $A = M$ , where the columns of $M$ are the constant function 1 and the identity function $x$ . I connect the two theorems explicitly in the Module Summary below, and used this specialisation directly for the worked example in Module Summary Question 2. |

### Abstract Vector Spaces

The following ten axioms, five for vector addition and five for scalar multiplication, define what it means for an arbitrary set  $V$ , together with two operations, to be a vector space over  $\mathbb{R}$  (Nicholson, Chapter 6, p. 328).

#### Axioms for vector addition

- A1. If  $\mathbf{u}$  and  $\mathbf{v}$  are in  $V$ , then  $\mathbf{u} + \mathbf{v}$  is in  $V$ .
- A2.  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$  for all  $\mathbf{u}$  and  $\mathbf{v}$  in  $V$ .
- A3.  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$  for all  $\mathbf{u}, \mathbf{v}$ , and  $\mathbf{w}$  in  $V$ .
- A4. An element  $\mathbf{0}$  in  $V$  exists such that  $\mathbf{v} + \mathbf{0} = \mathbf{v} = \mathbf{0} + \mathbf{v}$  for every  $\mathbf{v}$  in  $V$ .
- A5. For each  $\mathbf{v}$  in  $V$ , an element  $-\mathbf{v}$  in  $V$  exists such that  $-\mathbf{v} + \mathbf{v} = \mathbf{0}$  and  $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ .

#### Axioms for scalar multiplication

- S1. If  $\mathbf{v}$  is in  $V$ , then  $a\mathbf{v}$  is in  $V$  for all  $a$  in  $\mathbb{R}$ .
- S2.  $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$  for all  $\mathbf{v}$  and  $\mathbf{w}$  in  $V$  and all  $a$  in  $\mathbb{R}$ .
- S3.  $(a+b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$  for all  $\mathbf{v}$  in  $V$  and all  $a$  and  $b$  in  $\mathbb{R}$ .
- S4.  $a(b\mathbf{v}) = (ab)\mathbf{v}$  for all  $\mathbf{v}$  in  $V$  and all  $a$  and  $b$  in  $\mathbb{R}$ .
- S5.  $1\mathbf{v} = \mathbf{v}$  for all  $\mathbf{v}$  in  $V$ .

| Textbook Ref                        | Theorem/Definition statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Learning Notes                                                                                                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Example 6.1.2</b><br>Matrices    | <p>The set <math>\mathbf{M}_{mn}</math> of all <math>m \times n</math> matrices is a vector space using matrix addition and scalar multiplication.</p> <p>The <u>zero element</u> in this vector space is the zero matrix of size <math>m \times n</math>, and the vector space negative of a matrix (required by axiom A5) is the usual matrix negative discussed in Section 2.1. Note that <math>\mathbf{M}_{mn}</math> is just <math>\mathbb{R}^{mn}</math> in different notation.</p>                                                                                                                                                                                                                                                  | I checked all ten axioms for this example directly in Module Summary Question 3 below, and used $M_{2,2}(\mathbb{R})$ throughout the digital learning evidence for N7.3.1 and N7.3.5.                       |
| <b>Example 6.1.6</b><br>Polynomials | <p>Given <math>n \geq 1</math>, let <math>\mathbf{P}_n</math> denote the set of all polynomials of degree at most <math>n</math>, together with the zero polynomial. That is</p> $\mathbf{P}_n = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_0, a_1, a_2, \dots, a_n \text{ in } \mathbb{R}\}$ <p>Then <math>\mathbf{P}_n</math> is a vector space. Indeed, sums and scalar multiples of polynomials in <math>\mathbf{P}_n</math> are again in <math>\mathbf{P}_n</math>, and the other vector space axioms are inherited from <math>\mathbf{P}</math> (the set of all polynomials). In particular, the zero vector and the negative of a polynomial in <math>\mathbf{P}_n</math> are the same as those in <math>\mathbf{P}</math>.</p> | I also checked all ten axioms for this example in Question 3, and then constructed a counterexample for Question 4 to show why restricting to polynomials of degree exactly $n$ fails to be a vector space. |

## Module Summary

Module 7 draws together two applications that, on the surface, look quite different but both rely on the same underlying strategy of reframing a hard problem as a more tractable one using matrix algebra. Best Approximation and Least Squares (5.6) takes an inconsistent system, one with no exact solution, and replaces it with a related consistent system whose solution is provably the closest possible approximation. Examples and Basic Properties of Abstract Vector Spaces (6.1) then steps back from  $\mathbb{R}^n$  entirely and asks which other mathematical objects, such as matrices and polynomials, satisfy the same ten axioms that make  $\mathbb{R}^n$  behave the way it does, so that every tool built for  $\mathbb{R}^n$  in earlier modules, including span, independence, basis and dimension, becomes available in these new settings as well.

## 7.2 Best Approximation and Least Squares

### Relationship between Theorem 5.6.1 and Theorem 5.6.2 (Module Summary

**Question 1).** Theorem 5.6.1, the Best Approximation Theorem, is the general result: for any  $m \times n$  matrix  $A$  and any column  $\mathbf{b}$  in  $\mathbb{R}^m$ , the vector  $\mathbf{z}$  that minimises  $\|\mathbf{b} - A\mathbf{x}\|$  is obtained by solving the normal equations  $(A^T A)\mathbf{z} = A^T \mathbf{b}$ , and this  $\mathbf{z}$  is unique whenever the columns of  $A$  are independent. Theorem 5.6.2 is not a separate result but a direct specialisation of Theorem 5.6.1 to the specific problem of fitting a straight-line  $y = z_0 + z_1x$  through  $n$  data points. In this specialisation, the general matrix  $A$  is replaced by the particular  $n \times 2$  matrix  $M$  whose first column is entirely 1s and whose second column holds the  $x$ -values, and the general column  $\mathbf{b}$  becomes the column  $\mathbf{y}$  of observed  $y$ -values. The condition that the columns of  $A$  be independent, required for uniqueness in Theorem 5.6.1, becomes the more concrete condition that at least two of the  $x$ -values be



distinct, since this is precisely what stops the second column of  $M$  from being a multiple of the first. Everything else, the construction of the normal equations, the interpretation of  $\mathbf{z}$  as the minimiser of a distance, and the uniqueness argument through invertibility of  $M^T M$ , carries over unchanged. I found it useful to think of Theorem 5.6.2 as Theorem 5.6.1 with the answer already half-filled in: the matrix  $A$  is no longer arbitrary but is fixed by the shape of a straight line, which is what makes the line fitting problem so mechanical once the general theorem is understood.

**Worked example of the least squares line (Module Summary Question 2).** To illustrate Theorem 5.6.2 concretely, I chose five points in the  $xy$ -plane,  $(1, 2)$ ,  $(2, 3)$ ,  $(3, 5)$ ,  $(4, 4)$  and  $(5, 6)$ , selected so that no straight line passes through all five exactly, which is the situation the theorem is designed for. Figure 1 below shows these points together with the fitted line obtained from the working that follows.

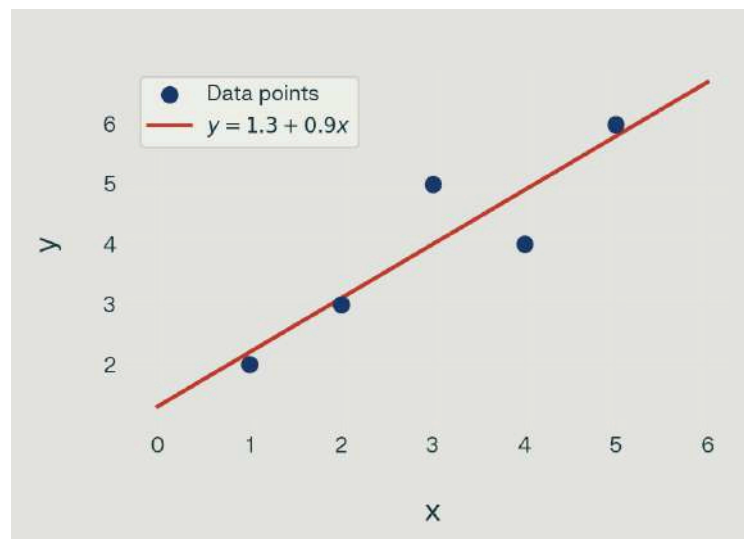


Figure 1. The five chosen data points together with the least squares approximating line  $y = 1.3 + 0.9x$ , obtained from Theorem 5.6.2 (own worked example, created using Python and Matplotlib).

Following the steps of Theorem 5.6.2 exactly, the matrix  $M$  and column  $\mathbf{y}$  are  $M =$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 4 \\ 6 \end{bmatrix}^T, \text{ where each row of } M \text{ pairs the constant 1 with one of the five } x\text{-}$$

values. Computing  $M^T M$  gives the  $2 \times 2$  matrix  $\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}$ , since the top left entry is simply  $n = 5$ , the off-diagonal entries are the sum of the  $x$ -values ( $1 + 2 + 3 + 4 + 5 = 15$ ), and the bottom right entry is the sum of the squared  $x$ -values ( $1 + 4 + 9 + 16 + 25 = 55$ ).

Computing  $M^T \mathbf{y}$  gives  $\begin{bmatrix} 20 \\ 69 \end{bmatrix}^T$ , since the sum of the  $y$ -values is  $2 + 3 + 5 + 4 + 6 = 20$ , and the sum of  $x_i y_i$  is  $(1)(2) + (2)(3) + (3)(5) + (4)(4) + (5)(6) = 2 + 6 + 15 + 16 + 30 = 69$ . The normal equations  $(M^T M)\mathbf{z} = M^T \mathbf{y}$  therefore become  $5z_0 + 15z_1 = 20$  and  $15z_0 + 55z_1 = 69$ . Solving the first equation for  $z_0$  gives  $z_0 = 4 - 3z_1$ , and substituting into the second equation gives  $15(4 - 3z_1) + 55z_1 = 69$ , which simplifies to  $10z_1 = 9$ , so  $z_1 = 9/10 = 0.9$

and, back substituting,  $z_0 = 4 - (27/10) = 13/10 = 1.3$ . Since two of the  $x$ -values are certainly distinct,  $M^T M$  is invertible by Theorem 5.6.2, confirming that this solution is the unique best approximation. The resulting least squares line is therefore  $y = 1.3 + 0.9x$ , exactly as drawn in Figure 1, and this matches the same normal-equation procedure I applied to every N7.2.1 and **N7.2.2** NUMBAS practice question in the digital learning evidence, where I additionally verified each solution by confirming that the residual  $M^T(\mathbf{y} - M\mathbf{z})$  is the zero vector, as required by the orthogonality property underlying Theorem 5.6.1.

Beyond the line-fitting case, Theorem 5.6.1 extends naturally to more general approximating functions built from any chosen basis functions rather than just 1 and  $x$ , which I explored directly in Nicholson Exercise 5.6.4(a) [1] in the handwritten learning evidence by fitting a function of the form  $f(x) = r_0x + r_1x^2 + r_2x^3$  to four data points, using the same normal-equation machinery with a  $4 \times 3$  matrix  $M$  in place of the usual 2 column line-fitting matrix. Nicholson Exercise 5.6.11 [1] then examines the boundary case where the number of data points exactly equals the number of unknown coefficients, showing that  $M$  becomes a square Vandermonde matrix whose determinant is non-zero whenever the  $x$ -values are distinct, so that the least squares polynomial collapses to the ordinary interpolating polynomial and the approximation error becomes exactly zero. Finally, Nicholson Exercise 5.6.12 [1] proves that whenever a best approximation is not unique, because the columns of  $A$  are not independent, every minimising vector differs from any particular one by an element of the null space of  $A^T A$ , which connects the least squares problem directly back to the null space and rank ideas first introduced in Module 1 and formalised further in Module 6.

### 7.3 Examples and Basic Properties of Abstract Vector Spaces

**Checking the ten axioms for Example 6.1.2, the matrix space  $M_{mn}(\mathbb{R})$  (Module Summary Question 3).** The set of all  $m \times n$  real matrices satisfies every one of the ten vector space axioms because matrix addition and scalar multiplication are both defined entry by entry, and every axiom reduces to the corresponding axiom for real numbers applied one entry at a time. Closure (A1 and S1) holds because adding two  $m \times n$  matrices, or multiplying one by a real scalar, produces another matrix of the same size, as I confirmed directly in N7.3.1 by representing four given  $2 \times 2$  matrices as vectors in  $\mathbb{R}^4$  and checking that every linear combination of them remained a  $2 \times 2$  matrix. Commutativity and associativity (A2 and A3) follow immediately since real-number addition is itself commutative and associative in every entry, so, for example, swapping two  $2 \times 2$  matrices  $A$  and  $B$  in  $A + B$  simply swaps the order of addition in each of the four entries, which does not change the result. The zero vector (A4) is the  $m \times n$  zero matrix, since adding it to any matrix  $A$  leaves every entry of  $A$  unchanged, and the additive inverse (A5) is the usual matrix negative  $-A$ , obtained by negating every entry, since  $A + (-A)$  then has every entry equal to  $a_{ij} + (-a_{ij}) = 0$ . For scalar multiplication, distributivity over vector addition (S2) and over scalar addition (S3) both reduce to the distributive law for real numbers applied entry by entry, associativity of scalars (S4) reduces to

associativity of real multiplication, and the scalar identity (S5) holds because multiplying every entry by 1 leaves it unchanged. I found this example particularly clarifying because, as the textbook itself notes,  $M_{mn}$  is simply  $\mathbb{R}^{mn}$  written in a different shape, so every axiom that already felt intuitive for column vectors in Module 5 transfers across without any real change in the underlying argument, only in how the entries happen to be arranged on the page.

**Checking the ten axioms for Example 6.1.6, the polynomial space  $P_n$  (Module Summary Question 3, continued).** The set  $P_n$  of all polynomials of degree at most  $n$ , including the zero polynomial, satisfies the same ten axioms for a closely related reason: once a polynomial is identified with the ordered list of its coefficients  $(a_0, a_1, \dots, a_n)$ , it becomes, in every respect that matters for the axioms, an  $(n + 1)$ -tuple, so  $P_n$  behaves exactly like  $\mathbb{R}^{n+1}$  in different notation. Closure (A1 and S1) holds because adding two polynomials of degree at most  $n$ , or scaling one by a real number, can never produce a polynomial of degree greater than  $n$ ; at worst, the leading coefficients cancel and the degree decreases, but it can never increase. Commutativity and associativity (A2 and A3) hold because polynomial addition is defined coefficient by coefficient, and each coefficient obeys the ordinary commutative and associative laws of real-number addition. The zero vector (A4) is the zero polynomial itself, all of whose coefficients are zero, and the additive inverse (A5) of a polynomial  $p(x)$  is simply  $-p(x)$ , obtained by negating every coefficient. Distributivity (S2 and S3), scalar associativity (S4) and the scalar identity (S5) all reduce, exactly as for matrices, to the corresponding properties of real-number arithmetic applied to each coefficient in turn. I applied this coefficient-vector correspondence directly in the digital learning evidence for **N7.3.2** and **N7.3.4**, where questions about the basis and dimension of a subspace of polynomials were solved by converting each polynomial into its coefficient vector in  $\mathbb{R}^{n+1}$  and then applying the same rank and row-reduction methods used for ordinary vectors since Module 2.

**Counterexample for polynomials of degree exactly  $n$  (Module Summary Question 4).** If  $P_n$  is redefined to mean only the polynomials whose degree is exactly  $n$ , excluding both lower degree polynomials and the zero polynomial, then this set fails to be a vector space, and two of the ten axioms break down simultaneously. Taking  $n = 2$  as a concrete case, consider  $p(x) = x^2 + x$  and  $q(x) = -x^2$ , both of which have degree exactly 2 and therefore both lie in this restricted set. Their sum is  $p(x) + q(x) = x$ , which has degree 1, not degree 2, so the sum does not belong to the set at all. This directly violates closure under addition, axiom A1, since the set is not closed under the very operation the axiom requires it to be closed under. A second, equally immediate failure occurs at axiom A4: the zero polynomial, whose degree is conventionally undefined or taken to be negative infinity rather than equal to any fixed  $n$ , is not a member of this restricted set by its own definition, so no element of the set can serve as the required zero vector, and the axiom fails before any calculation is even needed. These two failures show that the degree at most  $n$  condition in the genuine Example 6.1.6 is doing essential work: it is precisely the inclusion of the zero polynomial and of every lower degree polynomial that keeps the

set closed under addition and guarantees the existence of a zero element, and removing that flexibility by insisting on degree exactly  $n$  breaks the vector space structure immediately.

Beyond the two set examples given in the template, I also investigated a genuinely non-standard vector space in the handwritten learning evidence, based on Nicholson Exercise 6.1.4 [1], where  $V = \mathbb{R}^2$  is equipped with the unusual operations  $(x, y) + (x_1, y_1) = (x + x_1, y + y_1 + 1)$  and  $a(x, y) = (ax, ay + a - 1)$ . Working through all ten axioms directly from these definitions, rather than assuming the familiar componentwise rules, showed that the zero vector is not  $(0, 0)$  but  $(0, -1)$ , and that the additive inverse of  $(x, y)$  is  $(-x, -y - 2)$  rather than the expected  $(-x, -y)$ . This example, together with Nicholson Exercise 6.1.2(g) [1], where the set of  $2 \times 2$  matrices with zero determinant fails closure under addition since a matrix such as

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

gives the identity matrix with determinant 1, and Nicholson Exercise 6.1.15(a) [1], which proves that  $av = \mathbf{0}$  implies  $a = 0$  or  $v = \mathbf{0}$  using only the axioms and the existence of a scalar inverse, reinforced for me that the vector space axioms are structural conditions on whatever operations are actually defined, not properties that follow automatically just because a set happens to look like  $\mathbb{R}^n$ .

### Observations and Connections Across Sub-modules

Although 7.2 and 7.3 look unrelated at first glance, one dealing with approximating a line through scattered data and the other with abstract algebra, both sub-modules rest on the same idea of transferring familiar  $\mathbb{R}^n$  machinery to a new setting. Least squares approximation works because the columns of  $A$ , or of  $M$ , are themselves vectors in  $\mathbb{R}^m$ , so the entire theory of projections, normal equations and invertibility developed for ordinary vectors in earlier modules applies without modification. Abstract vector spaces work in the reverse direction: instead of vectors being mapped into  $\mathbb{R}^n$ , whole new sets, matrices and polynomials, are shown to already satisfy the axioms that make  $\mathbb{R}^n$  useful in the first place, so that once the coefficient-vector or entry-vector correspondence is established, questions about span, independence, basis and dimension can be answered using exactly the same rank and row-reduction techniques from Module 2 onward. This connection was concrete throughout the NUMBAS practice: N7.3.1, N7.3.2, N7.3.4 and N7.3.5 all reduced abstract basis and dimension questions to ordinary matrix rank calculations the moment the matrices or polynomials were rewritten as coordinate vectors, which is the same underlying idea that makes Theorem 5.6.2 a specific instance of Theorem 5.6.1 rather than an independent result.

### Extra Effort and Prior Knowledge

Beyond the prescribed unit-site material, I consulted a short GeeksforGeeks article on the method of least squares [3] to check that my explanation of the normal equations in

this summary matched standard terminology used outside this unit, and the Klemm supplementary background resource on vector spaces [4], originally provided for Module 5, which gave a slightly different informal framing of the ten axioms that helped me phrase the axiom-checking arguments above more precisely. In terms of prior knowledge, the idea of minimising a sum of squared errors was already familiar to me in an informal sense from basic statistics, but this module was the first time I saw the underlying linear algebra, the normal equations and their derivation from Theorem 5.6.1, that actually explains why the least squares line is the one that minimises this error. Similarly, the notion of a vector space had, until this module, been tied entirely to  $\mathbb{R}^n$  in my mind; working through Examples 6.1.2 and 6.1.6, and especially the non-standard operations in Nicholson Exercise 6.1.4, was what made the axioms feel like genuine structural conditions rather than properties that come for free.

## Reflecting on the Content

The idea I found most satisfying in this module was realising that Theorem 5.6.2 required no new mathematics of its own once Theorem 5.6.1 was properly understood; the entire line-fitting procedure is just the general best approximation argument with a specific, very simple matrix  $A$ . Working through the five-point example in Question 2 made this concrete for me in a way that reading the theorem alone did not, since setting up  $M$  and  $y$  by hand and watching the same  $2 \times 2$  normal equations appear that I had already solved several times in the NUMBAS practice confirmed that I actually understood the method rather than just recognising its final formula. The most conceptually demanding part of the module was constructing the counterexample for Question 4, since it required me to think carefully about exactly which axiom fails and why, rather than simply asserting that the restricted set is not a vector space. Finding that both closure and the existence of a zero element fail simultaneously, for what is essentially the same underlying reason, that the zero polynomial and lower degree polynomials are needed to keep the set well behaved under addition, was the moment this module's abstract material felt genuinely connected to the concrete vector space intuition built up since Module 5. Completing the self-assessment with a full 10 out of 10 across all six questions confirmed that treating least squares approximation and abstract vector spaces as two applications of the same underlying  $\mathbb{R}^n$  toolkit, rather than as two unrelated topics, was the right way to approach this comparatively lighter module.

## Tools and Originality Statement

The five data points used in the worked example for Module Summary Question 2 were chosen independently for this submission and do not reproduce any example from the textbook or unit site. All matrix calculations throughout this document, including the normal equations, matrix products and the coefficients of the fitted line, were independently verified using Python (NumPy and SymPy) before being written up, and

Figure 1 was generated using Python and Matplotlib. The counterexample constructed for Module Summary Question 4 was developed independently, and the axiom-checking arguments for Examples 6.1.2 and 6.1.6 were written in my own words after working through the definitions in Nicholson's textbook. The handwritten and digital learning evidence referenced throughout Sections 7.2 and 7.3, and the NUMBAS self-assessment, on which I scored 10 out of 10, were completed and submitted separately as part of this module's Learning Evidence and Self-Assessment documents. An AI writing assistant (Perplexity) was used to help structure this summary, refine explanations of the theorems and check written English; the mathematical reasoning, worked examples, and self-assessment results referenced throughout are my own.

## References

- [1] W. K. Nicholson, Linear Algebra with Applications, Open Edition, Version 2021A, Lyryx Learning Inc., Calgary, AB, Canada, 2021, Sec. 5.6 (Best Approximation and Least Squares) and Sec. 6.1 (Examples and Basic Properties of Vector Spaces), including Exercises 5.6.4(a), 5.6.11, 5.6.12, 6.1.2(g), 6.1.4 and 6.1.15(a).
- [2] Deakin University, "Topic 7: Least Squares Approximation and Abstract Vector Spaces - 7.1 Introduction, 7.2 Best Approximation and Least Squares, 7.3 Examples and Basic Properties of Abstract Vector Spaces," SIT292 Linear Algebra for Data Analysis, CloudDeakin Unit Site, Deakin University, Trimester 2, 2026.
- [3] GeeksforGeeks, "Least Square Regression Line," GeeksforGeeks, accessed Sep. 2026. [Online]. Available: <https://www.geeksforgeeks.org>
- [4] A. D. Klemm, "Background Summary - Vector Spaces in Linear Algebra," supplementary resource, SIT292 Linear Algebra for Data Analysis, CloudDeakin Unit Site, Deakin University, 2026.
- [5] Deakin University, SIT292 Module 7 Self-Assessment, Numbas, developed by Newcastle University, accessed 11 Sep. 2026.

# Module 7: Handwritten Learning Evidence

## Section 7.2 – Best Approximation and Least Squares

### Question 1 – Nicholson Exercise 5.6.4(a)

#### Question.

Find a least-squares approximating function of the form

$$f(x) = r_0x + r_1x^2 + r_22^x$$

for the data pairs

$$(-1, 1), \quad (0, 3), \quad (1, 1), \quad (2, 0).$$

#### Concepts / properties applied:

Theorem 5.6.1 (Best Approximation Theorem); Theorem 5.6.4 (general least-squares approximation); normal equations.

#### Method.

Use the three basis functions

$$x, \quad x^2, \quad 2^x$$

to form the matrix  $M$ , then solve the normal equations

$$(M^T M)z = M^T y.$$

#### Step-by-step working

Let

$$z = \begin{bmatrix} r_0 \\ r_1 \\ r_2 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 0 \end{bmatrix}.$$

Evaluating the basis functions at the given  $x$ -values gives

$$M = \begin{bmatrix} -1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 4 & 4 \end{bmatrix}.$$

Hence,

$$(M^T M)z = M^T y \quad (\text{Theorem 5.6.1}).$$

First,

$$M^T = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ \frac{1}{2} & 1 & 2 & 4 \end{bmatrix}.$$

Therefore,



$$\begin{aligned}
M^T M &= \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ \frac{1}{2} & 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & \frac{1}{2} \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 4 & 4 \end{bmatrix} \\
&= \begin{bmatrix} 6 & 8 & \frac{19}{2} \\ 8 & 19 & \frac{39}{2} \\ \frac{19}{2} & \frac{39}{2} & \frac{85}{4} \end{bmatrix} \quad (\text{Matrix multiplication}).
\end{aligned}$$

Also,

$$M^T y = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 4 \\ \frac{1}{2} & 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ \frac{11}{2} \end{bmatrix} \quad (\text{Matrix multiplication}).$$

Thus,

$$\begin{bmatrix} 6 & 8 & \frac{19}{2} \\ 8 & 19 & \frac{39}{2} \\ \frac{19}{2} & \frac{39}{2} & \frac{85}{4} \end{bmatrix} \begin{bmatrix} r_0 \\ r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ \frac{11}{2} \end{bmatrix}.$$

Multiplying through by 4,

$$\begin{aligned}
24r_0 + 32r_1 + 38r_2 &= 0, \\
32r_0 + 76r_1 + 78r_2 &= 20, \quad (\text{Equivalent system}) \\
38r_0 + 78r_1 + 85r_2 &= 22.
\end{aligned}$$

Eliminating  $r_0$  by combining the first two equations gives

$$100r_1 + 82r_2 = 60 \quad (\text{Gaussian elimination}).$$

Similarly, eliminating  $r_0$  using the first and third equations gives

$$164r_1 + 149r_2 = 132 \quad (\text{Gaussian elimination}).$$

From the first equation,

$$r_1 = \frac{30 - 41r_2}{50}.$$

Substituting into the second equation,

$$164 \left( \frac{30 - 41r_2}{50} \right) + 149r_2 = 132,$$

so

$$r_2 = \frac{280}{121}.$$

Therefore,

$$r_1 = \frac{30 - 41 \left( \frac{280}{121} \right)}{50} = -\frac{157}{121}.$$

Finally, using

$$24r_0 + 32r_1 + 38r_2 = 0,$$

we obtain

$$r_0 = -\frac{234}{121}.$$

### Verification

Using these coefficients,

$$Mz = \begin{bmatrix} \frac{217}{121} \\ \frac{123}{121} \\ \frac{121}{169} \\ \frac{121}{24} \\ \frac{24}{121} \end{bmatrix}.$$

Hence,

$$y - Mz = \begin{bmatrix} -\frac{96}{121} \\ \frac{240}{121} \\ \frac{48}{121} \\ -\frac{24}{121} \\ -\frac{24}{121} \end{bmatrix}.$$

Now,

$$M^T(y - Mz) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (\text{Normal-equation verification}).$$

Thus the residual satisfies the required orthogonality condition.

### Final answer

$$f(x) = -\frac{234}{121}x - \frac{157}{121}x^2 + \frac{280}{121}2^x$$

#### Key Learning Point.

The least-squares method is not restricted to straight-line fitting. The same normal-equation framework applies to a linear combination of different basis functions, with the columns of  $M$  representing those functions.

## Question 2 – Nicholson Exercise 5.6.11

### Question.

Assume  $n = m + 1$  in Theorem 5.6.3, so that  $M$  is square. If the  $x_i$  are distinct, show that  $M$  is invertible. Deduce that

$$z = M^{-1}y$$

and that the least-squares polynomial is the interpolating polynomial and actually passes through all the data points.

**Concepts / properties applied:**

Theorem 3.2.6 (interpolating polynomial); Theorem 5.6.1 (Best Approximation Theorem); Theorem 5.6.3 (least-squares polynomial); Vandermonde determinant; invertibility criterion.

**Method.**

Since  $n = m + 1$ ,  $M$  is a square Vandermonde matrix. Show that its determinant is non-zero for distinct  $x_i$ , then simplify the normal equations.

**Step-by-step working**

The matrix is

$$M = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^m \\ 1 & x_2 & x_2^2 & \cdots & x_2^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^m \end{bmatrix}.$$

Since the  $x_i$  are distinct,

$$\det(M) = \prod_{1 \leq i < j \leq n} (x_j - x_i) \neq 0 \quad (\text{Vandermonde determinant}).$$

Therefore,

$$M \text{ is invertible} \quad (\text{Invertibility criterion}).$$

The normal equations are

$$(M^T M)z = M^T y \quad (\text{Theorem 5.6.1}).$$

Since  $M$  and  $M^T$  are invertible,

$$\begin{aligned} z &= (M^T M)^{-1} M^T y \\ &= M^{-1} (M^T)^{-1} M^T y \quad (\text{Inverse-matrix property}) \\ &= M^{-1} y. \end{aligned}$$

Thus,

$$\boxed{z = M^{-1} y}.$$

**Verification**

Multiplying by  $M$ ,

$$Mz = y \quad (\text{Matrix multiplication}).$$

But

$$Mz = \begin{bmatrix} p(x_1) \\ p(x_2) \\ \vdots \\ p(x_n) \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

Therefore,

$$p(x_i) = y_i, \quad i = 1, \dots, n \quad (\text{Interpolation condition}).$$

Hence,

$$\|y - Mz\| = 0.$$

The least-squares polynomial therefore passes exactly through every data point.

**Final answer**

$$\det(M) \neq 0 \implies M \text{ is invertible}$$

$$z = M^{-1}y$$

and

$$p(x_i) = y_i \quad \text{for every } i$$

so the least-squares polynomial is exactly the interpolating polynomial.

**Key Learning Point.**

With exactly  $m + 1$  data points and distinct  $x_i$ -values, the least-squares problem becomes exact interpolation: the minimum possible error is zero.

### Question 3 – Nicholson Exercise 5.6.12

**Question.**

Let  $A$  be any  $m \times n$  matrix and define

$$K = \{x : A^T A x = 0\}.$$

Let  $b$  be an  $m$ -column. Show that, if  $z$  is an  $n$ -column such that  $\|b - Az\|$  is minimal, then all such vectors have the form

$$z + x$$

for some  $x \in K$ .

**Concepts / properties applied:**

Theorem 5.6.1 (Best Approximation Theorem); normal equations; linearity of matrix multiplication; definition of  $K$ .

**Method.**

Take two vectors that both produce the minimum residual. Show that their difference belongs to  $K$ , then prove the converse by showing that adding any vector from  $K$  preserves the normal equations.

**Step-by-step working**

Since  $z$  gives the minimum value of

$$\|b - Az\|,$$

it satisfies

$$A^T Az = A^T b \quad (\text{Theorem 5.6.1}).$$

Now let  $y$  be another vector giving the same minimum. Then

$$A^T Ay = A^T b \quad (\text{Normal-equation characterisation}).$$

Therefore,

$$A^T Ay = A^T Az.$$

Subtracting gives

$$A^T A(y - z) = 0 \quad (\text{Linearity}).$$

Let

$$x = y - z.$$

Then

$$A^T Ax = 0,$$

so

$$x \in K \quad (\text{Definition of } K).$$

Therefore,

$$y = z + x, \quad x \in K.$$

**Converse**

Now take any  $x \in K$ . Then

$$A^T Ax = 0 \quad (\text{Definition of } K).$$

Since  $z$  satisfies the normal equations,

$$A^T Az = A^T b.$$

Therefore,

$$\begin{aligned} A^T A(z + x) &= A^T Az + A^T Ax \\ &= A^T b + 0 && \text{(Linearity)} \\ &= A^T b. \end{aligned}$$

Thus  $z + x$  also satisfies the normal equations and is therefore also a best approximation.

(Theorem 5.6.1)

### Verification

We have established both directions:

$$y \text{ is a minimiser} \implies y = z + x, \quad x \in K,$$

and

$$x \in K \implies z + x \text{ is a minimiser.}$$

Therefore, the complete set of best-approximation vectors is

$$z + K \quad \text{(Solution-set structure).}$$

### Final answer

$$\boxed{\{y : \|b - Ay\| \text{ is minimal}\} = z + K}$$

where

$$\boxed{K = \{x : A^T Ax = 0\}.$$

Equivalently,

$$\boxed{y = z + x, \quad x \in K.}$$

#### Key Learning Point.

The normal equations describe the complete family of minimisers when uniqueness is not guaranteed. Any two best-approximation vectors differ by a vector in  $K$ , linking least-squares approximation with null-space structure.

## Section 7.3 – Examples and Basic Properties of Abstract Vector Spaces

---

### Question 1 – Nicholson Exercise 6.1.4

#### Question.

Let  $V$  be the set of ordered pairs  $(x, y) \in \mathbb{R}^2$ , with the operations

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1 + 1)$$

and

$$a(x, y) = (ax, ay + a - 1).$$

Show that  $V$  is a vector space and determine its zero vector.

#### Concepts / properties applied:

A1–A5; S1–S5; closure; zero vector; additive inverse; field properties.

#### Method.

Because the operations are non-standard, verify the vector-space axioms directly using the operations given.

#### Step-by-step working

##### A1 – Closure under addition

For  $(x, y), (x_1, y_1) \in V$ ,

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1 + 1) \in \mathbb{R}^2. \quad (\text{A1})$$

Hence the sum remains in  $V$ .

##### A2 – Commutativity

$$\begin{aligned} (x, y) + (x_1, y_1) &= (x + x_1, y + y_1 + 1) \\ &= (x_1 + x, y_1 + y + 1) \quad (\text{A2}) \\ &= (x_1, y_1) + (x, y). \end{aligned}$$

##### A3 – Associativity

$$\begin{aligned} ((x, y) + (x_1, y_1)) + (x_2, y_2) &= (x + x_1, y + y_1 + 1) + (x_2, y_2) \\ &= (x + x_1 + x_2, y + y_1 + y_2 + 2), \end{aligned}$$

while

$$\begin{aligned} (x, y) + ((x_1, y_1) + (x_2, y_2)) &= (x, y) + (x_1 + x_2, y_1 + y_2 + 1) \\ &= (x + x_1 + x_2, y + y_1 + y_2 + 2). \end{aligned}$$

Therefore,



$$\boxed{\left((x, y) + (x_1, y_1)\right) + (x_2, y_2) = (x, y) + \left((x_1, y_1) + (x_2, y_2)\right)}. \quad (\text{A3})$$

**A4 – Zero vector**

Let the zero vector be  $(p, q)$ . Then

$$(x, y) + (p, q) = (x, y).$$

Hence,

$$(x + p, y + q + 1) = (x, y),$$

so

$$p = 0, \quad q = -1.$$

Thus,

$$\boxed{\mathbf{0} = (0, -1)}. \quad (\text{A4})$$

A direct check gives

$$(x, y) + (0, -1) = (x, y - 1 + 1) = (x, y).$$

**A5 – Additive inverse**

Let

$$-(x, y) = (r, s).$$

Then

$$(x, y) + (r, s) = (0, -1),$$

so

$$x + r = 0, \quad y + s + 1 = -1.$$

Therefore,

$$r = -x, \quad s = -y - 2.$$

Hence,

$$\boxed{-(x, y) = (-x, -y - 2)}. \quad (\text{A5})$$

Indeed,

$$(x, y) + (-x, -y - 2) = (0, -1).$$

**S1 – Closure under scalar multiplication**

For  $a \in \mathbb{R}$ ,

$$a(x, y) = (ax, ay + a - 1) \in \mathbb{R}^2. \quad (\text{S1})$$

Thus scalar multiplication remains in  $V$ .

### S2 – Distributivity over vector addition

$$\begin{aligned} a((x, y) + (x_1, y_1)) &= a(x + x_1, y + y_1 + 1) \\ &= (a(x + x_1), a(y + y_1 + 1) + a - 1) \\ &= (ax + ax_1, ay + ay_1 + 2a - 1). \end{aligned}$$

On the other hand,

$$\begin{aligned} a(x, y) + a(x_1, y_1) &= (ax, ay + a - 1) + (ax_1, ay_1 + a - 1) \\ &= (ax + ax_1, ay + ay_1 + 2a - 1). \end{aligned}$$

Therefore,

$$\boxed{a(u + v) = au + av}. \quad (\text{S2})$$

### S3 – Distributivity over scalar addition

$$(a + b)(x, y) = ((a + b)x, (a + b)y + a + b - 1).$$

Also,

$$\begin{aligned} a(x, y) + b(x, y) &= (ax, ay + a - 1) + (bx, by + b - 1) \\ &= ((a + b)x, (a + b)y + a + b - 1). \end{aligned}$$

Hence,

$$\boxed{(a + b)v = av + bv}. \quad (\text{S3})$$

### S4 – Compatibility of scalar multiplication

$$\begin{aligned} a(b(x, y)) &= a(bx, by + b - 1) \\ &= (abx, a(by + b - 1) + a - 1) \\ &= (abx, aby + ab - 1). \end{aligned}$$

But

$$(ab)(x, y) = (abx, aby + ab - 1).$$

Therefore,

$$\boxed{a(bv) = (ab)v}. \quad (\text{S4})$$

### S5 – Scalar identity

$$1(x, y) = (x, y + 1 - 1) = (x, y). \quad (\text{S5})$$

Thus,

$$1v = v.$$

### Verification

All ten vector-space properties have been checked under the specified operations:

$$\boxed{\text{A1–A5 and S1–S5 all hold in } V}.$$

In particular, the zero vector and additive inverse satisfy their required identities.

### Final answer

$$\boxed{V \text{ is a vector space over } \mathbb{R}}$$

with

$$\boxed{\mathbf{0} = (0, -1)}.$$

Also,

$$\boxed{-(x, y) = (-x, -y - 2)}.$$

#### Key Learning Point.

The vector-space axioms depend on the structure of the operations rather than on their familiar appearance. The zero vector and additive inverse must therefore be determined from the operations actually defined.

## Question 2 – Nicholson Exercise 6.1.2(g)

### Question.

Determine whether the set  $V$  of all  $2 \times 2$  matrices with zero determinant, using the usual matrix operations, is a vector space. If not, give a reason.

#### Concepts / properties applied:

Closure under addition; determinant; counterexample.

### Method.

To disprove that  $V$  is a vector space, it is sufficient to find two matrices in  $V$  whose sum is not in  $V$ .

### Step-by-step working

Choose

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\det(A) = 1(0) - 0(0) = 0,$$

and

$$\det(B) = 0(1) - 0(0) = 0.$$

Therefore,

$$A, B \in V. \quad (\text{Given condition})$$

Now add the two matrices:

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

But

$$\det(A + B) = \det(I_2) = 1 \neq 0. \quad (\text{Determinant})$$

Therefore,

$$A + B \notin V.$$

Hence  $V$  is not closed under addition.

### Verification

We have explicitly found

$$A, B \in V$$

but

$$A + B \notin V.$$

Therefore the closure requirement fails.

### Final answer

The set of  $2 \times 2$  matrices with determinant 0 is not a vector space.

More specifically,

it is not closed under addition.

#### Key Learning Point.

A subset of a vector space is not automatically a vector space. The zero-determinant condition is not preserved by addition, so one valid counterexample is enough to rule out the vector-space structure.

### Question 3 – Nicholson Exercise 6.1.15(a)

#### Question.

Prove that, for a vector  $v$  and scalar  $a$ ,

$$av = 0$$

implies

$$a = 0 \quad \text{or} \quad v = 0.$$

#### Concepts / properties applied:

Vector-space axioms; scalar inverse; scalar associativity; cancellation; zero-vector property.

#### Method.

Start from  $av = 0$ . The case  $a = 0$  is immediate. For  $a \neq 0$ , multiply by  $a^{-1}$  and simplify using the vector-space properties.

#### Step-by-step working

Suppose

$$av = 0.$$

If

$$a = 0,$$

then the required conclusion is immediate.

Now consider the remaining case,

$$a \neq 0.$$

Therefore  $a^{-1}$  exists. Multiplying both sides by  $a^{-1}$ ,

$$a^{-1}(av) = a^{-1}0. \quad (\text{Scalar multiplication})$$

Using scalar associativity,

$$(a^{-1}a)v = a^{-1}0. \quad (\text{S4})$$

Since

$$a^{-1}a = 1,$$

we obtain

$$v = a^{-1}0. \quad (\text{S5})$$

Now, because

$$0 = 0 + 0,$$

we have

$$\begin{aligned} a^{-1}0 &= a^{-1}(0 + 0) \\ &= a^{-1}0 + a^{-1}0. \end{aligned} \quad (\text{S2})$$

Adding the additive inverse of  $a^{-1}0$  to both sides gives

$$a^{-1}0 = 0. \quad (\text{A5, cancellation})$$

Hence,

$$v = 0.$$

Thus,

$$a \neq 0 \implies v = 0.$$

Combining this with the case  $a = 0$ ,

$$\boxed{av = 0 \implies a = 0 \text{ or } v = 0}.$$

### Verification

The two possible cases are

$$a = 0,$$

or

$$a \neq 0 \implies v = 0.$$

Therefore the required implication holds for every scalar  $a$ .

### Final answer

$$\boxed{av = 0 \implies a = 0 \text{ or } v = 0}$$

#### Key Learning Point.

This property follows from the vector-space axioms rather than from the usual coordinate interpretation of vectors. The non-zero scalar case is resolved by its multiplicative inverse, together with the scalar multiplication axioms.

# Question 1

The system of equations,

$$\begin{aligned} -3x - 2y &= 3 \\ x - y &= -6 \\ -2x - 2y &= -5 \end{aligned}$$

where  $x, y \in \mathbb{R}$ , has no solution.

Find the vector  $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$  that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{-3} \\ \boxed{4.111} \end{pmatrix} \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

---

**Score: 1/1** 

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).



# Question 1

The system of equations,

$$\begin{aligned} -6x + 4y &= 2 \\ 5x - y &= 4 \\ 4x - y &= 4 \end{aligned}$$

where  $x, y \in \mathbb{R}$ , has no solution.

Find the vector  $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$  that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{1.455} \\ \boxed{2.667} \end{pmatrix} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

---

**Score: 1/1** 

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

The system of equations,

$$\begin{aligned}x - 4y &= -3 \\ 6x - 5y &= -6 \\ x + 2y &= 5\end{aligned}$$

where  $x, y \in \mathbb{R}$ , has no solution.

Find the vector  $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$  that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \boxed{0.195} \\ \boxed{1.294} \end{pmatrix} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Your answer is correct. You were awarded **1** mark. 

You scored **1** mark for this part.

---

**Score: 1/1** 

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

Find the least squares line of best fit  $y = a + bx$  for the data points:

$$\{(-4, -6), (-6, 0), (4, 4), (-5, -1)\}$$

$a =$   ✓

$b =$   ✓

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

**Gap 0**

Your answer is correct. You were awarded **1** mark. ✓

**Gap 1**

Your answer is correct. You were awarded **1** mark. ✓

You scored **2** marks for this part.

---

**Score: 2/2** ✓

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

Find the least squares line of best fit  $y = a + bx$  for the data points:

$$\{(5, 2), (-5, -2), (-4, 3), (-2, 3)\}$$

$$a = \boxed{1.77} \quad \checkmark$$

$$b = \boxed{0.18} \quad \checkmark$$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

**Gap 0**

Your answer is correct. You were awarded **1** mark. 

**Gap 1**

Your answer is correct. You were awarded **1** mark. 

You scored **2** marks for this part.

---

**Score: 2/2** 

---

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University  
(<http://www.newcastle.ac.uk>).

# Question 1

Find a basis and calculate the dimension for the following subspace of  $2 \times 2$  matrices (with standard addition and multiplication by a scalar operations)

$$\text{span} \left\{ \begin{pmatrix} 3 & 4 \\ 6 & 7 \end{pmatrix}, \begin{pmatrix} 2 & 3 \\ 2 & 2 \end{pmatrix}, \begin{pmatrix} -3 & -6 \\ 0 & -6 \end{pmatrix}, \begin{pmatrix} 3 & 9 \\ -7 & 10 \end{pmatrix} \right\}$$

Enter up to 4 basis matrices. For any matrices not required in the basis, enter

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} \boxed{3} & \boxed{4} \\ \boxed{6} & \boxed{7} \end{pmatrix}, \begin{pmatrix} \boxed{2} & \boxed{3} \\ \boxed{2} & \boxed{2} \end{pmatrix}, \begin{pmatrix} \boxed{-3} & \boxed{-6} \\ \boxed{0} & \boxed{-6} \end{pmatrix}, \begin{pmatrix} \boxed{3} & \boxed{9} \\ \boxed{-7} & \boxed{10} \end{pmatrix}$$

What is the dimension of the subspace?

Correct. Your basis matrices are independent and span the entire subspace.   
You were awarded **5** marks.

You scored **5** marks for this part.

**Score: 5/5** 

# Question 1

The vector space spanned by the following set of polynomials is a subspace of polynomials of degree  $n \leq 5$ .

$$\{3x^3 - 2x^2 - 2x^4 + 2x^5, \quad 2x^2 - 3x^3 + 4x^4 + 2x^5, \quad 3x^2 - 6x^3 - 2x^4 - 1\}$$

Identify a basis of the subspace. Enter the coefficients in a matrix such that

Rows:  Columns:

$$\begin{pmatrix} \begin{matrix} -2 & 3 & -2 & 2 \end{matrix} \\ \begin{matrix} 2 & -3 & 4 & 2 \end{matrix} \\ \begin{matrix} 3 & -6 & -2 & -10 \end{matrix} \end{pmatrix} \begin{pmatrix} x^2 \\ x^3 \\ x^4 \\ x^5 \end{pmatrix} = \begin{pmatrix} \text{basis polynomials} \end{pmatrix}$$

(reduce the size of the matrix if the dimension is less than 4)

What is the dimension of the subspace?

Correct. Your basis polynomials are independent and span the entire subspace. ✓  
You were awarded **2** marks.

You scored **2** marks for this part.

**Score: 2/2** ✓

Consider the set of ordered pairs defined on the Cartesian product

$V = \{0, 1, 2\} \times \{0, 1, 2\}$ , i.e. pairs of the form  $\mathbf{x} = (x_1, x_2) \in V$  with  $x_1 \in \{0, 1, 2\}$  and  $x_2 \in \{0, 1, 2\}$ . We can also refer to the pairs as 2-dimensional vectors.

We define scalar multiplication of elements in  $V$  such that for a scalar  $a \in \{0, 1, 2\}$ , we have  $a \otimes \mathbf{x} = (a \otimes x_1, a \otimes x_2)$ . Furthermore, vector addition between elements in  $V$  is defined such that  $\mathbf{x} \oplus \mathbf{y} = (x_1 \oplus y_1, x_2 \oplus y_2)$ . The addition ( $\oplus$ ) and multiplication ( $\otimes$ ) operations for scalars are defined according to the following tables.

| $\oplus$ | 0 | 1 | 2 | $\otimes$ | 0 | 1 | 2 |
|----------|---|---|---|-----------|---|---|---|
| 0        | 0 | 1 | 2 | 0         | 0 | 0 | 0 |
| 1        | 1 | 2 | 0 | 1         | 0 | 1 | 2 |
| 2        | 2 | 0 | 1 | 2         | 0 | 2 | 1 |

The set of vectors in  $V$  along with  $\otimes$  and  $\oplus$  defines a vector space. The following will ask you to provide examples for each property (see p328 of the text). Note that the examples you'll provide don't **prove** that the properties are satisfied, but rather allow you to get an intuitive feel for how the properties work. The standard symbol '+' will be used to mean  $\oplus$  in each of the axioms.

a)

First, give three distinct examples for vectors in  $V$ . Do not include  $(0, 0)$ .

$$\mathbf{u} = \begin{pmatrix} \boxed{1} & \boxed{0} \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix}$$

$$\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$$



Proceed to the next part. You were awarded **3** marks. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

b)

### Axioms for Vector Addition

For each of the following, fill the gaps for the examples. Note that  $\mathbf{u}$ ,  $\mathbf{v}$ ,  $\mathbf{w}$  should correspond with the vectors you provided in part a (write these down on paper to save yourself scrolling up and down).

A1. If  $\mathbf{u}$  and  $\mathbf{v}$  are in  $V$ , then  $\mathbf{u} + \mathbf{v}$  is in  $V$ .

*Example:*  $\mathbf{u}$  and  $\mathbf{v}$  are in  $V$ , the vector  $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$  ✓ is also in  $V$ .

A2.  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$  for all  $\mathbf{u}, \mathbf{v}$  in  $V$ .

*Example:*  $\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$  ✓,  $\mathbf{v} + \mathbf{u} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix}$  ✓

A3.  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$  for all  $\mathbf{u}, \mathbf{v}$  and  $\mathbf{w}$  in  $V$ .

*Example:*  $\mathbf{v} + \mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix}$  ✓,  $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = \begin{pmatrix} \boxed{2} & \boxed{2} \end{pmatrix}$  ✓

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix} \checkmark, \quad (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \begin{pmatrix} \boxed{2} & \boxed{2} \end{pmatrix} \checkmark$$

A4. An element  $\mathbf{0}$  in  $V$  exists such that  $\mathbf{v} + \mathbf{0} = \mathbf{v} = \mathbf{0} + \mathbf{v}$  for every  $\mathbf{v}$  in  $V$ .

*Example:* The zero element is  $\mathbf{0} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$ .

A5. For each  $\mathbf{v}$  in  $V$ , an element  $-\mathbf{v}$  in  $V$  exists such that  $-\mathbf{v} + \mathbf{v} = \mathbf{0}$  and  $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$ .

*Example:*  $-\mathbf{u} = \begin{pmatrix} \boxed{2} & \boxed{0} \end{pmatrix} \checkmark$ , and  $-\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$ .

### Gap 0

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. 

### Gap 1

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. 

### Gap 2

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded 1 mark. 

### Gap 3

Your answer is correct. You were awarded 1 mark. 

**Gap 4**

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

**Gap 5**

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

**Gap 6**

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

**Gap 7**

Your answer is correct. You were awarded **1** mark. ✓

**Gap 8**

This part was marked using your answers to previous parts.

Your answer is correct. You were awarded **1** mark. ✓

**Gap 9**

Your answer is correct. You were awarded **1** mark. ✓

You scored **10** marks for this part.

---

**Score: 10/10** ✓

c)

Now enter 2 values,  $a$ ,  $b$ , with both taken from  $\{0, 1, 2\}$ . They do not have to be different but ensure that  $a \neq 0$ .

$a =$

$b =$

Proceed to the next part. You were awarded **2** marks. ✓

You scored **2** marks for this part.

d)

**Axioms for Scalar Multiplication**

As with the addition axioms, fill out the examples below using your values of  $a$  and  $b$ , as well as the same  $\mathbf{u}$ ,  $\mathbf{v}$ ,  $\mathbf{w}$  from part a. (write them all down to save you scrolling up)

Note that  $a + b$  is defined the same as for  $a \oplus b$  given in the table, and that in contrast to the axioms of page 328 of the text, here we assume  $a, b$  are in  $\{0, 1, 2\}$  rather than in  $\mathbb{R}$  (this structure is called a 'Field').

S1. If  $\mathbf{v}$  is in  $V$ , then  $a\mathbf{v}$  is in  $V$  for all  $a$  in  $\{0, 1, 2\}$ .

Example:  $a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$ , which is in  $V$ .

S2.  $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$  for all  $\mathbf{v}$  and  $\mathbf{w}$  in  $V$  and all  $a$  in  $\{0, 1, 2\}$ .

Example:  $\mathbf{v} + \mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix} \checkmark$ ,  $a(\mathbf{v} + \mathbf{w}) = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix} \checkmark$ .

$a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$ ,  $a\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{1} \end{pmatrix} \checkmark$ ,  $a\mathbf{v} + a\mathbf{w} = \begin{pmatrix} \boxed{1} & \boxed{2} \end{pmatrix} \checkmark$ .

S3.  $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$  for all  $\mathbf{v}$  in  $V$  and all  $a, b$  in  $\{0, 1, 2\}$ .

Example:  $(a + b) = \boxed{0} \checkmark$ ,  $(a + b)\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$ .

$a\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$ ,  $b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$   
 $, a\mathbf{v} + b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{0} \end{pmatrix} \checkmark$ .

S4.  $a(b\mathbf{v}) = (ab)\mathbf{v}$  for all  $\mathbf{v}$  in  $V$  and all  $a, b$  in  $\{0, 1, 2\}$ .

Example:  $b\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$ ,  $a(b\mathbf{v}) = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$ .

$ab = \boxed{2} \checkmark$ ,  $(ab)\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{2} \end{pmatrix} \checkmark$ .

S5.  $1\mathbf{v} = \mathbf{v}$  for all  $\mathbf{v}$  in  $V$ .

Example:  $1\mathbf{v} = \begin{pmatrix} \boxed{0} & \boxed{1} \end{pmatrix} \checkmark$ .

**Gap 0**

Your answer is correct. You were awarded 1 mark.

**Gap 1**

Your answer is correct. You were awarded 1 mark.

**Gap 2**

Your answer is correct. You were awarded 1 mark.

**Gap 3**

Your answer is correct. You were awarded 1 mark.

**Gap 4**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 5**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 6**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 7**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 8**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 9**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 10**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 11**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 12**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 13**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 14**

Your answer is correct. You were awarded 1 mark. ✓

**Gap 15**

Your answer is correct. You were awarded 1 mark. ✓

You scored **16** marks for this part.

---

**Score: 16/16** ✓

Created using Numbas (<https://www.numbas.org.uk>), developed by Newcastle University (<http://www.newcastle.ac.uk>).



The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 & 1 \\ 1 & 2 & -2 & 3 & 0 \\ 2 & 4 & 1 & 1 & -2 \\ 1 & 2 & -2 & 3 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$S = \{1 + x + 2x^2 + x^3, 2 + 2x + 4x^2 + 2x^3, 1 - 2x + x^2 - 2x^3\}$$

Decide whether or not  $S$  is linearly independent. If **not**, write the **second** polynomial as a linear combination of the others. If it *is* linearly independent, write 0 for both coefficients.

$$\underbrace{2 + 2x + 4x^2 + 2x^3}_{\text{polynomial 2}} = \boxed{2} \checkmark \underbrace{(1 + x + 2x^2 + x^3)}_{\text{polynomial 1}} + \boxed{0} \checkmark$$

$(1 - 2x + x^2 - 2x^3)$   
polynomial 3

Your coefficients are correct. You were awarded **2** marks. ✓

You scored **2** marks for this part.

**Score: 2/2** ✓

b)

(i) Find a subset of  $S$  that is a basis for  $W$ . (Select the polynomials to include in the subset)

☒  $1 + x + 2x^2 + x^3$     ☐  $2 + 2x + 4x^2 + 2x^3$

☒  $1 - 2x + x^2 - 2x^3$

(ii) Hence find the dimension of  $W$ :

Correct. Dimension is 2 and polynomial 3 along with either 1 or 2 can make a basis. You were awarded **3** marks. ✓

You scored **3** marks for this part.

Score: 3/3 ✓

c)

Find a basis for the **row space** of  $A$ . Enter your basis as a matrix comprised of your basis row vectors. You can increase the number of rows as required.

Rows:  Columns:

|   |   |   |    |   |
|---|---|---|----|---|
| 1 | 2 | 0 | 1  | 0 |
| 0 | 0 | 1 | -1 | 0 |
| 0 | 0 | 0 | 0  | 1 |

Basis given for the row space of  $A$  is correct. You were awarded **1** mark. ✓

You scored **1** mark for this part.

Score: 1/1 ✓

# Question 1

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 3 & -1 & 1 & 1 & 3 \\ 2 & 1 & -2 & -1 & 7 \\ 2 & -2 & -3 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$$

Select a subset of matrices from the following which can be used to give a basis that spans  $\mathbf{M}_{2,2}(\mathbb{R})$ ?

- ☒  $\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$ 
☒  $\begin{pmatrix} 1 & -1 \\ 1 & -2 \end{pmatrix}$ 
☒  $\begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$ 
☒  $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$
- ☐  $\begin{pmatrix} 1 & 3 \\ 7 & 0 \end{pmatrix}$ 
☐ Not possible

Correct. The matrices selected will form a basis that spans the set of real 2x2 matrices. You were awarded **4** marks. ✓

You scored **4** marks for this part.

**Score: 4/4** ✓

# SIT292 Module 7 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

| Question Number            | Score               |
|----------------------------|---------------------|
| <a href="#">Question 1</a> | 1/1                 |
| <a href="#">Question 2</a> | 1/1                 |
| <a href="#">Question 3</a> | 2/2                 |
| <a href="#">Question 4</a> | 2/2                 |
| <a href="#">Question 5</a> | 3/3                 |
| <a href="#">Question 6</a> | 1/1                 |
| <b>Total</b>               | <b>10/10 (100%)</b> |

**Congratulations! You have achieved the minimum threshold for this module's self-assessment.**

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

## Performance Summary

|                    |                                 |
|--------------------|---------------------------------|
| <b>Exam Name:</b>  | SIT292 Module 7 Self-Assessment |
| <b>Session ID:</b> | 1042648011765624                |
| <b>Exam Start:</b> | Fri Sep 11 2026 21:33:26        |
| <b>Exam Stop:</b>  | Fri Sep 11 2026 22:06:41        |
| <b>Time Spent:</b> | 00:33:15                        |

# Question 1

The system of equations,

$$\begin{aligned} -2x - 4y &= -5 \\ -6x + 5y &= -2 \\ -2x + y &= 3 \end{aligned}$$

where  $x, y \in \mathbb{R}$ , has no solution.

Find the vector  $\mathbf{z} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$  that best approximates a solution.

$$\mathbf{z} = \begin{pmatrix} \underline{0.774} \\ \underline{0.752} \end{pmatrix} \quad \checkmark$$

Expected answer:  $\begin{pmatrix} \underline{41/53} \\ \underline{239/318} \end{pmatrix}$

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

✓ Your answer is correct. You were awarded 1 mark.

Score: 1/1 ✓

## Question 2

Find the least squares line of best fit  $y = a + bx$  for the data points:

$$\{(6, 4), (6, -2), (4, 5), (-4, -6)\}$$

$$a = \underline{-2.088} \quad \checkmark$$

Expected answer: -2.0882352941

$$b = \underline{0.779} \quad \checkmark$$

Expected answer: 0.7794117647

(answers within 0.0005 for each value will be marked as correct, i.e., correct to 3 decimal places)

Gap 0

Score: 1/1

- Your answer is correct. You were awarded **0.5** marks.

Gap 1

- Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

## Question 3

Find a basis and calculate the dimension for the following subspace of  $2 \times 2$  matrices (with standard addition and multiplication by a scalar operations)

$$\text{span} \left\{ \begin{pmatrix} -3 & 9 \\ -3 & -13 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ -2 & 0 \end{pmatrix}, \begin{pmatrix} -2 & 5 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} -3 & 4 \\ 10 & 4 \end{pmatrix} \right\}$$

Enter up to 4 basis matrices. For any matrices not required in the basis, enter

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} \frac{-3}{-3} & \frac{9}{-13} \\ \frac{-3}{-3} & \frac{9}{-13} \end{pmatrix}$$

Expected answer:  $\begin{pmatrix} \frac{1}{-2} & \frac{-2}{0} \\ \frac{1}{-2} & \frac{-2}{0} \end{pmatrix},$

$$\begin{pmatrix} \frac{1}{-2} & \frac{-2}{0} \\ \frac{1}{-2} & \frac{-2}{0} \end{pmatrix}$$

Expected answer:  $\begin{pmatrix} \frac{0}{-2} & \frac{1}{-2} \\ \frac{0}{-2} & \frac{1}{-2} \end{pmatrix},$

$$\begin{pmatrix} \frac{-2}{3} & \frac{5}{0} \\ \frac{-2}{3} & \frac{5}{0} \end{pmatrix}$$

Expected answer:  $\begin{pmatrix} \frac{0}{1} & \frac{0}{2} \\ \frac{0}{1} & \frac{0}{2} \end{pmatrix},$

$$\begin{pmatrix} \frac{-3}{10} & \frac{4}{4} \\ \frac{-3}{10} & \frac{4}{4} \end{pmatrix}$$

Expected answer:  $\begin{pmatrix} \frac{0}{0} & \frac{0}{1} \\ \frac{0}{0} & \frac{0}{1} \end{pmatrix}$

What is the dimension of the subspace?

4

Expected answer: 4

✓ Correct. Your basis matrices are independent and span the entire subspace. You were awarded 2 marks.

Score: 2/2 ✓



## Question 4

The vector space spanned by the following set of polynomials is a subspace of polynomials of degree  $n \leq 3$ .

$$\{3 + 6x + 2x^2 - 10x^3, \quad -2 - 4x - 2x^2 + 10x^3, \quad -3 - 6x - 3x^2 + 9x^3, \quad -2 -$$

Identify a basis of the subspace. Enter the coefficients as rows of a matrix such that

Rows: 3 Columns: 4

$$\begin{pmatrix} \underline{3} & \underline{6} & \underline{2} & \underline{-10} \\ \underline{-3} & \underline{-6} & \underline{-3} & \underline{9} \\ \underline{-2} & \underline{-4} & \underline{0} & \underline{4} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{1} & \underline{2} & \underline{0} & \underline{-2} \\ \underline{0} & \underline{0} & \underline{1} & \underline{-2} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ x \\ x^2 \\ x^3 \end{pmatrix} = \left( \text{basis polynomials} \right)$$

(reduce the size of the matrix if the dimension is less than 4)

What is the dimension of the subspace?

3

Expected answer: 3

✓ Correct. Your basis polynomials are independent and span the entire subspace. You were awarded 2 marks.

Score: 2/2 ✓

## Question 5

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & -2 & -2 & 1 & 1 \\ 2 & -4 & -4 & -2 & 1 \\ 2 & -4 & -4 & 2 & 0 \\ 2 & -4 & -4 & -2 & -3 \end{pmatrix} \quad B = \begin{pmatrix} 1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Let  $W$  be a subspace of  $\mathbf{P}_3(\mathbb{R})$  spanned by the set

$$S = \{1 + 2x + 2x^2 + 2x^3, -2 - 4x - 4x^2 - 4x^3, -2 - 4x - 4x^2 - 4x^3\}$$

a)

Decide whether or not  $S$  is linearly independent. If **not**, write the **second** polynomial as a linear combination of the others. If it *is* linearly independent, write 0 for both coefficients.

$$-2 - 4x - 4x^2 - 4x^3 = \underline{-2} \quad \checkmark$$

polynomial 2

Expected answer: -2

$$(1 + 2x + 2x^2 + 2x^3) + \underline{0} \quad \checkmark$$

polynomial 1

Expected answer: 0

$$(-2 - 4x - 4x^2 - 4x^3)$$

polynomial 3

## ▼ Feedback

Score: 1/1 ✓

- ✓ Your coefficients are correct. You were awarded 1 mark.

b)

(i) Find a subset of  $S$  that is a basis for  $W$ . (Select the polynomials to include in the subset)

☒  $1 + 2x + 2x^2 + 2x^3$     ☐  $-2 - 4x - 4x^2 - 4x^3$

☐  $-2 - 4x - 4x^2 - 4x^3$

Expected answer:

☒  $1 + 2x + 2x^2 + 2x^3$     ☐  $-2 - 4x - 4x^2 - 4x^3$

☐  $-2 - 4x - 4x^2 - 4x^3$

(ii) Hence find the dimension of  $W$ : 1

Expected answer: 1

## ▼ Feedback

Score: 1/1 ✓

- ✓ Correct. Dimension is 1 and so any of the polynomials in  $S$  can be used as a basis. You were awarded 1 mark.

c)

Find a basis for the **row space** of  $A$ . Enter your basis as a matrix comprised of your basis row vectors. You can increase the number of rows as required.

Rows: 3 Columns: 5

$$\begin{pmatrix} \underline{1} & \underline{-2} & \underline{-2} & \underline{0} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

Expected answer:

$$\begin{pmatrix} \underline{1} & \underline{-2} & \underline{-2} & \underline{0} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} \end{pmatrix}$$

▼ Feedback

Score: 1/1 ✓

- ✓ Basis given for the row space of A is correct. You were awarded 1 mark.

## Question 6

The following two matrices are related by a sequence of elementary row operations:

$$A = \begin{pmatrix} 1 & -1 & -2 & 1 & 1 \\ 2 & -2 & -4 & -2 & 1 \\ 2 & -2 & -4 & 1 & -1 \\ 1 & -1 & -2 & -1 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Select a subset of matrices from the following which can be used to give a basis that spans  $\mathbf{M}_{2,2}(\mathbb{R})$ ?

- ☐  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ 
☐  $\begin{pmatrix} -1 & -2 \\ -2 & -1 \end{pmatrix}$ 
☐  $\begin{pmatrix} -2 & -4 \\ -4 & -2 \end{pmatrix}$ 
☐  $\begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$
- ☐  $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ 
☒ Not possible

Expected answer:

- ☐  $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ 
☐  $\begin{pmatrix} -1 & -2 \\ -2 & -1 \end{pmatrix}$ 
☐  $\begin{pmatrix} -2 & -4 \\ -4 & -2 \end{pmatrix}$ 
☐  $\begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$
- ☐  $\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ 
☒ Not possible

✓ Correct. It is not possible to form a basis that spans the set of real 2x2 matrices with the elements given. You were awarded 1 mark.

Score: 1/1 ✓

# Module 7 Self-Assessment

## Working & Annotations

*Detailed mathematical working supporting the completed Numbas self-assessment questions*

### Question 1

#### Question.

The system

$$\begin{aligned} -2x - 4y &= -5, \\ -6x + 5y &= -2, \\ -2x + y &= 3 \end{aligned}$$

has no solution. Find the vector

$$z = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}$$

that best approximates a solution.

#### Concepts / properties applied:

Matrix representation of an inconsistent system; transpose; normal equations; Theorem 5.6.1; Gaussian elimination.

#### Method.

Write the system as  $Az = b$ , form

$$(A^T A)z = A^T b,$$

and solve the resulting  $2 \times 2$  system.

#### Step-by-step working

The coefficient matrix and right-hand side are

$$A = \begin{bmatrix} -2 & -4 \\ -6 & 5 \\ -2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} -5 \\ -2 \\ 3 \end{bmatrix}.$$

Let

$$z = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix}.$$

The normal equations are

$$(A^T A)z = A^T b. \quad (\text{Theorem 5.6.1})$$

First,

$$A^T = \begin{bmatrix} -2 & -6 & -2 \\ -4 & 5 & 1 \end{bmatrix}.$$

Hence,

$$\begin{aligned}
 A^T A &= \begin{bmatrix} -2 & -6 & -2 \\ -4 & 5 & 1 \end{bmatrix} \begin{bmatrix} -2 & -4 \\ -6 & 5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 4 + 36 + 4 & 8 - 30 - 2 \\ 8 - 30 - 2 & 16 + 25 + 1 \end{bmatrix} \\
 &= \begin{bmatrix} 44 & -24 \\ -24 & 42 \end{bmatrix}. \quad (\text{Matrix multiplication})
 \end{aligned}$$

Also,

$$A^T b = \begin{bmatrix} -2 & -6 & -2 \\ -4 & 5 & 1 \end{bmatrix} \begin{bmatrix} -5 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 + 12 - 6 \\ 20 - 10 + 3 \end{bmatrix} = \begin{bmatrix} 16 \\ 13 \end{bmatrix}. \quad (\text{Matrix multiplication})$$

Therefore,

$$\begin{bmatrix} 44 & -24 \\ -24 & 42 \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} 16 \\ 13 \end{bmatrix},$$

so

$$44x_0 - 24y_0 = 16,$$

$$-24x_0 + 42y_0 = 13.$$

Divide the first equation by 4:

$$11x_0 - 6y_0 = 4.$$

Divide the second equation by 3:

$$-8x_0 + 14y_0 = \frac{13}{3}.$$

Multiply the first simplified equation by 7:

$$77x_0 - 42y_0 = 28.$$

Multiply the second simplified equation by 3:

$$-24x_0 + 42y_0 = 13.$$

Adding,

$$53x_0 = 41,$$

hence

$$\boxed{x_0 = \frac{41}{53}}.$$

Substituting into

$$11x_0 - 6y_0 = 4,$$



$$11 \left( \frac{41}{53} \right) - 6y_0 = 4,$$

so

$$6y_0 = \frac{451}{53} - \frac{212}{53} = \frac{239}{53}.$$

Therefore,

$$y_0 = \frac{239}{318}.$$

### Verification

Numerically,

$$x_0 = \frac{41}{53} \approx 0.773585, \quad y_0 = \frac{239}{318} \approx 0.751572.$$

Thus

$$z \approx \begin{bmatrix} 0.774 \\ 0.752 \end{bmatrix}.$$

These are the values entered in Numbas and were accepted as correct.

### Final answer

$$z = \begin{bmatrix} \frac{41}{53} \\ \frac{239}{318} \end{bmatrix} \approx \begin{bmatrix} 0.774 \\ 0.752 \end{bmatrix}$$

#### Key Learning Point.

For an inconsistent system, the normal equations convert the original problem into a consistent system whose solution gives the least-squares best approximation.

## Question 2

### Question.

Find the least-squares line of best fit

$$y = a + bx$$

for the data points

$$\{(6, 4), (6, -2), (4, 5), (-4, -6)\}.$$

### Concepts / properties applied:

Least-squares line; Theorem 5.6.2; normal equations; matrix multiplication; Gaussian elimination.

### Method.

Write the line as

$$Mz = y, \quad z = \begin{bmatrix} a \\ b \end{bmatrix},$$

then solve

$$(M^T M)z = M^T y.$$

### Step-by-step working

The data give

$$M = \begin{bmatrix} 1 & 6 \\ 1 & 6 \\ 1 & 4 \\ 1 & -4 \end{bmatrix}, \quad y = \begin{bmatrix} 4 \\ -2 \\ 5 \\ -6 \end{bmatrix}.$$

Let

$$z = \begin{bmatrix} a \\ b \end{bmatrix}.$$

The normal equations are

$$(M^T M)z = M^T y. \quad (\text{Theorem 5.6.2})$$

First,

$$M^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6 & 6 & 4 & -4 \end{bmatrix}.$$

Hence,

$$\begin{aligned} M^T M &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6 & 6 & 4 & -4 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 1 & 6 \\ 1 & 4 \\ 1 & -4 \end{bmatrix} = \begin{bmatrix} 4 & 12 \\ 12 & 36 + 36 + 16 + 16 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 12 \\ 12 & 104 \end{bmatrix}. \quad (\text{Matrix multiplication}) \end{aligned}$$

Also,

$$\begin{aligned} M^T y &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 6 & 6 & 4 & -4 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 5 \\ -6 \end{bmatrix} = \begin{bmatrix} 4 - 2 + 5 - 6 \\ 24 - 12 + 20 + 24 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 56 \end{bmatrix}. \quad (\text{Matrix multiplication}) \end{aligned}$$

Therefore,

$$\begin{bmatrix} 4 & 12 \\ 12 & 104 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 56 \end{bmatrix}.$$

Hence,

$$4a + 12b = 1,$$

$$12a + 104b = 56.$$

Divide the first equation by 4:

$$a + 3b = \frac{1}{4}.$$

Thus,

$$a = \frac{1}{4} - 3b.$$

Substitute into the second equation:

$$12 \left( \frac{1}{4} - 3b \right) + 104b = 56.$$

Therefore,

$$3 - 36b + 104b = 56,$$

so

$$68b = 53.$$

Hence,

$$\boxed{b = \frac{53}{68}}.$$

Now,

$$a = \frac{1}{4} - 3 \left( \frac{53}{68} \right) = \frac{17}{68} - \frac{159}{68} = -\frac{142}{68} = -\frac{71}{34}.$$

Thus,

$$a = -\frac{71}{34}.$$

**Verification**

Numerically,

$$a = -\frac{71}{34} \approx -2.088235,$$

$$b = \frac{53}{68} \approx 0.779412.$$

Therefore the fitted line is

$$y \approx -2.088 + 0.779x.$$

The values  $a = -2.088$  and  $b = 0.779$  were accepted by Numbas.

**Final answer**

$$a = -\frac{71}{34} \approx -2.088, \quad b = \frac{53}{68} \approx 0.779$$

**Key Learning Point.**

A least-squares line is obtained by treating the intercept and slope as unknown coefficients and solving the corresponding normal equations.

### Question 3

#### Question.

Find a basis and calculate the dimension for the following subspace of  $2 \times 2$  matrices:

$$\text{span} \left\{ \begin{bmatrix} -3 & 9 \\ -3 & -13 \end{bmatrix}, \begin{bmatrix} 1 & -2 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 5 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} -3 & 4 \\ 10 & 4 \end{bmatrix} \right\}.$$

#### Concepts / properties applied:

Matrix-vector correspondence; linear independence; determinant; basis; dimension.

#### Method.

Represent each  $2 \times 2$  matrix by a vector in  $\mathbb{R}^4$ , then test the four resulting vectors for linear independence.

#### Step-by-step working

Let

$$M_1 = \begin{bmatrix} -3 & 9 \\ -3 & -13 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 1 & -2 \\ -2 & 0 \end{bmatrix},$$

$$M_3 = \begin{bmatrix} -2 & 5 \\ 3 & 0 \end{bmatrix}, \quad M_4 = \begin{bmatrix} -3 & 4 \\ 10 & 4 \end{bmatrix}.$$

Represent these matrices by their entries in the order

$$(a_{11}, a_{12}, a_{21}, a_{22}).$$

Thus,

$$v_1 = \begin{bmatrix} -3 \\ 9 \\ -3 \\ -13 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -2 \\ -2 \\ 0 \end{bmatrix},$$

$$v_3 = \begin{bmatrix} -2 \\ 5 \\ 3 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} -3 \\ 4 \\ 10 \\ 4 \end{bmatrix}.$$

Place these vectors as columns:

$$C = \begin{bmatrix} -3 & 1 & -2 & -3 \\ 9 & -2 & 5 & 4 \\ -3 & -2 & 3 & 10 \\ -13 & 0 & 0 & 4 \end{bmatrix}.$$

To test linear independence, calculate

$$\det(C) = 2 \neq 0. \quad (\text{Determinant})$$

Therefore,

$$\boxed{\text{rank}(C) = 4}.$$

Hence the four matrices are linearly independent.

Since there are four independent matrices in a 4-dimensional matrix space,

$$\boxed{\{M_1, M_2, M_3, M_4\}}$$

is a basis for the subspace.

### Verification

The determinant is non-zero, so no non-trivial linear combination of the four matrices can produce the zero matrix.

Thus all four original matrices are needed.

### Final answer

$$\mathcal{B} = \left\{ \begin{bmatrix} -3 & 9 \\ -3 & -13 \end{bmatrix}, \begin{bmatrix} 1 & -2 \\ -2 & 0 \end{bmatrix}, \begin{bmatrix} -2 & 5 \\ 3 & 0 \end{bmatrix}, \begin{bmatrix} -3 & 4 \\ 10 & 4 \end{bmatrix} \right\}$$

and

$$\boxed{\dim = 4}.$$

#### Key Learning Point.

A set of matrices can be tested for linear independence by converting each matrix into its coefficient vector. A non-zero determinant then immediately confirms that all four matrices form a basis.

## Question 4

### Question.

The vector space spanned by

$$\{3 + 6x + 2x^2 - 10x^3, -2 - 4x - 2x^2 + 10x^3, -3 - 6x - 3x^2 + 9x^3, -2 - 4x + 4x^3\}$$

is a subspace of polynomials of degree  $n \leq 3$ .

Identify a basis of the subspace and determine its dimension.

#### Concepts / properties applied:

Coefficient-vector representation; linear dependence; row reduction; pivot columns; basis; dimension.

### Method.

Represent the polynomials by their coefficient vectors in the order  $(1, x, x^2, x^3)$ , row-reduce the coefficient matrix, and use the pivot columns to select a basis from the original polynomials.

### Step-by-step working

Let

$$p_1 = 3 + 6x + 2x^2 - 10x^3,$$

$$p_2 = -2 - 4x - 2x^2 + 10x^3,$$

$$p_3 = -3 - 6x - 3x^2 + 9x^3,$$

$$p_4 = -2 - 4x + 4x^3.$$

Their coefficient vectors are

$$v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \\ -10 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -2 \\ -4 \\ -2 \\ 10 \end{bmatrix},$$

$$v_3 = \begin{bmatrix} -3 \\ -6 \\ -3 \\ 9 \end{bmatrix}, \quad v_4 = \begin{bmatrix} -2 \\ -4 \\ 0 \\ 4 \end{bmatrix}.$$

Form the coefficient matrix using these vectors as columns:

$$C = \begin{bmatrix} 3 & -2 & -3 & -2 \\ 6 & -4 & -6 & -4 \\ 2 & -2 & -3 & 0 \\ -10 & 10 & 9 & 4 \end{bmatrix}.$$

Row-reducing,



$$C \sim \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -\frac{2}{3} \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (\text{Gaussian elimination})$$

Therefore,

$$\boxed{\text{rank}(C) = 3}.$$

The pivot columns are columns 1, 2, 3, so the corresponding original polynomials  $p_1, p_2, p_3$  are linearly independent and form a basis.

Equivalently, the fourth polynomial is dependent on these three. From the RREF,

$$v_4 = -2v_1 - v_2 - \frac{2}{3}v_3. \quad (\text{Pivot/free-column relation})$$

Hence,

$$p_4 = -2p_1 - p_2 - \frac{2}{3}p_3.$$

### Verification

Substituting the three selected basis polynomials,

$$\begin{aligned} -2p_1 - p_2 - \frac{2}{3}p_3 &= -2(3 + 6x + 2x^2 - 10x^3) \\ &\quad - (-2 - 4x - 2x^2 + 10x^3) \\ &\quad - \frac{2}{3}(-3 - 6x - 3x^2 + 9x^3) \\ &= -2 - 4x + 4x^3 \\ &= p_4. \end{aligned}$$

Thus  $p_4$  is redundant, while  $p_1, p_2, p_3$  are independent.

### Final answer

A valid basis is

$$\boxed{\mathcal{B} = \{3 + 6x + 2x^2 - 10x^3, -2 - 4x - 2x^2 + 10x^3, -3 - 6x - 3x^2 + 9x^3\}}$$

with coefficient matrix

$$\boxed{\begin{bmatrix} 3 & 6 & 2 & -10 \\ -2 & -4 & -2 & 10 \\ -3 & -6 & -3 & 9 \end{bmatrix}}$$

and

$$\boxed{\dim = 3}.$$

**Key Learning Point.**

Polynomial basis questions can be handled exactly like ordinary vector basis questions by working with coefficient vectors. The rank identifies the dimension, while the pivot columns identify an independent subset of the original polynomials.

## Question 5

### Question.

The matrices

$$A = \begin{bmatrix} 1 & -2 & -2 & 1 & 1 \\ 2 & -4 & -4 & -2 & 1 \\ 2 & -4 & -4 & 2 & 0 \\ 2 & -4 & -4 & -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

are related by elementary row operations.

Let

$$W = \text{span} \{1 + 2x + 2x^2 + 2x^3, -2 - 4x - 4x^2 - 4x^3, -2 - 4x - 4x^2 - 4x^3\}.$$

(a)

Determine whether the set is linearly independent. If not, write the second polynomial as a linear combination of the others.

(b)

Find a subset forming a basis for  $W$ , and hence determine  $\dim W$ .

(c)

Find a basis for the row space of  $A$ .

#### Concepts / properties applied:

Linear dependence; scalar multiples; basis; dimension; elementary row operations; row-space invariance.

### Part (a) – Linear dependence

Let

$$p_1 = 1 + 2x + 2x^2 + 2x^3,$$

$$p_2 = -2 - 4x - 4x^2 - 4x^3,$$

$$p_3 = -2 - 4x - 4x^2 - 4x^3.$$

Compare  $p_2$  with  $p_1$ :

$$-2(1 + 2x + 2x^2 + 2x^3) = -2 - 4x - 4x^2 - 4x^3.$$

Therefore,

$$\boxed{p_2 = -2p_1}. \quad (\text{Scalar multiple})$$

Since  $p_3$  is identical to  $p_2$ ,

$$p_3 = p_2 = -2p_1.$$

Hence the set is linearly dependent.

The question specifically asks for the second polynomial as a linear combination of the first and third:

$$p_2 = -2p_1 + 0p_3.$$

Therefore the two required coefficients are

$$-2, \quad 0.$$

### Part (b) – Basis and dimension

Since

$$p_2 = -2p_1, \quad p_3 = -2p_1,$$

both  $p_2$  and  $p_3$  are redundant.

Therefore,

$$\mathcal{B}_W = \{p_1\}$$

is a basis for  $W$ .

Thus,

$$\dim W = 1.$$

### Part (c) – Basis for the row space

Since  $A$  and  $B$  are related by elementary row operations,

$$\text{row}(A) = \text{row}(B). \quad (\text{Elementary row-operation invariance})$$

The non-zero rows of  $B$  are

$$(1, -2, -2, 0, 0),$$

$$(0, 0, 0, 1, 0),$$

$$(0, 0, 0, 0, 1).$$

These therefore form a basis for the row space of  $A$ .

### Verification

There are three non-zero rows in the echelon-form matrix  $B$ , and each contains a leading entry in a different column. Hence the three rows are linearly independent.

Therefore,

$$\mathcal{B}_{\text{row}(A)} = \{(1, -2, -2, 0, 0), (0, 0, 0, 1, 0), (0, 0, 0, 0, 1)\}.$$

**Final answer**

For part (a),

$$\begin{bmatrix} -2, & 0 \end{bmatrix}.$$

For part (b),

$$\mathcal{B}_W = \{1 + 2x + 2x^2 + 2x^3\},$$

$$\dim W = 1.$$

For part (c),

$$\mathcal{B}_{\text{row}(A)} = \{(1, -2, -2, 0, 0), (0, 0, 0, 1, 0), (0, 0, 0, 0, 1)\}.$$

Equivalently, the row-basis matrix is

$$\begin{bmatrix} 1 & -2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

**Key Learning Point.**

A basis retains only the independent directions in a spanning set. Here the polynomial space has dimension one because all three given polynomials lie on the same scalar direction, while the non-zero rows of a row-equivalent matrix provide a basis for the original row space.

## Question 6

### Question.

The matrices

$$A = \begin{bmatrix} 1 & -1 & -2 & 1 & 1 \\ 2 & -2 & -4 & -2 & 1 \\ 2 & -2 & -4 & 1 & -1 \\ 1 & -1 & -2 & -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

are related by elementary row operations.

Determine whether a subset of the following matrices can form a basis for  $M_{2,2}(\mathbb{R})$ :

$$M_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad M_2 = \begin{bmatrix} -1 & -2 \\ -2 & -1 \end{bmatrix},$$

$$M_3 = \begin{bmatrix} -2 & -4 \\ -4 & -2 \end{bmatrix}, \quad M_4 = \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix},$$

$$M_5 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

#### Concepts / properties applied:

Dimension of  $M_{2,2}(\mathbb{R})$ ; scalar multiples; linear dependence; span; basis.

### Method.

Since  $M_{2,2}(\mathbb{R})$  has dimension 4, a spanning basis must contain four linearly independent matrices. Check whether the given matrices can supply four independent directions.

### Step-by-step working

First observe that

$$M_2 = \begin{bmatrix} -1 & -2 \\ -2 & -1 \end{bmatrix} = - \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = -M_1. \quad (\text{Scalar multiple})$$

Also,

$$M_3 = \begin{bmatrix} -2 & -4 \\ -4 & -2 \end{bmatrix} = -2 \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = -2M_1. \quad (\text{Scalar multiple})$$

Thus,

$$\text{span}\{M_1, M_2, M_3\} = \text{span}\{M_1\}.$$

So the first three matrices provide only one independent direction.

The remaining matrices  $M_4$  and  $M_5$  can contribute at most two additional independent directions.

Therefore,

$$\dim \text{span}\{M_1, M_2, M_3, M_4, M_5\} \leq 1 + 2 = 3.$$

But

$$\dim M_{2,2}(\mathbb{R}) = 4. \quad (\text{Dimension of matrix space})$$

Hence,

$$\text{span}\{M_1, M_2, M_3, M_4, M_5\} \neq M_{2,2}(\mathbb{R}).$$

Therefore no subset of these matrices can form a basis spanning the whole space.

### Verification

A basis for  $M_{2,2}(\mathbb{R})$  must contain four linearly independent matrices. However,

$$M_2 = -M_1, \quad M_3 = -2M_1,$$

so three of the five candidates collapse to a single independent direction.

Even if both  $M_4$  and  $M_5$  are independent of  $M_1$  and of one another, the largest possible independent subset has size only 3.

Thus four independent matrices cannot be obtained from the supplied candidates.

### Final answer

Not possible

#### Key Learning Point.

To span a 4-dimensional matrix space, four independent directions are essential. Detecting scalar-multiple matrices early can reveal that a proposed spanning set is too small in dimension, making a basis impossible.